

Learning Axioms and Axiom Schemas from Their Instances

Claude (Anthropic)*

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1 Introduction

2 Setting

This section fixes the data format, templates and their classes, matching, the learning protocol with the cautious verifier and anchors, refutation oracles, and the running examples. The learning framework is that of the report Claude (Anthropic) (2026), called *the parent report* below; we restate what we use from it. Technical definitions and longer proofs are in Appendix A.

2.1 Data in closure-normal form

We use two first-order languages: arithmetic $L_A = \{0, S, +, \cdot, =, <\}$ and set theory $L_\in = \{\in, =\}$, with connectives $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$ and quantifiers $\forall, \exists$; bounded quantifiers such as $\forall y < t \varphi$ and $\exists y \in t \varphi$ are abbreviations. The general results hold for every first-order signature; where a result needs a richness assumption it says so.

Closure-normal form. Formulas may contain *parameters* $a, b, c, p, q, w_0, w_1, \dots$, an infinite supply of extra constants. A *datum* is a formula whose free variables are parameters, read under its universal closure (*closure-normal form*). This removes the outer prefix of variable length that schema instances with parameters carry: a Separation instance $\forall \bar{w} \forall z \exists y \forall x (x \in y \leftrightarrow x \in z \wedge \varphi(x, z, \bar{w}))$ becomes $\forall z \exists y \forall x (x \in y \leftrightarrow x \in z \wedge \varphi(x, z, \bar{w}))$ with parameters $\bar{w}$, so all instances share one frame. The quantifiers of the frame stay; in particular, *leading universal quantifiers are not stripped*. Thus $\forall x (x + 0 = x)$ and $p + 0 = p$ are different data: their closures are equivalent, but only the second is an instance of the instance schema $z + 0 = z$ (Section 4).

Bound variables. In the theory and the code, bound variables are de Bruijn indices (de Bruijn, 1972): at a position with b binders above it, index $k < b$ refers to the $(k+1)$ -th binder going up, and $k \geq b$ is *free*. So formulas are equal iff they are α -equivalent. In the text we write named variables. A *sentence* has no free index; parameters are allowed. Positions p (with prefix order $p \leq q$ and incomparability $p \perp q$), subtrees $u|_p$, the number $b(p)$ of binders above p , free indices $\text{FV}(u)$ and shifting are defined in Appendix A.1. The *root symbol* of a term or formula is its head symbol (an index, or a hole z_m , counts as its own symbol).

2.2 Templates and four classes

Definition 2.1 (Bodies, templates, classes). brief §3; single §1; cases §1 A *body* of type $\iota^n \rightarrow s$ ($s \in \{\iota, o\}$) is $\lambda z_1 \dots z_n. \beta$ with β an object term ($s = \iota$) or formula ($s = o$) that has no free indices but may contain *holes* $z_1, \dots, z_n$ (leaves of sort ι), parameters and binders. *Plugging* $\beta[u_1, \dots, u_n]$ replaces each z_m by u_m , with the free indices of u_m shifted by the binders of β above that hole.

A *metavariable* M has type $\iota^n \rightarrow o$ (*formula metavariable*: $P, Q, F, \dots$) or $\iota^n \rightarrow \iota$ (*term metavariable*: $f, g, \dots$; z if 0-ary). A *template* T is built like a sentence, except that at a position of sort s there may be an *occurrence* $M(t_1, \dots, t_n)$ of some $M : \iota^n \rightarrow s$, whose arguments t_i are *metavariable-free* terms (they may contain indices bound above the occurrence, and parameters). $\text{Occ}(T)$ is the set of occurrence positions; it is an antichain. The *rigid skeleton* $\text{skel}(T)$ is the set of the other positions outside argument lists. An occurrence is a *pattern occurrence* if its arguments are pairwise distinct bound variables (for $n = 0$: always). A template without metavariables is *ground*, i.e. a single sentence. The classes are:

template	example of	FO	PAT	DT [°]	SO [°]
$z + 0 = z$	instance schema	yes	yes	yes	yes
$T_{\text{Sep}} = \forall z \exists y \forall x (x \in y \leftrightarrow x \in z \wedge P(x, z))$	Separation	no	yes	yes	yes
$T_{\text{Ind}} = (P(0) \wedge \forall x (P(x) \rightarrow P(Sx))) \rightarrow \forall x P(x)$	PA induction	no	no	yes	yes
$f(S0) + f(0) = f(S0)$	—	no	no	no	yes

Table 1: The hierarchy $\text{FO} \subseteq \text{PAT} \subseteq \text{DT}^\circ \subseteq \text{SO}^\circ$ on the templates of Example 2.3. All ZF(C) schemas lie in PAT; PA induction lies in $\text{DT}^\circ \setminus \text{PAT}$ (Section 5).

- SO° : all such templates (rigid occurrences, metavariable-free arguments);
- DT° (*determinate templates*): every metavariable has at least one pattern occurrence; other occurrences may have any metavariable-free arguments, e.g. $P(0)$, $P(Sx)$;
- DT_F° : members of DT° whose term metavariables are 0-ary (the class implemented, Section 7);
- PAT (higher-order *pattern templates* (Miller, 1991)): every occurrence is a pattern occurrence;
- FO (*first-order patterns*): every metavariable is 0-ary.

So $\text{FO} \subseteq \text{PAT} \subseteq \text{DT}^\circ \subseteq \text{SO}^\circ$ and $\text{FO} \subseteq \text{DT}_F^\circ \subseteq \text{DT}^\circ$.

Since bodies have no free indices and their own binders are de Bruijn binders, plugging cannot capture: in named notation, values of metavariables are closed λ -terms whose only free names are parameters. We call this the λ *convention*; it is the default.

Definition 2.2 (Instances, generality, minimal covering templates). ^{single §1} A *substitution* θ maps each metavariable of T to a body of its type, and the *instance* $T\theta$ replaces each occurrence $M(\bar{t})$ by $\theta(M)[\bar{t}]$ (one-step β -reduction; no new redex arises). $\text{inst}(T) = \{T\theta : \theta\}$. A *template substitution* σ maps each M to a *template body* $\lambda \bar{z}.\gamma$, where γ may contain occurrences $N(\bar{u})$ with metavariable-free $\bar{u}$. $T \succeq T'$ (T is *at least as general as* T') iff $T' = T\sigma$ for some such σ ; $T \equiv T'$ iff $T \succeq T' \succeq T$. T *covers* D if $D \subseteq \text{inst}(T)$. For a class $\mathcal{C}$ and finite nonempty D , $\text{Min}_{\mathcal{C}}(D)$ is the set of $\succeq$ -minimal covering templates in $\mathcal{C}$, one per $\equiv$ -class; $\text{Min}(D) := \text{Min}_{\text{DT}^\circ}(D)$.

The relation $\succeq$ is transitive, and $T \succeq T'$ implies $\text{inst}(T) \supseteq \text{inst}(T')$. Whether the converse holds in DT° is open and not needed: minimality is syntactic, while the verifier uses only instance sets ^{single §13}. In DT° , $T \equiv T'$ iff T' is T up to renaming of metavariables and a permutation of each metavariable’s argument places (Lemma 2.7).

Example 2.3 (The four classes). ^{[proved]^{single Cor D.3; cases A2’; prior induction C1} See Table 1. (1) The instance schema $z + 0 = z$ of $\forall x (x + 0 = x)$ is a first-order pattern; under the λ convention its instances are $t + 0 = t$ for terms t over L_A and parameters, e.g. $S0 + 0 = S0$ and $p + 0 = p$. (2) T_{Sep} is Separation in the form of Kunen (1980) (wording recalled, not re-checked against the book; Section 5). Its one occurrence $P(x, z)$ is a pattern occurrence, so $T_{\text{Sep}} \in \text{PAT} \setminus \text{FO}$. The side condition “ y not free in φ ” is built in: a body for P has no free variables but holes and parameters, and y is not an argument. (3) In T_{Ind} the two occurrences $P(x)$ are pattern occurrences, while $P(0)$ and $P(Sx)$ are not. So $T_{\text{Ind}} \in \text{DT}_F^\circ \setminus \text{PAT}$: induction is not a Miller pattern. (4) $f(S0) + f(0) = f(S0)$, with $f : \iota \rightarrow \iota$, has no pattern occurrence. It covers $0 + 0 = 0$ ($f := \lambda h.0$) and $S0 + 0 = S0$ ($f := \lambda h.h$), but not $SS0 + 0 = SS0$: a body with $\beta[0] = 0$ is h}

or 0, so $\beta[S0] \in \{S0, 0\}$. These two data anchor $z + 0 = z$ in DT° (Section 4), but not in SO° . Matching is ambiguous in SO° , too: $P(0)$, with $P : \iota \rightarrow o$, matches a sentence with k occurrences of 0 in 2^k ways.

Arguments must be metavariable-free. With metavariables inside arguments (class DT of the prior session, e.g. $M(F(x))$) the set of minimal covering templates can be infinite, and the prior enumeration-based learner was unsound for such a target (Remark A.2). In DT° it is finite (Section 3).

2.3 First-order encodings, capture and guards

Classical anti-unification works on first-order trees, where a metavariable is a leaf that any subtree may replace. Under binders this differs from the λ convention. Besides the template encoding (λ) we therefore use two *first-order encodings*, compared in Sections 4 and 5. In the *named* encoding (N), variables are names (constants of the tree signature) and binders are nodes $\text{all}(x, \cdot)$; in the *de Bruijn* encoding (dB), bound variables are indices treated as constants, and only well-formed results count. A *first-order schema* is a tree with metavariables at some positions, and in (N) and (dB) its instances may *capture*: a value may mention a variable bound by the frame. For example, $\exists y \neg(y = z)$ has the capture instance $\exists y \neg(y = y)$, false in every structure, and the unguarded first-order form $\forall z \exists y \forall x (x \in y \leftrightarrow x \in z \wedge A)$ of Separation has Russell's instance $A := \neg(x \in y)$ cases A4', D1. Side conditions are therefore expressed by *guards*, decidable predicates of the values from a fixed family, such as $\text{fresh}(v, A)$ (v not free in the value of A), $\text{loose}(A) \subseteq I$ (in (dB)), $\text{closed}(z)$ and $\text{nocap}(z)$. The *most-specific-guard rule* of the parent report (its `lem:setting:guard`) keeps every guard of the family that holds on all data; one datum that violates a guard deletes it.

For first-order trees every finite nonempty D has a *least general generalization* $\text{lgg}(D)$, unique up to renaming: the common prefix of the data, with one metavariable per distinct disagreement column (Plotkin, 1970; Reynolds, 1970).

Lemma 2.4 (When the lgg recovers a schema). *[known]^{parent lem:setting:recover} Let σ be a first-order schema and $D = \{\sigma\theta_1, \dots, \sigma\theta_N\}$. Then $\text{lgg}(D) \preceq \sigma$, and $\text{lgg}(D) \equiv \sigma$ iff (R) for each metavariable z of σ the roots of $\theta_1(z), \dots, \theta_N(z)$ are not all equal, and (D) for distinct metavariables $z \neq z'$ some j has $\theta_j(z) \neq \theta_j(z')$.*

Section 3 gives the analogue for DT° : (R) becomes root variation at every occurrence, (R*); (D) becomes the no-coincidence event (U); and a third event (N) (every argument place is used) appears.

2.4 Matching

Lemma 2.5 (Preservation). *[proved]^{single Lemma 1.1} Let $T \in \text{SO}^\circ$ and let θ be a substitution or a template substitution. If $p \in \text{skel}(T)$, $(T\theta)|_p$ has the root symbol of $T|_p$; if $p \in \text{Occ}(T)$ carries $M(\bar{t})$, then $(T\theta)|_p = \theta(M)[\bar{t}]$.*

Theorem 2.6 (Matching). *[proved; computed]^{single Thm A; prior induction C1} Let $T \in \text{DT}^\circ$ and let s be a sentence.*

- (a) *At most one substitution θ satisfies $T\theta = s$.*
- (b) *The following decides $s \in \text{inst}(T)$ and returns θ in time $O(|T| + |s|)$ (unit-cost index arithmetic): walk $\text{skel}(T)$ against s , failing on a clash; for each metavariable M pick a pattern occurrence $M(y_1, \dots, y_n)$ at p and set $\theta(M) := \lambda \bar{z}. (s|_p)[z_m/y_m]$, failing if $s|_p$ has a free*

index outside $\{y_1, \dots, y_n\}$; check every other occurrence $M(\bar{t})$ at q by comparing $s|_q$ with $\theta(M)[\bar{t}]$ lazily.

- (c) For $T, T' \in \text{DT}^\circ$: $T \succeq T'$ iff $\text{freeze}(T') \in \text{inst}(T)$, where freeze turns each metavariable of T' into a fresh rigid symbol of the same type. So generality is decidable in linear time.
- (d) For $T \in \text{SO}^\circ$, membership $s \in \text{inst}(T)$ is decidable in polynomial time, but there can be $2^{\Omega(|s|)}$ matchers.

Proof idea. At a pattern occurrence $M(\bar{y})$ at p , Lemma 2.5 gives $s|_p = \theta(M)[\bar{y}]$; as the y_m are distinct indices and $\theta(M)$ has no free indices, $\theta(M) = \lambda \bar{z}.(s|_p)[z_m/y_m]$ is forced. The rest of (b) is evaluation, and it is linear because occurrence positions form an antichain, so the parts of s inspected at different occurrences are disjoint. For (c), $T\sigma = T'$ yields a ground matcher of $\text{freeze}(T')$, and conversely. For (d), distinct metavariables do not interact, and for each a projection/imitation recursion (cf. Huet and Lang, 1978) decides whether a common body exists; $P(0)$ gives 2^k matchers. Full proof and computed checks in Appendix A.3. $\square$

Part (a) is the special case of Miller’s pattern unification (Miller, 1991) with one side ground and each metavariable read off one occurrence. General second-order matching has finitely many matchers (Huet and Lang, 1978); its NP-hardness, usually attributed to Baxter’s 1977 thesis (not checked by us), needs arguments that contain metavariables, which SO° excludes. Part (d) was proved in the prior session.

Lemma 2.7 (Equivalence is renaming). *[proved]^{single Lemma R} For $T, T' \in \text{DT}^\circ$: $T \equiv T'$ iff T' arises from T by a bijective renaming of metavariables and a permutation of each metavariable’s argument places.*

The proof (Appendix A.4) shows that a template substitution mapping T to itself fixes every pattern occurrence, hence is the identity.

2.5 Protocol, cautious verifier and anchors

Targets, data, hypotheses. A *target* is a set R^* of sentences, typically $\text{inst}(\sigma)$ or $\bigcup_{i \leq k'} \text{inst}(\sigma_i)$ for templates σ_i . The practice is assumed correct, i.e. all data lie in R^* (mistakes: Section 6). Data come as a *text* (finite $D_1 \subseteq D_2 \subseteq \dots \subseteq R^*$ with union R^*) or i.i.d.: each datum picks schema i with probability π_i , then an instance $\sigma_i\Theta$ with $\Theta \sim \Lambda_i$. *Tagged* data carry the index i (as a proof assistant records the axiom it uses); *untagged* data do not. The hypothesis classes are $H_1(\mathcal{C}) = \{\text{inst}(T) : T \in \mathcal{C}\}$; tagged calculi $H_k^{\text{tag}}(\mathcal{C})$, k -tuples of templates, one per tag; and $H_k(\mathcal{C}) = \{\bigcup_{l \leq k} \text{inst}(T_l) : T_l \in \mathcal{C}\}$.

Definition 2.8 (Protocol; cautious verifier; anchors). parent def:setting:protocol, def:caution:vs In rounds $t = 1, 2, \dots$ a *prover* submits a sentence q_t and the *verifier* answers ACCEPT, REJECT or ESCALATE; on ESCALATE an oracle says whether $q_t \in R^*$, and the labelled query joins the evidence. The prover may be randomized, adaptive and computationally unbounded; it knows the target, the verifier’s code, the data and the history, but not future data or coins. It models any proof search, the learner’s own included. A deterministic verifier is *sound* for $S^\dagger$ if no prover ever gets a sentence outside $S^\dagger$ accepted; *target-sound* if $S^\dagger = R^*$; *truth-sound* if $S^\dagger = \text{Tr}$, the set of data whose closure is true in the intended structure ($\mathbb{N}$, or the universe V of sets, assumed to satisfy ZFC). For a class $\mathcal{H}$, positive evidence D and negative evidence N , the *version space* is $\text{VS}(D, N) = \{h \in \mathcal{H} : D \subseteq h, h \cap N = \emptyset\}$. The *version-space verifier* accepts q iff $q \in \bigcap \text{VS}(D, N)$, rejects iff $q \notin \bigcup \text{VS}(D, N)$, and escalates otherwise. Its

acceptance set $\text{Acc}_{\mathcal{H}}(D, N) = \bigcap \text{VS}(D, N)$ (with $\bigcap \emptyset = \text{all sentences}$) is the *cautious verifier*; $\text{Acc}_{\mathcal{H}}(D) = \text{Acc}_{\mathcal{H}}(D, \emptyset)$, $\text{Acc}(D) = \text{Acc}_{H_1(\text{DT}^\circ)}(D)$, $\text{Acc}_k(D, N) = \text{Acc}_{H_k(\text{DT}^\circ)}(D, N)$. A finite $A \subseteq h^*$ is an *anchor* for h^* in $\mathcal{H}$ if every $h \in \mathcal{H}$ with $A \subseteq h$ contains h^* . The *closure* of h^* is $\text{cl}_{\mathcal{H}}(h^*) = \bigcap \{h \in \mathcal{H} : h \supseteq h^*\}$; h^* is $\mathcal{H}$ -closed if $\text{cl}_{\mathcal{H}}(h^*) = h^*$.

For positive data, $\text{Acc}(D) = \bigcap \{\text{inst}(T) : T \in \text{DT}^\circ \text{ covers } D\}$. For nonempty D , every covering template lies above one of the finitely many minimal ones (Section 3), so $\text{Acc}(D) = \bigcap_{T \in \text{Min}(D)} \text{inst}(T)$, and Section 3 decides membership in polynomial time without listing $\text{Min}(D)$. Two remarks. $\text{Acc}(\emptyset) = \emptyset$, because two distinct ground templates have disjoint instance sets; so $\emptyset$ is never an anchor. And the 0-ary formula metavariable P has every sentence as an instance, so with positive data nothing is rejected outright; the verifier's cost is its number of escalations (Section 3).

Theorem 2.9 (Caution is sound). *[known]^{parent thm:caution:vs} (a) If $R^* \in \mathcal{H}$ and the data lie in R^* , then $R^* \in \text{VS}(D, N)$ at all times (positive labels lie in R^* , negative ones outside), so the version-space verifier is target-sound against every prover at every time. (b) A deterministic verifier that is sound for every target in $\mathcal{H}$ accepts, at each history it produces against a target in $\mathcal{H}$, only sentences in that history's $\bigcap \text{VS}$: the cautious verifier is the most permissive sound one.*

Realizability can be weakened to closedness; this is the sense in which soundness is relative to the class.

Lemma 2.10 (Sound iff closed). *[proved]^{prior induction C6.1; parent thm:imitation:anchor} Let $\mathcal{H}$ be any class of sets of sentences, h^* a target, and $D \subseteq h^*$ finite (positive data only).*

- (a) $\text{Acc}_{\mathcal{H}}(D) \subseteq \text{cl}_{\mathcal{H}}(h^*)$; so if h^* is $\mathcal{H}$ -closed (e.g. $h^* \in \mathcal{H}$), the cautious verifier is sound for every $D \subseteq h^*$. Also, $D \subseteq D'$ implies $\text{Acc}_{\mathcal{H}}(D) \subseteq \text{Acc}_{\mathcal{H}}(D')$.
- (b) If D contains an anchor, $\text{Acc}_{\mathcal{H}}(D) = \text{cl}_{\mathcal{H}}(h^*)$. So once the data contain an anchor, the cautious verifier is sound iff h^* is $\mathcal{H}$ -closed, and it is then exact.
- (c) If h^* is $\mathcal{H}$ -closed, $\text{Acc}_{\mathcal{H}}(D) = h^*$ iff D is an anchor; on a text for h^* the verifier is eventually exact iff h^* has a finite anchor, and then it stays exact.

Proof. (a) $\text{VS}(D) \supseteq \{h \in \mathcal{H} : h \supseteq h^*\}$; and a larger D gives a smaller version space. (b) An anchor makes $\text{VS}(D) = \{h \in \mathcal{H} : h \supseteq h^*\}$. (c) If $\text{Acc}_{\mathcal{H}}(D) = h^*$, every $h \in \text{VS}(D)$ contains h^* , so D is an anchor; the converse is (b). A text eventually contains any finite subset of h^* ; with monotonicity this gives the rest. $\square$

For example, PA induction is closed in the size-bounded class DT_s iff $s \geq 12$, two symbols below the size 14 of T_{Ind} ; for $7 \leq s \leq 11$ the closure is strictly larger, so on anchor data the verifier accepts a false sentence, whatever further data arrive ^{prior induction C6.2}. Identification in the limit is due to Gold (1967), with tell-tales characterizing it for indexed families (Angluin, 1980); up to effectivity, anchors are the $\subseteq$ -tell-tales of strong-monotonic learning (Lange and Zeugmann, 1992). Caution is reliable learning (Rivest and Sloan, 1988) and perfect selective classification (El-Yaniv and Wiener, 2010) applied to sentences. For tagged data the version space factors over tags: (i, q) is accepted iff $q \in \text{Acc}(D^{(i)})$, with $D^{(i)}$ the data of tag i . For untagged unions a bound k is unavoidable, since otherwise $\text{Acc}(D) = D$ (Gold, 1967); see Section 6.

2.6 Refutation oracles

Positive data never exclude a hypothesis that contains them. The second source of evidence is the *world*: sound but one-sided evidence that a sentence is false.

Definition 2.11 (Refutation oracle). untagged §1; parent lem:twotier:onesided A *refutation oracle* is a family $\text{Ref}_0 \subseteq \text{Ref}_1 \subseteq \dots$ of sets of data (refuted within budget or depth d) with (OS) $\text{Ref}_d \cap \text{Tr} = \emptyset$ for all d (one-sided soundness), and (Dec) membership in Ref_d , and whether $\text{inst}(T) \cap \text{Ref}_d \neq \emptyset$ (T is d -refuted), are decidable uniformly in d and (T, d) . $\text{Ref}_\infty = \bigcup_d \text{Ref}_d$. A fixed finite set of false sentences, used at every d , is an oracle; this is how the implementation is read (Section 7).

The oracles used are sound and none is complete untagged §1; experiments Prop E2. $R\text{-}\Delta_0$ instantiates parameters and leading universal quantifiers of an arithmetic sentence by numerals and evaluates within depth d (numeral counterexamples refute $\forall$, numeral witnesses verify $\exists$). $R\text{-HF}$ gives hereditarily finite counterexamples to Π_1 -shaped set-theoretic sentences, sound by Δ_0 -absoluteness. $R\text{-logic}$ certifies logical falsity by a propositionally unsatisfiable set of ground instances of the Skolem form. $R\text{-coh}$ (coherence) is $R\text{-logic}$ applied to the sentence together with designated true sentences. The implementation's evaluators for $\mathbb{N}$ and V are sound in the same sense (Section 7).

Lemma 2.12 (Refutation is one-sided). [proved]untagged Lemma 1.1; parent lem:twotier:onesided *Let the oracle satisfy (OS) and $R^* \subseteq \text{Tr}$. (a) A template T with $\text{inst}(T) \subseteq \text{Tr}$ is never refuted, at any depth; in particular no target instance is. So refutation cannot tell the target from a truth-sound over-generalization, and never certifies that a template is the target. (b) If $R^* \in \mathcal{H}$, $D \subseteq R^*$ and $N \subseteq \text{Ref}_\infty$, then $R^* \in \text{VS}(D, N)$, so $\text{Acc}_{\mathcal{H}}(D, N) \subseteq R^*$: refuted sentences can serve as negatives without loss of target-soundness. $\text{Acc}_{\mathcal{H}}(D, N)$ is monotone in D and N .*

Proof. (a) $\text{Ref}_d \cap \text{Tr} = \emptyset$. (b) R^* contains D and avoids N , since $N \cap \text{Tr} = \emptyset$ and $R^* \subseteq \text{Tr}$. Larger D or N shrink the version space. $\square$

Two consequences recur. A false $\forall x \varphi$ whose closed instances are all true is never refuted by evaluating closed instances (Section 4). And a truth-sound over-generalization is never exposed; the named first-order lgg of spelled-out Replacement data with textbook variable names is one (Section 5).

2.7 Running examples

Arithmetic. Robinson's arithmetic $\mathbf{Q}$ (Tarski et al., 1953) has the axioms Q1 $\forall x \neg(Sx = 0)$; Q2 $\forall x \forall y (Sx = Sy \rightarrow x = y)$; Q3 $\forall x (\neg x = 0 \rightarrow \exists y x = Sy)$; Q4 $\forall x (x + 0 = x)$; Q5 $\forall x \forall y (x + Sy = S(x + y))$; Q6 $\forall x (x \cdot 0 = 0)$; Q7 $\forall x \forall y (x \cdot Sy = x \cdot y + x)$. Each is a ground template. In closure-normal form an axiom may also be presented with parameters (Q4 as $a + 0 = a$) or only through closed instances (e.g. $\underline{3} + 0 = \underline{3}$, instances of $z + 0 = z$); these are different data sets, and Sections 4, 6 and 7 say which they use. $\mathbf{PA}$ is $\mathbf{Q}$ plus

$$\text{Ind}(\varphi) = (\varphi(0) \wedge \forall x (\varphi \rightarrow \varphi[Sx/x])) \rightarrow \forall x \varphi,$$

where the motive φ has at most x free besides parameters; $\text{inst}(T_{\text{Ind}})$ is the set of these instances ($P := \lambda x. \varphi$). For example $\text{Ind}(x + 0 = x)$ is $(0 + 0 = 0 \wedge \forall x (x + 0 = x \rightarrow Sx + 0 = Sx)) \rightarrow \forall x (x + 0 = x)$. Induction is our main schema outside PAT; Section 5 shows that first-order and pattern anti-unification are unsound on it while the DT° learner is exact.

Set theory. The ZFC axioms other than the two schemas (Ext, Pair, Union, Pow, Inf, Found, AC) are single sentences, hence ground templates, each its own anchor; e.g. Ext is $\forall a \forall b (\forall x (x \in a \leftrightarrow x \in b) \rightarrow a = b)$, or $\forall x (x \in a \leftrightarrow x \in b) \rightarrow a = b$ with parameters. Separation, Replacement and the theorem schemas Collection and $\in$ -induction are pattern templates, e.g. T_{Sep} and $T_{\text{EInd}} = \forall x (\forall y (y \in x \rightarrow P(y)) \rightarrow P(x)) \rightarrow \forall x P(x)$. Section 5 lists all formulations used (Kunen, 1980; Jech, 2003).

Universal axioms. Let $\varphi(x_1, \dots, x_k)$ be a formula in which each x_i occurs free. Its *instance schema* is $\sigma_\varphi = \varphi[z_1/x_1, \dots, z_k/x_k]$, with 0-ary term metavariables z_i at the free occurrences of the x_i ; it is a first-order pattern. Its *closed instances* $\text{inst}_c(\varphi) = \{\varphi(\bar{t}) : \bar{t} \text{ closed}\}$ and *open instances* $\text{inst}_o(\varphi)$ ($\bar{t}$ may contain parameters) are the two data regimes of Section 4. Under the λ convention $\text{inst}(\sigma_\varphi)$ is $\text{inst}_o(\varphi)$ if parameters are allowed and $\text{inst}_c(\varphi)$ otherwise; in (N) and (dB) it also contains capture instances $\text{Cap}(\varphi)$ cases §1.1, A4'.

2.8 Parameters in theory and implementation

The theory treats parameters as constants: $\text{inst}(T)$ is the set of *literal* instances, and a template that mentions a parameter rigidly matches only that parameter. This is sound but not invariant under renaming of parameters; it arises when data share a parameter at the same place single §13. The implementation canonicalizes each datum (parameters renamed $w_0, w_1, \dots$ by first occurrence) and matches up to an injective renaming of the template’s parameters. For parameter-free templates the two readings agree. For templates with rigid parameters, the minimal covering templates must then be computed over all parameter alignments of the data; Appendix A.6 defines this set $\text{Min}^{\text{al}}(D)$ and proves that it has the properties of $\text{Min}(D)$ (Proposition A.3), while the literal $\text{Min}(D)$ can miss them (Example A.4). The experiments keep leading quantifiers, so their targets are parameter-free, and they use Min^{al} throughout (Section 7).

3 Learning one schema

This section is the general theory of learning a *single* template of the class DT° from positive data. Fix a target $T^* \in \text{DT}^\circ$ and a finite set $D \subseteq \text{inst}(T^*)$ of its instances. The cautious verifier of Section 2 accepts

$$\text{Acc}(D) = \bigcap \{\text{inst}(T) : T \in \text{DT}^\circ, D \subseteq \text{inst}(T)\}.$$

Since T^* is one of the templates intersected, $\text{Acc}(D) \subseteq \text{inst}(T^*)$: the verifier is sound at all times, against every prover. The questions are about completeness and cost. We show that the version space is finitary and contains a template below the target (Section 3.2); that $\text{Acc}(D)$ is a polynomial-time test of “features” of the data (Section 3.3); that D is an anchor iff three witness events hold, which reduce to Plotkin’s events for first-order targets and to two simple events for PA induction and every ZF schema (Section 3.4); exponential rates for random data (Section 3.5); a tight quadratic escalation bound, which refutes an earlier linear conjecture (Section 3.6); finite elasticity and explicit anchors for unions (Section 3.7); and the complexity of the version space (Section 3.8). The cost of verifying unions and of the refutation test of DTRC is in Section 6.

The results come from the research track “single”; the gray tags name them, and “earlier” refers to the analysis of PA induction that preceded this work (**research/prior/induction**). Full proofs are in Appendix B. As in Section 2, parameters are constants, bound variables are de Bruijn indices in definitions and proofs, and examples use named variables.

3.1 Notation and data notions

We recall the notation of Section 2. Positions are ordered by the prefix order $\leq$; $p \perp q$ means that neither is a prefix of the other; $s|_p$ is the subterm or subformula at p ; $b(p)$ is the number of binders strictly above p , so the bound variables in scope at p are the indices $0, \dots, b(p) - 1$. The *root symbol* is the head symbol, an index counting as a symbol. $|s|$ is the number of positions. In a template, $\text{Occ}(T)$ is the set of positions of metavariable occurrences $M(\bar{t})$ (an antichain; the t_i are metavariable-free), and the *skeleton* $\text{skel}(T)$ is the set of the other positions outside argument lists. A body $\lambda \bar{z}. \beta$ is closed except for the holes z_m (parameters and binders may occur); plugging $\beta[\bar{u}]$ substitutes u_m for z_m with the usual shifting. For a map $\bar{u}$ on free indices, $s[\bar{u}]$ substitutes u_y for each free index y of s . $T \succeq T'$ if $T' = T\sigma$ for a substitution by template bodies; then $\text{inst}(T) \supseteq \text{inst}(T')$. By the matching theorem of Section 2, a sentence $s \in \text{inst}(T)$, $T \in \text{DT}^\circ$, has a unique matcher θ , read off pattern occurrences: $\theta(M) = \lambda \bar{z}. (s|_p)[z_m/y_m]$ for a pattern occurrence $M(\bar{y})$ at p ; membership and generality are decided in linear time. If T covers D (i.e. $D \subseteq \text{inst}(T)$), θ_d is the matcher of $d \in D$ and $\beta_M^d := \theta_d(M)$.

Definition 3.1 (Common prefix, slots, scopes, features). ^{single §1} Let D be a nonempty finite set of sentences.

- The *common prefix* $C(D)$: the root is in $C(D)$ iff all data have the same root symbol, and $p \cdot i \in C(D)$ iff $p \in C(D)$ and all $d|_{p \cdot i}$ have the same root symbol. The *slots* $\text{slots}(D)$ are the positions outside $C(D)$ whose parent is in $C(D)$ (or the root, if it is outside). $A(D) := C(D) \cup \text{slots}(D)$ (*admissible* positions). The *scope* of a slot is $Y_\sigma(D) := \bigcup_{d \in D} \text{FV}(d|_\sigma)$.
- A sentence q has the *feature* $\text{Sym}(p)$ ($p \in C(D)$) if $q|_p$ exists and has the root symbol common to the data at p ; $\text{Scope}(\sigma)$ (σ a slot) if $\text{FV}(q|_\sigma) \subseteq Y_\sigma(D)$; and $\text{Eq}(\sigma, r, \bar{u})$ (σ a slot, $r \in A(D)$, $\sigma \perp r$ of the same sort, $\bar{u}$ a map from $Y_\sigma(D)$ to metavariable-free terms in scope at r) if $\text{FV}(q|_\sigma) \subseteq Y_\sigma(D)$ and $q|_r = (q|_\sigma)[\bar{u}]$.
- A *D-feature* is a feature that every datum has; $\text{Feat}(D)$ is the set of sentences with every D -feature.

So a D -feature says: here all data have the same symbol; or no datum has a bound variable at this slot outside this set; or, in every datum, the content at r is the content at slot σ with its free variables replaced by $\bar{u}$.

Lemma 3.2 (Basic facts). ^{[proved] single §§1, 3}

- (a) Preservation. If $p \in \text{skel}(T)$, then $(T\theta)|_p$ has the root symbol of $T|_p$; if $p \in \text{Occ}(T)$ carries $M(\bar{t})$, then $(T\theta)|_p = \theta(M)[\bar{t}]$.
- (b) Substitution. For β closed except for holes: (i) $(\beta[\bar{t}])[\bar{u}] = \beta[\bar{t}[\bar{u}]]$; (ii) if z_m occurs in β and $\beta[\bar{a}] = \beta[\bar{a}']$, then $a_m = a'_m$; (iii) if the root of β is not a hole, $\beta[\bar{a}]$ has the same root. Formula bodies always have symbol roots.
- (c) Rigid prefix. If T covers D , then $\text{skel}(T) \subseteq C(D)$ with the same symbols, and $\text{Occ}(T) \subseteq A(D)$.
- (d) Forcing. Let $\sigma \perp r$ be positions of every datum and $Y = \bigcup_d \text{FV}(d|_\sigma)$. At most one $\bar{u}$ on Y satisfies $d|_r = (d|_\sigma)[\bar{u}]$ for all $d \in D$; one exists iff one exists for every pair $\{d, d'\} \subseteq D$; and it is found in time $O(\sum_d |d|)$.
- (e) Equivalence is renaming. For $T, T' \in \text{DT}^\circ$: $T \succeq T' \succeq T$ iff T' arises from T by renaming metavariables bijectively and permuting each metavariable's argument places.

Proofs in Appendix B.1. Part (d) is the second-order form of the fact behind Plotkin’s lgg: an equation between two positions is either violated by the data or determined by them.

3.2 Minimal covering templates

$\text{Min}(D)$ is the set of $\succeq$ -minimal covering templates in DT° . A covering $T \in \text{DT}^\circ$ is *saturated* if (S1) every argument place is used (z_m occurs in some β_M^d , for every M and m), and (S2) for every M the roots of the β_M^d , $d \in D$, are not all equal (a hole z_m counting as the root z_m).

Theorem 3.3 (Minimal covering templates). *[proved; computed]^{single Thm B} Let D be nonempty and finite.*

- (a) *Every covering $T \in \text{DT}^\circ$ satisfies $T \succeq T'$ for some saturated covering T' .*
- (b) *In a saturated covering template, $\text{skel} \subseteq C(D)$; every pattern occurrence of a metavariable sits at a slot σ and has argument set $Y_\sigma(D)$; every occurrence sits in $A(D)$; and the arguments of each occurrence are determined by D , its position and the pattern slot of its metavariable. So there are at most $2^{|A(D)|} |\text{slots}(D)|^{|A(D)|}$ saturated covering templates up to renaming.*
- (c) *Every minimal covering template is saturated. So $\text{Min}(D)$ is finite up to renaming, and every covering template is $\succeq$ some member of $\text{Min}(D)$.*
- (d) *If $D \subseteq \text{inst}(T^*)$, $T^* \in \text{DT}^\circ$, some $T_m \in \text{Min}(D)$ has $T^* \succeq T_m$, hence $\text{inst}(T_m) \subseteq \text{inst}(T^*)$.*

Proof idea. (a) Two specializations, each keeping coverage and determinacy: drop an argument place that no value uses; or, if all values of M have the same root, make it rigid ($M \mapsto \lambda \bar{z}.z_m$, or $M \mapsto \lambda \bar{z}.f(M_1(\bar{z}, \cdot), \dots)$ with fresh M_j that also take the variable bound by f , if f binds). Pattern occurrences stay pattern occurrences, and $(|C(D)| - |\text{skel}|, \sum_M \text{arity}(M))$ decreases lexicographically. (b) At a pattern occurrence $d|_\sigma = \beta_M^d[\bar{y}]$, so (S2) puts σ outside $C(D)$ and (S1) gives the argument set; other arguments are forced by Lemma 3.2(b)(ii). (c) uses Lemma 3.2(e). Appendix B.2. $\square$

Part (d) is what the clustering method of Section 6 needs: data of one schema always have a minimal covering template below it, hence a sound one; the theorem is the finiteness hypothesis used there. Metavariable-free arguments are essential. The earlier analysis conjectured finiteness (“C8.3”) for the class with nested arguments, where it fails: its referee found the infinite antichain $(0 = 0 \wedge \forall x(f(x) = x \rightarrow f^k(Sx) = Sx)) \rightarrow \forall x f(x) = x$, $k \geq 1$, of minimal covering templates of $\{\text{Ind}(x = x), \text{Ind}(0 = x)\}$. With metavariable-free arguments the arguments are forced, and C8.3 holds.

Example 3.4 (Induction data, no anchor). *[computed]^{single §4} For $D = \{\text{Ind}(x = x), \text{Ind}(0 = x)\}$ (both motives have root $=$, so D is no anchor), $\text{Min}(D)$ consists of exactly the four templates*

$$(\zeta_1 = \zeta_2 \wedge \forall x(f(x) = x \rightarrow f(Sx) = Sx)) \rightarrow \forall x f(x) = x, \quad \zeta_1, \zeta_2 \in \{0, f(0)\},$$

of which only $\zeta_1 = f(0)$, $\zeta_2 = 0$ lies below T_{Ind} . The earlier analysis reported eight, from a search bounded by size 23; the template with $\zeta_1 = \zeta_2 = f(0)$ has size 24, and five of the eight lie strictly above it. Brute force at size ≤ 24 and the referee’s independent enumerator confirm the four (Appendix B.2).

3.3 The cautious verifier is a feature test

Let $T_0(D)$ be the skeleton $C(D)$ with a fresh metavariable $M_\sigma(Y_\sigma)$ at every slot (arguments in increasing order). For an Eq-feature $\varphi = \text{Eq}(\sigma_0, r, \bar{u})$ of D , let T_φ be $C(D)$ cut at r , with $M_\sigma(Y_\sigma)$ at every slot not below r and the derived occurrence $M_{\sigma_0}(\bar{u})$ at r . Both are in DT° and cover D .

Theorem 3.5 (The cautious verifier checks features). *[proved; computed]^{single Thm C} For every nonempty finite set D of sentences,*

$$\text{Acc}(D) = \text{Feat}(D) = \text{inst}(T_0(D)) \cap \bigcap \{\text{inst}(T_\varphi) : \varphi \text{ an Eq-feature of } D\}.$$

Proof idea. $\text{inst}(T_0(D))$ consists of the sentences with all Sym and Scope features, and T_φ adds φ ; these templates cover D , so $\text{Acc}(D) \subseteq \text{Feat}(D)$. Conversely, a covering T is $\succeq$ a saturated T' (Theorem 3.3(a)), and by Theorem 3.3(b) $\text{inst}(T')$ is cut out by D -features: Sym on its skeleton, Scope at one pattern slot per metavariable, and an Eq feature for every other occurrence. So $\text{Feat}(D) \subseteq \text{inst}(T)$. Appendix B.3. $\square$

Corollary 3.6 (A polynomial verifier). *[proved]^{single Cor C.1} With $m = |A(D)| \leq \min_d |d|$ and $n = \sum_d |d|$, the D -features are listed in time $O(m^2 n)$, there are $O(m^2)$ of them, and a query q is decided in time $O(m^2 |q|)$. No enumeration of $\text{Min}(D)$ is needed.*

Reading. The verifier accepts q iff q has the data’s common skeleton, no slot of q mentions a bound variable that no datum mentions there, and every equation “content at r = content at slot σ with variables replaced by $\bar{u}$ ” that holds in all data holds in q . The last clause is the second-order analogue of Plotkin’s “equal columns get equal variables” (Plotkin, 1970; Reynolds, 1970). $T_0(D)$ is the higher-order pattern anti-unifier without merging; merging columns equal up to renaming, as the pattern lgg does (Pfenning, 1991; Baumgartner et al., 2017), is the case of an Eq feature whose $\bar{u}$ is a permutation. DT° adds *derived* occurrences, and the price is that the version space can have exponentially many minimal elements (Proposition 3.20); their intersection is still described by polynomially many equations. Second-order generalization beyond patterns is also studied by Hirata et al. (2004) and in the frameworks surveyed by Cerna and Kutsia (2023); we have not checked whether DT° or Theorem 3.5 appears there. The earlier induction learner enumerated $\text{Min}(D)$ and intersected, which is exponential in the worst case; the feature test replaces it.

Evidence. $\text{Min}(D)$ from saturated templates agreed with an independent brute-force enumerator in all 103 completed random cases, and $\text{Feat}(D)$ with the intersection of all enumerated covering templates on 7621 queries; the referee’s independent code, with bounds containing $T_0(D)$ and every T_φ , agreed on 125,109 queries over 519 data sets (Appendix B.9).

3.4 Anchors

Let $T^* \in \text{DT}^\circ$ and $D = \{s_1, \dots, s_N\}$, $s_i = T^* \theta_i$, $N \geq 1$, $\beta_M^i := \theta_i(M)$. For $\sigma \in \text{Occ}(T^*)$ carrying $M(\bar{t}_\sigma)$ let $Y_\sigma^* := \bigcup_m \text{FV}(t_{\sigma,m})$. Consider pairs (σ, r) with $\sigma \in \text{Occ}(T^*)$, $r \in \text{skel}(T^*) \cup \text{Occ}(T^*)$, $\sigma \perp r$, of the same sort; such a pair is *valid* if r carries an occurrence $M(\bar{t}_r)$ of the *same* metavariable with $\bar{t}_r = \bar{t}_\sigma[\bar{u}_0]$ for some $\bar{u}_0$. A valid pair gives an equation that holds on all of $\text{inst}(T^*)$.

Definition 3.7 (Witness events; richness). ^{single §6}

- (R^*) (root variation at every occurrence): for every $\sigma \in \text{Occ}(T^*)$ the roots of $s_1|_\sigma, \dots, s_N|_\sigma$ are not all equal. (At a pattern occurrence this is Plotkin’s (R) for the values; at a derived occurrence $M(\bar{t})$ it concerns the plugged values $\beta_M^i[\bar{t}]$.)

- (N) (non-vacuity): for every metavariable M and argument place m , some β_M^i contains z_m .
- (U) (no coincidence): for no non-valid pair (σ, r) is there a $\bar{u}$ on Y_σ^* with $s_i|_r = (s_i|_\sigma)[\bar{u}]$ for all i .
- (Rich): the signature has a relation symbol of arity ≥ 1 , and, if T^* has a term-valued metavariable, closed terms with two different root symbols exist (automatic with parameters). (Rich) holds for arithmetic and set theory.

Theorem 3.8 (Anchor theorem). *[proved; computed]^{single Thm D} If (R^*) , (N) and (U) hold, D is an anchor for $\text{inst}(T^*)$ in DT° ; this needs no assumption. Under (Rich), every anchor satisfies (R^*) , (N) and (U).*

Proof idea. $(\Leftarrow)$ (R^*) gives $C(D) = \text{skel}(T^*)$ and $\text{slots}(D) = \text{Occ}(T^*)$, and (N) gives $Y_\sigma(D) = Y_\sigma^*$; so instances of T^* have every Sym and Scope feature. By (U) every Eq-feature comes from a valid pair, and forcing identifies its $\bar{u}$ with $\bar{u}_0$; such equations hold on $\text{inst}(T^*)$. So $\text{inst}(T^*) \subseteq \text{Feat}(D) = \text{Acc}(D)$. $(\Rightarrow)$ When an event fails, the corresponding Sym, Scope or Eq property is a D -feature, and (Rich) supplies an instance of T^* violating it: a value with another root, a value that uses the unused hole, or values that separate the two sides of the coincidence. Appendix B.4. $\square$

So the candidate events of the brief are right in spirit but must be read occurrence-wise: root variation must hold at *derived* occurrences too, and Plotkin’s distinctness (D) is only the simplest coincidence.

Corollary 3.9 (Specializations). *[proved; computed]^{single Cor D.1–D.3}*

- First-order targets (*all metavariables 0-ary, possibly under binders*): $(R^*) \wedge (N) \wedge (U)$ is equivalent to Plotkin’s $(R) \wedge (D)$. So DT° has the same anchors on first-order targets as the class of first-order patterns (Plotkin, 1970; Reynolds, 1970): the larger class costs no data. For the instance schema of $\forall x \varphi(x)$ (Section 4), two instances whose terms have different roots are an anchor (with several variables, (D) is needed as well).
- Formula metavariables only: (R^*) is equivalent to (R) , and under $(R) \wedge (N)$, (U) is equivalent to (D^+) : for $M \neq M'$ and occurrences σ of M , r of M' , no $\bar{u}$ has $\beta_{M'}^i[\bar{t}_r] = (\beta_M^i[\bar{t}_\sigma])[\bar{u}]$ for all i . With a single formula metavariable, D is an anchor iff (R) and (N) hold.
- PA induction and ZF. Hence, in closure-normal form: $T_{\text{Ind}} = P(0) \wedge \forall x (P(x) \rightarrow P(Sx)) \rightarrow \forall x P(x)$ is anchored iff the motives do not all have the same main symbol and some motive has x free (the earlier anchor theorem for induction); Separation $\forall a \exists b \forall x (x \in b \leftrightarrow x \in a \wedge P(x, a))$ iff (R) , $(N)_x$, $(N)_a$ (b cannot occur in P : freshness is built in); spelled-out Replacement, whose three occurrences $P(x, y, a)$, $P(x, z, a)$, $P(x, y, a)$ are all pattern occurrences, iff (R) , $(N)_x$, $(N)_y$, $(N)_a$; $\in$ -induction iff (R) and (N) . Single axioms are ground templates, each its own anchor. One datum is never an anchor of a schema, two can be.

In (a)–(c) the “if” directions need no assumption and the “only if” directions use (Rich).

Section 5 treats the ZF schemas in all formulations, encodings and classes. *Evidence:* Theorem 3.8 agreed with feature-based anchor status on 248/248 random targets with one or two metavariables of all types (referee, independently: 540/540, and 519/519 against brute force), and (a), (c) on all of 571 and about 2000 random data sets (Appendix B.9).

The weaker candidates are not enough, and (Rich) cannot be dropped.

Proposition 3.10 (Weaker events do not suffice). *[proved; computed]^{single Prop D.4, D.5, Ex. D.6}*

- (a) (R)+(N)+(D): $T^* = \forall x(0 = f(x))$, $D = \{\forall x(0 = 0), \forall x(0 = x)\}$. (R*) and (N) hold and (D) is vacuous, but $\forall x(f(0) = f(x))$ covers D and misses $\forall x(0 = Sx)$.
- (b) (U) for pattern occurrences only: $T^* = \forall x \forall y(Sf(x, y) = 0) \wedge \forall x \forall y(f(x, Sy) = 0)$ with the values $\lambda_{z_1 z_2. z_1}$, $\lambda_{z_1 z_2. z_2}$, $\lambda_{z_1 z_2. Sz_1}$. (R*), (N), (D) hold, and (U) holds for every pair whose first component is the pattern occurrence. But the content at the rigid S above $f(x, y)$ equals the content at the derived occurrence $f(x, Sy)$ under $x \mapsto Sx$, so $\forall x \forall y(f'(y, Sx) = 0) \wedge \forall x \forall y(f'(y, x) = 0)$ covers D and misses $T^*[f := \lambda_{z_1 z_2. 0}]$.
- (c) Projections: $T^* = \forall x \forall y(f(x, y) = 0) \wedge \forall x(x = 0)$ with values $\lambda_{z_1 z_2. z_1}$, $\lambda_{z_1 z_2. z_2}$ satisfies (R*), (N), but $\forall x \forall y(f(x, y) = 0) \wedge \forall x(f(x, x) = 0)$ covers D . So (U) matters even for one metavariable with pattern occurrences only.

On random data (R) $\wedge$ (N) $\wedge$ (D) and (R*) $\wedge$ (N) $\wedge$ (D) were wrong in 27 and 17 of 248 cases (always “says anchor, is not”; referee: 45, 29 of 540), while (U) restricted to pattern occurrences was never wrong, so (b) needs special structure. (Rich) *cannot be dropped*.^{single Ex. D.7} in pure set theory without parameters the only term bodies are projections, so $T^* = \forall x(f(x) \in x)$ has $\text{inst}(T^*) = \{\forall x(x \in x)\}$, and one datum is an anchor although (R*) fails. The exact anchor condition without (Rich) is open.

3.5 Rates

Let Λ be a law with countable support on ground substitutions for T^* , $\theta_1, \theta_2, \dots$ i.i.d. $\sim \Lambda$, and $D_N = \{T^*\theta_1, \dots, T^*\theta_N\}$. Let $\rho_\sigma := 1 - \max_h \Lambda(\text{root}((T^*\theta)|_\sigma) = h)$ for $\sigma \in \text{Occ}(T^*)$, and $\nu_{M,m} := \Lambda(z_m \in \theta(M))$. For a non-valid pair (σ, r) (their set is $U(T^*)$) and a map $\bar{u}$ let $A_{\sigma,r,\bar{u}} := \{\theta : (T^*\theta)|_r = ((T^*\theta)|_\sigma)[\bar{u}]\}$, $\kappa_{\sigma,r} := \Lambda^{\otimes 2}\{(\theta, \theta') : \text{no } \bar{u} \text{ has } \theta, \theta' \in A_{\sigma,r,\bar{u}}\}$ and $\chi_{\sigma,r} := \sup_{\bar{u}} \Lambda(A_{\sigma,r,\bar{u}})$. Let $c(T^*) := |\text{Occ}(T^*)| + \sum_M \text{arity}(M) + |U(T^*)|$ and λ the minimum of all $\rho_\sigma, \nu_{M,m}, \kappa_{\sigma,r}$. For a ground target (e.g. a single ZFC axiom) $c(T^*) = 0$; we set $\lambda := 1$, and in general $\bar{c} := \max(c(T^*), 1)$. A ground T^* has $\text{Acc}(\{T^*\}) = \{T^*\}$, so D_N is an anchor iff $N \geq 1$.

Theorem 3.11 (Sample complexity). *[proved; computed]^{single Thm E} The upper bounds need no assumption; the lower bounds use (Rich).*

- (a) For $N \geq 1$,

$$\begin{aligned} \Pr[D_N \text{ is not an anchor}] &\leq \sum_{\sigma} (1 - \rho_\sigma)^{N-1} + \sum_{M,m} (1 - \nu_{M,m})^N \\ &\quad + \sum_{(\sigma,r) \in U(T^*)} (1 - \kappa_{\sigma,r})^{\lfloor N/2 \rfloor} \leq c(T^*)(1 - \lambda)^{\lfloor N/2 \rfloor}, \end{aligned}$$

and $\Pr[D_N \text{ is not an anchor}] \geq \max(\max_{\sigma} (1 - \rho_\sigma)^N, \max_{M,m} (1 - \nu_{M,m})^N, \max_{(\sigma,r)} \chi_{\sigma,r}^N)$.
($D_0 = \emptyset$ is never an anchor: $\text{Acc}(\emptyset) = \emptyset$.)

- (b) $\lambda > 0$ iff the set $\{T^*\theta : \theta \in \text{supp } \Lambda\}$ satisfies (R*), (N), (U). Then the cautious verifier is exact with probability $\geq 1 - \delta$ once $N \geq 1 + (2/\lambda) \ln(\bar{c}/\delta)$; for a ground T^* , once $N \geq 1$.
- (c) Tagged data. For k targets $T_i^* \in \text{DT}^\circ$ (ground ones allowed) cited by tags of probability π_i , with parameters $\bar{c}_i, \lambda_i$, the per-tag cautious verifier (the version space factors over tags, Claude (Anthropic), 2026) fails to be exact after N tagged data with probability at most $\sqrt{e} \sum_i \bar{c}_i \exp(-3N\pi_i\lambda_i/8)$. Order $1/(\pi_i\lambda_i)$ data are necessary in general: the failure probability is at least $(1 - \pi_i)^N$ (no tag- i datum; exact for a ground tag), and at least $(1 - \pi_i\rho)^N$ if some head at a pattern occurrence of T_i^* has probability $1 - \rho$.

Proof idea. By the “if” half of Theorem 3.8, failure lies in the union of the failures of the events. All roots at σ agree with probability $\sum_h p_h^N \leq (1 - \rho_\sigma)^{N-1}$; a coincidence on D_N implies one on each of $\lfloor N/2 \rfloor$ disjoint, independent pairs. A union bound gives (a); (c) averages over the binomial number of data per tag. Appendix B.5. $\square$

For one formula metavariable (induction, the ZF schemas) the coincidence terms vanish (Corollary 3.9(b)) and the bound is $(1 - \rho)^{N-1} + \sum_m (1 - \nu_m)^N$; for induction this is the earlier exact rate $\sum_f p_f^N + (1 - q)^N - \sum_f r_f^N$ up to its inclusion–exclusion term. The coincidence terms need not be tight:

Proposition 3.12 (Coincidence terms). *[proved; computed]^{single Prop E.2} For $(\sigma, r) \in U(T^*)$: (i) the probability of a coincidence at (σ, r) on D_N is $\geq \chi_{\sigma, r}^N$, and under (Rich) such a coincidence makes D_N a non-anchor; (ii) $\chi_{\sigma, r}^2 \leq 1 - \kappa_{\sigma, r}$; (iii) if Λ has finite support and $\mathcal{K}$ is the family of maximal support sets lying in a common $A_{\sigma, r, \bar{u}}$, the coincidence probability equals $\sum_{\emptyset \neq \mathcal{J} \subseteq \mathcal{K}} (-1)^{|\mathcal{J}|+1} \Lambda(\cap \mathcal{J})^N \in [\chi^N, |\mathcal{K}| \chi^N]$. So its exact rate is χ , which can be smaller than the rate $\sqrt{1 - \kappa}$ of the upper bound; they agree if $|\mathcal{K}| = 1$.*

For $T^* = \forall x(0 = f(x))$ with $f \sim \{\lambda z.0 : 0.4, \lambda z.z : 0.3, \lambda z.Sz : 0.3\}$, the exact non-anchor probability at $N = 8$ is 0.057714, between the coincidence lower bound 0.057648 and the upper bound 0.059942; the root and non-vacuity lower bounds alone give 0.000655. Exact computations by the referee, which decide anchor status by feature search rather than by the events of Theorem 3.8, confirm the bound (Appendix B.5).

3.6 Escalations

In the protocol of Section 2, a query the verifier does not accept is *escalated*: an oracle labels it, and a valid query joins the data. Against an honest prover (all queries are target instances) the cautious verifier’s escalated queries form a chain $q_1, \dots, q_m$ with $q_j \notin \text{Acc}(\{q_1, \dots, q_{j-1}\})$, and every such chain is forced by the honest prover that asks it in order (Claude (Anthropic), 2026). $\text{Esc}(\mathcal{H}; N)$ is the supremum of the lengths of such chains inside members of $\mathcal{H}$, from no data, on sentences of size $\leq N$; the first query is always escalated, as $\text{Acc}(\emptyset) = \emptyset$. For first-order schemas over a term signature, $\text{Esc}(H_1; N) \leq N + 1$ (Claude (Anthropic), 2026). Every chain is honest for the universal template (a 0-ary formula metavariable).

Let $\Pi(D) := (C(D), (Y_\sigma(D))_\sigma, E(D))$, where $E(D)$ is the set of pairs (σ, r) carrying an Eq-feature. As data grow, C shrinks, scopes grow, and while both are fixed E shrinks; and by Theorem 3.5 a query outside $\text{Acc}(D)$ changes Π (Lemma B.1).

Theorem 3.13 (Escalation bound). *[proved; computed]^{single Thm F} Let d have n positions and let $q_j \notin \text{Acc}(\{d, q_2, \dots, q_{j-1}\})$ for $j = 2, \dots, m$ (no honesty needed). Then*

$$m - 1 \leq n + \sum_p \text{b}(p) + \#\{(\sigma, r) : \sigma \perp r\} \leq n + n \text{b}_{\max} + n(n - 1),$$

with p, σ, r ranging over positions of d . Hence $\text{Esc}(\text{DT}^\circ; N) \leq 1 + N + 2N(N - 1) \leq 2N^2 + 1$.

Proof idea. Each escalation (α) shrinks C (at most n times), or (β) enlarges a scope (a position is a slot during one interval, and its scope grows inside $\{0, \dots, \text{b}(\sigma) - 1\}$), or (γ) removes a pair from E (a removed pair never returns). Appendix B.6. $\square$

First-order targets already give $\text{Esc}(\text{DT}^\circ; N) \geq N + 1$ ($N \geq 5$), via $S^k 0 = 0$, $S^k a = 0$, $S^{k-1} 0 = 0$, $\dots$, $0 = 0$, $0 = S0$, $\neg(0 = 0)$ with $k = N - 3$. The quadratic term is real:

Proposition 3.14 (Quadratic escalations). *[proved; computed]^{single Prop F.8} Let $d_1 = \forall y_1 \dots \forall y_m (0 = 0 \wedge \dots \wedge 0 = 0)$ with m atoms, d_2 the same with every left-hand side replaced by a parameter a , and $q_{j,k}$ ($j \leq m$, $k < m$) the same with the j -th left-hand side replaced by the bound variable of index k . (i) In the order $d_1, d_2, q_{1,0}, \dots, q_{1,m-1}, q_{2,0}, \dots, q_{m,m-1}$, each sentence lies outside the acceptance set of its predecessors (it violates a Scope feature). All have size $5m - 1$. Hence $2 + \lfloor (N + 1)/5 \rfloor^2 \leq \text{Esc}(\text{DT}^\circ; N) \leq 2N^2 + 1$ for $N \geq 4$: $\text{Esc}(\text{DT}^\circ; N) = \Theta(N^2)$. (ii) $\text{Ind}(d_1), \text{Ind}(d_2), \text{Ind}(q_{j,k})$, in the same order, are instances of T_{Ind} of size $20m + 1$, and each lies outside the acceptance set of its predecessors.*

The earlier analysis conjectured (“C11”) that escalations are at most *linear* in the size of the first escalated datum. By (ii), $1 + m^2$ escalations follow a first datum of size $n = 20m + 1$, which exceeds cn once $m > 20c + 1$. So *C11 is false*: for DT° , for the class with nested arguments (whose acceptance sets are smaller), for the universal template and for the induction target. The refutation is asymptotic (for $m = 2, \dots, 5$ the counts 5, 10, 17, 26 are below $n = 41, \dots, 101$). The quadratic bound holds and is tight. Its quadratic part comes from scope growth, which needs deep binder nesting:

Proposition 3.15 (Binder-free data). *[proved; computed]^{single Prop F.9} For a chain as in Theorem 3.13, at most $n - 1$ steps of kind (γ) remove a pair whose source has empty scope. If d has no binder, $m - 1 \leq 2n - 1$; on binder-free sentences $N + 1 \leq \text{Esc} \leq 2N$ ($N \geq 5$).*

Conjecture 3.16 (Equation kills are linear). *[conjecture]^{single Conj. F.7} The number of (γ) -steps is $O(n)$; hence $\text{Esc}(\text{DT}^\circ; N) = O(N(1 + b_{\max}))$ for binder depth $\leq b_{\max}$.*

The evidence is weak: greedy adversarial searches found at most about $1.4m$ kill steps on $m \leq 12$ slots, but they also missed the scope-growth chains of Proposition 3.14. Surviving equations need not form an equivalence relation; they compose like a preorder and obey a triangle rule (Lemma B.2, suggested by the referee), and chains of preorders on m points can have length of order m^2 .

3.7 Finite elasticity and unions

A class has *finite elasticity* if there are no infinite $w_0, w_1, \dots$ and members $L_1, L_2, \dots$ with $\{w_0, \dots, w_{n-1}\} \subseteq L_n \not\supseteq w_n$ for all n (Wright, 1989; Motoki et al., 1991). Let $H_k(\text{DT}^\circ) = \{\bigcup_{l \leq k} \text{inst}(T_l) : T_l \in \text{DT}^\circ\}$ and $\text{Acc}_k(D) = \bigcap \{U \in H_k(\text{DT}^\circ) : D \subseteq U\}$.

Corollary 3.17 (Elasticity and unions). *[proved; known]^{single Cor F.2–F.4, Prop F.3} (i) $\{\text{inst}(T) : T \in \text{DT}^\circ\}$ has finite elasticity (an elastic sequence is an escalation chain, bounded by Theorem 3.13), but not finite thickness: $\forall x(0 = 0) \wedge (0 = 0)$ is an instance of $\forall x P(x) \wedge P(t)$ for every closed t , and these instance sets differ. (ii) Finite elasticity survives unions of at most k members (Wright, 1989; Motoki et al., 1991). So $H_k(\text{DT}^\circ)$ has finite elasticity, every target in it has a finite anchor (Claude (Anthropic), 2026), and the cautious verifier over $H_k(\text{DT}^\circ)$ is sound at all times and exact from some finite time on, on every text. (iii) With a k -ary relation symbol, a unary function symbol and a constant, $\text{Esc}(H_k(\text{DT}^\circ); N) \geq \lfloor (N - 1)/k \rfloor^k$; over $=, +, S, 0$, $\text{Esc}(H_k(\text{DT}^\circ); N) \geq (\lfloor (N - 1)/k \rfloor - 1)^k$ for $N \geq 2k + 1$, already for one target.*

For an explicit condition, let the target be $\bigcup_{i \leq k'} \text{inst}(T_i^*)$, $k' \leq k$, $T_i^* \in \text{DT}^\circ$, and $\Theta_i = \{\theta : T_i^* \theta \in D\}$. Let $\text{Fail}(T^*)$ consist of the failure sets of the three events: $\{\theta : \text{root}((T^* \theta)|_\sigma) = h\}$, $\{\theta : z_m \notin \theta(M)\}$, and $\{\theta : (T^* \theta)|_r = ((T^* \theta)|_\sigma)[\bar{u}]\}$ for non-valid (σ, r) .

Proposition 3.18 (Explicit untagged anchors). *[proved; computed]^{single Prop F.5} If for each i the set Θ_i is nonempty and not covered by k members of $\text{Fail}(T_i^*)$, then D is an anchor for the union in $H_k(\text{DT}^\circ)$. For a ground T_i^* (a single axiom) $\text{Fail}(T_i^*) = \emptyset$, and the condition says that the axiom itself is a datum.*

Proof idea: a covering k -union splits each Θ_i into at most k nonempty parts; if no part were an anchor, each would lie in a failure set (“if” half of Theorem 3.8), so some part is an anchor (Appendix B.7). Nonemptiness was missing in the first version and matters for ZFC’s single axioms: with target $\text{inst}(T_{\text{Ind}}) \cup \{\text{Ext}\}$ and $k = 2$, the union verifier rejects Extensionality until it is a datum. For one formula metavariable, Fail reduces to “all roots equal f ” and “argument m unused”, the failure family of the earlier untagged PA analysis (Section 6).

Proposition 3.19 (The union verifier). *[proved; computed]^{single Prop F.6} $q \in \text{Acc}_k(D)$ iff every partition of D into at most k nonempty blocks has a block B with $q \in \text{Acc}(B)$. So Acc_k is decidable in time $(\text{number of such partitions}) \times \text{poly}$.*

Section 6 shows that this is polynomial for $k \leq 2$ but coNP-complete for each fixed $k \geq 3$. On unlabelled data from PA induction and $\in$ -induction (three instances each), $k = 1$ accepts the false $(0 = 0 \wedge \forall x(x = x \rightarrow Sx = Sx)) \rightarrow \forall x x = 0$, while $k = 2, 3$ accept all five fresh instances tried and reject it and two other non-instances (reproduced by the referee).^{single e9}

3.8 Complexity of the version space

Matching and generality are linear, $\text{Acc}(D)$ is polynomial (Corollary 3.6), and whether D is an anchor for a *given* T^* is polynomial (at most $|\text{Occ}(T^*)| |T^*|$ pairs, each decided by forcing). But $\text{Min}(D)$ itself can be huge:

Proposition 3.20 (Exponentially many minimal templates). *[proved; computed]^{single Prop G.2} Let C_n be the conjunction of n copies of $0 = 0$ and $D_n = \{\forall x(x = x) \wedge C_n, \forall x(0 = 0) \wedge C_n\}$, of total size $8n + 8$. Then $|\text{Min}(D_n)| = 4^n$ (each of the $2n$ zeros of C_n may be rigid or a derived occurrence $f(0)$), while $\text{Acc}(D_n) = D_n$.*

Proposition 3.21 (Existence of an lgg). *[proved; computed]^{single Prop G.3} D has a least covering template iff (i) no Eq-feature of D has $r \in C(D)$, and (ii) in the graph on $\text{slots}(D)$ with an edge $\sigma \rightarrow r$ for each slot-to-slot Eq-feature, every weakly connected component K has a source s with $s \rightarrow x$ for all $x \in K \setminus \{s\}$. Then the lgg is $C(D)$ with $M_K(Y_s)$ at each source s and $M_K(\bar{u}_{s,x})$ at the other slots x of K , where $\bar{u}_{s,x}$ is the forced map.*

Corollary 3.22 (An anchor determines its template). *[proved; computed]^{single Cor G.4} Under (Rich), if D is an anchor for $\text{inst}(T^*)$ in DT° , then T^* is the least covering template of D (up to renaming), and Proposition 3.21 computes it in polynomial time.*

So $\text{Min}(D)$ cannot be output in polynomial time and the verifier must not enumerate it; but existence of an lgg is decided, and the lgg computed, in polynomial time, and an anchor always has one. This extends the earlier result “ T_{Ind} is the lgg of every induction anchor” to all of DT° . Before an anchor the version space need not have a least element (Example 3.4). The referee’s brute force agreed with Proposition 3.21 on all 115 decided random cases, and in all 63 anchor cases the lgg was the target. Proofs in Appendix B.8.

result	content
Theorem 3.3	every covering template lies above a minimal one; $\text{Min}(D)$ finite; one minimal template lies below the target (earlier C8.3: true for DT° , false with nested arguments)
Example 3.4	4 minimal templates for $\{\text{Ind}(x = x), \text{Ind}(0 = x)\}$ (earlier count 8 corrected)
Theorem 3.5, Corollary 3.6	$\text{Acc}(D) = \text{Feat}(D)$, decided in polynomial time
Theorem 3.8, Corollary 3.9	anchor iff (R^*) , (N) , (U) (“only if” under (Rich)); first-order: $(R)+(D)$; one formula metavariable: $(R)+(N)$
Proposition 3.10	weaker candidate events fail; (Rich) cannot be dropped
Theorem 3.11,	exponential tails for one template and for tagged data; ground targets in-
Proposition 3.12	cluded
Theorem 3.13,	$\text{Esc}(\text{DT}^\circ; N) = \Theta(N^2)$; earlier C11 (linear) refuted; $\leq 2N$ without binders
Propositions 3.14 and 3.15	
Conjecture 3.16	equation kills linear (conjecture)
Corollary 3.17,	finite elasticity, infinite thickness; unions have anchors, explicit ones; union
Propositions 3.18 and 3.19	verifier via partitions
Propositions 3.20 and 3.21,	$ \text{Min}(D) $ up to 4^n ; lgg existence polynomial; an anchor’s lgg is the target
Corollary 3.22	

Table 2: Results of Section 3. All are proved except Conjecture 3.16 (conjecture) and Example 3.4 (computed); Corollary 3.17(ii) uses known results. All were also checked by computation, most also by an independent referee implementation (Appendix B.9).

3.9 Summary, caveats and open problems

For one target in DT° , positive data suffice and checking is cheap: the verifier is sound from the first datum, a polynomial feature test, exact after two well-chosen instances for PA induction and every ZF schema, and fails with exponentially small probability on random data. What one template does not give cheaply is bounded escalation (quadratic in the data size) and, for unions, more than the existence of anchors. Table 2 collects the results.

Caveats. (1) The “only if” half of Theorem 3.8 needs (Rich) , which always holds with parameters. (2) Minimality is with respect to $\succeq$; whether $\text{inst}(T) \supseteq \text{inst}(T')$ implies $T \succeq T'$ in DT° is open, and not needed, since the verifier and the anchor theorem concern instance sets. (3) Parameters are constants: data sharing a parameter at one place give templates that mention it, which is sound but not renaming-invariant; the implementation aligns parameters (Sections 2 and 7). (4) A ground axiom is its own anchor, but in unions it is learned only if it is a datum. (5) Propositions 3.12 and 3.15, Lemma B.2 and the remark on (Rich) were added after the referee report; their proofs were checked again for this paper and by computation, but not by the independent referee.

Open problems. The anchor condition without (Rich) ; Conjecture 3.16; polynomial-delay enumeration and counting of $\text{Min}(D)$; size-bounded subclasses of DT° (in the earlier induction analysis, templates of size ≤ 11 make the cautious verifier accept a false induction-shaped sentence after any anchor, size 12 suffices for soundness^{prior induction C6.2}); lower bounds for the coincidence terms with infinite support.

4 Universal axioms from their instances

The first question is: “does the same thing work for learning an axiom of the form $\forall x \phi(x)$ from a bunch of sentences of the form $\phi(x)$?” “The same thing” is the method developed for PA induction: a cautious verifier over a class of templates, sound at all times and exact once the data contain an anchor (Sections 2 and 3). We read the data as instances $\varphi(t)$ at terms t . (Read literally, with x a free variable, a datum $\varphi(x)$ is in closure-normal form already the axiom $\forall x \varphi$; Proposition 4.11.) The answer has four parts.

- (1) *The instance schema $\varphi(z)$ is learned, easily.* It is a first-order pattern. First-order anti-unification and the DT° learner both learn it, with Plotkin’s anchors (R) and (D); for one variable, two instances such as $\varphi(0)$, $\varphi(S0)$ suffice (Theorem 4.3 and Proposition 4.4). In SO° these anchors fail. Rates have exact formulas (Theorem 4.5). In a first-order syntax that lets a metavariable take a bound variable, *capture* instances, possibly false in every structure, must be excluded by a guard (Proposition 4.7).
- (2) *The instance schema is not the sentence.* “All closed instances are true, hence $\forall x \varphi$ ” is truth-safe for all φ in M iff the closed-term substructure of M is elementary (Proposition 4.8): in $\mathbb{N}$, not in $\mathbb{R}$, V or a nonstandard model of PA. It is not derivable: $\mathbb{Q} \vdash 0 + \underline{n} = \underline{n}$ for each n , but $\mathbb{Q} \not\vdash \forall x (0 + x = x)$ (Example 4.10).
- (3) *A cautious learner takes this ω -step silently* when parameters are admissible values (Proposition 4.6). A closedness guard prevents it; a datum at a parameter licenses it, as ordinary generalization. Evaluating closed instances never refutes a false universal whose closed instances are true (Proposition 4.13).
- (4) *Without labels*, $\forall x \varphi$ is learned by a sound merge of instances (Section 4.7).

The results come from the research record of the track “cases”, checked by an adversarial referee with independent code, and from the tracks “untagged” and “experiments”. Proofs are in Appendix C.

4.1 The target

Fix a first-order language L with at least one constant and an L -formula $\varphi(x_1, \dots, x_k)$, $k \geq 1$, without parameters, in which each x_i occurs free. A term is *closed* if it has neither variables nor parameters.

Definition 4.1 (Instance schema; witness events). ^{cases §1.1} The *instance schema* of φ has a 0-ary term metavariable z_i at each *free* occurrence of x_i : $\sigma_\varphi := \varphi[z_1/x_1, \dots, z_k/x_k]$, a first-order pattern. The *closed instances* are $\text{inst}_c(\varphi) = \{\varphi(\bar{t}) : \bar{t} \text{ closed}\}$, the *open instances* $\text{inst}_o(\varphi) = \{\varphi(\bar{t}) : \bar{t} \text{ terms over } L \cup \text{Par}\}$, with Par the parameters; data are read under universal closure. For data $D = \{\varphi(\bar{t}_1), \dots, \varphi(\bar{t}_N)\}$, $\bar{t}_j = (t_{j,1}, \dots, t_{j,k})$, Plotkin’s events (Section 2) read: (R_{*i*}): the roots of $t_{1,i}, \dots, t_{N,i}$ are not all equal; (D_{*ii'*}) ($i < i'$): some j has $t_{j,i} \neq t_{j,i'}$. (R) and (D) are their conjunctions.

Which sentences are instances of σ_φ depends on the admissible values (Section 2). Under the λ convention, the default, a 0-ary term metavariable takes a term without bound variables, so $\text{inst}(\sigma_\varphi) = \text{inst}_o(\varphi)$ if parameters are admissible and $\text{inst}_c(\varphi)$ if not. In the *named* (N) and *de Bruijn first-order* (dB) syntaxes a metavariable may also take a bound-variable name or index, and $\text{inst}(\sigma_\varphi)$ contains *capture instances* (Section 4.4). The question has content for languages

with closed terms, such as L_A ; in L_∞ every instance is at parameters, and a datum $\varphi(\bar{p})$ with distinct parameters already is the axiom. Several results need closed terms of different shapes:

(Rich_c) for every k there are pairwise distinct closed $t_1, \dots, t_k$ and pairwise distinct closed $t'_1, \dots, t'_k$ with $\text{root}(t_i) \neq \text{root}(t'_i)$ for all i

(in L_A : $t_i = S^i 0$, $t'_i = 0 + S^i 0$; the record calls this (H)).

4.2 Anchors

We use the classical facts recalled in Section 2 (Plotkin, 1970; Reynolds, 1970): a finite nonempty set D has a least general generalization $\text{lgg}(D)$, which covers D and satisfies $\tau \succeq \text{lgg}(D)$ for every first-order schema τ covering D ; and (Plotkin’s recovery lemma) for instances D of a first-order schema σ , $\text{lgg}(D) \equiv \sigma$ iff (R) and (D) hold. $H_1(\text{FO})$ denotes the single first-order schemas of the syntax at hand (λ , N or dB).

Proposition 4.2 (Instance sets determine schemas). *[proved; computed]^{cases A1} Assume (Rich_c). Let σ be a first-order schema with 0-ary term metavariables and τ any first-order schema. Then $\text{inst}(\sigma) \subseteq \text{inst}(\tau)$ iff every instance of σ at closed terms lies in $\text{inst}(\tau)$ iff $\tau \succeq \sigma$, in each syntax.*

Proof. Only the step to $\tau \succeq \sigma$ needs proof. With $\theta(z_i) := t_i$ and $\theta'(z_i) := t'_i$ from (Rich_c), the closed instances $\sigma\theta, \sigma\theta'$ satisfy (R) and (D), so $\text{lgg}(\sigma\theta, \sigma\theta') \equiv \sigma$; τ covers both, so $\tau \succeq \sigma$. $\square$

Theorem 4.3 (Anchors for the instance schema). *[proved; computed]^{cases Thm A2} Assume (Rich_c), and let $D = \{\varphi(\bar{t}_1), \dots, \varphi(\bar{t}_N)\} \subseteq \text{inst}_c(\varphi)$, $N \geq 1$. In each syntax the following are equivalent: (i) D is an anchor for $\text{inst}_c(\varphi)$ in $H_1(\text{FO})$, i.e. every first-order schema covering D contains $\text{inst}_c(\varphi)$; (ii) $\text{lgg}(D) \equiv \sigma_\varphi$; (iii) (R) and (D) hold. Without (Rich_c), (iii) $\Leftrightarrow$ (ii) $\Rightarrow$ (i).*

Proof. (ii) $\Leftrightarrow$ (iii) is the recovery lemma with $\theta_j(z_i) := t_{j,i}$. (ii) $\Rightarrow$ (i): a covering τ has $\tau \succeq \text{lgg}(D) \equiv \sigma_\varphi$. (i) $\Rightarrow$ (ii): $\text{lgg}(D)$ covers D , hence contains $\text{inst}_c(\varphi)$, hence $\text{lgg}(D) \succeq \sigma_\varphi$ by Proposition 4.2; and $\sigma_\varphi \succeq \text{lgg}(D)$ as σ_φ covers D . Only closed instances occur, so the syntax does not matter. $\square$

Corner cases ^{cases §1.3; experiments E1}. Repeated occurrences of x_i give equal columns and need no further event. Several variables need (D): on diagonal data $\varphi(t, t)$ the lgg for $x + y = y + x$ is $z + z = z + z$, a sound specialization. Closed subterms of φ and bound occurrences of the name x_i give constant columns, which merge with an x_i -column only if (R) fails: for $x + S0 = Sx$, data terms $S0$ and 0 recover σ_φ , while $S0$ and $SS0$ give $S(z) + S0 = SS(z)$. One datum is never an anchor; two with componentwise different roots and pairwise distinct components are. Numerals have roots 0 and S only, so from numeral data every anchor of $z + 0 = z$ contains $0 + 0 = 0$, which is also an instance of $0 + z = z$; this slows the rates and matters for unlabelled mixtures (Section 6). *Evidence* ^{cases q1_lgg (1), (2); referee r1}: on 141 736 data sets (pairs and triples from 12 closed terms, six formulas) “ $\text{lgg} \equiv \sigma_\varphi$ iff (R) $\wedge$ (D)” held throughout; the referee’s code agreed on 36 000 data sets from 600 random formulas with binders.

The induction method uses the much larger class DT° . For first-order targets it needs no more data; without determinacy it does.

Proposition 4.4 (Anchors in DT° and in SO°). *[proved; computed]^{cases A2'; single D.1; untagged 5.5} Work under the λ convention.*

- (a) Let $\sigma \in \text{FO}$ (0-ary metavariables of term or formula sort, possibly under binders) and $D = \{\sigma\theta_1, \dots, \sigma\theta_N\}$ with values without bound variables (parameters allowed). If (R) and (D) hold, every $T \in \text{DT}^\circ$ covering D satisfies $T \succeq \sigma$. In particular, data $D \subseteq \text{inst}_o(\varphi)$ with (R) and (D) are an anchor for $\text{inst}(\sigma_\varphi)$ in $H_1(\text{DT}^\circ)$, and $\text{Min}(D) = \{\sigma_\varphi\}$.

(b) Under (Rich_c) , if $D \subseteq \text{inst}_c(\varphi)$ violates (R) or (D), then $\text{lgg}(D) \in \text{FO} \subseteq \text{DT}^\circ$ covers D and misses a closed instance. So the anchors for $\text{inst}_c(\varphi)$ in $H_1(\text{DT}^\circ)$ are exactly those of Theorem 4.3.

(c) In SO° this fails: for $\varphi = (x + 0 = x)$ the data $0 + 0 = 0$, $S0 + 0 = S0$ satisfy (R), but $f(S0) + f(0) = f(S0)$, with $f : \iota \rightarrow \iota$ and no pattern occurrence, covers them ($f := \lambda h.0$, resp. $\lambda h.h$) and misses $SS0 + 0 = SS0$.

Proof idea. (a) By (R), the rigid skeleton of a covering T lies in that of σ (rigid-prefix lemma, Section 3). A pattern occurrence $M(\bar{u}^1)$ at c_1 determines a body B_M : $\sigma|_{c_1}$ with $\bar{u}^1$ abstracted. At another occurrence $M(\bar{u}^s)$ at c_s , $B_M[\bar{u}^s]$ and $\sigma|_{c_s}$ agree on all data; an identity lemma (Lemma C.1) shows that templates in which the metavariables of σ occur only as leaves and which agree at values satisfying (R) and (D) are equal. So $T[M := \lambda \bar{h}.B_M] = \sigma$. (b) A failed event survives in $\text{lgg}(D)$. (c) A body with $\beta[0] = 0$ is h or 0 , so $\beta[S0] \in \{S0, 0\}$. Details in Appendix C.1. $\square$

Part (c) has analogues with binders (e.g. $\forall y (G(0, S0) + G(y, y) = y + F(y))$ for $\forall y (x + y = y + x)$, Appendix C.1): a metavariable without a pattern occurrence can use different arguments on different data, the phenomenon that also makes SO° anchors for induction stronger (Section 5). For one variable, (a) is the *two-point lemma* used in Section 6; in general it is the “if” half of the first-order case of the anchor theorem of Section 3, proved directly. *Evidence*: a bounded search over covering DT° templates agreed with (R) $\wedge$ (D) on 99/99 pairs (five formulas, three with binders), the referee’s on 60/60, and every formula had an SO° non-anchor with the DT° events `cases q1_dt, q1_lgg (5); referee r1 (3), r5`. The search is bounded; the statement is proved.

4.3 Rates

Let $\bar{t} \sim \Lambda$ i.i.d., $p_{i,f} := \Lambda(\text{root}(t_i) = f)$, $c_{ii'} := \Lambda(t_i = t_{i'})$, and let P_N be the probability that N instances violate (R) or (D), i.e. (under (Rich_c)) are not an anchor in $H_1(\text{FO})$ or $H_1(\text{DT}^\circ)$.

Theorem 4.5 (Failure probabilities). *[proved; computed]^{cases Thm A3}* (a) For $k = 1$, $P_N = \sum_f p_f^N$. (b) For $k = 2$, $P_N = \sum_f a_f^N + \sum_g b_g^N + c^N - \sum_{f,g} d_{fg}^N - \sum_f e_f^N$, with $a_f = \Lambda(\text{root}(t_1) = f)$, $b_g = \Lambda(\text{root}(t_2) = g)$, $c = \Lambda(t_1 = t_2)$, $d_{fg} = \Lambda(\text{root}(t_1) = f, \text{root}(t_2) = g)$, $e_f = \Lambda(\text{root}(t_1) = f, t_1 = t_2)$. For independent components with term law μ and root law p : $P_N = 2S - S^2 + c^N - C$, $S = \sum_f p_f^N$, $C = \sum_f c_f^N$, $c_f = \sum_{\text{root}(t)=f} \mu(t)^2$. (c) In general $P_N \leq \sum_{i,f} p_{i,f}^N + \sum_{i < i'} c_{ii'}^N \leq (2k + \binom{k}{2})(1 - \rho)^N \leq (2k + \binom{k}{2})e^{-N\rho}$, where $\rho = \min(\min_i \rho_i, \min_{i < i'} \Lambda(t_i \neq t_{i'}))$ and $\rho_i = \max_G \min(\Lambda(\text{root}(t_i) \in G), \Lambda(\text{root}(t_i) \notin G))$ is the best split of the roots, as in the coupon-collector bound of Claude (Anthropic) (2026).

The proof is inclusion–exclusion over the events $\neg(R_i)$, $\neg(D_{ii'})$ and a union bound (Appendix C.2). For a Galton–Watson law with root law $0:.4, S:.3, +:.2, \cdot:.1$, the exact values at $N = 2, 4, 6, 10$ are .300, .035, .0049, .00011 ($k = 1$) and .52, .070, .0098, .00022 ($k = 2$), in agreement with 20 000 Monte Carlo runs; the exact failure probability is below 0.01 from $N = 6$, while the bound (c) needs $N = 11$ and 13 `cases q1_lgg (4); referee r1 (5)`.

Experiment E1 `experiments E1, Prop E7; code/results/e1_universal.md`. The implemented DT_F° verifier and first-order lgg, run on instances of $x + 0 = x$, $x \cdot 0 = 0$, $0 + x = x$, $\neg Sx = 0$, $x < Sx$ and $x + y = y + x$ (2000 runs each), became exact at the same N in every run. With mixed data (numerals 0..7, random closed terms, parameters) the mean N until exact (runs capped at 40 instances) was 3.27 against 3.25 predicted by (a); with numerals only, 8.42 against 8.11 (1.85 standard errors; the five one-variable targets share their runs, since exactness depends only on heads, so this is one

sample; the referee's simulation of the head process gives 8.117). For $x + y = y + x$: 4.18 and 11.85. The experiments section (Section 7) plots the curves. On $0 + 0 = 0$, $1 + 0 = 1$ the learned $z + 0 = z$ accepts $w_0 + 0 = w_0$, the axiom in parameter form, but not the string $\forall x (x + 0 = x)$, which is a different datum (Section 2).

4.4 Parameters, guards and capture

Write $\text{cl}_{\mathcal{H}}(h^*) = \bigcap \{h \in \mathcal{H} : h \supseteq h^*\}$. Once the data contain an anchor, the cautious verifier accepts exactly this closure, so it is sound iff the target is closed in the class (Section 2).

Proposition 4.6 (Open instances and the hidden ω -step). *[proved; computed]^{cases Prop A4} (a) For distinct parameters $\bar{p}$, $\varphi(\bar{p}) \in \text{inst}_o(\varphi)$ has universal closure $\forall \bar{x}\varphi$; $\text{inst}_o(\varphi)$ and $\forall \bar{x}\varphi$ are inter-derivable in first-order logic (instantiation, resp. generalization). (b) Assume (Rich_c) and the λ convention with parameters admissible. In $H_1(\text{FO})$ and in $H_1(\text{DT}^\circ)$, $\text{cl}(\text{inst}_c(\varphi)) = \text{inst}(\sigma_\varphi) = \text{inst}_o(\varphi) \supsetneq \text{inst}_c(\varphi)$. So once closed data contain an anchor, the cautious verifier accepts $\varphi(\bar{p})$: it takes the ω -rule step “all closed instances, hence $\forall \bar{x}\varphi$ ”, which no datum licenses. In (N) and (dB) the closure $\text{inst}(\sigma_\varphi)$ also contains the capture instances of Proposition 4.7. (c) With the guard $\text{closed}(z_i)$ (“the value contains no variable and no parameter”), $\text{inst}_c(\varphi)$ is realizable in every syntax. By the most-specific-guard rule (Section 2), closed data teach the guard, and the learner accepts neither $\varphi(\bar{p})$ nor any capture instance; one datum with a parameter at z_i deletes $\text{closed}(z_i)$.*

Proof in Appendix C.3; the key point of (b) is that every class member containing $\text{inst}_c(\varphi)$ contains two closed instances forming an anchor. *Computed*: from $0 + 0 = 0$, $0 + S0 = S0$ the guarded learner learns $0 + z = z$ with $\text{closed}(z)$ and rejects $0 + p = p$; after the datum $0 + (p + S0) = p + S0$ it accepts $0 + q = q$, i.e. $\forall x (0 + x = x)$ ^{cases q1_lgg (3)}. The class DT° has no guards, so the DT° learner on closed data takes the ω -step after the first anchor whenever parameters are admissible, as E1 shows.

In a first-order syntax the step can be worse ^{cases referee F1}. A value $\theta(z_i)$ is *captured* if it contains a variable bound at some free occurrence of x_i (in (N) , a name in the scope of a binder of that name; in (dB) , an index that refers to an enclosing binder). $\text{Cap}(\varphi)$ is the set of instances $\sigma_\varphi\theta$ with a captured value.

Proposition 4.7 (Capture). *[proved; computed]^{cases Prop A4'} Work in syntax (N) or (dB) .*

- (a) $\text{inst}(\sigma_\varphi) = \text{inst}_c(\varphi) \cup \text{inst}_o(\varphi) \cup \text{Cap}(\varphi)$ (the middle term if parameters are admissible), and $\text{Cap}(\varphi) \cap \text{inst}_o(\varphi) = \emptyset$.
- (b) $\text{Cap}(\varphi) \neq \emptyset$ iff: in (N) with sentences as data, some name is bound at every free occurrence of some x_i ; in (N) with free names read as parameters, some free occurrence of some x_i is in the scope of a binder; in (dB) , every free occurrence of some x_i lies under a binder.
- (c) Capture instances can be false in every structure while all genuine instances hold in every structure with two elements ($\exists y \neg(y = x)$ gives $\exists y \neg(y = y)$), or false in $\mathbb{N}$ while $\forall x\varphi$ is logically valid ($\exists y (y = Sx)$ gives $\exists y (y = Sy)$).
- (d) Hence, if $\text{Cap}(\varphi) \neq \emptyset$, the unguarded cautious verifier accepts after an anchor sentences outside $\text{inst}_o(\varphi)$, for suitable φ some false in every structure: it is unsound outright, not merely ω -incomplete.
- (e) With guards $\text{closed}(z_i)$ and $\text{nocap}(z_i)$ (no captured value: the free-for condition, i.e. the λ convention as a guard), closed data teach both, and after an anchor the accepted set is

$\text{inst}_c(\varphi)$. A datum at a parameter deletes $\text{closed}(z_i)$ but not $\text{nocap}(z_i)$, and the accepted set becomes $\text{inst}_o(\varphi)$. Capture instances are never accepted.

Proof in Appendix C.3. Under the λ convention $\text{Cap}(\varphi)$ does not arise. The de Bruijn *first-order* encoding does not prevent capture, which there happens by index $(\exists \neg(\#0 = z))$ with $z := \#0$. *Computed* cases q1_capture; referee F1, F2: for $\exists y \neg(y = x)$, $\text{lbg}(\varphi(0), \varphi(S0)) = \exists y \neg(y = z)$ has the instance $\exists y \neg(y = y)$, false in all equality structures of sizes 1–5; for seven formulas the decomposition of (a) and the criterion of (b) were checked on all substituted terms of size ≤ 3 ; the guards of (e) behave as stated, while the unguarded verifier accepts $\exists y \neg(y = y)$.

4.5 The ω -gap

For an L -structure M let $M_0 := \{t^M : t \text{ closed}\}$ be its *closed-term substructure*.

Proposition 4.8 (When closed instances decide universals). *[proved; known in substance]* cases Prop A5 For an L -structure M the following are equivalent: (i) for every L -formula $\varphi(\bar{x})$, $M \models \forall \bar{x} \varphi$ iff $M \models \varphi(\bar{t})$ for all closed $\bar{t}$; (ii) $M_0 \preceq M$. In particular (i) holds for term-generated M ($M_0 = M$), e.g. $\mathbb{N}$, and for every model of true arithmetic.

Proof. (ii) $\Rightarrow$ (i): if all $\varphi(\bar{t})$ hold in M , they hold in M_0 by elementarity, and the elements of M_0 are the t^M ; so $M_0 \models \forall \bar{x} \varphi$, hence $M \models \forall \bar{x} \varphi$, again by elementarity. The converse direction of (i) is $\forall$ -elimination. (i) $\Rightarrow$ (ii) is the Tarski–Vaught criterion (Tarski and Vaught, 1957): if $M \models \exists x \psi(x, \bar{s})$ with $\bar{s}$ closed but no closed t has $M \models \psi(t, \bar{s})$, then (i) for $\neg \psi(x, \bar{s})$ gives $M \models \forall x \neg \psi(x, \bar{s})$, a contradiction. In a model of true arithmetic $M_0 \cong \mathbb{N}$, and M and $\mathbb{N}$ satisfy the same sentences with numerals, so $M_0 \preceq M$. $\square$

Example 4.9 (Where the step fails). *[proved]* cases Ex. A6, A7, A7' (a) In $\mathbb{R}$ as an ordered field (with $0, 1, +, -, \cdot, <$), closed terms denote integers, so $\neg(t \cdot t = 1 + 1)$ holds for every closed t , but $\forall x \neg(x \cdot x = 1 + 1)$ fails ($\sqrt{2}$). The theory RCF of real closed fields refutes it; the ordered-field axioms OF do not ($\mathbb{Q}$). (b) Sets with $\emptyset$, $p(a, b) = \{a, b\}$, $u(a, b) = a \cup b$: closed terms denote exactly the hereditarily finite sets. With $\text{Ind}(x) := \emptyset \in x \wedge \forall y \in x u(y, p(y, y)) \in x$ (a Δ_0 formula), every closed instance of $\neg \text{Ind}(x)$ is true in V , and $\forall x \neg \text{Ind}(x)$ is refuted by Infinity. (c) If $M \models \text{PA} + \neg \text{Con}(\text{PA})$ (such M exist if PA is consistent) and Prf is a Δ_0 proof predicate, every closed instance $\neg \text{Prf}(t, \ulcorner \perp \urcorner)$ is a true Δ_0 sentence and holds in M , while $M \models \exists x \text{Prf}(x, \ulcorner \perp \urcorner)$. Even within arithmetic the step is truth-safe only in the right model.

Example 4.10 (Not derivable). *[proved; computed]* cases Ex. A8 $\mathbb{Q} \vdash 0 + \underline{n} = \underline{n}$ for every n (Q4, then Q5 and induction on n outside Q). But Q plus all closed instances of $0 + x = x$ does not prove $\forall x (0 + x = x)$: there is a model of Q on $\mathbb{N} \cup \{a, b\}$, standard on $\mathbb{N}$, with $Sa = a$, $Sb = b$ and $0 + a = b$ (Appendix C.4).

That $\mathbb{Q} \not\models \forall x (0 + x = x)$ is well known; the model is from the research record, checked by hand case by case and by machine cases q1_qmodel; referee r1 (4). With data at parameters the gap disappears:

Proposition 4.11 (Parameters and generalization). *[proved]* cases Prop A9 In closure-normal form a datum $\varphi(\bar{p})$ with distinct parameters (none occurs in φ) is the sentence $\forall \bar{x} \varphi$, its universal closure; with trusted generalization $\forall \bar{x} \varphi$ is derived from it in one step.

syntax, guards and data	after an anchor the cautious verifier accepts
λ convention, no parameters admissible	$\text{inst}_c(\varphi)$: exact
λ convention with parameters (FO or DT°), closed data	$\text{inst}_o(\varphi) \ni \varphi(\bar{p})$: the ω -step; truth-sound iff $\forall \bar{x}\varphi$ is true
first-order schemas with guard	$\text{inst}_c(\varphi)$: exact, no ω -step
closed(z_i), closed data	
a datum at a parameter (λ , or (N)/(dB) with nocap)	$\text{inst}_o(\varphi)$: exact for the target $\text{inst}_o(\varphi)$; $\forall \bar{x}\varphi$ by generalization
(N) or (dB) without guards, $\text{Cap}(\varphi) \neq \emptyset$	$\text{inst}(\sigma_\varphi) \supseteq \text{Cap}(\varphi)$: unsound; capture instances can be false in every structure

Table 3: What the cautious single-schema verifier accepts once genuine data contain an anchor ((Rich_c) assumed). Rows 2–5: Proposition 4.6(b), (c), Proposition 4.7(e), (d).

4.6 Consequences for a learner

Proposition 4.12 (What is soundly learnable). *[proved]^{cases A10} (1) From closed instances the soundly learnable object is $\text{inst}_c(\varphi)$, with anchors $(R) \wedge (D)$ and the rates of Theorem 4.5. If parameters are admissible values, a learner sound for it needs a closedness guard; in (N) and (dB) it also needs nocap whenever some free x_i lies under a binder (automatic under the λ convention). (2) The ω -step is truth-safe in the intended M for every φ iff $M_0 \preceq M$ (Proposition 4.8). (3) It is derivable from the instances only by an ω -rule (Example 4.10). (4) If the community uses instances at parameters, as proof-assistant practice does when it applies $\forall x, \varphi x$ to a variable, the target is $\text{inst}_o(\varphi)$ and $\forall \bar{x}\varphi$ is obtained soundly (Proposition 4.11); in (N) and (dB) nocap is still needed.*

Table 3 summarizes (1) and (4). Refutation by the world (Section 2) helps only partly:

Proposition 4.13 (Refutation and the ω -gap). *[proved]^{cases A11} Let $\forall \bar{x}\varphi$ be false in the intended M while all closed instances are true. (a) No refutation that ends by evaluating closed instances condemns $\forall \bar{x}\varphi$. (b) A coherence refutation from designated true premises Γ exists iff $\Gamma \cup \{\forall \bar{x}\varphi\}$ is inconsistent (RCF and ZF refute the examples of Example 4.9(a),(b); OF does not refute (a)). (c) In a term-generated M such as $\mathbb{N}$ this cannot happen (Proposition 4.8); if the channel decides the closed instances of φ (as a full Δ_0 evaluator does for Δ_0 formulas), search finds a false one. (d) Capture instances are refuted by pure logic when logically false; when false only in M , a designated theory must prove the negation: $\mathbb{Q} \not\models \forall y \neg(y = Sy)$ ($Sa = a$ in the model of Example 4.10), PA proves it. No evaluation of closed instances bears on them.*

So where the ω -step is truth-safe a false universal has a false closed instance; elsewhere ($\mathbb{R}$, V) a wrong step is caught only through the designated theory, and sometimes not at all. Against capture the world channel alone does not protect an unguarded first-order learner; the guard, or the λ convention, does.

4.7 Without labels: $\forall x\varphi$ as a sound merge

In the untagged setting of Section 6, unlabelled data from several targets are clustered by DTRC: two clusters merge iff their union is *d-coherent*, i.e. some minimal covering DT° template of it has no instance refuted by the world oracle within budget d ; each final cluster gets its own cautious verifier. Regard each instance $\varphi(t_j)$ as its own ground target; learning $\forall x\varphi$ then means merging them, across “targets”.

Proposition 4.14 ($\forall x\varphi$ as a sound merge). *[proved; computed]^{untagged Prop 5.6 (U7)} Let φ have one variable and $D = \{\varphi(t_1), \dots, \varphi(t_n)\}$ be closed instances, two with different heads. (a) $\text{Min}(D) = \{\sigma_\varphi\}$, so D is d -coherent iff σ_φ has no d -refuted instance. (b) $\text{inst}(\sigma_\varphi) \ni \varphi(a)$ for a parameter a , which is $\forall x\varphi$; so all instances are true iff $\forall x\varphi$ is. (c) If $\forall x\varphi$ is true, every subset of D is coherent, and DTRC forms one cluster under every merge policy and accepts $\text{inst}(\sigma_\varphi) \ni \varphi(a)$: the axiom is learned, nothing false is accepted. Soundness for the n ground “targets” fails, as it must: instances and axiom give the same data and oracle answers. (d) If $\forall x\varphi$ is false and some instance is d -refuted, D is not merged; data whose terms share a head may still merge into an unrefuted specialization such as $\varphi(Sz)$. (e) “Never refuted” means “ $\forall x\varphi$ true” if the oracle refutes every false instance at some depth. $R\text{-}\Delta_0$ does so for quantifier-free φ : a false instance of an arithmetic template without binders in its rigid part has a false quantifier-free instance, which $R\text{-}\Delta_0$ refutes (Lemma C.2). It need not for quantified φ , since it verifies universals only by equational derivations from true closed equations. At a finite budget a false universal with a large least counterexample can merge (the residue of Section 6). An oracle that evaluates only closed instances certifies “true at every closed term”: in $\mathbb{N}$ the ω -rule; in the field $\mathbb{R}$, $\neg(z \cdot z = 1 + 1)$ is an unsound merge it never refutes (a decision procedure for RCF would).*

Proofs in Appendix C.6. The record corrected (b) and (e) after the referee: a must be fresh, and (e) needs an oracle complete for the false instances. *Computed^{untagged u6}*: numeral instances of $x + 0 = x$, $x \cdot 0 = 0$, $x + Sy = S(x + y)$, as 12 ground targets, form three clusters, and DTRC accepts $a + 0 = a$, i.e. $\forall x(x + 0 = x)$; instances of the false $\forall x(x \cdot x = x)$ at 0, 1 stay apart ($z \cdot z = z$ is refuted at 2); instances of $\neg(x \cdot x = 2)$ at 0, 1, 2 merge. What certifies the universal is the absence of a counterexample, not a derivation.

4.8 Summary

For the instance schema the answer is yes: it is a first-order pattern, its anchors are Plotkin’s (R) and (D) in the first-order class and in DT° , two instances suffice, and rates have exact formulas. For the axiom the answer is “only where $M_0 \preceq M$ ”: the passage from all closed instances to $\forall \bar{x}\varphi$ is an ω -rule step, truth-safe exactly then and not derivable in general. A cautious learner takes it silently when parameters are admissible; a closedness guard stops it, and data at parameters license it as generalization. In a first-order syntax that lets metavariables take bound variables, capture makes the unguarded learner unsound outright, and a learned nocap guard repairs it. Without labels, $\forall x\varphi$ is learned by a sound merge. Open^{cases §6}: the exact SO° anchor condition for universal axioms over languages with closed terms, an analogue of the shielding condition for induction (Section 5); only counterexamples are known.

5 The ZF(C) schemas, and PA induction as the contrast

The second question asks whether the method developed for PA induction also learns each axiom schema of ZFC from a collection of its instances. The answer is yes, and more easily than for induction. Every ZFC axiom other than Separation and Replacement is a single sentence, its own anchor. Every ZF schema, in each textbook formulation considered here, is a higher-order pattern (Miller, 1991): its one metavariable is applied only to distinct bound variables. Two instances are an anchor iff their bodies have different main symbols and jointly use every argument place, in every class between PAT and SO° (Theorem 5.10), and pattern anti-unification recovers each schema from any anchor (Corollary 5.12). Plain first-order anti-unification works iff the metavariable is applied to literally the same argument tuple at every occurrence, which depends

on the formulation and on how bound variables are written (Theorem 5.3). Where it fails, no finite union of first-order schemas covers the schema without a false member (Theorem 5.7). PA induction is the contrast. It is not a pattern, and first-order and pattern anti-unification are both unsound on it. The determinate class DT° learns it, and there the anchor condition is again root variation plus non-vacuity (Theorem 5.18); without determinacy a stronger condition is needed (Proposition 5.20). The difference is structural: $L_\in$ has no function symbols, so an instance of a ZF schema only renames variables inside the body, while induction substitutes the terms 0 and Sx .

Section 5.1 fixes formulations and encodings, Sections 5.2 and 5.3 treat first-order learners, Section 5.4 gives anchor theorems and rates for pattern targets, Section 5.5 treats induction, and Table 6 summarizes. Proofs and the computational evidence are in Appendix D. Untagged mixtures of these schemas are treated in Section 6.

Conventions. Data are instances in closure-normal form (Section 2): the textbook outer prefix $\forall \bar{w}$ is dropped (its variables become parameters), while the frame’s own variables stay bound. For a class H , a finite $D \subseteq \text{inst}(T^*)$ is an *anchor in H* if every $T \in H$ with $D \subseteq \text{inst}(T)$ has $\text{inst}(T) \supseteq \text{inst}(T^*)$; the cautious verifier over H is then exact, and it is sound before whenever T^* is closed in H (Section 2). We use two facts about first-order anti-unification (Plotkin, 1970; Reynolds, 1970). Every first-order schema covering a finite set P is at least as general as $\text{lgg}(P)$, so “every covering schema has a false instance” means “ $\text{lgg}(P)$ has one”. And the lgg gives equal columns of disagreeing subterms equal metavariables. A *guarded* first-order schema (σ, G) admits only substitutions that satisfy the guards G . Here these are freshness guards $\text{fresh}(v, Z)$ (“ v is not free in the value of Z ”) or loose-index guards $\text{loose}(Z) \subseteq I$. The *most specific guard rule* learns, together with $\text{lgg}(D)$, every such guard that all data satisfy (Claude (Anthropic), 2026).

5.1 Formulations and encodings

ZFC consists of Extensionality, Pairing, Union, Power set, Infinity, the schemas of Separation and Replacement, Foundation and Choice (and Empty set in some texts). Collection and $\in$ -induction are theorem schemas of ZF [known]: Collection via ranks or reflection (see e.g. Lévy, 1979; we did not check the exact location), $\in$ -induction via transitive closures and Foundation. Conversely, $\in$ -induction with $\varphi(x) := x \notin S$ proves Foundation for S . We include both because texts and libraries state them as schemas. Table 4 lists the templates. The Kunen- and Jech-style wordings follow Kunen (1980, ch. I) and Jech (2003, ch. 1) from memory; an independent referee’s recollection agrees, but we did not re-check them against the books. As far as we know Kunen treats $\exists!$ as an abbreviation, so $\text{Repl}_{\exists!}$, with $\exists!$ as a primitive binder, is our encoding choice.

Proposition 5.1 (Single axioms). *[proved]^{cases §2.1; single Cor D.3} Each ZFC axiom other than the two schemas, Choice included, is a single sentence s . Its target $\{s\}$ lies in every class, $\{s\}$ is an anchor in every class, and matching is equality. Proof: every template covering $\{s\}$ contains the whole target.*

Definition 5.2 (Encodings of bound variables). [known]^{cases §1} (a) *Named first-order*: bound variables are names, treated as constants. Instantiation is textual, so a value may contain a name that a binder of the frame captures. Unless said otherwise the data use the textbook names of Table 4; under α -variation the frame’s binders get arbitrary distinct names. (b) *de Bruijn first-order*: bound variables are indices $\#k$ ($\#0$ is the innermost binder). A metavariable at binder depth d may take a formula with loose indices $< d$, and only well-formed results are instances.

name	template T^* (closure-normal form)	arguments of P
Sep (Kunen)	$\forall z \exists y \forall x (x \in y \leftrightarrow x \in z \wedge P(x, z))$	(x, z)
Sep _J (Jech)	$\forall X \exists Y \forall u (u \in Y \leftrightarrow u \in X \wedge P(u))$	(u)
EInd	$\forall x (\forall y (y \in x \rightarrow P(y)) \rightarrow P(x)) \rightarrow \forall x P(x)$	$(y), (x), (x)$
Coll	$\forall A (\forall x (x \in A \rightarrow \exists y P(x, y, A)) \rightarrow \exists Y \forall x (x \in A \rightarrow \exists y (y \in Y \wedge P(x, y, A))))$	(x, y, A) twice
Repl _{∃!}	as Coll, with $\exists!y$ in place of the first $\exists y$	(x, y, A) twice
Repl _{sp}	$\forall A (\forall x (x \in A \rightarrow \exists y (P(x, y, A) \wedge \forall u (P(x, u, A) \rightarrow u = y))) \rightarrow \exists Y \forall x (x \in A \rightarrow \exists y (y \in Y \wedge P(x, y, A))))$	$(x, y, A), (x, u, A), (x, y, A)$
Repl _J (Jech)	$\forall x \forall y \forall u (P(x, y) \wedge P(x, u) \rightarrow y = u) \rightarrow \forall X \exists Y \forall y (y \in Y \leftrightarrow \exists x (x \in X \wedge P(x, y)))$	$(x, y), (x, u), (x, y)$
Found	$\forall S (\exists x x \in S \rightarrow \exists x (x \in S \wedge \forall y (y \in x \rightarrow \neg y \in S)))$	ground

Table 4: The ZF schemas as templates with one formula metavariable P (closure-normal form). Plugging a body $\lambda \bar{z}. \varphi$ whose free variables are holes and parameters gives a sentence.

(c) λ (*pattern*): templates as in Section 2. Values are bodies whose free variables are holes and parameters, so capture cannot occur and freshness is built in.

5.2 Which ZF schemas are first-order patterns

Let $T^* = F[P(\bar{a}_1), \dots, P(\bar{a}_m)]$ be a pattern template over $L_{\in}$ with a metavariable-free, parameter-free frame F and one metavariable P of arity n , applied at its k -th occurrence to distinct bound variables $\bar{a}_k$. Let N_k be the tuple of *names* of $\bar{a}_k$ (named encoding) and I_k the tuple of their *indices* at occurrence k (de Bruijn).

Theorem 5.3 (Classification). *[proved]^{cases Thm B1} (a) $T^* \in \text{PAT} \subseteq \text{DT}^\circ$. (b) With textbook names, $\text{inst}(T^*)$ is the instance set of a single first-order schema with freshness guards iff $N_1 = \dots = N_m$. The schema is then $F[Z, \dots, Z]$, with $\text{fresh}(v, Z)$ for each frame-bound name v not in N_1 . (c) In de Bruijn notation the same holds with $I_1 = \dots = I_m$ and the guard $\text{loose}(Z) \subseteq I_1$. Well-formedness implies this guard iff $I_1 = \{0, \dots, d-1\}$, where d is the least binder depth of an occurrence.*

Proof idea. If the tuples are equal, every occurrence reads the same text. If two differ, take the instances with bodies $\bigwedge_i (z_i = z_i)$ and its negation. Their lgg has different metavariables at the two occurrences. Filling one with the closed $\top$ and the others with $\perp$ respects every freshness guard, but no body produces the result (Appendix D.2). $\square$

Table 5 applies the criterion. It corrects two natural expectations. $\in$ -induction is a de Bruijn first-order pattern, since all its occurrences read $\#0$. And freshness is automatic only in the λ encoding, not in the de Bruijn first-order one.

Proposition 5.4 (α -variation). *[proved]^{cases Prop B1 α} Under α -variation (with name metavariables and guards fresh and distinct), the α -variants of instances of T^* form the instance set of a single guarded first-order schema iff all occurrences of P are applied to the same tuple of binders. This is true for Sep and Sep_J, and false for EInd, Coll, Repl_{∃!}, Repl_{sp} and Repl_J.*

Proposition 5.5 (What the first-order lgg does). *[proved]^{cases Prop B2} Suppose the bodies' main symbols are not all equal. Then the first-order lgg (either encoding, textbook names) is F with*

schema	named first-order (textbook names)	de Bruijn first-order	named, α -varied
Sep	yes, guard $\text{fresh}(y, Z)$	yes, guard “#1 not loose” (not implied)	yes
Sep _J	yes, guards $\text{fresh}(X, Z)$, $\text{fresh}(Y, Z)$	yes, guard $\text{loose}(Z) \subseteq \{\#0\}$ (not implied)	yes
EInd	no (y vs. x)	yes , guard implied by well-formedness	no
Coll, Repl _{∃!}	yes, guard $\text{fresh}(Y, Z)$	no (A is #2 vs. #3)	no
Repl _{sp} , Repl _J	no (y vs. u)	no	no
Coll, Repl _{∃!} , $\varphi(x, y)$ only	yes, guards $\text{fresh}(Y, Z)$, $\text{fresh}(A, Z)$	yes, guard $\text{loose}(Z) \subseteq \{\#0, \#1\}$ (not implied)	no

Table 5: Is the schema a guarded first-order pattern? (By Theorem 5.3 and Proposition 5.4; every schema is in PAT.) Without its guard, every guarded entry has a false instance (Proposition 5.6).

a metavariable at each P -position, and positions k, l share it iff no body uses an argument place at which $\bar{a}_k$ and $\bar{a}_l$ differ. If all positions share one metavariable, the lgg with its most specific guards accepts only instances of T^* .

So on anchor data the first-order lgg decouples exactly the occurrences that Theorem 5.3 cannot tie.

5.3 Where first-order anti-unification fails

Let $\top := \forall w (w = w)$ and $\perp := \neg \top$.

Proposition 5.6 (Two instances give a false lgg instance). *[proved]^{cases §2.4} In each of the following cases two genuine instances give an lgg with a false sentence instance that respects the most specific guards and is refuted in ZF. (The complete list is in Appendix D.3. For Repl_{sp} in the named encoding with textbook names there is no such pair; see Proposition 5.8.) (i) EInd, named. From $\text{EInd}(x = x)$ and $\text{EInd}(\neg x \in a)$ the lgg is $\forall x (\forall y (y \in x \rightarrow Z_1) \rightarrow Z_2) \rightarrow \forall x Z_2$. Its instance $\forall x (\forall y (y \in x \rightarrow \neg y = y) \rightarrow \forall w \neg w \in x) \rightarrow \forall x \forall w \neg w \in x$ has a valid antecedent and says that every set is empty. (ii) Repl_J, named. From $\text{Repl}_J(x \in y)$ and $\text{Repl}_J(\neg x = a)$ the occurrence $P(x, u)$ gets its own metavariable. Filling it with $\neg u = u$ and the others with $y = y$ gives a valid antecedent and a universal set at $X = \{\emptyset\}$. (iii) Coll, Repl_{∃!}, Repl_{sp}, de Bruijn or α -varied. From the bodies $y = x$ and $\neg x = A$ the consequent’s occurrence is decoupled. With $y = x$ in the antecedent ($\perp$ at Repl_{sp}’s uniqueness occurrence) and $\perp$ in the consequent, the instance is equivalent to $\forall A \neg \exists x x \in A$. (iv) Guard violations. Without its guard, Separation’s lgg has Russell’s instance (Proposition 5.9). Without $\text{fresh}(Y, \cdot)$, the named lgg of Coll, Repl_{∃!} and Repl_{sp} have the instance with body $y = Y$ (for Repl_{sp}, $\neg u = u$ at the uniqueness occurrence), which forces $Y \in Y$.*

Theorem 5.7 (No finite union of first-order schemas). *[proved]^{cases Lemma C1, Thm C2} In each case below, every finite set of guarded first-order schemas whose instances include all instances of the schema has a false instance that is refutable in ZF: (i) EInd, named; (ii) Repl_J, both encodings; (iii) Coll, Repl_{∃!}, Repl_{sp}, de Bruijn; (iv) Coll, Repl_{∃!}, Repl_{sp}, named under α -variation.*

Proof idea. Suppose a schema covers a sentence and has fewer nodes than the depth of one of its leaves ℓ . Then it has a formula metavariable Z at a proper prefix q of ℓ . If the subterm at q is unique, the schema also covers the sentence with $\top$ or $\perp$ at q , and closed values respect every guard (Lemma D.2). Take instances with deep bodies, such as $\text{EInd}(\neg^{2n}\forall w \neg w \in x)$, and a leaf in the occurrence that cannot be tied. For each prefix, one of $\top, \perp$ gives a false sentence (Appendix D.3). $\square$

With textbook names, the one non-first-order case not covered is Repl_{sp} in the named encoding. It is a genuine exception:

Proposition 5.8 (A truth-sound over-generalization). *[proved]^{cases Props C3, C3'} Let*

$$L_S := \forall A (\forall x (x \in A \rightarrow \exists y (Z_1 \wedge \forall u (Z_2 \rightarrow u = y))) \rightarrow \exists Y \forall x (x \in A \rightarrow \exists y (y \in Y \wedge Z_1))).$$

This is the named lgg of Repl_{sp} data on which the bodies' roots vary and some body uses y . Take it with the guards learned from anchor data: $\text{fresh}(Y, Z_1)$, $\text{fresh}(u, Z_1)$, $\text{fresh}(Y, Z_2)$, $\text{fresh}(y, Z_2)$. (a) Every guarded instance of L_S is a theorem of ZF, and $\text{inst}(L_S) \supsetneq \text{inst}(\text{Repl}_{\text{sp}})$. (b) For any theory B , all guarded instances of L_S are theorems of B iff B proves the Collection schema.

Proof idea. The antecedent implies $\forall x \in A \exists y Z_1$, and Collection gives the consequent. Conversely, $Z_1 := \varphi$ and $Z_2 := \perp$ give a formula equivalent to $\text{Coll}(\varphi)$, since $\forall u (\perp \rightarrow u = y)$ is valid. This formula is not an instance of Repl_{sp} (Appendix D.3). $\square$

So here the first-order learner is truth-sound but not exact, and no refutation can ever detect its over-generalization. With Jech's image consequent the decoupling gives false instances instead, and under α -variation the phenomenon disappears. By (b), the lgg is truth-sound over ZF without Foundation exactly if ZF without Foundation proves Collection. We do not know whether it does, and leave this open.

Proposition 5.9 (Separation: the guard and Russell's instance). *[proved]^{cases Prop D1} (a) The unguarded lgg of Separation data (either encoding) has Russell's instance $\forall z \exists y \forall x (x \in y \leftrightarrow x \in z \wedge \neg x \in y)$. It is refuted in pure logic from $\exists z \exists x x \in z$. (b) The most specific guard rule always learns $\text{fresh}(y, Z)$, since no genuine body mentions y . The learned guards equal the target's iff some body uses x and some body uses z . For Coll and $\text{Repl}_{\exists!}$ (named), the learned guards equal the target's, $\{\text{fresh}(Y, Z)\}$, iff every argument place is used.*

Proof. (a) For $x_0 \in z$ and the y given by the sentence, $x_0 \in y \leftrightarrow \neg x_0 \in y$. (b) A freshness guard for v is learned iff no datum's value mentions v . $\square$

5.4 Pattern targets: anchors, pattern anti-unification and rates

Let T^* be as in Section 5.2 and $D = \{T^*[P := \lambda \bar{z}. \varphi_j] : j \leq N\}$. Let (R) say that the main symbols of the bodies φ_j are not all equal, and (N_i) ($i \leq n$) that some φ_j has the hole z_i free.

Theorem 5.10 (Anchors for the ZF schemas). *[proved]^{cases Thm E; single Cor D.3} For every class H with $\text{PAT} \subseteq H \subseteq \text{SO}^\circ$, a finite $D \subseteq \text{inst}(T^*)$ is an anchor in H iff (R) and (N_i) hold for all $i \leq n$. In particular one instance never suffices, and two can.*

Proof idea. "If", for $H = \text{SO}^\circ$. By (R), a covering template's rigid part is a prefix of the frame, and its metavariables sit at frame positions or slot roots. Let a metavariable M have several occurrences. Then (R) forces the parts of T^* below them to have slots and inner frame symbols

at the same places. Each variable bound above the first occurrence is recovered as a hole of M 's body, either from a frame leaf or, via (N_i) , from a slot argument. This gives a body B_M with $T[M := \lambda \bar{h}. B_M] = T^*$. “Only if”: replace each $P(\bar{a}_k)$ by a pattern with the common root (e.g. $\neg Q(\bar{a}_k)$), or drop an unused argument (Appendix D.4). $\square$

Minimal anchors, as bodies, are: $x \in z$ and $\neg(x = a)$ for Sep; $x \in x$ and $\neg(x = x)$ for EInd; $y = x$ and $\neg(x = A)$ for Coll, $\text{Repl}_{\exists!}$, Repl_{sp} ; $x \in y$ and $\neg(x = a)$ for Repl_J . If a community never lets φ mention the bounding set z , the learner stays at the sound specialization $P(x)$, which is Jech's form. Since $L_{\in}$ has no function symbols, the “shielding” that separates SO° from DT° for induction (Proposition 5.20) and for universal axioms over arithmetic (Section 4) cannot occur. For $H = \text{DT}^\circ$ the theorem is also the one-metavariable case of the general anchor theorem of Section 3.

Several metavariables. ZF needs one metavariable, but untagged mixtures behave like templates with several (Section 6). Then a Plotkin-(D)-type event appears, and it depends on the class. Let T^* be a pattern template over $L_{\in}$ with metavariables $P, Q, \dots$ and bodies $\varphi_{P,j}$. For a map $\pi : [n_P] \rightarrow [n_Q] \cup \text{Par}$, let $\varphi \circ \pi$ replace each hole z_i by $z_{\pi(i)}$ (or by the parameter $\pi(i)$). The events are: $(D_{P \rightarrow Q}^{\text{PAT}})$: no bijection π has $\varphi_{Q,j} = \varphi_{P,j} \circ \pi$ for all j ; $(D_{P \rightarrow Q}^{\text{DT}})$: no map π has this property; $(D_{P \rightarrow Q}^{\text{root}})$: some j has $\text{root}(\varphi_{P,j}) \neq \text{root}(\varphi_{Q,j})$. Then $(D_{P \rightarrow Q}^{\text{root}})$ implies $(D_{P \rightarrow Q}^{\text{DT}})$, which implies $(D_{P \rightarrow Q}^{\text{PAT}})$.

Theorem 5.11 (Several metavariables). *[proved]^{cases Thm E'} Assume (R_P) and all $(N_{P,i})$, for every metavariable P . Then D is an anchor in PAT iff $(D_{P \rightarrow Q}^{\text{PAT}})$ holds for all $P \neq Q$. It is an anchor in DT° iff $(D_{P \rightarrow Q}^{\text{DT}})$ holds for all ordered pairs. In SO° , (D^{root}) for all pairs implies that D is an anchor, which implies (D^{DT}) for all ordered pairs; the second implication cannot be reversed. Without (R_P) or $(N_{P,i})$, D is an anchor in no class between PAT and SO° .*

Proof idea. The proof of Theorem 5.10 works once each slot of M 's first occurrence faces a slot of the *same* metavariable at every other occurrence. If it faces a Q -slot instead, there is a fixed argument map from P 's bodies to Q 's, which the events exclude. Conversely, a violating map lets one rewrite every $Q(\bar{b})$ as an occurrence of P (Appendix D.4). $\square$

The three classes differ. Take $T_B := \forall x \forall y (P(x, y) \rightarrow Q(x))$ with bodies $(x \in y, x \in x)$ and $(\neg x = y, \neg x = x)$. Pattern anti-unification returns T_B , but $\forall x \forall y (P(x, y) \rightarrow P(x, x)) \in \text{DT}^\circ$ covers the data and misses $\forall x \forall y (x \in y \rightarrow \neg x = x)$. Take $T_A := \forall x \forall y (P(x, y) \rightarrow Q(x, y))$ with bodies $(x \in y, y \in x)$ and $(\neg x = y, \neg x = x)$. Here D is a DT° anchor, but $\forall x \forall y (G(x, y, x) \rightarrow G(x, x, y)) \in \text{SO}^\circ$ covers it and misses the same sentence. Whether (D^{root}) is necessary in SO° is open; we conjecture that it is not.

Corollary 5.12 (Pattern anti-unification recovers every ZF schema). *[proved]^{cases Cor F, F'} The least general pattern generalization $L_{\text{pat}}(D)$ exists and is unique (Pfenning, 1991; Baumgartner et al., 2017). For $D \subseteq \text{inst}(T^*)$, $L_{\text{pat}}(D) \preceq T^*$, and $L_{\text{pat}}(D) \equiv T^*$ iff (R) and every (N_i) hold. With several metavariables, $L_{\text{pat}}(D) \equiv T^*$ iff the PAT events of Theorem 5.11 hold.*

Proof. $T^* \in \text{PAT}$ covers D . Under the events, Theorem 5.10 (resp. 5.11) for $H = \text{PAT}$ gives $L_{\text{pat}}(D) \succeq T^*$. Otherwise its PAT witnesses lie above $L_{\text{pat}}(D)$ and miss an instance. $\square$

The merge rule must identify columns that are equal up to a permutation of the arguments: Replacement's $P(x, y)$ and $P(x, u)$ columns agree only up to $y \leftrightarrow u$.

Theorem 5.13 (Rates). *[proved]^{cases Thm G} Let the bodies be i.i.d. from μ . Let $C_{F,I}$ be the set of bodies with root F (no condition if $F = *$) in which no z_i , $i \in I$, is free. Then*

$$\Pr[D_N \text{ is not an anchor}] = \sum_{(F,I) \neq (*,\emptyset)} (-1)^{[F \neq *] + |I| + 1} \mu(C_{F,I})^N \leq \sum_f p_f^N + \sum_i (1 - q_i)^N,$$

where $p_f = \mu(\text{root} = f)$ and $q_i = \mu(z_i \text{ free})$.

Proof. Inclusion–exclusion over the events “all roots are f ” (pairwise disjoint) and “ z_i is never free”. $\square$

5.5 PA induction: the contrast

A *motive* is a formula φ of L_A with $\text{FV}(\varphi) \subseteq \{x\}$, and $\text{Ind}(\varphi) = \varphi(0) \wedge \forall x(\varphi \rightarrow \varphi[Sx/x]) \rightarrow \forall x\varphi$. The target is $\text{inst}(T_{\text{Ind}})$ with $T_{\text{Ind}} = P(0) \wedge \forall x(P(x) \rightarrow P(Sx)) \rightarrow \forall x P(x)$. Since $P(0)$ and $P(Sx)$ are not pattern occurrences, $T_{\text{Ind}} \in \text{DT}^\circ \setminus \text{PAT}$. Here (R) says that the motives’ main symbols are not all equal, and (N) that some motive has x free. The results are from the prior record, for parameter-free motives unless said otherwise.

First-order learners. A first-order schema is *sound* if its sentence instances are true in $\mathbb{N}$. The false instances below are closed and capture-free, so the results hold in the named and in the de Bruijn encoding, with fixed or varying bound-variable names.

Proposition 5.14 (Two raw instances suffice for unsoundness). *[proved]^{prior induction B1, B3} (a) $\text{lgg}(\text{Ind}(x+0=x), \text{Ind}(0+x=x)) = (0+0=0) \wedge \forall x(z_1+z_2=x \rightarrow z_3+z_4=Sx) \rightarrow \forall x(z_1+z_2=x)$. It has the false instance $(0+0=0) \wedge \forall x(0+0=x \rightarrow 0+S0=Sx) \rightarrow \forall x(0+0=x)$. One raw instance never forces unsoundness, and some pairs, such as $\text{Ind}(x=x)$ and $\text{Ind}(Sx=Sx)$, have a sound lgg. (b) If (R) and (N) hold, then $\text{lgg}(D) = L_\infty := Z_1 \wedge \forall x(Z_2 \rightarrow Z_3) \rightarrow \forall x Z_2$, which has the false instance $(0=0) \wedge \forall x(0=S0 \rightarrow 0=S0) \rightarrow \forall x(0=S0)$. For i.i.d. motives, $\Pr[\text{lgg}(D_N) \neq L_\infty] \leq p_{\max}^{N-1} + (1-q)^N$.*

So the diversity that identifies T_{Ind} among determinate templates (Theorem 5.18) destroys the first-order lgg, at the same coupon-collector rate.

Theorem 5.15 (No finite union of first-order schemas). *[proved]^{prior induction B2} Let $g_0(t) = t$, $g_{n+1}(t) = 0 + g_n(t)$ and $F' = \{\text{Ind}(g_n(x) = x) : n \geq 0\}$. Every first-order schema covering two members of F' has a false instance that is refutable in $\mathbb{Q}$. Hence k sound first-order schemas cover at most k members of F' , and no finite union of first-order schemas covers the raw induction instances (or those of open induction) soundly.*

Proof idea. For $n < m$ the lgg descends through n copies of $0 + \cdot$ and stops at mixed roots. It is $L_n = (g_n(a) = 0) \wedge \forall x(g_n(b) = x \rightarrow g_n(c) = Sx) \rightarrow \forall x(g_n(b) = x)$, which contains the false sentence $I_M^* = (g_M(0) = 0) \wedge \forall x(g_M(0) = x \rightarrow g_M(S0) = Sx) \rightarrow \forall x(g_M(0) = x)$ for every $M \geq n$ (Appendix D.5). $\square$

The occurrence and freshness guards learnable from F' do not help (Appendix D.5). A guard “is an induction instance” trivially would, and that is what the Sub encoding below implements. Worse, some unsound lggs are never caught by evaluating closed sentences.

Theorem 5.16 (An unsound lgg that closed truths never refute). *[proved]^{prior induction referee R1} Let $K_0 := (z_1 + 0 = z_1) \wedge \forall x(z_1 + x = z_2 \rightarrow z_1 + Sx = Sz_2) \rightarrow \forall x(z_1 + x = z_2)$, the lgg of $\text{Ind}(0 + x = x)$ and $\text{Ind}(S0 + x = Sx)$. Its instance $(0 + 0 = 0) \wedge \forall x(0 + x = 0 \rightarrow 0 + Sx = S0) \rightarrow \forall x(0 + x = 0)$ is false. Yet the true closed quantifier-free sentences together with all sentence instances of K_0 are consistent. So no sound derivation from true closed quantifier-free premises (or Δ_0 ones, with $<$ primitive) refutes an instance of K_0 .*

Proof idea. Take an instance that is not trivially true. A convergent rewrite system over the numerals and a new constant g gives an algebra containing $\mathbb{N}$ in which the step clause fails at g . Amalgamating these algebras over $\mathbb{N}$ gives a model of all instances and all true closed quantifier-free sentences (Appendix D.5). $\square$

K_0 is refuted once the recursion axiom $\forall x \forall y (x + Sy = S(x + y))$ is a designated premise ^{prior induction B5(d)}. That PA is not finitely axiomatizable (Ryll-Nardzewski, 1952) is a different matter: Theorem 5.15 is about the syntax of substitution, and PA is Q plus one guarded first-order pattern (Proposition 5.22).

Pattern anti-unification and determinate templates.

Proposition 5.17 (Pattern anti-unification is unsound on induction). *[proved]^{prior induction C9} Under (R) and (N), the least general pattern generalization of $D \subseteq \text{inst}(T_{\text{Ind}})$ is $T_{\text{pat}} := Z \wedge \forall x(P(x) \rightarrow Q(x)) \rightarrow \forall x P(x)$. It has the false non-instance $(0 = 0) \wedge \forall x(x = 0 \rightarrow x = 0) \rightarrow \forall x(x = 0)$.*

Theorem 5.18 (Anchors for induction in DT°). *[proved]^{prior induction C3; single Cor D.3} A finite $D \subseteq \text{inst}(T_{\text{Ind}})$ is an anchor in DT° iff (R) and (N). The same holds in the size-bounded classes DT_s° for $s \geq 7$, and, by the general anchor theorem of Section 3, in closure-normal form with parameters. Two instances suffice; one never does.*

Proof idea. By (R) and root preservation the motive slots disagree, so a covering template's rigid part is a prefix of the five-node frame of Ind . Its metavariables are formula-valued. Each is either 0-ary, and then occurs once (the closed slots have different logical sizes), or unary with a pattern occurrence $M(x)$. Fix a motive with x free; injectivity of substitution forces the other arguments to be x , Sx or 0 , as in T_{Ind} . Witnesses for “only if” are $Z_1 \rightarrow \forall x Z_2$ and root-specialized templates such as $(\neg Z_1 \wedge Z_2) \rightarrow Z_3$ (Appendix D.5). $\square$

Here $|T|$ counts symbols, an occurrence $M(\bar{t})$ counting $1 + \sum_i |t_i|$; so $|T_{\text{Ind}}| = 14$.

Proposition 5.19 (Generalizations, lgg, size threshold). *[proved]^{prior induction C4, C6.2} (a) Up to renaming, exactly 26 templates of DT° contain $\text{inst}(T_{\text{Ind}})$. Each takes a prefix of the frame, puts fresh metavariables at the cut slots, and partitions the motive slots present, where $P(0)$ and $P(Sx)$ join a class only together with a $P(x)$ slot. For every anchor these 26 are exactly the covering templates, so $\text{lgg}_{\text{DT}^\circ}(D) = T_{\text{Ind}}$. (b) $\text{inst}(T_{\text{Ind}})$ is the intersection of the members of DT_s° that contain it iff $s \geq 12$. For $7 \leq s \leq 11$, the cautious verifier over DT_s° accepts on every anchor the false sentence $(0 = 0 \wedge \forall x(x = x \rightarrow Sx = Sx)) \rightarrow \forall x(x = 0)$.*

Proposition 5.17 reads T_{pat} off this list: it is the least generalization without non-pattern occurrences. Part (b) is the closure criterion of Section 2 at work: soundness needs the target to be an intersection of class members, not a member.

Proposition 5.20 (Without determinacy). *[proved]^{prior induction C7} Call a free occurrence of x unshielded if it lies inside no proper term $t \neq x$ with $\text{FV}(t) \subseteq \{x\}$, and let (B) say that some motive has one. In SO° , D is an anchor iff (R) and (B). For example, $\{\text{Ind}(Sx = S0), \text{Ind}(\neg Sx = 0)\}$ satisfies (R) and (N), but $Z \rightarrow \forall x P(Sx)$ covers it.*

Before an anchor the DT° verifier is sound, but the version space can lack a least element: $\{\text{Ind}(x = x), \text{Ind}(0 = x)\}$ has four minimal covering templates in DT° , and one of them is below T_{Ind} single Thm B; corrects prior C8.2.

Encodings that make induction first-order.

Proposition 5.21 (Sub encoding). *[proved]^{prior induction C5, P1, P2, P6} Record each induction as the step from $\text{Sub}(\varphi, x, 0, \alpha)$ and $\text{Sub}(\varphi, x, Sx, \beta)$ to $\alpha \wedge \forall x(\varphi \rightarrow \beta) \rightarrow \forall x\varphi$, where the decidable judgment $\text{Sub}(\varphi, x, t, \psi)$ says that $\psi = \varphi[t/x]$ (Claude (Anthropic), 2026). With x fixed, this is a first-order pattern in P, α, β . The lgg of genuine data recovers it iff (R) and (N), with $\Pr[\text{no anchor}] = \sum_f p_f^N + (1 - q)^N - \sum_f r_f^N$, where r_f is the probability of root f with x not free. If the induction variable varies, the variables must also not all be equal (or must be α -normalized). With the decidable guard “all Sub premises are true”, the cautious verifier accepts only genuine inductions.*

Proof. These are Plotkin’s events: φ , $\varphi[0/x]$ and $\varphi[Sx/x]$ share their root, and they are pairwise distinct iff x is free in φ . $\square$

So the second-order learner on raw sentences loses nothing against a first-order learner on Sub-annotated data.

Proposition 5.22 (PA is Q plus one first-order pattern). *[proved]^{prior induction B6} Let $\text{IndEq}(P) := \forall x(x = 0 \rightarrow P) \wedge \forall y(\forall x(x = y \rightarrow P) \rightarrow \forall x(x = Sy \rightarrow P)) \rightarrow \forall x P$, with P a 0-ary metavariable and x, y fixed (Tarski, 1965). (a) First-order logic proves $\text{IndEq}(\varphi) \leftrightarrow \text{Ind}(\varphi)$ for every motive φ , up to renaming a bound variable. With parameters, the guard $y \notin \text{FV}(P)$ is needed. (b) $\text{lgg}(\text{IndEq}(\psi_1), \dots, \text{IndEq}(\psi_N)) = \text{IndEq}(\text{lgg}(\psi_1, \dots, \psi_N))$, which is $\text{IndEq}(P)$ iff (R) holds. (c) $\text{Q} + \text{inst}(\text{IndEq}) \vdash \text{PA}$.*

Proof idea. (a) $\forall x(x = t \rightarrow \varphi) \leftrightarrow \varphi[t/x]$ for $x \notin \text{FV}(t)$; take $t = 0, y, Sy$. (c) Parameter-free induction for all formulas yields induction with parameters, by inducting on $\forall \bar{p}[\varphi(0, \bar{p}) \wedge \forall z(\varphi(z, \bar{p}) \rightarrow \varphi(Sz, \bar{p})) \rightarrow \varphi(x, \bar{p})]$; compare Kaye et al. (1988) (Appendix D.5). $\square$

So the obstruction in Theorem 5.15 is the meta-level substitution in $\varphi(0)$ and $\varphi(Sx)$, not induction. Proof assistants record or recompute the motive. Lean 4’s `Nat.rec` takes it as an implicit argument, and the `induction` tactic records it as a λ (de Moura and Ullrich, 2021). Metamath’s `set.mm` states induction with implicit-substitution hypotheses and distinct-variable conditions (Megill and Wheeler, 2019); as a first-order pattern this needs freshness guards (Appendix D.5). In DT° terms the motive is the unique matcher of T_{Ind} , so for closed $\forall$ -form steps recording it changes neither anchors nor rates.

Experiment. The implemented DT_F° learner of Section 7 was given tagged instances with random bodies. It was exact in 23, 27 and 30 of 30 runs at $N = 2, 3$ and 8 for Separation; 23, 24, 30 for Replacement ($\exists!$ spelled out, image consequent); 27, 29, 30 for EInd; and 25, 30, 30 for induction. It accepted no non-instance in 927 probes. The genuine pattern lgg did as well on Separation and Replacement, slightly worse on EInd (24/30 at $N = 2$), and was never exact on

target	first-order			PAT	DT ^o	anchor condition
	named	dB	named, α			
ground axioms (incl. Found, AC)	yes	yes	yes	yes	yes	the sentence itself
Sep	g	g	g	yes	yes	(R), (N _x), (N _z)
Sep _J	g	g	g	yes	yes	(R), (N _u)
EInd	no [†]	yes	no	yes	yes	(R), (N _x)
Coll, Repl _{∃!}	g	no [†]	no [†]	yes	yes	(R), (N _x), (N _y), (N _A)
Repl _{sp}	no [†]	no [†]	no [†]	yes	yes	(R), (N _x), (N _y), (N _A)
Repl _J	no [†]	no [†]	no	yes	yes	(R), (N _x), (N _y)
Ind, raw	no [†]	no [†]	no [†]	no [§]	yes	DT ^o : (R), (N); SO ^o : (R), (B)
Ind, Sub-encoded	yes	–	–	–	–	first-order lgg: (R), (N)
IndEq	g	–	–	–	–	first-order lgg: (R)

Table 6: All targets of this section. “yes”: a first-order pattern (resp. in the class). “g”: a first-order pattern with a freshness or loose-index guard; it has false instances without the guard. “no[†]”: no finite union of guarded first-order schemas covers the target without a false member (Theorems 5.7 and 5.15). “no[†]”: not first-order, but the first-order lgg is truth-sound iff the base theory proves Collection (Proposition 5.8). “no[§]”: not a pattern, and the pattern lgg is unsound (Proposition 5.17). “–”: not analysed. For the ZF schemas the anchor condition is the same in every class between PAT and SO^o (Theorem 5.10). Two instances suffice in every row except the ground axioms, where one does.

induction, where 224 of 229 probes at $N = 2$ were non-instances. First-order lggs were unsound except on EInd in de Bruijn indices, consistent with Table 5 ^{experiments E2}.

5.6 Summary

Table 6 collects the answers. All ZF schemas lie in the pattern fragment. Pattern anti-unification and the DT^o learner each learn them from two suitable instances, and are sound at all times. For first-order learners the robust statement is Theorem 5.3 and Proposition 5.4: a schema is a first-order pattern iff its metavariable’s argument tuple is literally the same at every occurrence in the chosen syntax. PA induction lies outside the pattern fragment. The determinate class handles it, with the same anchor events as the first-order learner on Sub-annotated data; without determinacy the anchors need the stronger event (B).

6 Many axioms and schemas without labels

The third question asks for one method that also learns “many axiom(scheme)s at once, even if they do not come with labels for instances of which axiom scheme they are”. The data now come from several targets $\sigma_1, \dots, \sigma_{k'}$, and nothing says which datum is an instance of which target. This section develops the theory of this setting and the method DTRC (Determinate Templates with Refutation Clustering). The results come from the research record of the “untagged” track, checked by an adversarial referee with independent code; the complexity results of Section 6.9 come from the “single” track. The record states several results under the hypothesis that minimal covering templates are finitely many and lie below every covering template; that hypothesis is the finitariness theorem of Section 3, so we state the results without it. In brief:

- From positive data alone a bound k on the number of targets is necessary, and spare slots are paid for in data diversity, which no sound negative evidence can replace (Section 6.2).
- With the exact bound, negatives that refute every cross-target merge make untagged learning as easy as tagged learning, and the learner can find them itself (Theorem 6.9).
- DTRC needs no bound. Under *refutation separation* it recovers the hidden labels, is sound at all times, and is exact once every target’s data contain an anchor, with the tagged failure probability (Theorem 6.11). Without separation it is sound relative to a residue of unrefuted cross merges (Theorem 6.12). Learning $\forall x\varphi$ from instances is a sound cross merge (Proposition 6.15). The refutation depth cannot be eliminated by any computable learner (Theorem 6.17).
- PA, ZF and ZFC are separated at small depth by Δ_0 evaluation in $\mathbb{N}$, hereditarily finite counterexamples and logic; Russell’s paradox separates the comprehension-shaped axioms. Weak forms of Union and Power merge into a false universal-set schema that only coherence refutes (Section 6.11).

6.1 Setting

Data are formulas in closure-normal form with de Bruijn bound variables (Section 2), and templates are in DT° . Parameters are treated as constants; the one place where sentences are identified up to renaming of parameters (head classes, Appendix E.2) says so. Let $\mathcal{S}$ be the set of such formulas and $\text{Tr} \subseteq \mathcal{S}$ the set of those whose universal closure is true in the intended structure ($\mathbb{N}$ or V).

Definition 6.1 (practice, data, oracle, k -union learner). [definition]^{untagged §1} A *practice* is a finite set $\Sigma^* = \{\sigma_1, \dots, \sigma_{k'}\} \subseteq \text{DT}^\circ$ of targets (ground templates, i.e. single axioms, allowed), with $R^* := \bigcup_i \text{inst}(\sigma_i)$; it is *valid* if $R^* \subseteq \text{Tr}$, which we assume unless said otherwise. The data form a finite multiset $D \subseteq R^*$ with *hidden* labels $D = D_1 \sqcup \dots \sqcup D_{k'}$, $D_i \subseteq \text{inst}(\sigma_i)$ (a datum in two instance sets carries one label). In the random model each datum picks target i with probability π_i and then an instance $\sigma_i\theta$, $\theta \sim \Lambda_i$. A *refutation oracle* is a family $\text{Ref}_0 \subseteq \text{Ref}_1 \subseteq \dots \subseteq \mathcal{S}$ with (OS) $\text{Ref}_d \cap \text{Tr} = \emptyset$ for all d , and (Dec) membership in Ref_d and the property “ $\text{inst}(T) \cap \text{Ref}_d \neq \emptyset$ ” (“ T is d -refuted”) are decidable uniformly in d , resp. (T, d) ; $\text{Ref}_\infty := \bigcup_d \text{Ref}_d$. A fixed finite $N \subseteq \mathcal{S} \setminus \text{Tr}$, used as every Ref_d , is an oracle. For a finite set N of negatives, $\text{VS}_k(D, N) := \{h \in H_k : D \subseteq h, h \cap N = \emptyset\}$, where $H_k = \{\bigcup_{l \leq k} \text{inst}(\tau_l) : \tau_l \in \text{DT}^\circ\}$, and the *cautious k -union learner* accepts $\text{Acc}_k(D, N) := \bigcap \text{VS}_k(D, N)$. For finite nonempty X , $\text{Cov}(X) := \{T \in \text{DT}^\circ : X \subseteq \text{inst}(T)\}$ and $\text{Min}(X)$ is the set of its $\succeq$ -minimal members up to renaming.

The oracles are the refutation channels of Section 2, in these forms (all sound, none complete, all *target-blind*: their answers are a fixed function of the query). **R- Δ_0** at depth d , for sentences of size $\leq d$: instantiate parameters by numerals $\leq d$, evaluate closed atoms and connectives, refute $\forall$ by a numeral counterexample $\leq d$, verify $\exists$ by a witness $\leq d$, and verify $\forall$ (refute $\exists$) only by equality reasoning from the true closed equations of $\mathbb{N}$. **R-HF**: hereditarily finite counterexamples to Π_1 -shaped set-theoretic sentences, sound by Δ_0 absoluteness. **R-logic**: a propositionally unsatisfiable set of ground instances of the Skolem form. **R-coh** (coherence): R-logic applied together with designated true sentences.

Proposition 6.2 (minimal covering templates). [proved]^{single Thm B} For finite nonempty $X \subseteq \mathcal{S}$: (a) $\text{Min}(X)$ is finite up to renaming and computable from X ; (b) every $T \in \text{Cov}(X)$ satisfies

$T \succeq T_0$, hence $\text{inst}(T) \supseteq \text{inst}(T_0)$, for some $T_0 \in \text{Min}(X)$; (c) if $X \subseteq \text{inst}(\sigma)$ with $\sigma \in \text{DT}^\circ$, some $T_0 \in \text{Min}(X)$ has $\text{inst}(T_0) \subseteq \text{inst}(\sigma)$.

A finite $A \subseteq \text{inst}(\sigma)$ is an *anchor* for σ if every $T \in \text{Cov}(A)$ has $\text{inst}(T) \supseteq \text{inst}(\sigma)$. Section 3 characterizes anchors by witness events ((R)+(N) for T_{Ind} and the ZF schemas, Plotkin's (R)+(D) for first-order patterns); a ground σ has the anchor $\{\sigma\}$, and an anchor of a non-ground target has at least two elements.

Lemma 6.3 (cautious soundness; finite form). *[proved; known in substance]^{untagged} Lemmas 1.1, 1.2* (a) If $k \geq k'$ and $N \cap R^* = \emptyset$, then $R^* \in \text{VS}_k(D, N)$, so $\text{Acc}_k(D, N) \subseteq R^*$; Acc_k is monotone in D and in N . (b) With $\text{Min}_N(B) := \{T \in \text{Min}(B) : \text{inst}(T) \cap N = \emptyset\}$, $\text{Acc}_k(D, N)$ is the intersection of the sets $\bigcup_{B \in \pi} \text{inst}(T_B)$ over all partitions π of D into at most k nonempty blocks and all choices $T_B \in \text{Min}_N(B)$. So for finite N it is decidable (in time exponential in $|D|$).

Part (b) assigns each datum to the first slot covering it and applies Proposition 6.2(b) to each block (Appendix E.1).

6.2 Positive data: a bound is necessary, and spare slots cost diversity

Proposition 6.4 (a bound is necessary). *[proved; known (Gold)]^{untagged} Prop 4.3* Let $H_\infty := \bigcup_k H_k$. For every finite $D \subseteq R^*$ and every sound N , $\text{Acc}_\infty(D, N) = D$. No learner, however it uses a fixed sound oracle, identifies H_∞ in the limit from text.

Proof. D is a finite union of ground templates and avoids N . For the second claim take a true non-ground σ : the class contains $\text{inst}(\sigma)$ and all its finite subsets, all inside Tr , on which the oracle answers identically. So the learner is a function of the text, and Gold's locking-sequence argument for superfinite classes applies (Gold, 1967); it needs no computability. $\square$

Negatives cannot replace a bound: every finite set of true data is itself a consistent hypothesis. Negatives remove over-general *lumps*; a bound forbids *splits*, which replace a target by finitely many specializations that together cover its data. Splits are sound, so one-sided refutation never removes them.

Theorem 6.5 (splits; threshold). *[proved]^{untagged} Thm 3.2* (a) For every N with $N \cap R^* = \emptyset$ and $k \geq k'$, $\text{VS}_k(D, N)$ contains every $h \in H_k$ with $D \subseteq h \subseteq R^*$. So if such an h misses $q \in R^*$, no sound negatives make the learner accept q . (b) Suppose moreover that every D_j is nonempty and that N cross-separates D : every template covering data of two targets meets N . Put $m := k - k' + 1$. If every family of at most m templates that covers D_i and avoids N has union $\supseteq \text{inst}(\sigma_i)$, then $\text{inst}(\sigma_i) \subseteq \text{Acc}_k(D, N)$. If some such family misses a $q \in \text{inst}(\sigma_i)$ lying in no other target's instance set, then $q \notin \text{Acc}_k(D, N)$. With pairwise disjoint instance sets this is an exact threshold.

Proof idea. (a) $h \cap N \subseteq R^* \cap N = \emptyset$. (b) Cross-separation makes the slots used by different targets disjoint, so target i has at most m ; conversely the bad family plus the other $k' - 1$ targets is a k -slot hypothesis (Appendix E.2).

In general a target's slots may be shared with others, provided every shared slot contains the whole target.

Theorem 6.6 (slot accounting). *[proved]^{untagged} Thm 4.1* Fix a target i , a bound k and a sound N , and let $\bar{D}_i := D \setminus D_i$. Assume (Abs): every N -avoiding template covering a datum of D_i and one of $\bar{D}_i$ contains $\text{inst}(\sigma_i)$. Let c_i be the least number of N -avoiding templates not containing

$\text{inst}(\sigma_i)$ that cover $\bar{D}_i$. (a) If every family of at most $k - c_i$ N -avoiding templates covering D_i has union $\supseteq \text{inst}(\sigma_i)$, then $\text{inst}(\sigma_i) \subseteq \text{Acc}_k(D, N)$. (b) If some such family and some optimal cover of $\bar{D}_i$ both miss $q \in \text{inst}(\sigma_i)$, then $q \notin \text{Acc}_k(D, N)$.

Corollary 6.7 (slot accounting for PA). *[proved; computed (consistency check)]^{untagged Cor 4.2} Let the targets be $\mathbf{Q}$'s axioms Q1–Q7 as ground sentences in closure-normal form ($\neg Sp = 0$; $Sp = Sp' \rightarrow p = p'$; $p + 0 = p$; $p + Sp' = S(p + p')$; $p \cdot 0 = 0$; $p \cdot Sp' = p \cdot p' + p$; $\neg p = 0 \rightarrow \exists y p = Sy$) and T_{Ind} , so $k' = 8$. (Abs) holds for induction for every motive: a template covering a $\mathbf{Q}$ -axiom and an induction instance is F_0 , or $F_0 \rightarrow F_1$ for Q2 and Q7, and both contain $\text{inst}(T_{\text{Ind}})$. On positive data $c = 4$ (Q3–Q6 share $z_0 = z_1$; Q1, Q2, Q7 stay alone), so at $k = 8$ induction gets four slots: an induction instance whose motive has x free and a main connective unseen in the data is accepted iff the induction data cannot be split into four blocks each root-homogeneous or all-vacuous (without vacuous motives: iff at least five main connectives occur). With the three negatives $0 = S0$, $(p + 0) = 0$, $(p \cdot 0) = S0$, $c = 7$, induction gets one slot, and the tagged anchor condition (R)+(N) suffices. (In the Sub-encoded presentation of induction analysed in the prior work of this project, $c = 1$.)*

The computed check of $c = 4$ and $c = 7$ (24/24 cases) holds by construction given (Abs) and the anchor theorem for T_{Ind} , so it is a consistency check of the implementation, not independent evidence (as the referee noted).

Remark 6.8 (head classes and numerals). *[proved; computed]^{untagged U17} Because data are read under universal closure, a one-variable instance schema $\varphi(z)$ is covered by its $h = |F| + p + 1$ head specializations (F the function symbols, p the number of parameters of φ ; $5 + p$ in arithmetic), one per head class of the value of z , a parameter outside φ being one class (Proposition E.3). So with $m \geq h$ spare slots the “failure family” covers everything. (An earlier version of the record claimed the contrary, counting renamed parameters as different sentences; the referee’s counterexample led to this correction.) For quantifier-free, parameter-free φ the exact threshold is known for $m \leq h$ (Proposition E.4) and, over a unary signature, in closed form (Proposition E.5); in particular *numeral-only data never make the cautious k -union learner accept $\forall x \varphi$ with one spare slot*: the sound split $\{\varphi(0)\} \cup \text{inst}(\varphi(Su))$ misses $\varphi(a)$. Brute-force checks agree with all three propositions (2000/2000, 60/60, 254/254, 2509/2509 cases). Beyond this the spare-slot threshold is open (binary symbols with $m > h$, several metavariables, templates with parameters, T_{Ind} with $m \geq 8$, where the eight root specializations already cover $\text{inst}(T_{\text{Ind}})$).*

6.3 The pigeonhole theorem

Theorem 6.9 (pigeonhole). *[proved]^{untagged Thm 2.1} Let $k = k'$, $D \subseteq R^*$, $A_i \subseteq D_i$ a nonempty anchor for σ_i , and $N \cap R^* = \emptyset$ with (X): for $i \neq j$, $a \in A_i$, $b \in A_j$, every $T \in \text{DT}^\circ$ with $a, b \in \text{inst}(T)$ meets N . Then $\text{Acc}_k(D, N) = R^*$.*

Proof. The inclusion $\subseteq$ is Lemma 6.3. Let $h = \bigcup_{l \leq k} \text{inst}(\tau_l)$ be in $\text{VS}_k(D, N)$, and let L_i be the set of slots l such that τ_l covers an element of A_i . The L_i are nonempty and, by (X), pairwise disjoint; k' of them in k' slots are singletons $\{l_i\}$, so τ_{l_i} covers A_i and $\text{inst}(\tau_{l_i}) \supseteq \text{inst}(\sigma_i)$. $\square$

Only anchor data matter, and (X) forces the anchors to be disjoint. The negatives need no labels: let $N(D, d)$ contain the first d -refuted instance of every d -refuted $T \in \text{Min}(\{a, b\})$, over all pairs $a, b \in D$. It is sound, and if every minimal template of every cross pair of anchor data is d -refuted (*anchor-pair separation*), it satisfies (X) by Proposition 6.2(b). Hence $\text{Acc}_{k'}(D, N(D, d))$ is sound at all times, decidable, and exact as soon as D contains anchors; at most $\mu \sum_{i < j} |A_i| |A_j|$

negatives are needed, μ the largest $|\text{Min}(\{a, b\})|$ (Corollary E.6). For PA (raw DT° encoding), every cross pair among Q1–Q7 and T_{Ind} has exactly one minimal covering template ($F_0, F_0 \rightarrow F_1, z_0 = z_1, p + z_0 = z_1$ or $p \cdot z_0 = z_1$), and four sentences refute them all: $0 = S0, 0 = 0 \rightarrow 0 = S0, p + 0 = 0$ and $p \cdot 0 = S0$ (computed; the referee’s independent enumeration of all 68 covering templates of size ≤ 14 of the 49 PA cross pairs agrees). So at $k = 8$ the untagged learner is exact after one occurrence of each Q-axiom and two induction instances with different main connectives, one of them non-vacuous.

6.4 dtrc: Determinate Templates with Refutation Clustering

Definition 6.10 (coherence, DTRC, separation). [definition]^{untagged §§5.1–5.2} A finite $X \subseteq \mathcal{S}$ is *d-coherent* if some $T \in \text{Min}(X)$ is not *d-refuted* (equivalently, by Proposition 6.2, some covering template is not). **Algorithm dtrc_d**: (1) normalize the data (closure-normal form, canonical parameter names, de Bruijn indices); (2) starting from singleton clusters, repeatedly merge two clusters whose union is *d-coherent*, chosen by a *merge policy* (e.g. online: each datum joins the first cluster it is coherent with, then pairwise merges to a fixpoint), until no pair has a coherent union; (3) for each final cluster C let $\text{Min}_d^+(C)$ be its unrefuted minimal templates and $\text{Acc}_d(C) := \bigcap_{T \in \text{Min}_d^+(C)} \text{inst}(T)$ ($\emptyset$ if there are none); (4) assert $\text{Acc}_d := \bigcup_C \text{Acc}_d(C)$. The practice is *d-separated on D*, $\text{RS}_d(D)$, if every template covering data of two different targets is *d-refuted*, and *globally d-separated* if every template covering instances of two different targets is.

$\text{Acc}_d(C)$ is the cautious single-template verifier of C with negatives Ref_d . Coherence is downward closed and antitone in d (Lemma E.1). By Proposition 6.2, $\text{RS}_d(D)$ holds iff every minimal template of every cross pair $(a, b) \in D_i \times D_j$ is *d-refuted*, a finite check that the implementation performs after each run.

Theorem 6.11 (DTRC). [proved]^{untagged Thm 5.2} *Let the practice be valid and the oracle satisfy (OS), (Dec). (i) Within-target merges are never refuted: every $X \subseteq \text{inst}(\sigma_i)$ is d-coherent for every d. (ii) Separation recovers the labels: under $\text{RS}_d(D)$ the final clustering is $\{D_i : D_i \neq \emptyset\}$ for every merge policy. (iii) Soundness at all times, exactness from anchors: under $\text{RS}_d(D)$, $\text{Acc}_d \subseteq R^*$, and $\text{Acc}_d = R^*$ once each D_i contains an anchor of σ_i . (iv) Rates: under global separation and i.i.d. data, $\Pr[\text{Acc}_d \neq R^*] \leq \sum_i \Pr[D_i \text{ contains no anchor}]$, the failure probability of the tagged learner, bounded explicitly in Section 3. (For T_{Ind} with n induction data and i.i.d. motives this is $\sum_f p_f^n + (1 - q)^n - \sum_f r_f^n$, with p_f the probability of main connective f , q that of x occurring free, r_f that of root f with x not free; see Section 5.) (v) Tests: with n distinct data the pairwise fixpoint makes $O(n^2)$ coherence tests; under $\text{RS}_d(D)$ the online policy makes at most $nk' + k'(k' - 1)/2$.*

Proof idea. (i) Some $T_0 \in \text{Min}(X)$ has $\text{inst}(T_0) \subseteq \text{inst}(\sigma_i) \subseteq \text{Tr}$ (Proposition 6.2(c)), and (OS) never refutes it. (ii) Pure clusters of different targets never merge, by separation; two final clusters of one target would merge, by (i). (iii) In the cluster D_i the template T_0 of (i) is intersected; with an anchor, every member of $\text{Min}(D_i)$ contains $\text{inst}(\sigma_i)$. Details in Appendix E.4. $\square$

Under separation, refutation does the job that citations do in tagged data: it says which instances belong together. Within each recovered cluster the learner *is* the tagged cautious learner, so tagged anchors and rates (Section 3) transfer verbatim, and the number of targets is discovered, not given. The inequality in (iv) can be strict, since refuted non-target templates drop out before an anchor arrives. When a bound $k \geq k'$ is known, accepting only what both DTRC and $\text{Acc}_k(D, N(D, d))$ accept is target-sound at all times, with or without separation, and exact under

$\text{RS}_d(D)$ with $k = k'$ once every D_i contains an anchor, because $\text{RS}_d(D)$ implies (X) for $N(D, d)$ (Remark E.7).

6.5 Without separation: residue, sound merges, fragmentation

Theorem 6.12 (soundness relative to a residue). *[proved; computed]^{untagged} Thm 5.4* Call a final cluster mixed if it contains data of two or more targets, and let $\text{Res}_d(D) := \bigcup \{\text{inst}(T) : T \in \text{Min}(X) \text{ not } d\text{-refuted, } X \subseteq D \text{ } d\text{-coherent and mixed}\}$. (a) $\text{Acc}_d \subseteq R^* \cup \text{Res}_d(D)$. (b) If every mixed final cluster C has a truth-sound $T \in \text{Min}_d^+(C)$ (a sound merge), then $\text{Acc}_d \subseteq \text{Tr}$: truth-sound, though not target-sound. (c) Otherwise Acc_d is sound only relative to $\text{Res}_d(D)$, which decreases in d ; $\text{Res}_\infty(D)$ consists of templates with no refutable instance. (d) Exactness can fail even on clean data with an anchor in every D_i and disjoint instance sets: a sound cross merge can absorb part of an anchor (fragmentation), depending on the merge policy.

For (d), let $\sigma_1 = (z + 0 = z)$, $g = (0 + S0 = S0)$ a ground target and $D = \{0 + 0 = 0, S0 + 0 = S0, g\}$; the first two data are an anchor of σ_1 . In the order $g, 0 + 0 = 0, S0 + 0 = S0$, the pair $\{g, 0 + 0 = 0\}$ merges through the *true* template $0 + z = z$; then $S0 + 0 = S0$ cannot join, since the three data have the single minimal template $z_0 + z_1 = z_2$, refuted by $0 + 0 = S0$. The output rejects $p + 0 = p$ and accepts the true non-target $0 + p = p$; in the order $0 + 0 = 0, S0 + 0 = S0, g$ it is exact (computed). This failure is intrinsic:

Proposition 6.13 (ambiguous practices). *[proved; computed]^{untagged} Rem 5.11* For the valid practices $P = \{z + 0 = z, 0 + S0 = S0\}$ and $P' = \{0 + z = z, S0 + 0 = S0\}$, the data $D = \{0 + 0 = 0, S0 + 0 = S0, 0 + S0 = S0\}$ lie in $R_P^* \cap R_{P'}^*$ and contain anchors for both. No learner (computable or not) with a target-blind oracle is on D both exact for P and target-sound for P' .

Proof. Anchors: $\{0 + 0 = 0, S0 + 0 = S0\}$, $\{0 + S0 = S0\}$ for P ; $\{0 + 0 = 0, 0 + S0 = S0\}$, $\{S0 + 0 = S0\}$ for P' . The learner's output on D is the same under both; exactness for P needs $p + 0 = p$, which is not in $R_{P'}^*$. $\square$

Only separating data such as $SS0 + 0 = SS0 \in R_P^* \setminus R_{P'}^*$ resolve this; the k' -union learner with self-generated negatives is not exact on this D either (computed), since the true template $0 + z = z$ violates (X). Residual merges come in three kinds: *target-sound* ones (a redundant axiom that is an instance of a target schema, e.g. Empty Set as an instance of Separation); *sound* ones (true templates below no target, as for $\forall x\varphi$ below); and *unsound unrefuted* ones (the K_0 -shaped templates of induction and the universal-set schema, Section 6.11).

6.6 Learning $\forall x\varphi$ from instances as a sound merge

Regard each instance $\varphi(t)$ as its own ground target. The universal sentence is learned when DTRC merges the instances.

Lemma 6.14 (two-point anchors). *[proved]^{untagged} Lem 5.5* Let $\varphi(z)$ have a single metavariable, a 0-ary z (term or formula sort) occurring in it. If closed t_1, t_2 have different head symbols, $\{\varphi(t_1), \varphi(t_2)\}$ is an anchor for $\varphi(z)$ in DT° . So for every set of instances of $\varphi(z)$ containing two with different heads, $\text{inst}(\varphi(z))$ is contained in the instance set of every covering template.

Proof idea. If $G(z), K(z)$ contain z only as a leaf and agree at two closed values with different heads, then $G = K$. This forces every occurrence of a covering template to agree with $\varphi(z)$ (Appendix E.4). It is the case of the anchor theorem of Section 3 in which the no-coincidence event (U) holds automatically.

Proposition 6.15 (universals from instances). *[proved; computed]^{untagged Prop 5.6} Let z be a term metavariable and $D = \{\varphi(t_1), \dots, \varphi(t_n)\}$ closed instances, two with different heads. (a) D is d -coherent iff $\varphi(z)$ is not d -refuted. (b) $\text{inst}(\varphi(z))$ contains $\varphi(a)$ for a parameter a fresh for φ , which under the closure reading is $\forall x\varphi(x)$ (in parameter form; the string $\forall x\varphi$ is a different datum, Section 2); so $\varphi(z)$ is truth-sound iff $\forall x\varphi$ is true. (c) If $\forall x\varphi$ is true, every subset of D is coherent, DTRC forms one cluster under every policy, and $\text{Acc}_d \ni \varphi(a)$: the axiom is learned and nothing false accepted. (Target-soundness for the n ground “targets” fails, as it must: the instances and the axiom give the same data and oracle answers.) (d) If $\forall x\varphi$ is false and some instance of $\varphi(z)$ is d -refuted, the merge is refused, though root-homogeneous subsets may still merge into an unrefuted specialization such as $\varphi(Sz)$. (e) If the oracle refutes every false instance of $\varphi(z)$ at some depth, “never refuted” means “ $\forall x\varphi$ true”; $R\text{-}\Delta_0$ does so for quantifier-free φ (Proposition 6.16) but not for quantified φ . At a finite budget, false universals with large least counterexamples merge (part of Res_d). An oracle that only evaluates closed instances certifies “true at every closed term”: in $\mathbb{N}$ this is the ω -rule, sound but not derivable ($\mathbb{Q} \vdash 0 + \underline{n} = \underline{n}$ for each n , $\mathbb{Q} \not\vdash \forall x(0 + x = x)$); in the field $\mathbb{R}$ with $0, 1, +, \cdot$, where closed terms denote natural numbers, $\neg(z \cdot z = 1 + 1)$ is an unsound unrefuted merge.*

Section 4 characterizes when closed instances decide universals (the Tarski–Vaught criterion). Computed: numeral instances of Q3–Q5, as 12 ground targets, form three clusters with templates $z_0 + 0 = z_0$, $z_0 \cdot 0 = 0$, $z_0 + Sz_1 = S(z_0 + z_1)$, and DTRC accepts $p + 0 = p$; instances of the false $n \cdot n = n$ ($n = 0, 1$) stay apart (refuted at 2); $\neg(n \cdot n = 2)$, $n \leq 2$, merge and the true $\neg(p \cdot p = 2)$ is accepted.

Proposition 6.16 (quantifier-free templates). *[proved]^{untagged Prop 5.13} Let $T \in \text{DT}^\circ$ be an arithmetic template with no binder in its rigid part (so all its metavariables are 0-ary; formula metavariables may still take quantified closed formulas). If T has a false instance, it has a false quantifier-free instance, which $R\text{-}\Delta_0$ refutes at some depth. Hence every such template in Res_∞ (for $R\text{-}\Delta_0$) is truth-sound.*

Proof. Let $T\theta$ be false at numerals $\bar{n}$ for its parameters. Keep the term values of θ and replace each formula value $\theta(F)$ by $0 = 0$ or $0 = S0$ according to the truth value of $\theta(F)[\bar{n}]$. The result $T\theta'$ is quantifier-free and false at $\bar{n}$, since the rigid part is a Boolean combination of atoms over unchanged terms and of metavariable occurrences with unchanged truth values. $R\text{-}\Delta_0$ at depth $\geq \max(\bar{n}, |T\theta'|)$ refutes it. $\square$

So in the untagged setting the first question has a positive answer by a sound merge. What certifies the universal is the absence of a counterexample, not a derivation; that is truth-sound in $\mathbb{N}$ for quantifier-free φ with $R\text{-}\Delta_0$, and otherwise only relative to the strength of the oracle. Unsound permanent residues in arithmetic need quantifiers in the template.

6.7 Depth relativization is necessary

Theorem 6.17 (depth relativization). *[proved]^{untagged Thm 5.7} There is a uniformly computable family of practices (finite sets of DT° targets in arithmetic, each with a data law, all with the oracle $R\text{-}\Delta_0$) such that no computable learner achieves, for every valid practice $\mathcal{P}$ of the family and some $\delta < \frac{1}{2}$, both (a) $\Pr[\exists t : \text{Acc}_t \not\subseteq R_{\mathcal{P}}^* \cup \text{Res}_\infty(\mathcal{P})] \leq \delta$, where $\text{Res}_\infty(\mathcal{P})$ is the union of the instance sets of templates that cover instances (in the support of the data law) of two different targets of $\mathcal{P}$ and have no refutable instance; and (b) if $\mathcal{P}$ is separated at some finite depth and each target’s data law yields an anchor with positive probability, then $\Pr[\exists t \forall t' \geq t : \text{Acc}_{t'} = R_{\mathcal{P}}^*] \geq 1 - \delta$.*

Proof idea. By MRDP (Davis et al., 1961; Matiyasevich, 1993), $e \in K$ (the halting set) iff $\exists \bar{x} P(e, \bar{x}) = Q(e, \bar{x})$ for fixed polynomials. Let $T_e := \neg \exists \bar{x} (z = SS(x_1 + \dots + x_k) \wedge P_e(\bar{x}) = Q_e(\bar{x}))$, $\mathcal{P}_e := \{T_e\}$, and $\mathcal{P}'_e := \{T_e[z := 0], T_e[z := S0]\}$ (two true ground targets), both with data uniform on $\mathcal{P}'_e$, an anchor of T_e . If $e \in K$, $R\text{-}\Delta_0$ refutes an instance of T_e by exhibiting the witnesses, so $\mathcal{P}'_e$ is separated with empty residue and (a) forbids accepting $q_e := T_e[z := SS0]$. If $e \notin K$, $\mathcal{P}_e$ is valid and separated, so (b) forces eventual acceptance of q_e . Then “ q_e is accepted by some time with probability $> \frac{1}{2}$ ” would be a Σ_1 definition of $\omega \setminus K$ (Appendix E.4). $\square$

The obstruction is the Π_1 -completeness of “this merge is never refuted”, not lack of information. DTRC with a depth schedule $d_t \rightarrow \infty$ meets (b), is sound relative to Res_{d_t} at time t and to Res_∞ in the limit, and changes its mind finitely often on every separated practice. This is the form of soundness of the parent report (Claude (Anthropic), 2026), and the theorem shows it cannot be improved.

6.8 Whole-cluster tests and noise

Proposition 6.18 (whole-cluster versus pairwise tests). *[proved; computed]^{untagged Prop 5.8} Coherence is downward closed but not transitive. Every cluster built by DTRC is coherent at all times (except a refutable singleton, which contributes nothing), whereas single linkage on the pairwise-coherence graph can build clusters with no unrefuted minimal template. Under $\text{RS}_d(D)$, the coherent subsets of D are exactly the subsets of the D_i , and single linkage, complete linkage and DTRC agree. Without separation, DTRC returns some maximal partition into coherent blocks, which may depend on the policy.*

Example: $a = (0 + 0 = 0)$, $b = (S0 + 0 = S0)$, $c = (0 + S0 = S0)$. The pairs $\{a, b\}$, $\{a, c\}$ have the unrefuted templates $z_0 + 0 = z_0$, $0 + z_0 = z_0$, while $\{b, c\}$ and $\{a, b, c\}$ have only $z_0 + z_1 = S0$, resp. $z_0 + z_1 = z_2$, refuted by $0 + 0 = S0$. Single linkage lumps all three; DTRC returns $\{\{a, b\}, \{c\}\}$ or $\{\{a, c\}, \{b\}\}$.

Noise. Let $D = D_{\text{clean}} \cup E$ with mistakes $E \cap R^* = \emptyset$.

Proposition 6.19 (noise). *[proved; computed]^{untagged Prop 5.9} (a) A refutable mistake stays an incoherent singleton. (b) Under $\text{RS}_d(D_{\text{clean}})$ no coherent set contains clean data of two targets, so mistakes never bridge targets. (c) An unrefutable mistake can be absorbed into a target’s cluster and make it unsound, and (d) it can fragment the target: $C \cup \{m\} \cup C'$ can be incoherent although $C \cup C'$ is coherent. (e) For $|C| > e$ the trimmed verifier $\text{Acc}_d^e(C) := \bigcap \{\text{inst}(T) : T \text{ not } d\text{-refuted}, |C \setminus \text{inst}(T)| \leq e\}$ is contained in $\text{inst}(\sigma_i)$ when C has at most e mistakes and its clean part lies in D_i , and equals $\text{inst}(\sigma_i)$ when moreover every subset of C of size $\geq |C| - e$ contains an anchor.*

Robust DTRC asserts only clusters of multiplicity $\geq s > e$, verified with Acc_d^e . If $\text{RS}_d(D_{\text{clean}})$ holds, every mistake is refutable or *noise-separated* (every template covering it and a clean datum is refuted), and every coherent set of mistakes has multiplicity $< s$, then the asserted clusters are exactly the D_i of multiplicity $\geq s$, verified soundly, and exactly once every subset of D_i of size $\geq |D_i| - e$ contains an anchor. Fragmentation is not prevented, and a frequent coherent family of mistakes is a systematic error that no positive-data method can tell from a rule. A computed example on PA (Example E.8) shows absorption of the false instance $J_0^* = (0 + 0 = 0 \wedge \forall x(0 + x = 0 \rightarrow 0 + Sx = S0)) \rightarrow \forall x(0 + x = 0)$ into a non-diverse induction cluster, its rejection by the trimmed verifier, and order-dependent fragmentation.

6.9 Cost: the union verifier and the refutation test

Theorem 6.11(v) counts coherence tests. A test needs at most $|\text{Min}(X)|$ refutation checks, and $|\text{Min}(X)|$ can be 4^n for two data of size $O(n)$ (Section 3). It is 1, and the test polynomial, when X is quantifier-free (Plotkin’s lgg) or contains an anchor (under (Rich)), and for a single negative n the test is “ $n \notin \text{Acc}(X)$ ”, polynomial by the feature verifier of Section 3 (Proposition E.9). In every test of the PA and ZF runs below, $|\text{Min}(X)| = 1$. In general both the exact union verifier and the refutation test are hard; write $\text{Acc}_k(D) := \text{Acc}_k(D, \emptyset)$.

Theorem 6.20 (complexity). *[proved; computed]^{single Thm H} (a) For $k \leq 2$, “ $q \in \text{Acc}_k(D)$ ” is decidable in polynomial time (2-SAT). (b) For every fixed $k \geq 3$ it is coNP-complete (from graph k -colourability). (c) With k in the input it is coNP-complete already for quantifier-free data over $\wedge, =, S, 0$ (from Set Cover); for binder-free data and fixed k it is polynomial. (d) The refutation test of DTRC step 2, “does some $T \in \text{Min}(X)$ avoid a given finite set R ?”, is NP-complete, already for $|X| = 2$ with one binder (from CNF-SAT); it is polynomial when X has a least covering template, in particular when X contains an anchor (under (Rich)). $R \cap \text{Acc}(X) \neq \emptyset$ is a polynomial sufficient condition for blocking a merge.*

The proofs (Appendix E.5) describe the sets B with $q \notin \text{Acc}(B)$ by polynomially many *failure types*. These results were added to the single track’s record after its referee round and were not refereed independently; they are worst-case results. In the hardness proof of (d) one datum is false; whether the test stays hard when X is true and R false in a fixed structure, as in DTRC, is open.

6.10 Without negatives: MDL and the size principle

Could compression replace refutation? Determinacy makes a two-part code well defined: each datum is coded by a covering template and its unique matcher. For $\mathbf{Q}$ plus induction, the question is whether MDL keeps T_{Ind} whole or splits it by the main connective of the motive. A naive code (symbols at $\log_2$ of the alphabet size, two-part code for the labels) prefers the split by a margin that grows linearly in the number of induction data (Proposition E.11). A well-specified code gains at most $O(\log n)$ from splitting (Proposition E.12), which does not show that it selects the target partition. Computations with adaptive codes (Table 11) show that under a natural usage law even depth-aware codes split for $n \geq 1.6 \cdot 10^4$, because usage really differs across connectives, while under a law for which the code is well specified it keeps T_{Ind} whole. **MDL tracks the statistics of usage, not the logical boundaries of schemas.** For schemas it errs towards incompleteness (a split never accepts induction on an unused connective); for rarely used ground axioms it errs towards unsoundness (two ground axioms used m times each merge into their lgg once m times the body cost is below the template saving, and $0 + 0 = 0$, $0 \cdot 0 = 0$ merge into a false template). The refutation test of DTRC catches the second error, and the anchor condition controls the first. This is the size principle (Tenenbaum and Griffiths, 2001) and MDL (Rissanen, 1978) meeting a hypothesis space whose boundaries are logical rather than statistical.

6.11 Natural examples: PA, ZF and ZFC

The computations use the untagged track’s prototype implementation of DTRC (separate from the package of Section 7). After the referee found that its first version decided refutation by template-specific searches, which is not monotone in $\succeq$, it was changed to keep a global set of refuted sentences and to audit each run; a run that passes the audit *is* DTRC_d with a fixed finite

oracle, so the theorems above apply to it verbatim (Proposition E.13). Every run passed on the first pass, and no result changed.

PA. Every cross merge among Q1–Q7 and T_{Ind} , and among the instance schemas of Q’s axioms (the alternative presentation of Q by closed instances), is refuted by R- Δ_0 (Table 12); for Q1–Q7 and T_{Ind} this global separation is proved, for the instance schemas it is supported by samples. On three seeds of 40 shuffled unlabelled data, DTRC returned one pure cluster per target present (7 or 8); the induction cluster’s only unrefuted minimal template was T_{Ind} ; 200/200 held-out induction instances and 0 of 91 non-instances (near misses) were accepted; separation on the data was verified for all 147, 201 and 175 cross pairs. Induction is never at risk of a cross merge, because every template covering a Q-axiom and an induction instance is absorbing and refuted; it is learned incompletely only for lack of diversity in its own data, as in the tagged case.

ZF and ZFC. The targets are Ext, Pair, Union, Power, Inf, Found, Separation $\exists b \forall c (c \in b \leftrightarrow c \in p \wedge P(c))$, Replacement with $\exists!$ spelled out, and $\in$ -induction, in closure-normal de Bruijn form; the three schemas are Miller patterns (Section 5). All 36 cross pairs have a unique minimal covering template, refuted by an HF counterexample (22 pairs: F_0 or $F_0 \rightarrow F_1$), by pure logic (14 pairs, e.g. $\exists x F_0(x)$), and in three of the latter by **Russell’s paradox**: Union–Power, Union–Separation and Power–Separation all have the minimal template $\exists x \forall y (y \in x \leftrightarrow F_0(y))$, naive comprehension, whose instance $\exists x \forall y (y \in x \leftrightarrow \neg y \in y)$ is logically false (Example E.14; the referee’s independent enumeration of 598 covering templates agrees). On three seeds of 45 data DTRC returned 9 pure clusters, each schema cluster exact (180/180 held-out instances), and 0/30 instances of naive comprehension accepted: Separation without its guard is not learned. The Axiom of Choice is one more ground axiom: its 12 cross templates are all refuted, and on ZFC data (three seeds of 50) DTRC returned 10 pure clusters with AC accepted, its anchor being {AC} (Example E.15). So the ZFC axioms are separated by the two refutation channels of the setting, and Russell’s paradox does the work exactly where the axioms look alike. Section 7 reports an independent implementation with the same picture on PA and ZF mixtures.

Where separation fails. *Sound merges*— $\forall x \varphi$ from instances, redundant targets, instances shared by two true generalizations—cannot and should not be prevented; they are why DTRC is truth-sound rather than target-sound without separation. *An unsound merge that only coherence refutes*: in bounding form, $\text{Union}_W = \exists b \forall c (\exists d (d \in p \wedge c \in d) \rightarrow c \in b)$ and $\text{Power}_W = \exists b \forall c (\forall d (d \in c \rightarrow d \in p) \rightarrow c \in b)$ have as unique minimal covering template the *universal-set schema* $U(F) = \exists x \forall y (F(y) \rightarrow y \in x)$, whose instance $\exists x \forall y (y = y \rightarrow y \in x)$ is false.

Proposition 6.21 (the universal-set schema). *[proved; computed]^{untagged Prop 7.1} No instance of U is refuted by R-logic, by R-HF, or by any oracle sound for the structure $M_0 = \text{HF} \cup \{u\}$ with $\in$ as in HF on HF and $y \in u$ for all $y \in M_0$. The template is refuted by coherence, together with the true Π_1 sentence $\forall x \neg x \in x$ or with the closure of one Separation instance.*

Proof. Every instance holds in M_0 with $x := u$; HF is a transitive part of M_0 , so R-HF’s steps (Δ_0 facts about HF, HF witnesses and counterexamples) are sound in M_0 ; R-logic refutes only logical falsehoods. A witness b of $\exists x \forall y (y = y \rightarrow y \in x)$ gives $b \in b$; or $r = \{y \in b : y \notin y\}$ gives $r \in r \leftrightarrow r \notin r$. $\square$

With logic and HF only, DTRC merges Union_W and Power_W and *accepts the false universal-set sentence* (computed), an unsoundness inside the residue. The learned Separation cluster accepts

Russell’s instance of Separation; designating its closure as true (*bootstrapped coherence*) and re-running gives 9 pure clusters, with the universal set rejected. Bootstrapping is sound exactly when the trusted clusters are pure, and circular otherwise. K_0 : the first-order lgg K_0 of $\text{Ind}(0 + x = x)$ and $\text{Ind}(S0 + x = Sx)$ has only instances that are jointly consistent with the true quantifier-free sentences of $\mathbb{N}$ (a result of the prior work’s referee), so $\text{R-}\Delta_0$ and logic refute none of them, although J_0^* is one. In DT° the pair has two minimal templates, the T_{Ind} -specialization with motive $\lambda x.(f(0) + x = f(x))$ and a K_0 -shaped one (confirmed over 6258 covering templates); because the verifier intersects both, J_0^* is rejected. The K_0 -shaped template harms only when the sound alternative disappears (absorption of J_0^* , Example E.8); coherence with designated or learned $\mathbf{Q}$ refutes it. *Permanent residue*: templates whose false instances are all consistent with the designated truths and Δ_0 -correct (e.g. families containing $\neg\text{Con}(\text{PA})$ -like Σ_1 sentences) are never refuted, and by Theorem 6.17 no computable learner can tell in general whether a merge will be. In arithmetic they need quantifiers in the template (Proposition 6.16); a natural merge of this kind was not found (open). Section 7 reports a budget effect of the same type: with injected mistakes and a small refutation budget, two mistakes merged into a false template and a false sentence was accepted; a larger budget refuted the merge.

6.12 Summary: the answer to the third question

DTRC answers the third question, with four amendments to the method first proposed, each forced by a result above: (1) test *whole clusters* and intersect *all* unrefuted minimal templates (coherence is not transitive, and DT° is not unitary: a K_0 -shaped template can sit next to the sound one); (2) keep the refutation depth as a parameter and re-cluster as it grows (Theorem 6.17); (3) when a bound $k \geq k'$ is known, intersect with the k -union learner fed with DTRC’s own negatives, which restores target-soundness without separation and, at $k = k'$ under separation, loses no exactness once anchors are present, while spare slots cost diversity (Remark 6.8) and ambiguous data defeat every learner (Proposition 6.13); (4) implement refutation as a growing set of refuted sentences and audit the run (Proposition E.13). Under refutation separation the untagged problem is exactly as hard as the tagged one, and PA, ZF and ZFC are separated at small depth. Open: spare-slot thresholds beyond one-variable schemas; the refutation test for true data and false negatives; usable bounds on the separation depth for natural targets (there is no uniform computable one); a natural permanent residue; a fragmentation-free robust merge rule (which needs more than anchors); a theorem that some natural universal code selects the target partition; and guarantees for bootstrapped coherence.

7 Implementation and experiments

We implemented the determinate-template learner and DTRC to check the theory of Sections 3 to 6 on data it did not choose, and to see what goes wrong when its hypotheses fail: when refutation has a budget, when targets look alike, and when the data contain mistakes. The package `dtrc` (pure Python, `axiom-schemas/code/`) is independent of the prototype of Section 6. An adversarial referee re-ran every experiment with independent checks and found one false empirical claim and one false proposition in the first version of the record (the *first record* below); both are corrected here and listed in Appendix G. *Computed* means that a seeded script was run; re-running reproduces every number except timings. Proofs, further numbers and commands are in Appendix F.

In brief: on Q1 and Q2 the implemented learner behaves as the theory predicts, and on Q2 it

accepted no non-instance in 927 probes. On untagged mixtures of the PA and ZF axioms, DTRC recovered the hidden labels in every run, and with a sharing pass for ambiguous data it matched, target by target, a tagged learner given every label a datum deserves; for these mixtures we prove refutation separation. But with injected mistakes and a small refutation budget, DTRC *accepted a false sentence*; similar targets can merge into valid templates that no world-based refutation separates; and the evidence for separation beyond the mixtures is narrow.

7.1 What was implemented

Data and the class. Formulas over L_A and $L_\in$ use de Bruijn indices and named parameters. Each datum is in closure-normal form and is canonicalized on its own (parameters renamed $w_0, w_1, \dots$ by first occurrence). Leading universal quantifiers are *not* stripped: the axiom Q4 is the datum $\forall x (x + 0 = x)$, and the schema $z + 0 = z$ accepts $w_0 + 0 = w_0$ but not Q4. With this choice the PA-mixture targets below have disjoint instance sets except for the sentence $0 + 0 = 0$, and the ZF-mixture targets are pairwise disjoint (Proposition 7.9). (With leading quantifiers stripped, Q4 becomes an instance of $z + 0 = z$ and Q6 of $z \cdot 0 = 0$; the referee re-ran DTRC so and found ARI 1 on unambiguous data, Q4 and Q6 absorbed into the clusters of these schemas, and no non-target probe ^{referee R13}.) The matcher decides membership up to an injective renaming of the template’s parameters; in this section *inst*, “covers” and $\succeq$ are read in this sense (*inst* $^\sim$, $\succeq^\sim$ of Appendix A), and anchors are taken in the class $\{\text{inst}(T) : T \in \text{DT}_F^\circ\}$.

The implemented class is DT_F° (Section 2). Every target used below lies in it, so the cautious verifier stays sound; positive-arity term metavariables blow up the minimal covering sets (E7); and every DT° anchor is a DT_F° anchor, so the classes differ only before an anchor arrives.

Minimal covering templates. Let $C(D)$ be the *common-prefix tree* of a finite set D of data: the positions p such that all data carry the same label and arity at p and above it (parameters compared by name), and the *slots*, the maximal positions at which they disagree. The *column* at p is $(d|_p)_{d \in D}$; it is *open* if some $d|_p$ contains a variable bound above p . In $\mathcal{T}(D)$, each common atom node whose subtree contains a slot with an open column is cut to a leaf slot (an *F-slot*); so every term slot of $\mathcal{T}(D)$ has a closed column.

Definition 7.1 (Normal configurations). [definition]^{experiments §1.3} For a slot σ of $\mathcal{T}(D)$ let $\bar{y}_\sigma$ list the variables free in its column, M_σ a fresh metavariable of the slot’s sort and arity $|\bar{y}_\sigma|$, and $\beta_d(\sigma)$ the abstraction of $d|_\sigma$ over $\bar{y}_\sigma$. A node π is *derivable* from σ , $\sigma \rightarrow \pi$, if metavariable-free terms $\bar{t} =: \bar{t}(\sigma, \pi)$ satisfy $\beta_d(\sigma)[\bar{t}] = d|_\pi$ for all $d \in D$ ($\bar{t}$ is then unique). A *normal configuration* is a set S' of slots, none derivable from another, an antichain $A \supseteq S'$ of nodes of $\mathcal{T}(D)$ with every slot at or below a member of A , and a choice of $c(\pi) \in S'$ with $c(\pi) \rightarrow \pi$ for each $\pi \in A \setminus S'$. Its template has $M_\sigma(\bar{y}_\sigma)$ at $\sigma \in S'$, $M_{c(\pi)}(\bar{t}(c(\pi), \pi))$ at $\pi \in A \setminus S'$, and the common labels elsewhere.

Proposition 7.2 (Normal form). [proved; computed]^{experiments Props E1-lit, E1-al; E0, E0b} Let D be finite and nonempty, and compare parameters by name. (a) Every normal configuration lies in DT_F° and covers D . (b) Every $T \in \text{DT}_F^\circ$ covering D satisfies $T \succeq N$ for a normal configuration N . (c) Hence the minimal normal configurations form a finite set $\text{Min}_F^{\text{lit}}(D)$; every covering $T \in \text{DT}_F^\circ$ lies above a member; members are minimal among all covering templates in DT_F° ; and if $D \subseteq \text{inst}(T^*)$ with $T^* \in \text{DT}_F^\circ$, some member lies below T^* .

Proof idea. For (b): rigid symbols survive instantiation, so the skeleton of T lies in $C(D)$, and no occurrence sits strictly below an F-slot (it would be a 0-ary term metavariable with a closed value containing an open column). Push pattern occurrences down through common labels; drop

argument places no datum uses; the remaining arguments are forced; replace an own slot derivable from another by the derived occurrence. Each step specializes T and keeps it covering. Details in Appendix F.1. $\square$

With parameters compared up to renaming, the learner uses $\text{Min}_F(D)$, the minimal elements of the union of $\text{Min}_F^{\text{lit}}(\alpha(D))$ over all *alignments* α of the data’s parameters; Appendix A proves (a)–(c) for it, given the finiteness that Proposition 7.2(c) supplies. The literal set is not enough: the first record stated (c) for renaming-closed covering, and the referee found a counterexample with a rigid parameter (Appendix A). The aligned and literal sets differed on none of the 720 data sets of E2 and in one call of all DTRC runs of E3; no cap on alignments or configurations was hit. Computed checks against the prior referee’s enumerator (E0) and on 800 random data sets in both languages (E0b) found no failure of the aligned set; the literal one failed exactly when the generating template had a rigid parameter (Appendix F.1).

World oracles. Two three-valued evaluators return True, False or unknown. *PA* (truth in $\mathbb{N}$) evaluates numerals exactly, decides atoms with symbolic variables by polynomial normal forms, enumerates bounded quantifiers with numeral bounds, searches numeral counterexamples and witnesses, and case-splits an unbounded quantifier into $x \in \{0, 1\}$ and $x = x' + 2$. *ZF* (truth in V) uses hereditarily finite sets of rank ≤ 3 and *generic objects*: an unbounded quantifier is split into x in the transitive closure E of the concrete values in scope, x an in-scope generic, or x a fresh generic. Both use Kleene’s strong connectives (Kleene, 1952) with supervaluation over unknown leaves (van Fraassen, 1966). Both first simplify by *constant propagation*: propositional laws valid in every nonempty structure, after replacing by $\top$ or $\perp$ the atoms whose value does not depend on the variables ($t = t$; closed arithmetic atoms; $t < t$; $x \in x$, false by Foundation).

Proposition 7.3 (The oracles are sound). *[proved; computed]^{experiments Prop E2; recheck R3, R4, R11} If the PA evaluator returns True (False) on a datum, its universal closure is true (false) in $\mathbb{N}$. If $V \models \text{ZF}$ and the ZF evaluator returns True (False) on a datum, its universal closure is true (false) in V .*

Proof idea. Induction on the formula. For ZF the claim is made for every *realization* of the generics: each denotes a set outside the transitive closure fixed when it was introduced, distinct from the generics then in scope. The atom rules are sound (e.g. $G \in G$ is false by Foundation), the three cases of a quantifier cover every set, Kleene’s tables are monotone, and supervaluation enumerates a superset of the realizable assignments. Details in Appendix F.2. $\square$

The referee’s independent evaluators found 0 contradictions on 2197 PA and 592 ZF verdicts logged during DTRC runs and on 3600 random sentences (Appendix F.2). Neither oracle is complete: no witness is constructed from a generic or a term, so $\forall a \exists b \forall x (x \in b \leftrightarrow a \in x)$ is not refuted and $\forall x (x = 0 \vee \exists y x = Sy)$ stays undecided. Both are truth oracles, not provability oracles.

Template refuter and dtrc. A template is *refuted* once the oracle returns False on one of its instances. The refuter tries, within a budget (default 80 instances): all $\top/\perp$ assignments to the formula metavariables; one metavariable running through a pool of small bodies while the others are $\top$ (or $\perp$); enumerations of small bodies; and bodies read off the data. It is *sound*, *budgeted*, and *history-dependent* (it caches verdicts and accumulates data-guided tries), so it is not a fixed oracle Ref_d (Section 2); the guarantees below are stated for it directly. Every “refuted” count is a lower bound on what an *ideal* refuter, one that refutes every template with a false instance, would refute.

DTRC as implemented: (1) canonicalize and discard data the oracle refutes; (2) agglomerate from singletons, trying pairs of clusters in decreasing order of common-prefix size, where the merge of A and B succeeds iff some member of $\text{Min}_F(A \cup B)$ is not refuted, and a failed pair stays failed for clusters containing its parts; (3) optionally a *sharing pass*: for each final cluster C and each kept datum $d \notin C$, in data order, add d to C if the test on $C \cup \{d\}$ succeeds (clusters may then overlap); (4) per cluster, $\text{Acc}(C) = \bigcap \{\text{inst}(T) : T \in \text{Min}_F(C) \text{ not refuted}\}$, or C if all are refuted; (5) assert $\bigcup_C \text{Acc}(C)$. This is DTRC of Section 6 with the class DT_F° , a fixed merge order and a budgeted refuter.

Baselines. (i) First-order lgg per tag (Plotkin, 1970; Reynolds, 1970), with binders named by level or with de Bruijn indices; instantiation may capture. (ii) Higher-order pattern lgg per tag, in our reading of the merge rule of Baumgartner et al. (2017) (their implementation was not run): the *genuine* version generalizes a term disagreement containing a bound variable by a term variable $f(x, \dots)$; the *formula-level* version of the first record generalizes the whole atom instead. (iii) The cautious k -union verifier over DT_F° , with and without refutation negatives. (iv) *Skeleton clustering*: DTRC without refutation, stopped at the true number of targets. (v) The tagged refutation-filtered verifier with *gold labels* (the target a datum was generated from) or *membership labels* (every target it instantiates).

Data sets and metrics. *PA-mix*: Q1–Q7; 30 induction instances with random motives; 8, 6, 4 instances of $U_{\text{add}0} = (z + 0 = z)$, $U_{\text{mul}0} = (z \cdot 0 = 0)$, $U_{0\text{add}} = (0 + z = z)$. In the main regime z is a numeral uniform in 0..7 (numerals written as digits); in the *mixed* regime of the first record it is a numeral (0.6), a random closed term of depth ≤ 2 (0.25) or a parameter (0.15). *ZF-mix*: Ext, Pair, Union, Pow, Inf, Found; 12 instances of T_{Sep} , 10 of Replacement Repl_{sp} ($\exists!$ spelled out, image consequent), 10 of T_{EInd} , with random small bodies. Stress sets are described with E4 and E9. Held-out sets contain schema instances only, disjoint from the training data. Metrics: the adjusted Rand index (ARI; Hubert and Arabie, 1985) on unambiguous data; *exactness*, by a sufficient syntactic test for $\text{Acc} = \text{inst}(T^*)$ (every unrefuted minimal template is $\succeq T^*$ and one is $\preceq T^*$), reported separately for ground axioms, where it only means that the axiom stayed a singleton; held-out acceptance; and *probes*: sampled accepted sentences, how many are instances of no target, and how many of those the oracle refutes. Probe counts are lower bounds on unsoundness (Proposition 7.12).

7.2 Guarantees for the implemented learner

The following are the analogues of the theorems of Section 6 for DT_F° , Min_F and the budgeted, history-dependent refuter. The world is $\mathbb{N}$ or V , and the refuter is sound.

Proposition 7.4 (Merge monotonicity; within-target merges succeed). *[proved]^{experiments Prop E3}*
(a) Let $D \subseteq D'$ and the refuter be ideal. If every member of $\text{Min}_F(D)$ is refuted, so is every member of $\text{Min}_F(D')$. (b) If $\emptyset \neq X \subseteq \text{inst}(T^*)$ with $T^* \in \text{DT}_F^\circ$ and all instances of T^* true, then some $T_0 \in \text{Min}_F(X)$ with $T_0 \preceq T^*$ is never refuted, so the merge test on X succeeds for every sound refuter and budget.

Proof. (a) A member of $\text{Min}_F(D')$ covers D , so it lies above a member of $\text{Min}_F(D)$ and contains that member’s false instance. (b) Proposition 7.2(c), with alignments, gives T_0 ; its instances are true. $\square$

Theorem 7.5 (DTRC under refutation separation). *[proved]^{experiments Thm E4} Assume (T) targets $T_1, \dots, T_k \in \text{DT}_F^\circ$ with all instances true; (D) data $D = D_1 \sqcup \dots \sqcup D_k$, $D_i \subseteq \text{inst}(T_i)$, each datum an instance of exactly one target; (R) a sound refuter; and (S) every merge test on $A \cup B$ with $\emptyset \neq A \subseteq D_i$, $\emptyset \neq B \subseteq D_j$, $i \neq j$, fails. Then DTRC without sharing discards no datum, its final clusters are the nonempty D_i , and $\text{Acc}(D_i) \subseteq \text{inst}(T_i)$ (target-soundness), with equality whenever D_i contains an anchor of T_i . If moreover (H-ref) verdicts do not depend on the call history, $\text{Acc}(D_i)$ equals the tagged verifier’s acceptance set on D_i .*

Proof idea. Clusters stay inside one D_i : cross merges fail by (S), within-label merges succeed by Proposition 7.4(b). So inherited failures never block a within-label merge, and agglomeration ends with one cluster per label. The never-refuted $T_0 \preceq T_i$ is intersected, which gives soundness; with an anchor every covering template contains $\text{inst}(T_i)$. Details in Appendix F.3. $\square$

The first record claimed exactness *exactly* where the tagged learner is exact; the “only if” half fails when instance sets overlap, hence the per-cluster statement. The implemented refuter violates (H-ref) ^{referee F8}.

Proposition 7.6 (Per-cluster soundness does not depend on the budget). *[proved]^{experiments Prop E5} If $C \subseteq \text{inst}(T^*)$ with $T^* \in \text{DT}_F^\circ$ and all instances of T^* true, then $\text{Acc}(C) \subseteq \text{inst}(T^*)$ for every sound refuter; more refutations only enlarge $\text{Acc}(C)$.*

Proof. T_0 of Proposition 7.4(b) is intersected, so $\text{Acc}(C) \subseteq \text{inst}(T_0) \subseteq \text{inst}(T^*)$; fewer intersected templates give a larger set. $\square$

So a budgeted refuter harms DTRC only through clustering: an unrefuted merge across targets or with a mistake. Such a mixed cluster is truth-sound if one of its unrefuted minimal templates is valid, and may accept false sentences otherwise (Proposition 7.12).

Proposition 7.7 (The sharing pass). *[proved]^{experiments Prop E4-share} Assume (T), (R), data $D \subseteq \bigcup_l \text{inst}(T_l)$ (a datum may instantiate several targets), (S^+) every merge or sharing test on a set contained in no $\text{inst}(T_l)$ fails, and (P) each final cluster C of the agglomeration lies in exactly one $\text{inst}(T_l)$. Then distinct final clusters of the agglomeration have distinct targets; after the sharing pass the cluster of T_l is $C^+ = D \cap \text{inst}(T_l)$, in any order; and $\text{Acc}(C^+) \subseteq \text{inst}(T_l)$, with equality whenever C^+ contains an anchor of T_l . Under (H-ref) $\text{Acc}(C^+)$ is the acceptance set of the tagged learner with membership labels.*

The proof is in Appendix F.3. In the numerals regime every anchor of $U_{\text{add}0}$ or $U_{0\text{add}}$ contains $0 + 0 = 0$, by the following proposition; a partition gives this datum to at most one of them, the sharing pass to both.

Proposition 7.8 (One-variable instance schemas). *[proved; computed]^{experiments Prop E7; referee R2} Let T^* have one 0-ary term metavariable z , at rigid positions, no other metavariable and no parameter, and let $D \subseteq \text{inst}(T^*)$ contain a parameter-free datum. Then $\text{Min}_F(D) = \{T^*\}$ iff the terms substituted for z do not all have the same root symbol (0 and each parameter count as root symbols); otherwise $\text{Acc}(D) \neq \text{inst}(T^*)$. So for i.i.d. terms with root law (p_h) , $\Pr[\text{not exact after } N] = \sum_h p_h^N$, and numerals uniform in 0..7 form an anchor iff they include 0.*

This is Plotkin’s (R) for the implemented class (Section 4); the referee found no violation in 3000 random data sets over 9 templates. Proof in Appendix F.3.

case	target pairs	members of $\text{Min}_F(\{a, b\})$	false instance
A	root clash. PA: $Q_i\text{--}T_{\text{Ind}}, Q_i\text{--}U, T_{\text{Ind}}\text{--}U$ (31 pairs); ZF: $\text{Inf}\text{--}\text{any}, T_{\text{EInd}}\text{--}\text{any}$ (15)	P	$\perp$
B	clash below a common $\forall$ -prefix. PA: $Q_i\text{--}Q_j$ (21); ZF: the other 17 pairs among $\text{Ext}, \text{Pair}, \text{Union}, \text{Pow}, \text{Found}, T_{\text{Sep}}, \text{Repl}_{\text{sp}}$	$\forall x P(x)$ or $\forall x \forall y P(x, y)$	$\forall \bar{x} \perp$
C	comprehension clash. ZF: $\text{Union}\text{--}\text{Pow}, \text{Union}\text{--}T_{\text{Sep}}, \text{Pow}\text{--}T_{\text{Sep}}$	$\forall x \exists y \forall z (z \in y \leftrightarrow P(z, x))$	$P := \neg z \in z$
D	ZF: $\text{Found}\text{--}\text{Repl}_{\text{sp}}$	$\forall x (A(x) \rightarrow \exists y B(y, x))$	$A := \top, B := \perp$
E	PA: two of $U_{\text{add0}}, U_{\text{mul0}}, U_{\text{0add}}$ (3)	$z_0 = z_1, z_0 = 0, \text{ or } z_0 + z_1 = R$	$1 = 0; S(r) + 0 = r$

Table 7: Minimal covering templates of cross pairs of mixture targets (Proposition 7.9). In case E, z_0, z_1 occur only where shown, R is built from the common prefix of the two substituted terms, and r is R with all metavariables set to 0. Russell’s instance in C fails at $z := y$.

Proposition 7.9 (Separating the mixture targets). *[proved; computed]^{experiments Prop E8; E10} Let $T_i \neq T_j$ both be PA-mix targets ($Q1\text{--}Q7, T_{\text{Ind}}, U_{\text{add0}}, U_{\text{mul0}}, U_{\text{0add}}$) or both ZF-mix targets ($\text{Ext}, \text{Pair}, \text{Union}, \text{Pow}, \text{Inf}, \text{Found}, T_{\text{Sep}}, \text{Repl}_{\text{sp}}, T_{\text{EInd}}$), and $a \in \text{inst}(T_i) \setminus \text{inst}(T_j), b \in \text{inst}(T_j) \setminus \text{inst}(T_i)$. Then every member of $\text{Min}_F(\{a, b\})$ has a form in Table 7 and the false instance shown there.*

Proof idea. In cases A–D the targets differ at a position above all their metavariables (in B possibly an F-slot), and agree rigidly elsewhere. A slot where two data have different root symbols derives only nodes with the same pair of root symbols (Lemma F.1), so the only normal configuration has an own metavariable at each slot. In case E derivation between the closed term slots would need identical columns, which do not occur. Details in Appendix F.3. $\square$

Corollary 7.10 (Separation of the mixtures). *[proved]^{experiments Cors E8.2, E8.3} With an ideal refuter, (S) holds iff every member of $\text{Min}_F(\{a, b\})$ is refuted for all cross pairs $a \in D_i, b \in D_j$. Hence (S), and with it Theorem 7.5, holds for PA-mix and ZF-mix data in which no datum instantiates two targets; $0 + 0 = 0$ is the only sentence that does.*

Proof. “Only if” is the case $|A| = |B| = 1$; “if” follows from Proposition 7.4(a) with $\{a, b\} \subseteq A \cup B$. Then apply Proposition 7.9; disjointness is read off its proof. $\square$

A computed check (E10) of 9100 random data pairs found the predicted shape and a certified false instance in every case.

7.3 Results

E1: universal axioms (Q1). On six universal axioms (2000 runs each) the DT_F° verifier and the first-order lgg became exact at the same N in every run. For the five one-variable axioms the mean N was 3.27 (predicted 3.25, mixed regime) and 8.42 (predicted 8.11, numerals); these rows share seeds and form one sample, in which 8.42 is 1.85 standard errors off, and the referee’s 200 000-run simulation gives 8.117. Details and curves are in Section 4 ^{experiments E1}.

schema	DT _F ^o , exact		pattern lgg, exact		first-order lgg, complete	
	plain	+refutation	genuine	formula-level	named	de Bruijn
T_{Sep}	23/27/30	23/27/30	23/27/30	23/27/30	25/28/30	25/28/30
Repl_{sp}	23/24/30	23/24/30	23/24/30	23/24/30	28/28/30	28/28/30
T_{EInd}	27/29/30	27/29/30	24/29/30	27/29/30	24/29/30	24/29/30
T_{Ind}	25/30/30	27/30/30	0/0/0	0/0/0	25/30/30	25/30/30

Table 8: E2: runs out of 30 at $N = 2/3/8$ tagged instances. For first-order lggs only completeness (all 30 held-out instances accepted) is shown; their soundness fails (text) ^{experiments E2}.

E2: each schema from tagged instances (Q2). Table 8 gives exactness. No learner is exact from one instance: a datum is covered by itself as a ground template. Probes ($N = 2$ and 6, 8 seeds): the DT_F^o learner accepted 0 non-instances in 927 probes. Both pattern lggs accepted none on the three ZF schemas; on induction 224 of 229 probes at $N = 2$ were non-instances, 58 of them refuted (genuine; formula-level 213/220, 46), in line with the prior result that pattern anti-unification generalizes induction data to $(A \wedge \forall x(P(x) \rightarrow Q(x))) \rightarrow \forall xP(x)$, which has false instances (Section 5) ^{prior induction C9}. First-order lggs were unsound on Separation (capture of the comprehension variable), Replacement and induction in both encodings, and on $\in$ -induction in the named encoding; on $\in$ -induction in de Bruijn indices, a first-order pattern there (Section 5), they were sound and complete (0 non-instances in 182 probes). An exhibit certified false: $\forall x\exists y\forall z(z \in y \leftrightarrow z \in x \wedge \neg w_0 \in y)$ (Separation, both encodings). More in Appendix F.5.

The ZF schemas, Miller patterns in de Bruijn form (Section 5), are learned from 2–3 instances by DT_F^o and by the genuine pattern lgg, which is slightly later on $\in$ -induction because it keeps atoms that DT_F^o lifts to formula metavariables. The first record’s claim that the pattern lgg equals DT_F^o on the ZF schemas was an artifact of the formula-level baseline and is withdrawn ^{referee F6}.

E3: dtrc on untagged mixtures (Q3). Table 9 gives totals over seeds 0–4 for PA-mix with numerals. T_{Ind} is exact for every learner in every seed, $U_{\text{mul}0}$ in seeds 1–4 (seed 0 has no $0 \cdot 0 = 0$). The first four learners of the table differ only on $U_{\text{add}0}$ and $U_{0\text{add}}$; seed 3 has no $0 + 0 = 0$ and hence no anchor of either. In the mixed regime DTRC is exact on 19/20 schema targets, DTRC+share and both tagged learners on 20/20; skeleton clustering gets 17/20 with 197 non-target probes. On ZF-mix, DTRC, skeleton clustering and both tagged learners have ARI 1, all 15 schema targets and 30 axioms exact, and no non-target probe; the pattern lgg is also exact, the first-order lggs give 223 (named) and 128 (de Bruijn) non-target probes. A DTRC run took at most 0.8 s (PA) or 6 s (ZF) ^{experiments E3}.

Reading. DTRC recovers the labels in every run. Its exactness differs from the gold-labelled learner’s only through the ambiguous $0 + 0 = 0$ (the totals coincide, with different targets exact); with sharing it matches the membership-labelled learner target by target and seed by seed, as Proposition 7.7 predicts. With numerals the instance schemas are often not exact for any learner: n numerals form an anchor with probability $1 - (7/8)^n$ (Proposition 7.8). The first record’s headline 54/55 and 45/45 counted ground axioms (35 and 30 of them) ^{referee F4}. Skeleton clustering with the true k is unsound on PA (it lumps Q4 and Q6 into $\forall x P(x)$) and perfect on ZF, whose targets have distinct skeletons.

learner	ARI	schemas exact	axioms exact	held-out	probes / non-target / refuted
DTRC	1.000	13/20	35/35	451/500	226 / 0 / 0
DTRC+share	1.000	17/20	35/35	474/500	227 / 0 / 0
tagged, gold labels	—	13/20	35/35	—	—
tagged, membership labels	—	17/20	35/35	—	—
skeleton clustering (k given)	0.993	9/20	29/35	425/500	282 / 68 / 64
first-order lgg per tag	—	—	—	453/500	180 / 100 / 28
pattern lgg per tag	—	8/20	—	453/500	221 / 99 / 33

Table 9: E3, PA-mix, universal axioms through numerals only; totals over 5 seeds. Schemas: T_{Ind} and the three instance schemas; axioms: Q1–Q7. Both encodings, resp. both pattern versions, gave identical rows experiments E3.

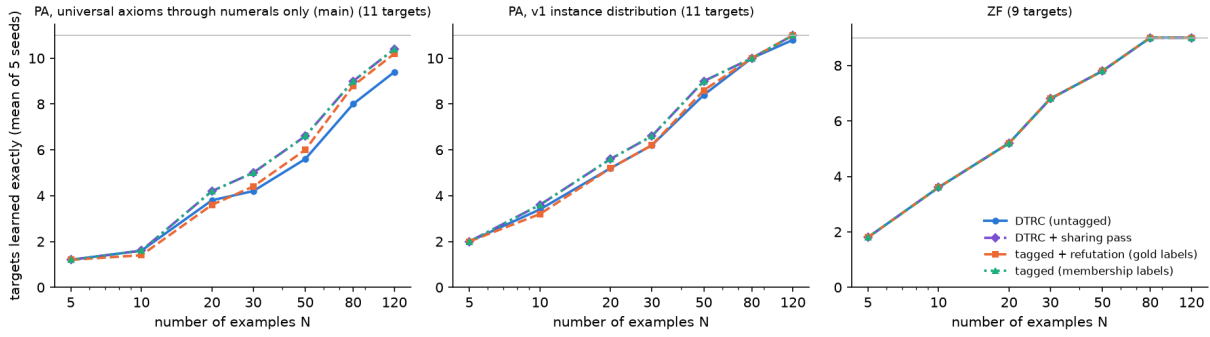


Figure 1: E5: targets learned exactly after N examples, mean over seeds 0–4 (rows of one stream share seeds). Left: PA, numerals (11 targets); middle: PA, mixed regime; right: ZF (9 targets) experiments E5.

E5: examples to exactness. Streams are drawn i.i.d. from a mixture (PA: each Q axiom 0.02, induction 0.50, $U_{\text{add}0}$ 0.16, $U_{\text{mul}0}$ 0.12, $U_{0\text{add}}$ 0.08; ZF: each axiom 0.03, Separation 0.30, Replacement 0.27, $\in$ -induction 0.25); see Figure 1. DTRC+share coincides with the membership-labelled tagged learner in every stream, in the means and in the first N from which each target stays exact, per target and seed. All differences between DTRC and the gold-labelled learner concern $U_{\text{add}0}$, $U_{0\text{add}}$ and come from $0 + 0 = 0$. The curves are dominated by the first appearance of each ground axiom; the schemas are exact after 5–20 examples experiments E5; referee R9.

E6 and E7. *The k -union verifier on Q plus induction, without clustering.* Without negatives, $k = 8$ and $k = 9$ accept 0/8 held-out induction instances: one slot $\forall x P(x)$ takes all of Q, and each induction datum gets its own slot. With refutation negatives, $k = 8$ accepts 8/8 from the anchor pair alone; $k = 9$ accepts 0/8 from two induction instances (the spare slot keeps them apart) and 8/8 from three. No false non-instance was accepted. This reproduces the prior untagged result (Section 6) experiments E6; prior pa-untagged. *Why DT_F° .* Over all pairs of mixture data, full DT° gives up to 64 (PA) and 32 (ZF) minimal templates and 3 timeouts (over 2s) on ZF pairs; DT_F° gives at most 4 and 8, consistent with the exponential lower bound of Section 3 experiments E7.

E9: refutation separation, and how narrow the evidence is. Table 10 tests cross merges at budget 80. On the mixtures every merge is refuted, but the thousands of merges reduce to 15 (PA) and 5 (ZF) distinct templates, refuted by trivial instances (such as $1 = 0$, $1 + 0 = 0$, $\forall \bar{x} \perp$,

targets	merges	refuted	distinct minimal templates	distinct unrefuted
PA-mix	2200 / 2200	2200 / 2200	15 / 13	0 / 0
ZF-mix	540 / 540	540 / 540	5 / 5	0 / 0
PA near-miss	1080 / 1080	1073 / 1080	61 / 36	5 / 0
ZF near-miss	225 / 225	225 / 225	5 / 5	0 / 0
ZF hard	432 / 432	431 / 430	14 / 15	1 / 2

Table 10: E9: cross-target merges, pairs of single data / sets of one or two data per target; data instantiating both targets excluded; budget 80 ^{experiments E9}.

$\top \rightarrow \perp$) or a comprehension instance in the first two passes. That is Proposition 7.9, and no evidence of separation between *similar* targets ^{referee F3}. The *PA near-miss* set adds $z + 1 = Sz$, $z \cdot 1 = z$, the nested $z + 0 + 0 = z + 0$ and two reorderings of induction; its unrefuted merges are valid, e.g. $n + z = S^n z$ from $n + 0 = n$ and $n + 1 = \underline{n + 1}$ (Lemma 7.13), so no world-based refuter separates them. The *hard ZF* set has T_{Sep} , Separation from $\bigcup a$, Repl_{sp} , Collection, T_{EInd} , $\in$ -induction along $z \in y \in x$ (EInd2), the minimal-element schema, and the bounding forms Union_W , Power_W of Section 6.

Proposition 7.11 (Two hard ZF pairs). *[proved; computed]^{experiments Prop E9; E9, E9b} (a) The unique minimal covering template of $\text{Union}_W = \forall a \exists b \forall x (\exists y (y \in a \wedge x \in y) \rightarrow x \in b)$ and $\text{Power}_W = \forall a \exists b \forall x (\forall y (y \in x \rightarrow y \in a) \rightarrow x \in b)$ is the universal-set schema $U = \forall a \exists b \forall x (P(x, a) \rightarrow x \in b)$. The ZF oracle refutes its instance $P := \top$, whereas no oracle sound for HF plus a universal element refutes any instance (Section 6). (b) Let $\text{EInd2}(\psi) = \forall x (\forall y \in x \forall z \in y \psi(z) \rightarrow \psi(x)) \rightarrow \forall x \psi(x)$. When the motives’ root symbols differ, the $\in$ -induction motive does not begin with $\forall$, and some motive uses its variable (the most frequent case), the minimal template of an EInd datum and an EInd2 datum is $T_{EE} = (\forall x (\forall y (y \in x \rightarrow P_0(y)) \rightarrow P_1(x))) \rightarrow \forall x P_1(x)$, which is false. Other merges of this pair are valid, and some false ones are refuted only at budget 400.*

Proof idea. (a) After simplification the instance is $\exists b \forall x (x \in b)$; the only case for b is a fresh generic G , and $x := G$ fails because $G \in G$ is false: generics obey Foundation. (b) $P_0 := \perp$, $P_1(x) := \forall z \neg z \in x$ give the true premise $\forall x (x = \emptyset \rightarrow x = \emptyset)$ and a conclusion that fails at $\{\emptyset\}$. Details in Appendix F.4. $\square$

Constant propagation is needed for (b): without it $\forall y (y \in x \rightarrow \perp)$ and $\forall z \neg z \in x$ are different unknown leaves, and 0 of 12 random EInd/EInd2 merges are refuted, against 10 of 12 with it (E9b). The one survivor in E9 is false but is refuted only at budget 400. Valid merges also occur (EInd($x = x$) with EInd2($x = x$); Separation with Separation from $\bigcup a$ on the body $x \in a$), so (S) fails for these pairs in general.

E4: stress sets. (a) *Near-miss targets* (3 seeds; the PA set of E9, and on ZF Separation, its conjunct-swapped form, Repl_{sp} , T_{EInd} , Found and the minimal-element schema). On ZF all 6 targets are exact in every seed. On PA the three forms of induction are always separated, $z + 0 + 0 = z + 0$ merges into $z + 0 = z$, and in seed 0 data of $z + 0 = z$ and $z + 1 = Sz$ merge into $5 + z = S^5 z$ and $4 + z = S^4 z$; every accepted template of a mixed cluster is valid (Lemma 7.13).

(b) *Injected mistakes.* PA: 8 mistakes (induction-like sentences with a wrong step $\varphi(x) \rightarrow \varphi(x)$, a wrong base $\varphi(S0)$, or the true conclusion $\forall x \varphi(Sx)$; and $n + 0 = 0$) and 3 true non-targets. ZF: 6 mistakes (Separation with the comprehension variable free in the body, $\forall x (\varphi \rightarrow \varphi) \rightarrow \forall x \varphi$, complement comprehension). Clean targets stay exact (PA 11/11 mixed and 9–10/11 numerals,

ZF 9/9). Refuted mistakes are discarded; the others become singletons or merge. Every mixed cluster was checked with a post-hoc refuter and by hand (Appendix F.5).

Proposition 7.12 (A false sentence is accepted). *[proved; computed]^{experiments Prop E10; referee F1, R15} In $E_4(b)$, PA-mix seed 0, both regimes, the wrong-step and the $\forall x \varphi(Sx)$ mistakes merge at refuter budgets 80 and 400 into a cluster whose only accepted template is*

$$T_F = (\exists x(P_0(x, 0) \wedge \neg P_1(x, 0)) \wedge \forall x(\exists y(P_0(y, x) \wedge \neg P_1(y, x)) \rightarrow \exists y(y < Sx \wedge \neg P_1(y, Sx)))) \\ \rightarrow \forall x \exists y (y < Sx \wedge \neg P_1(y, Sx)).$$

Its instance q_F with $P_0(y, x) := (x = 0)$ and $P_1(y, z) := (z = 2)$,

$$q_F = (\exists x(0 = 0 \wedge \neg 0 = 2) \wedge \forall x(\exists y(x = 0 \wedge \neg x = 2) \rightarrow \exists y(y < Sx \wedge \neg Sx = 2))) \\ \rightarrow \forall x \exists y (y < Sx \wedge \neg Sx = 2),$$

is false in $\mathbb{N}$ and an instance of no PA-mix target. DTRC accepts q_F at budgets 80 and 400, not at 2000.

Proof. The first premise is true. The second is true: at $x = 0$ take $y = 0$, since $1 \neq 2$; at $x \geq 1$ its antecedent is false. The conclusion fails at $x = 1$. The rest is computed (E4(b); recheck R15). $\square$

Refuting q_F needs a case split on $x = 0$ for the Π_1 premise: the first record’s oracle returns unknown, the revised one False, but the refuter does not reach q_F within 400 tries, and the probes, which never sampled q_F , reported “0 refuted”. The first record’s claim that PA mistakes merge only into valid templates is withdrawn; the referee found this exhibit. So *with mistakes in the data, DTRC’s union is not truth-sound even in arithmetic, and probe counts are lower bounds on unsoundness*. The other false templates met (two induction-step templates and five Separation-capture templates, all at budget 80) disappear at budget 400, and at 2000 no false merge remains; the remaining residual merges are valid:

Lemma 7.13 (Valid residual templates). *[proved]^{experiments Lemma V} Every instance of the following is true in $\mathbb{N}$ (1–4) or V (5): (1) $S^n 0 + z = S^n z$; (2) $z + 1 = Sz$, $z + 0 = z$, $0 + z = z$; (3) $(\alpha \wedge \dots) \rightarrow \dots$ with first conjunct α one of $z < 0$, $0 = Sz$, $\exists x(x < t \wedge \dots)$ for $t \in \{0, 0+0, 0 \cdot 0\}$; (4) $(P(0) \wedge \forall x(P(x) \rightarrow P(Sx))) \rightarrow \forall x P(Sx)$; (5) $\forall x(P(x) \rightarrow x = x) \rightarrow \forall x(x = x)$ and $\forall x(\forall y(y \in x \rightarrow P(y)) \rightarrow x = x) \rightarrow \forall x(x = x)$.*

Proof. (1) $n + t = S^n t$. (2) True schemas. (3) α is false. (4) Induction gives $\forall x P(x)$. (5) The conclusion is true. $\square$

(c) *Unequal frequencies* (PA: induction 60, $U_{\text{add}0}$ 20, $U_{\text{mul}0}$ 3, $U_{0\text{add}}$ 1; ZF: Separation 30, Replacement 4, $\in$ -induction 1; 3 seeds). Induction and Separation are exact in 3/3, Replacement and $U_{\text{mul}0}$ in 2/3, $U_{\text{add}0}$ in 1/3 (2/3 with sharing; $0 + 0 = 0$ is its only anchor). Targets with one instance are never generalized. No non-target probe ^{experiments E4}.

7.4 Limitations

(1) Budgeted refutation can accept false sentences (Proposition 7.12); whether a natural refuter refutes every false minimal template of a cross pair within a budget polynomial in the data is open. (2) Valid residual merges cannot be separated by any world-based method; DTRC is then truth-sound, not target-sound (Section 6). (3) The sharing pass needs (S^+) and (P) ; a

guarantee without (P) is open. (4) The oracles construct no witnesses. (5) A one-instance target is never generalized, and an unrefuted singleton mistake looks like a ground axiom. (6) DT_F° excludes positive-arity term metavariables, and the feature test of Section 3 is not implemented. (7) Without (S) the partition can depend on the merge order. (8) Separation is proved for the mixtures only; characterizing it for similar targets is open. (9) Scale: pure Python, about ten minutes for all experiments.

8 Discussion

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A Details for the setting

This appendix fixes the de Bruijn representation (Appendix A.1), proves the basic lemmas and the matching theorem (Appendices A.2 and A.3), proves that equivalence of determinate templates is renaming (Appendix A.4), explains why arguments must be metavariable-free (Appendix A.5), and treats parameter alignment (Appendix A.6).

A.1 The de Bruijn representation

Trees. Terms, formulas, bodies and templates are finite ordered trees. A node is labelled by a function symbol or constant of the signature, a parameter, an index $\#k$ ($k \in \mathbb{N}$), a hole z_m (only in bodies), a relation symbol or $=$, a connective, a binder $\forall$ or $\exists$ (one child; it binds index 0 in its child), or a metavariable (only in templates; its children are the arguments). Every node has a sort, ι (terms) or o (formulas). Terms contain no binders. $|u|$ is the number of nodes of u ; an occurrence $M(t_1, \dots, t_n)$ counts $1 + \sum_i |t_i|$.

Positions. A position is a finite sequence of positive integers: ε is the root and $p \cdot i$ is the i -th child of p . $u|_p$ is the subtree at p ; $p \leq q$ means that p is a prefix of q , and $p \perp q$ that neither is a prefix of the other. $b_u(p)$, or $b(p)$, is the number of binder nodes at proper prefixes of p .

Free indices. An occurrence of $\#k$ at position q of u is *bound in u* if $k < b_u(q)$; it is then bound by the $(k+1)$ -th binder above q . Otherwise it is *free* and denotes the free index $k - b_u(q)$ of u , counted from the root of u . $\text{FV}(u)$ is the set of free indices of u . A sentence s has $\text{FV}(s) = \emptyset$, and for a position p of s , $\text{FV}(s|_p) \subseteq \{0, \dots, b_s(p) - 1\}$: the free index j of $s|_p$ refers to the $(j+1)$ -th binder above p in s . In a template, the arguments of an occurrence at p are read relative to p : their indices refer to binders of T above p .

Shifting, substitution, abstraction, plugging.

- $\uparrow^e u$ adds e to every free index occurrence of u and leaves bound ones unchanged.
- For a term or formula u and a map $\bar{u}$ from a set $Y \supseteq \text{FV}(u)$ of indices to terms, $u[\bar{u}]$ replaces each free occurrence of the index y , at a position q (where it is written $\#(y + b_u(q))$), by $\uparrow^{b_u(q)} \bar{u}_y$.
- For pairwise distinct indices $y_1, \dots, y_n$ with $\text{FV}(u) \subseteq \{y_1, \dots, y_n\}$, the *abstraction* $u[z_m/y_m]$ replaces each free occurrence of y_m by the hole z_m . The result has no free indices, so $\lambda \bar{z}.u[z_m/y_m]$ is a body.
- For a body $\lambda \bar{z}.\beta$, plugging $\beta[u_1, \dots, u_n]$ replaces each occurrence of z_m , at position q of β , by $\uparrow^{b_\beta(q)} u_m$. Since β has no free indices, $\text{FV}(\beta[\bar{u}]) = \bigcup \{\text{FV}(u_m) : z_m \text{ occurs in } \beta\}$.

Abstraction inverts plugging at distinct indices: $(u[z_m/y_m])[y_1, \dots, y_n] = u$, since a hole at depth e is replaced by $\uparrow^e \#y_m = \#(y_m + e)$, which is the original occurrence. Index arithmetic is counted at unit cost.

Instances. For a template T and a substitution θ , $T\theta$ replaces the subtree at each $p \in \text{Occ}(T)$, which is $M(\bar{t})$, by $\theta(M)[\bar{t}]$. The indices of $\bar{t}$ are shifted under the binders of $\theta(M)$, so they keep referring to the binders of T above p . For a template substitution σ , $T\sigma$ is defined in the same way; inside a template body γ , plugging also replaces the holes that occur in the arguments of occurrences $N(\bar{u})$ of γ , so these arguments stay metavariable-free.

A.2 Proofs of the basic lemmas

Proof of Lemma 2.5 (preservation). Positions strictly below an occurrence lie in its argument list, so a position $p \in \text{skel}(T)$ is not below any occurrence, and no occurrence position is a prefix of p . $T\theta$ arises from T by replacing the subtrees at the occurrence positions only; hence the label at p is unchanged. If $p \in \text{Occ}(T)$ carries $M(\bar{t})$, the replacement at p is $\theta(M)[\bar{t}]$, and the replacements at the other occurrences, which are incomparable with p ($\text{Occ}(T)$ is an antichain), do not affect it. No further reduction takes place: bodies contain no metavariables, and arguments contain none either. For template substitutions the argument is the same. $\square$

Lemma A.1 (Substitution facts). *[proved]^{single Lemma 1.2} Let $\lambda\bar{z}.\beta$ be a body. (i) Composition: $(\beta[\bar{t}])[\bar{u}] = \beta[\bar{t}[\bar{u}]]$. (ii) Injectivity: if z_m occurs in β and $\beta[\bar{a}] = \beta[\bar{a}']$, then $a_m = a'_m$. (iii) If the root of β is not a hole, $\beta[\bar{a}]$ has the same root; formula bodies always have such roots.*

Proof. (i) Fix an occurrence of z_m at depth $e = b_\beta(q)$ in β . On the left, $\beta[\bar{t}]$ has $\uparrow^e t_m$ there, and $[\bar{u}]$ replaces a free index y of t_m , sitting at depth e' inside t_m and hence written $\#(y + e + e')$, by $\uparrow^{e+e'} \bar{u}_y$. On the right, $t_m[\bar{u}]$ has $\uparrow^{e'} \bar{u}_y$ at that place, and plugging shifts it by e more. The indices of β itself are all bound in β , so $[\bar{u}]$ does not touch them on the left. Both sides agree everywhere else. (ii) At an occurrence of z_m at depth e , $\beta[\bar{a}]$ has $\uparrow^e a_m$ and $\beta[\bar{a}']$ has $\uparrow^e a'_m$; shifting is injective. (iii) Plugging changes only hole positions, and the root is not a hole. Holes have sort ι , so the root of a formula body is never a hole. $\square$

A.3 Proof of the matching theorem

Proof of Theorem 2.6. (a) Uniqueness. Let $T\theta = s$, let $M(y_1, \dots, y_n)$ be a pattern occurrence at p , and write $\theta(M) = \lambda\bar{z}.\beta$. By Lemma 2.5, $s|_p = \beta[y_1, \dots, y_n]$. β has no free indices, so the free occurrences of indices in $s|_p$ are exactly the images of the holes of β , and the free index y_m comes from z_m ; since the y_m are pairwise distinct, $\beta = (s|_p)[z_m/y_m]$, and $\text{FV}(s|_p) \subseteq \{y_1, \dots, y_n\}$ is necessary. Every metavariable has a pattern occurrence, so θ is determined by s .

(b) *Correctness.* If a matcher exists, the skeleton walk succeeds (Lemma 2.5), step 2 computes it by (a), and the checks succeed. Conversely, suppose all steps succeed, and let θ be the computed substitution. The skeleton walk compares labels and arities at every position of $\text{skel}(T)$, so s and $T\theta$ agree there, and every occurrence position of T is a position of s of the right sort. At the chosen pattern occurrence of M , $\theta(M)[\bar{y}] = ((s|_p)[z_m/y_m])[\bar{y}] = s|_p$ (Appendix A.1); at the other occurrences the check establishes $\theta(M)[\bar{t}] = s|_q$. Hence $T\theta = s$.

(b) *Time.* The skeleton walk costs $O(|\text{skel}(T)|)$. Abstraction at the chosen pattern occurrence p_M is one pass over $s|_{p_M}$, which replaces each free y_m by z_m and fails on any other free index: $O(|s|_{p_M}|)$. The check at another occurrence $M(\bar{t})$ at q walks the body β of $\theta(M)$ and $s|_q$ in lock step, counting the binders of β passed. A non-hole node of β is compared with the current node of $s|_q$ and consumes it. A hole z_m at depth e is compared with $\uparrow^e t_m$ by walking t_m and the current subtree of $s|_q$ in lock step, shifting on the fly; on success it consumes $|t_m| \geq 1$ nodes of $s|_q$, and the walk stops at the first mismatch. So at most $|s|_q| + 1$ nodes of β are visited, and the check costs $O(|s|_q| + |\bar{t}|)$. Occurrence positions are pairwise incomparable, so the subtrees $s|_q$ at distinct occurrences are disjoint and $\sum_q |s|_q| \leq |s|$; and $\sum_q |\bar{t}_q| \leq |T|$. The total is $O(|T| + |s|)$.

(c) *Generality by freezing.* Let freeze replace each occurrence $N(\bar{u})$ of a metavariable of T' by $\hat{N}(\bar{u})$, where $\hat{N}$ is a fresh function symbol (if N is term-valued) or relation symbol (if N is formula-valued) of the same arity. Then $\text{freeze}(T')$ is a sentence over the extended signature. If $T' = T\sigma$, let $\theta(M) := \lambda\bar{z}.\text{freeze}(\gamma_M)$, where $\sigma(M) = \lambda\bar{z}.\gamma_M$. This is a ground body over the extended signature, and $T\theta = \text{freeze}(T\sigma) = \text{freeze}(T')$, because freezing commutes with plugging.

Conversely, let $T\theta = \text{freeze}(T')$, with $\theta(M) = \lambda\bar{z}.\beta_M$. In $\text{freeze}(T')$ every frozen symbol is applied to arguments that contain no frozen symbol (the arguments of T' are metavariable-free). By (a), β_M is an abstraction of a subtree of $\text{freeze}(T')$, so in β_M , too, every frozen symbol is applied to arguments without frozen symbols, built from holes, indices bound in β_M , and parameters. Replacing each $\hat{N}$ by N therefore turns β_M into a template body γ_M , and $\sigma(M) := \lambda\bar{z}.\gamma_M$ gives $T\sigma = T'$. The test is an instance of (b) over the extended signature, so it takes linear time.

(d) *Rigid templates without determinacy* (prior C1(b), proved in the prior session). Let $T \in \text{SO}^\circ$. Walk $\text{skel}(T)$ against s as in (b). Distinct metavariables have independent bodies, so $s \in \text{inst}(T)$ iff the walk succeeds and, for each metavariable M with occurrences $M(\bar{t}_1), \dots, M(\bar{t}_k)$ at $p_1, \dots, p_k$, there is a body β with $\beta[\bar{t}_j] = s|_{p_j}$ for all j . Decide this by recursion over a relative position r (a position of $s|_{p_1}$), with the column $c_j := (s|_{p_j})|_r$ and the number e of binders above r inside the body (the same for all j , because the body is common). A body subtree at r is either

- a hole z_m (*projection*), possible iff $c_j = \uparrow^e t_{j,m}$ for all j ; or
- a node with label h and independent subtrees at the children (*imitation*), possible iff all c_j have the label h with the same arity, h is a signature symbol, a parameter, or an index $\#i$ with $i < e$ (bound inside the body), and the recursion succeeds at every child.

This is the projection/imitation analysis of second-order matching (Huet and Lang, 1978), restricted to metavariable-free arguments. The children's subproblems share no unknowns, so the recursion visits each position of $s|_{p_1}$ at most once, and each projection test is a comparison of polynomial size. Hence membership is polynomial. The number of bodies satisfies $\#(r) = \#\{m : \text{projection to } z_m \text{ works at } r\} + [\text{imitation works at } r] \cdot \prod_i \#(r \cdot i)$. For $T = P(0)$ and a sentence s with k occurrences of the constant 0, such as a conjunction of $k/2$ copies of $0 = 0$, both projection and imitation work at each 0, and only imitation works elsewhere, so there are 2^k matchers, with $k = \Theta(|s|)$. $\square$

Evidence (single, `e1_matching.out`). On 1500 random DT° templates (one or two metavariables of six types, derived occurrences with random arguments) and random substitutions, a full projection/imitation count finds exactly one matcher and the algorithm of (b) returns the generating substitution in every case; on 1500 perturbed queries it agrees with the SO° test of (d); and 400/400 random pairs $T \succeq T\sigma$ are recognized by freezing. The single-track referee's independent matcher, used in all of that referee's checks, produced no covering or recovery error
single referee, item A.

A.4 Proof that equivalence is renaming

Proof of Lemma 2.7. “If” is clear. Conversely, let $T' = T\sigma$ and $T = T'\sigma'$. Then $T = T\tau$ with $\tau = \sigma; \sigma'$ (apply σ' to the bodies of σ).

Step 1: τ is the identity. Let $M(\bar{y})$ be a pattern occurrence of M in T , at p , and write $\tau(M) = \lambda\bar{z}.\gamma$. By Lemma 2.5, $\gamma[\bar{y}] = T|_p = M(\bar{y})$. The root of γ is not a hole (a hole plugs to an index), and not a rigid symbol (it would survive plugging). So $\gamma = N(\gamma_1, \dots, \gamma_k)$, and plugging gives $N = M$, $k = n$ and $\gamma_j[\bar{y}] = y_j$. Each γ_j is a metavariable-free term at the root level of γ , so it contains no binder and no index (an index there would be free in the body). Since $\gamma_j[\bar{y}]$ is the index y_j , γ_j is a hole z_i with $y_i = y_j$, so $i = j$ because the y 's are distinct. Hence $\tau(M) = \lambda\bar{z}.M(\bar{z})$ for every M .

Step 2: the shape of σ . Write $\sigma(M) = \lambda\bar{z}.\gamma'$. Applying σ' to γ' gives $M(\bar{z})$ by Step 1. The root of γ' is not a rigid symbol (it would survive σ') and not a hole (it would stay a hole). So it is an occurrence $N(\bar{u})$ of a metavariable N of T' ; since $\bar{u}$ is metavariable-free, $\gamma' = N(\bar{u})$. Write

$\sigma'(N) = \lambda \bar{w}.\delta$; then $\delta[\bar{u}] = M(\bar{z})$, and as in Step 1, $\delta = M(\delta_1, \dots, \delta_n)$ with each δ_j a term over holes, function symbols and parameters, and $\delta_j[\bar{u}] = z_j$. A non-hole δ_j would give a non-hole root, so $\delta_j = w_{\pi(j)}$ and $u_{\pi(j)} = z_j$; π is injective because the z_j are distinct. Distinct M give distinct N , since $\sigma'(N)$ has root M . Every metavariable of $T' = T\sigma$ occurs in some $\sigma(M)$, and each $\sigma(M)$ is the single occurrence $N(\bar{u})$, so $M \mapsto N$ is a bijection.

Step 3: the arguments are permuted. N has a pattern occurrence in T' . Occurrences of T' arise only by plugging, so it is $N(\bar{u}[\bar{t}])$ for some occurrence $M(\bar{t})$ of T . Each $u_l[\bar{t}]$ is an index, and each u_l is a term over holes, function symbols and parameters, so u_l is a hole $z_{\rho(l)}$. The $u_l[\bar{t}]$ are pairwise distinct, so ρ is injective, and $u_{\pi(j)} = z_j$ gives $\rho \circ \pi = \text{id}$. Hence the arity of N equals n and ρ is a permutation, and $\sigma(M) = \lambda \bar{z}.N(z_{\rho(1)}, \dots, z_{\rho(n)})$. So T' is T with M renamed to N and its argument places permuted by ρ . $\square$

A.5 Why arguments are metavariable-free

Remark A.2 (Nested arguments). [proved; computed]^{prior induction referee; single §4} The prior session also considered the class DT in which arguments may contain metavariables, as in $M(F(x))$. For $D = \{\text{Ind}(x = x), \text{Ind}(0 = x)\}$ the templates

$$T_k = (0 = 0 \wedge \forall x (f(x) = x \rightarrow f^k(Sx) = Sx)) \rightarrow \forall x f(x) = x, \quad k \geq 1,$$

are in DT ($f(x)$ in the conclusion is a pattern occurrence), cover D (via $f := \lambda z.z$ and $f := \lambda z.0$), are pairwise incomparable (the instance with $f := \lambda z.z + 0$ lies in T_k only), and are each minimal (the prior referee's hand proof). So $\text{Min}_{\text{DT}}(D)$ is infinite. The prior enumeration-based learner also failed to compute the cautious verifier of DT for targets with nested arguments, and was unsound for one such target, $\forall x (f(x) = f(x)) \wedge f(f(0)) = f(f(0))$. In DT° , $\text{Min}(D)$ is finite for every D (Section 3); for this D it has exactly four elements (computed by the single track, by saturation and by brute force up to size 24).

A.6 Parameter alignment

Renaming-closed instances. Let $\text{par}(u)$ be the set of parameters occurring in u . A *renaming* is an injective map on parameter names, applied to every parameter occurrence. For a template T , let

$$\text{inst}^\sim(T) = \{(\rho T)\theta : \rho \text{ injective on } \text{par}(T), \theta \text{ a substitution}\}.$$

Renamings are required to be injective because a bijective renaming of parameters leaves the universal closure unchanged up to bound names, while identifying two parameters changes it. T *covers D up to renaming* if $D \subseteq \text{inst}^\sim(T)$. Let $T \succeq^\sim T'$ iff $T' = (\rho T)\sigma$ for a renaming ρ injective on $\text{par}(T)$ and a template substitution σ ; equivalently, $\text{freeze}(T') \in \text{inst}^\sim(T)$. This is a preorder that contains $\succeq$, and $T \succeq^\sim T'$ implies $\text{inst}^\sim(T) \supseteq \text{inst}^\sim(T')$. (Compose: if $T' = (\rho T)\sigma$ and $T'' = (\rho' T')\sigma'$, then $T'' = ((\rho'\rho)T)(\rho'\sigma)\sigma'$, and $\rho'\rho$ is injective on $\text{par}(T)$ because $\text{par}(T') \supseteq \rho(\text{par}(T))$ by Lemma 2.5.) For parameter-free T , $\text{inst}^\sim(T) = \text{inst}(T)$.

Alignments. Let D be finite and nonempty, and let $r \in D$ be a datum with fewest parameters. An *alignment* $\alpha = (G, (\iota_d)_{d \in D \setminus \{r\}})$ consists of a set $G \subseteq \text{par}(r)$ and injections $\iota_d : G \rightarrow \text{par}(d)$. The aligned data $\alpha(D)$ rename, in r , each $q \in G$ to a global name g_q ; in each $d \neq r$, each $\iota_d(q)$ to g_q ; and every other parameter of every datum to a name of its own, used nowhere else. Write α_d for the bijective renaming applied to d . $\text{Align}(D)$ is finite. For a class $\mathcal{C}$, let

$$\text{Min}_{\mathcal{C}}^{\text{al}}(D) = \text{the } \succeq^\sim \text{-minimal elements (one per } \equiv^\sim \text{-class) of } \bigcup_{\alpha \in \text{Align}(D)} \text{Min}_{\mathcal{C}}(\alpha(D)),$$

where $\text{Min}_{\mathcal{C}}$ is the literal minimal set of Definition 2.2. If some datum has no parameter there is one alignment, and $\text{Min}_{\mathcal{C}}^{\text{al}}(D)$ is $\text{Min}_{\mathcal{C}}(D)$ up to $\equiv^{\sim}$.

Literal finitariness. The proof below uses one property of $\mathcal{C}$:

- (F) for every finite nonempty X , $\text{Min}_{\mathcal{C}}(X)$ is finite (up to $\equiv$), and every $T \in \mathcal{C}$ that covers X literally satisfies $T \succeq T_0$ for some $T_0 \in \text{Min}_{\mathcal{C}}(X)$.

For $\mathcal{C} = \text{DT}^{\circ}$, with parameters treated as constants, (F) is the finiteness theorem for minimal covering templates in Section 3 (Theorem B of the single track). For $\mathcal{C} = \text{DT}_F^{\circ}$ it is Proposition E1-lit of the experiments track, proved there through a normal form for literal covering. Both classes are closed under renaming of parameters.

Proposition A.3 (Alignment-complete minimal templates). *[proved; computed]^{experiments Prop E1-al} Let $\mathcal{C} \in \{\text{DT}^{\circ}, \text{DT}_F^{\circ}\}$ and let D be finite and nonempty. (a) Every member of $\text{Min}_{\mathcal{C}}^{\text{al}}(D)$ lies in $\mathcal{C}$ and covers D up to renaming. (b) Every $T \in \mathcal{C}$ that covers D up to renaming satisfies $T \succeq^{\sim} M$ for some member M . (c) $\text{Min}_{\mathcal{C}}^{\text{al}}(D)$ is finite; its members are $\succeq^{\sim}$ -minimal among all templates in $\mathcal{C}$ that cover D up to renaming; and if $D \subseteq \text{inst}^{\sim}(T^*)$ with $T^* \in \mathcal{C}$, some member M satisfies $T^* \succeq^{\sim} M$.*

For DT_F° this is Proposition E1-al of the experiments track (computed check: E0b, below). The same proof gives it for DT° , with (F) supplied by the finiteness theorem of Section 3 instead of its DT_F° analogue.

Proof. Let $\mathcal{C} \in \{\text{DT}^{\circ}, \text{DT}_F^{\circ}\}$; it satisfies (F).

(a) *Members cover D up to renaming.* Let $M \in \text{Min}_{\mathcal{C}}(\alpha(D))$. It covers $\alpha(D)$ literally: $\alpha_d(d) = M\theta_d$ for every d . The rigid parameters of M occur in $M\theta_d = \alpha_d(d)$ (Lemma 2.5), and α_d is a bijection from $\text{par}(d)$ onto $\text{par}(\alpha_d(d))$. So α_d^{-1} is injective on $\text{par}(M)$, and $d = (\alpha_d^{-1}M)(\alpha_d^{-1}\theta_d)$, where $\alpha_d^{-1}\theta_d$ renames the parameters in the bodies. Hence $d \in \text{inst}^{\sim}(M)$.

(b) *Every template covering D up to renaming lies above a member.* Let $T \in \mathcal{C}$ with $d = (\rho_d T)\theta_d$ for every $d \in D$, each ρ_d injective on $\text{par}(T)$. By Lemma 2.5, $\rho_d(\text{par}(T)) \subseteq \text{par}(d)$. Put $G := \rho_r(\text{par}(T)) \subseteq \text{par}(r)$ and $\iota_d := \rho_d \circ \rho_r^{-1} : G \rightarrow \text{par}(d)$, which is injective; so $\alpha := (G, (\iota_d)) \in \text{Align}(D)$. Let T' be T with each parameter p renamed to $g_{\rho_r(p)}$; then $T \succeq^{\sim} T' \succeq^{\sim} T$. For every d and $p \in \text{par}(T)$, $\alpha_d(\rho_d(p)) = g_{\rho_r(p)}$: for $d = r$ by the definition of α , and for $d \neq r$ because $\rho_d(p) = \iota_d(\rho_r(p))$. Hence $\alpha_d(d) = (\alpha_d \rho_d T)(\alpha_d \theta_d) = T'(\alpha_d \theta_d)$, so T' covers $\alpha(D)$ literally. By (F), $T' \succeq M'$ for some $M' \in \text{Min}_{\mathcal{C}}(\alpha(D))$, and $M' \succeq^{\sim} M$ for some $\succeq^{\sim}$ -minimal element M of the finite union. So $T \succeq^{\sim} M \in \text{Min}_{\mathcal{C}}^{\text{al}}(D)$.

(c) *Finiteness, minimality, and a member below the target.* $\text{Align}(D)$ is finite and each $\text{Min}_{\mathcal{C}}(\alpha(D))$ is finite, so $\text{Min}_{\mathcal{C}}^{\text{al}}(D)$ is finite. If some $T \in \mathcal{C}$ covering D up to renaming satisfied $M \succ^{\sim} T$ for a member M , then by (b) $T \succeq^{\sim} M''$ for a member M'' , so $M \succ^{\sim} M''$ with both in the union, contradicting the minimality of M there. If $D \subseteq \text{inst}^{\sim}(T^*)$ with $T^* \in \mathcal{C}$, apply (b) to T^* . $\square$

Example A.4 (Literal minimal templates are not enough). *[proved; computed]^{experiments E1, E0b} Let $D = \{0 = 0 \wedge w_0 = w_0, w_0 = 0 \wedge w_1 = w_1\}$ (each datum canonicalized on its own). Literally, $\text{Min}_{\text{DT}_F^{\circ}}(D) = \{f_0 = 0 \wedge f_1 = f_1\}$, with 0-ary term metavariables. But $T = (f = 0 \wedge w_0 = w_0)$ covers D up to renaming (in the second datum, $f := w_0$ and the template's w_0 is renamed to w_1), and $f_0 = 0 \wedge f_1 = f_1 \succ T$ (map f_1 to w_0). So no literal minimal template lies below T . The alignment that identifies w_0 of the first datum with w_1 of the second produces T (up to renaming) as a member of $\text{Min}_{\text{DT}_F^{\circ}}^{\text{al}}(D)$. Version 1 of the experiments record stated the last*

property of Proposition A.3 for Min_C ; the referee’s counterexample above refutes it, and the alignment-complete set is the repair.

Computed (experiments, E0b): for random generating templates T^* with a rigid parameter (2–3 data each), no literal minimal template lies below T^* in 17/200 (arithmetic) and 36/200 (set theory) data sets; with Min^{al} there are no failures, with or without a rigid parameter. The implementation caps the number of alignments at 4096 and reports truncation; none occurred. On the data of experiment E2 the two sets agree on all 720 data sets (the referee: 360 of 360, with independent code).

B Proofs for Section 3 (one schema)

This appendix gives the full proofs of Section 3, in the notation of Section 3.1. Bound variables are de Bruijn indices. A datum d in $\text{inst}(T)$ has the unique matcher θ_d , and $\beta_M^d = \theta_d(M)$. All proofs were transcribed from the research record and checked again; we found no error, and some steps are spelled out in more detail than there.

B.1 Basic lemmas

Proof of Lemma 3.2. (a) $T\theta$ replaces each occurrence $M(\bar{t})$ by $\theta(M)[\bar{t}]$ and leaves every other position outside argument lists unchanged. No new redex arises, because bodies are first-order and arguments are metavariable-free (one-step plugging).

(b) (i) β is closed except for holes, so the free indices of $\beta[\bar{t}]$ are those of the plugged copies of the t_m (shifted under the binders of β). Substituting $\bar{u}$ for them is the same as substituting into each t_m before plugging. (ii) At a position of β that carries z_m , $\beta[\bar{a}]$ carries a_m shifted by the binders of β above it; shifting is injective. (iii) Plugging replaces only holes. A formula body is a formula, and holes are terms, so its root is a symbol.

(c) The skeleton is prefix-closed: the parent of a skeleton position is a skeleton position, because the children of an occurrence are its arguments. By (a), every datum carries the symbol $T|_p$ at each $p \in \text{skel}(T)$; by induction on the depth of p , $p \in C(D)$. The parent of an occurrence is in $\text{skel}(T) \subseteq C(D)$, so the occurrence lies in $C(D)$ or is a slot.

(d) Match $d|_\sigma$ against $d|_r$, treating the free indices of $d|_\sigma$ as first-order variables. Walk both in lock step. At a free-index occurrence y of $d|_\sigma$ under e local binders, read off the subterm of $d|_r$ at the same place and lower it by e ; fail if it mentions one of those e local binders, or if two occurrences of y disagree; fail on any other symbol clash. The walk either fails or returns a partial map e_d defined exactly on $\text{FV}(d|_\sigma)$, and $d|_r = (d|_\sigma)[\bar{u}]$ holds iff $\bar{u}$ extends e_d . So a common $\bar{u}$ on Y exists iff no e_d fails and the e_d agree on common arguments. This is a pairwise condition, and $\bar{u}$ is then determined on $\bigcup_d \text{FV}(d|_\sigma) = Y$. The walks take time $O(\sum_d |d|)$. The argument does not use finiteness of D .

(e) “If” is clear. Let $T' = T\sigma$ and $T = T'\sigma'$, and let $\tau := \sigma; \sigma'$ (apply σ' to the template bodies of σ), so $T = T\tau$. Let $M(\bar{y})$ be a pattern occurrence of M at p . By (a), applied to template substitutions, $\tau(M)[\bar{y}] = T|_p = M(\bar{y})$. Write $\tau(M) = \lambda \bar{z}. \gamma$. Then γ is not a hole (a hole plugs to an index), and its root is not a rigid symbol (it would survive), so $\gamma = M(\gamma_1, \dots, \gamma_n)$ with $\gamma_j[\bar{y}] = y_j$. Each γ_j is a metavariable-free term at depth 0 of the body, so it contains no index of its own; $\gamma_j[\bar{y}] = y_j$ forces γ_j to be a hole $z_{j'}$ with $y_{j'} = y_j$, and $j' = j$ because the y ’s are distinct. Every metavariable has a pattern occurrence, so τ is the identity.

Now write $\sigma(M) = \lambda \bar{z}. \gamma'$. Since $\gamma'\sigma' = M(\bar{z})$, the root of γ' is an occurrence $N(\bar{u})$ of a metavariable of T' (a rigid root or a hole would survive σ'), and $\sigma'(N) = \lambda \bar{w}. M(\bar{\delta})$ with $\delta_j[\bar{u}] = z_j$.

As before, δ_j has no index of its own and is not a symbol, so $\delta_j = w_{\pi(j)}$ and $u_{\pi(j)} = z_j$. Distinct M give distinct N , since the root of $\sigma'(N)$ is M . The only metavariable in $\gamma' = N(\bar{u})$ is N , and every metavariable of $T' = T\sigma$ occurs in some $\sigma(M)$; so $M \mapsto N$ is a bijection. Finally, N has a pattern occurrence in T' ; it is the image $N(\bar{u}[\bar{t}])$ of some occurrence $M(\bar{t})$ of T . Each $u_l[\bar{t}]$ is an index, and a non-hole u_l would plug to a term with a symbol root, so $u_l = z_{\rho(l)}$; distinctness of the $u_l[\bar{t}]$ makes ρ injective, and $\rho \circ \pi = \text{id}$ makes it surjective. So $\sigma(M) = \lambda \bar{z}. N(z_{\rho(1)}, \dots, z_{\rho(n)})$ with ρ a permutation. $\square$

B.2 Minimal covering templates

Proof of Theorem 3.3. (a) We use two reduction steps. Each turns a covering $T \in \text{DT}^\circ$ into a covering $T' = T\sigma \in \text{DT}^\circ$, so $T \succeq T'$.

Step V (vacuous place). Suppose z_m occurs in no β_M^d . Put

$$\sigma(M) := \lambda \bar{z}. M'(z_1, \dots, z_{m-1}, z_{m+1}, \dots, z_n) \quad (M' \text{ fresh}).$$

A pattern occurrence becomes an occurrence with a sub-list of distinct indices, again a pattern occurrence. D is covered with $\theta'_d(M') := \beta_M^d$ (holes renumbered).

Step H (common root). Suppose all β_M^d ($d \in D$) have the same root.

- If the root is a hole z_m (M is term-valued), put $\sigma(M) := \lambda \bar{z}. z_m$: each occurrence $M(\bar{t})$ becomes t_m . D is still covered, since $d|_q = z_m[\bar{t}] = t_m$ at each occurrence q .
- If the root is a symbol f with children $1, \dots, k$, child j under $e_j \in \{0, 1\}$ binders of f , put $\sigma(M) := \lambda \bar{z}. f(M_1(\bar{z}, 0^{e_1}), \dots, M_k(\bar{z}, 0^{e_k}))$, with fresh M_j of arity $n + e_j$; here 0^e is the index 0 (the variable bound by f) if $e = 1$ and nothing if $e = 0$, and the holes are shifted under f by plugging. A pattern occurrence $M(\bar{y})$ becomes $f(\dots, M_j(\bar{y}\uparrow, 0), \dots)$, whose arguments are distinct indices: a pattern occurrence. Other occurrences get metavariable-free arguments. D is covered with $\theta'_d(M_j) :=$ the j -th child of β_M^d , with the variable bound by f replaced by the new hole. (If f is a constant or a parameter, $k = 0$ and M disappears.)

Let $\mu(T) = (|C(D)| - |\text{skel}(T)|, \sum_M \text{arity}(M)) \in \mathbb{N}^2$, ordered lexicographically; the first component is ≥ 0 by Lemma 3.2(c). Step H strictly increases $|\text{skel}|$: the root symbol, or the index y_m , becomes rigid at least at the pattern position. Step V keeps the skeleton and lowers $\sum_M \text{arity}(M)$. So the process stops, and it stops exactly when (S1) and (S2) hold, at a saturated template. Generality is transitive.

(b) $\text{skel} \subseteq C(D)$ is Lemma 3.2(c). Let $M(\bar{y})$ be a pattern occurrence at σ . By Lemma 3.2(a), $d|_\sigma = \beta_M^d[\bar{y}]$; its root is the root of β_M^d if that is a symbol, and y_m if $\beta_M^d = z_m$ (a body has no free index, so an index root comes from a hole). Since the y_m are distinct, the roots of the $d|_\sigma$ are all equal iff the roots of the β_M^d are. By (S2) they are not, so $\sigma \notin C(D)$; its parent lies in $\text{skel} \subseteq C(D)$, so σ is a slot. By (S1) each y_m is free in some $d|_\sigma$, so $\{\bar{y}\} \subseteq Y_\sigma(D)$; coverage forces $\text{FV}(d|_\sigma) \subseteq \{\bar{y}\}$, so $Y_\sigma(D) = \{\bar{y}\}$. Every occurrence lies in $A(D)$ by Lemma 3.2(c). For another occurrence $M(\bar{t})$ at q , the value $\beta_M^d = (d|_\sigma)[\bar{z}/\bar{y}]$ is determined by D and the pattern slot σ ; by (S1) each hole z_m occurs in some β_M^d , and by Lemma 3.2(b)(ii), t_m is determined by $d|_q = \beta_M^d[\bar{t}]$.

So, up to renaming metavariables and permuting argument places (Lemma 3.2(e)), a saturated covering template is determined by its set $Q \subseteq A(D)$ of occurrence positions and by the map that sends each $q \in Q$ to the pattern slot of the metavariable occurring there (one pattern occurrence fixed per metavariable). The skeleton is then determined too: it consists of the positions of $C(D)$ that are neither in Q nor below an element of Q . The sort of each metavariable is the sort of its positions, and its arity is $|Y_\sigma(D)|$. This gives the bound $2^{|A(D)|} \cdot |\text{slots}(D)|^{|A(D)|}$.

(c) Let T be minimal. By (a), $T \succeq T'$ with T' saturated; minimality gives $T' \succeq T$, and by Lemma 3.2(e) T is a renaming of T' , hence saturated. Now let T cover D arbitrarily, and let S_T be the set of saturated covering templates T' with $T \succeq T'$. It is finite up to renaming by (b) and nonempty by (a), and $\succeq$ is a partial order on renaming classes (Lemma 3.2(e)). Take $T_m \succeq$ -minimal in S_T . If U covers D and $T_m \succeq U$, then $U \succeq U'$ for some saturated U' , by (a), and $T \succeq T_m \succeq U \succeq U'$; so $U' \in S_T$ and $T_m \succeq U'$, and minimality in S_T gives $U' \equiv T_m$, hence $U \equiv T_m$. So T_m is minimal among all covering templates, and $T \succeq T_m$.

(d) is (c) with $T = T^*$. $\square$

Example 3.4 in detail. The D -features are: the Sym features of the common frame (including the whole antecedent $0 = 0$ and the right-hand sides x, Sx, x of the three atoms in the motive copies under $\forall x$); Scope with scope $\{x\}$ at the three slots; and the Eq features $\sigma_b \leftrightarrow \sigma_d$ (identity), $\sigma_b \rightarrow \sigma_c$ and $\sigma_d \rightarrow \sigma_c$ ($x \mapsto Sx$), and $\sigma_b, \sigma_d \rightarrow$ each of the two zeros of the antecedent ($x \mapsto 0$). (No feature leaves σ_c : its column $(Sx, 0)$ cannot be mapped onto $(x, 0)$, nor onto a rigid 0 or Sx .) A minimal template therefore merges σ_b and σ_d into $f(x)$, puts $f(Sx)$ at σ_c , and chooses at each zero of the antecedent between the rigid 0 and $f(0)$: four incomparable templates, separated by the instance $f := \lambda z.Sz$. Computed: the saturated templates number 35, of which 4 are minimal; brute force at size ≤ 24 finds the same 4, and every one of 2749 enumerated covering templates lies above one of them; at size ≤ 23 there are 8 apparent minima, as reported earlier. single e2; referee t1, t1b

B.3 The feature verifier

Proof of Theorem 3.5. The templates. $T_0(D)$ and every T_φ are in DT° : every M_σ has the pattern occurrence $M_\sigma(Y_\sigma)$, and in T_φ the metavariable M_{σ_0} keeps its pattern occurrence at σ_0 , which is not below r and differs from r (as $\sigma_0 \perp r$). Both cover D with $\theta_d(M_\sigma) := (d|_\sigma)[\bar{z}/Y_\sigma]$, which is closed except for holes because $\text{FV}(d|_\sigma) \subseteq Y_\sigma(D)$; at r , $\theta_d(M_{\sigma_0})[\bar{u}] = (d|_{\sigma_0})[\bar{u}] = d|_r$ because φ is a D -feature.

(1) $\text{Acc}(D) \subseteq \text{Feat}(D) = \text{inst}(T_0(D)) \cap \bigcap_\varphi \text{inst}(T_\varphi)$. A sentence lies in $\text{inst}(T_0(D))$ iff it has every Sym and every Scope feature: such a q has the skeleton $C(D)$, and $q = T_0(D)\theta$ with $\theta(M_\sigma) := (q|_\sigma)[\bar{z}/Y_\sigma]$; conversely an instance has the skeleton, and $\text{FV}(\theta(M_\sigma)[Y_\sigma]) \subseteq Y_\sigma$. If q has every D -feature, then it has the skeleton of T_φ , Scope at every slot, and $q|_r = (q|_{\sigma_0})[\bar{u}]$, so $q = T_\varphi\theta$ with the same θ . Conversely every instance $q = T_\varphi\theta$ has φ : $q|_{\sigma_0} = \theta(M_{\sigma_0})[Y_{\sigma_0}]$ and $q|_r = \theta(M_{\sigma_0})[\bar{u}] = (q|_{\sigma_0})[\bar{u}]$ by Lemma 3.2(b)(i). So $\text{Feat}(D)$ equals the displayed intersection, and $\text{Acc}(D) \subseteq \text{Feat}(D)$ because $T_0(D)$ and the T_φ cover D .

(2) $\text{Feat}(D) \subseteq \text{Acc}(D)$. Let T cover D . By Theorem 3.3(a), $T \succeq T'$ with T' saturated and covering, so $\text{inst}(T) \supseteq \text{inst}(T')$. Fix for every metavariable M of T' a pattern occurrence $M(\bar{y})$, at the slot σ_M (Theorem 3.3(b)). A sentence q lies in $\text{inst}(T')$ if it has (i) $\text{Sym}(p)$ for every $p \in \text{skel}(T')$, (ii) $\text{Scope}(\sigma_M)$ for every M , and (iii) $\text{Eq}(\sigma_M, r, \bar{u}_r)$, with $\bar{u}_r := (y_m \mapsto t_{r,m})_m$, for every other occurrence $M(\bar{t}_r)$ at r . Indeed, by (ii) $\theta(M) := (q|_{\sigma_M})[\bar{z}/\bar{y}]$ is a body, and (i), (iii) give $T'\theta = q$, since $\theta(M)[\bar{t}_r] = (q|_{\sigma_M})[\bar{u}_r] = q|_r$. Each of these is a D -feature: (i) by Lemma 3.2(c); (ii) because σ_M is a slot with $Y_{\sigma_M}(D) = \{\bar{y}\}$; (iii) because $r \in A(D)$ (Lemma 3.2(c)), $r \perp \sigma_M$ (distinct occurrences are incomparable), the sorts agree, $\bar{u}_r$ maps $Y_{\sigma_M}(D)$ to metavariable-free terms in scope at r , and T' covers D : $d|_r = \beta_M^d[\bar{t}_r] = (\beta_M^d[\bar{y}])[\bar{u}_r] = (d|_{\sigma_M})[\bar{u}_r]$. So $\text{Feat}(D) \subseteq \text{inst}(T') \subseteq \text{inst}(T)$ for every covering T , i.e. $\text{Feat}(D) \subseteq \text{Acc}(D)$. $\square$

Proof of Corollary 3.6. Positions in $A(D)$ are positions of every datum (a slot's parent carries the same symbol in all data, hence has the same arity), so $m = |A(D)| \leq \min_d |d|$. $C(D)$, the slots and the scopes are computed in one simultaneous walk, in time $O(n)$. For each of the at

most m^2 ordered pairs (σ, r) with σ a slot, $r \in A(D)$, $\sigma \perp r$, of the same sort, Lemma 3.2(d) decides in time $O(n)$ whether an Eq-feature exists and returns its unique $\bar{u}$. So there are at most m^2 Eq-features, plus at most m Sym and m Scope features. Each feature is checked on q in time $O(|q|)$: Sym and Scope directly; Eq($\sigma, r, \bar{u}$) by walking $q|_\sigma$ and $q|_r$ in lock step, comparing each free-index occurrence y of $q|_\sigma$ with the corresponding subterm of $q|_r$ against u_y (shifted), which consumes that subterm. Hence $O(m^2|q|)$ per query. $\square$

B.4 Anchors

Proof of Theorem 3.8. ($\Leftarrow$; no assumption.) By Lemma 3.2(c), $\text{skel}(T^*) \subseteq C(D)$, and by (R*) no occurrence lies in $C(D)$; since $C(D)$ is prefix-closed and the data positions are the skeleton, the occurrences and the positions below occurrences, $C(D) = \text{skel}(T^*)$, $\text{slots}(D) = \text{Occ}(T^*)$ and $A(D) = \text{skel}(T^*) \cup \text{Occ}(T^*)$. For σ carrying $M(\bar{t}_\sigma)$,

$$Y_\sigma(D) = \bigcup_i \text{FV}(\beta_M^i[\bar{t}_\sigma]) = \bigcup_i \bigcup_{m: z_m \in \beta_M^i} \text{FV}(t_{\sigma,m}) = Y_\sigma^*$$

by (N). Let $q = T^*\theta$. Then q has every Sym feature (Lemma 3.2(a)) and every Scope feature ($\text{FV}(q|_\sigma) \subseteq Y_\sigma^*$). Let Eq($\sigma, r, \bar{u}$) be a D -feature; (σ, r) is one of the pairs of Definition 3.7. By (U) it is valid: $\bar{t}_r = \bar{t}_\sigma[\bar{u}_0]$. Then $s_i|_r = \beta_M^i[\bar{t}_\sigma[\bar{u}_0]] = (s_i|_\sigma)[\bar{u}_0]$ for all i (Lemma 3.2(b)(i)), and by forcing (Lemma 3.2(d)) $\bar{u} = \bar{u}_0$ on $Y_\sigma(D) = Y_\sigma^*$. So $q|_r = \theta(M)[\bar{t}_\sigma[\bar{u}]] = (\theta(M)[\bar{t}_\sigma])[\bar{u}] = (q|_\sigma)[\bar{u}]$. Hence $q \in \text{Feat}(D) = \text{Acc}(D)$ (Theorem 3.5): $\text{inst}(T^*) \subseteq \text{Acc}(D)$, i.e. D is an anchor.

($\Rightarrow$; under (Rich).) If D is an anchor, every instance of T^* has every D -feature (Theorem 3.5). We exhibit an instance violating a D -feature whenever an event fails; metavariables not mentioned get arbitrary values.

- (R*) fails at $\sigma \in \text{Occ}(T^*)$, carrying M , with common root h . The ancestors of σ lie in $\text{skel}(T^*) \subseteq C(D)$, so $\sigma \in C(D)$ and $\text{Sym}(\sigma)$ is a D -feature. By (Rich) choose $\theta(M)$ with a symbol root $\neq h$ (a closed term, resp. a formula body); by Lemma 3.2(b)(iii), $(T^*\theta)|_\sigma$ has that root.
- (R*) holds and (N) fails at (M, m) . Take a pattern occurrence $M(\bar{y})$ at σ ; σ is a slot. Since the y 's are distinct and the β_M^i are closed except for holes, y_m is free in $s_i|_\sigma = \beta_M^i[\bar{y}]$ only if z_m occurs in β_M^i ; so $y_m \notin Y_\sigma(D)$, and $\text{Scope}(\sigma)$ is a D -feature. Choose $\theta(M)$ containing z_m : z_m itself, or $R(z_m, \dots, z_m)$ for a relation symbol R , which exists by (Rich). Then $y_m \in \text{FV}((T^*\theta)|_\sigma)$.
- (R*) and (N) hold and (U) fails at a non-valid pair (σ, r) with witness $\bar{u}$. As in the first part, σ is a slot, $r \in A(D)$ and $Y_\sigma(D) = Y_\sigma^*$, so Eq($\sigma, r, \bar{u}$) is a D -feature. If $r \in \text{skel}(T^*)$ has rigid root g , choose $\theta(M)$ with a symbol root $\neq g$; then $((T^*\theta)|_\sigma)[\bar{u}]$ has that root (substituting for free indices keeps a symbol root), while $(T^*\theta)|_r$ has root g . If r carries an occurrence of $M' \neq M$ (of the same sort), choose $\theta(M)$ and $\theta(M')$ with different symbol roots. If r carries $M(\bar{t}_r)$, then $\bar{t}_r \neq \bar{t}_\sigma[\bar{u}]$ because the pair is not valid; pick m with $t_{r,m} \neq t_{\sigma,m}[\bar{u}]$ and put $\theta(M) := \lambda \bar{z}. z_m$ (term-valued) or $\lambda \bar{z}. R(z_m, \dots, z_m)$ (formula-valued). The two sides then differ in that argument. $\square$

Proof of Corollary 3.9. (a) Every occurrence of a 0-ary metavariable is a pattern occurrence, and $s_i|_\sigma = \beta_M^i$; so (R*) is (R). (N) is vacuous. $Y_\sigma^* = \emptyset$, so (U) at (σ, r) asks whether $s_i|_r = \beta_M^i$ for all i . If $r \in \text{skel}(T^*)$, its root is fixed, and (R) excludes the equation. If r carries $M' \neq M$ (necessarily of the same sort), the equation is the failure of (D) for M, M' . If r carries M , the pair is valid. So $(R^*) \wedge (N) \wedge (U) \Leftrightarrow (R) \wedge (D)$, and Theorem 3.8 applies.

(b) Formula bodies have symbol roots, and they survive plugging (Lemma 3.2(b)(iii)); so the root of $s_i|_\sigma$ is the root of β_M^i at every occurrence: $(R^*) \Leftrightarrow (R)$. Assume (R) and (N). For $r \in \text{skel}(T^*)$ (of formula sort) a coincidence would make all roots of the β_M^i equal to the rigid root at r , contradicting (R). For r an occurrence of the same M , $\beta_M^i[\bar{t}_r] = \beta_M^i[\bar{t}_\sigma[\bar{u}]]$ for all i forces $\bar{t}_r = \bar{t}_\sigma[\bar{u}]$ by (N) and Lemma 3.2(b)(ii); so either the pair is valid or there is no coincidence. All occurrences have formula sort. What remains of (U) is (D^+) .

(c) T_{Ind} and $\in$ -induction have one unary formula metavariable; Separation (arguments x, a) and spelled-out Replacement (arguments x, y, a) one binary resp. ternary one. In de Bruijn notation all occurrences in Separation, Replacement and $\in$ -induction are pattern occurrences (for example $P(x, y, a)$ and $P(x, z, a)$ in Replacement apply P to distinct indices), while T_{Ind} has the non-pattern occurrences $P(0)$ and $P(Sx)$. Part (b) gives $(R) \wedge (N)$, where (N) splits into one condition per argument place. A ground template T^* has $C(\{T^*\}) = \text{all positions}$ and no slot, so $\text{Feat}(\{T^*\}) = \{T^*\}$ by Theorem 3.5. One instance of a schema never satisfies (R); two instances whose bodies have different main symbols and together use every argument place do (for Replacement, for example, the bodies $y = x$ and $\neg(x = a)$). If no datum mentions a in Separation, the learner stays at the strictly more special template with $P(x, a)$ replaced by $P'(x)$, which is sound. $\square$

Verification of Proposition 3.10 and of the remark on (Rich). (a) $\forall x(f(0) = f(x))$ is in DT° ($f(x)$ is a pattern occurrence) and covers $\forall x(0 = 0)$ ($f := \lambda z.0$) and $\forall x(0 = x)$ ($f := \lambda z.z$, $f(0) = 0$). An instance with $f(x) = Sx$ has $f(0) = S0 \neq 0$. The brute-force minimal covering templates are exactly $\forall x(0 = f(x))$ and $\forall x(f(0) = f(x))$.

(b) The data are

$$\forall x \forall y (Sx = 0) \wedge \forall x \forall y (x = 0), \quad \forall x \forall y (Sy = 0) \wedge \forall x \forall y (Sy = 0), \quad \forall x \forall y (SSx = 0) \wedge \forall x \forall y (Sx = 0).$$

Write σ for the left-hand side of the second conjunct (the derived occurrence $f(x, Sy)$) and r for the left-hand side of the first (the rigid S above $f(x, y)$). The data have $(d|_r, d|_\sigma) = (Sx, x), (Sy, Sy), (SSx, Sx)$, and $d|_r = (d|_\sigma)[x \mapsto Sx, y \mapsto y]$ in all three. For the covering template, the pattern occurrence $f'(y, x)$ at σ reads $f' = \lambda z_1 z_2. z_2, \lambda z_1 z_2. Sz_1, \lambda z_1 z_2. Sz_2$, and then $f'(y, Sx)$ gives Sx, Sy, SSx at r . For $f := \lambda z_1 z_2. 0$, T^* gives $\forall x \forall y (S0 = 0) \wedge \forall x \forall y (0 = 0)$, whereas the covering template forces $f' = \lambda z_1 z_2. 0$ from the second conjunct and then 0, not $S0$, in the first. (U) holds for the pairs with first component $f(x, y)$ (column (x, y, Sx)): the right-hand zeros have the wrong root, and the only coincidence, with the derived occurrence $f(x, Sy)$ under $x \mapsto x, y \mapsto Sy$, belongs to a valid pair. Brute force (size ≤ 18 , arguments ≤ 2 , arity ≤ 2) finds exactly two minimal covering templates, T^* (up to argument order) and the displayed one; the referee's code agrees. single e2, e8; referee t2

(c) With $d_1 = \forall x \forall y (x = 0) \wedge \forall x (x = 0)$ and $d_2 = \forall x \forall y (y = 0) \wedge \forall x (x = 0)$, the derived occurrence $f(x, x)$ reads x from both values. The rigid x of the second conjunct coincides with the slot's content under $x \mapsto x, y \mapsto x$; the instance $f := \lambda z_1 z_2. Sz_1$ of T^* is missed.

The remark on (Rich). Over $\{\in, =\}$ without constants, function symbols or parameters, a term over holes is a hole, so every body of type $\iota \rightarrow \iota$ is $\lambda z. z$ and $\text{inst}(\forall x(f(x) \in x)) = \{\forall x(x \in x)\}$; every template covering this datum contains the whole target.

B.5 Rates

Proof of Theorem 3.11. (a) Let $N \geq 1$. By the “if” half of Theorem 3.8, which needs no assumption, “not an anchor” is contained in $\neg(R^*) \cup \neg(N) \cup ((R^*) \wedge (N) \wedge \neg(U))$. For a ground T^* all

three events are empty; consistently, every nonempty $D \subseteq \text{inst}(T^*)$ equals $\{T^*\}$ and is an anchor. In general:

- $\Pr[\text{all roots at } \sigma \text{ equal}] = \sum_h p_h^N \leq p_{\max}^{N-1} \sum_h p_h = (1 - \rho_\sigma)^{N-1}$, where p_h is the probability of root h at σ .
- $\Pr[z_m \text{ unused by all } \theta_i(M)] = (1 - \nu_{M,m})^N$.
- A coincidence at a non-valid pair (σ, r) on D_N means that some $\bar{u}$ has all $\theta_i \in A_{\sigma,r,\bar{u}}$. Then each of the $\lfloor N/2 \rfloor$ disjoint pairs $(\theta_1, \theta_2), (\theta_3, \theta_4), \dots$ has a common $\bar{u}$; these pairs are independent, each with probability $1 - \kappa_{\sigma,r}$.

The union bound gives the first inequality. Each term is at most $(1 - \lambda)^{\lfloor N/2 \rfloor}$, since $N - 1 \geq \lfloor N/2 \rfloor$ for $N \geq 1$; there are $c(T^*)$ terms. Lower bounds: under (Rich) each single failure event is a non-anchor event (Theorem 3.8, “only if”). All N roots at σ equal the most likely head with probability $(1 - \rho_\sigma)^N$; z_m is unused with probability $(1 - \nu_{M,m})^N$; and a coincidence at (σ, r) has probability $\geq \chi_{\sigma,r}^N$ (Proposition 3.12(i)).

(b) Each failure event has positive probability for some N iff some support element, or some pair of support elements, witnesses it. For (U) this uses Lemma 3.2(d): a common $\bar{u}$ for the whole support exists iff one exists for every pair; the lemma does not need finiteness. So $\lambda > 0$ iff the support satisfies the three events. Then $c(T^*)(1 - \lambda)^{\lfloor N/2 \rfloor} \leq \bar{c} e^{-\lambda \lfloor N/2 \rfloor} \leq \bar{c} e^{-\lambda(N-1)/2}$, which is $\leq \delta$ iff $N \geq 1 + (2/\lambda) \ln(\bar{c}/\delta)$. For a ground T^* the failure probability is 0 for $N \geq 1$.

(c) The version space over tagged calculi factors over tags (Claude (Anthropic), 2026), so the verifier is exact iff the tag- i data are an anchor for every i , and

$$\Pr[\text{not exact}] \leq \sum_i \Pr[\text{the tag-}i \text{ data are not an anchor}].$$

Condition on the number $n_i \sim \text{Bin}(N, \pi_i)$ of tag- i data. For all $n_i \geq 0$,

$$\Pr[\text{not an anchor} \mid n_i] \leq \bar{c}_i (1 - \lambda_i)^{\lfloor n_i/2 \rfloor} :$$

for $n_i \geq 1$ this is (a); for $n_i = 0$ the right side is $\bar{c}_i \geq 1$; for a ground tag the left side is $[n_i = 0]$ and the right side is $0^{\lfloor n_i/2 \rfloor} \geq [n_i = 0]$ (with $0^0 = 1$). Next, $(1 - \lambda)^{\lfloor n/2 \rfloor} \leq (1 - \lambda)^{(n-1)/2} \leq e^{\lambda/2} e^{-\lambda n/2} \leq \sqrt{e} e^{-\lambda n/2}$, as $\lambda \leq 1$ (for $\lambda = 1$ check $n = 0, 1$ directly). And $\mathbb{E}[e^{-a n_i}] = (1 - \pi_i(1 - e^{-a}))^N \leq \exp(-N \pi_i(1 - e^{-a}))$, with $1 - e^{-a} \geq 3a/4$ for $a = \lambda_i/2 \leq 1/2$ (the left side is concave, and the inequality holds at $a = 1/2$). This gives $\sqrt{e} \bar{c}_i \exp(-3N \pi_i \lambda_i/8)$ per tag. Lower bounds: with probability $(1 - \pi_i)^N$ there is no tag- i datum, and then Acc of the tag- i data is empty; this is the exact failure probability of a ground tag. Under (Rich), all tag- i data share the head h at the pattern occurrence, or there are none, with probability $(1 - \pi_i + \pi_i(1 - \rho))^N = (1 - \pi_i \rho)^N$, and then they are no anchor. $\square$

The (U)-terms vanish for a single formula metavariable because, under $(R) \wedge (N)$, (U) holds automatically (Corollary 3.9(b)), and all occurrence events coincide with (R); so the bound is $(1 - \rho)^{N-1} + \sum_m (1 - \nu_m)^N$.

Proof of Proposition 3.12. (i) For each $\bar{u}$, $\Pr[\text{all } \theta_i \in A_{\bar{u}}] = \Lambda(A_{\bar{u}})^N$; take the supremum. Under (Rich), a coincidence at a non-valid pair is a failure of (U), so D_N is not an anchor (by Theorem 3.8 “only if” if (R*) and (N) hold, and also otherwise). (ii) For every $\bar{u}$, $1 - \kappa = \Lambda \otimes \Lambda\{(\theta, \theta') : \exists \bar{u}' \theta, \theta' \in A_{\bar{u}'}\} \geq \Lambda(A_{\bar{u}})^2$. (iii) The event says that the set of sampled support points is compatible, i.e. lies in some $A_{\bar{u}}$, i.e. in some $K \in \mathcal{K}$; inclusion-exclusion over $\mathcal{K}$ gives the formula. Each K lies in some $A_{\bar{u}}$, so $\Lambda(K) \leq \chi$ and the probability is at most $|\mathcal{K}| \chi^N$. Conversely $A_{\bar{u}} \cap \text{supp } \Lambda$ is compatible for

every $\bar{u}$, hence contained in a maximal compatible set; so $\chi = \max_K \Lambda(K)$, and the probability is at least χ^N . If $\mathcal{K} = \{K\}$ is a singleton, a pair (or a single point) is compatible iff it lies in K , so $1 - \kappa = \Lambda(K)^2 = \chi^2$. $\square$

Two worked laws. *First law:* $T^* = \forall x(0 = f(x))$ with $f \sim \{\lambda z.0 : 0.4, \lambda z.z : 0.3, \lambda z.Sz : 0.3\}$, the referee's example. The only non-valid pair is (slot of $f(x)$, rigid 0), and its compatible set is $\{\lambda z.0, \lambda z.z\}$ (with $\bar{u} : x \mapsto 0$); $\lambda z.Sz$ is in no $A_{\bar{u}}$. (R^*) fails iff all values are equal, (N) fails iff all values are $\lambda z.0$, and (U) fails iff all values lie in $\{0, z\}$. So $\Pr[\text{not an anchor}] = 0.7^N + 0.3^N$. The upper bound of Theorem 3.11(a) at $N = 8$ is $0.4^7 + 0.4^8 + 0.49^4 = 0.059942$. *Second law:* $T^* = \forall x \forall y(f(x) = g(y))$ with $f \sim U\{z, Sz\}$ and $g \sim U\{z, 0, Sz\}$ independent; write θ as the pair (value of f , value of g). At the pair $(f(x), g(y))$ the maximal compatible sets are $\{(z, z), (Sz, Sz)\}$ (with $\bar{u} : x \mapsto y$), $\{(z, 0)\}$ ($x \mapsto 0$) and $\{(z, Sz)\}$ ($x \mapsto Sy$), of masses $1/3, 1/6, 1/6$; so $|\mathcal{K}| = 3$, $\chi = 1/3$, and $1 - \kappa = 1/9 + 2/36 = 1/6$, so $\sqrt{1 - \kappa} = 0.408 > \chi$. At $N = 8$ the exact coincidence probability is $(1/3)^8 + 2(1/6)^8 = 1.54 \cdot 10^{-4}$, against $\chi^N = 1.52 \cdot 10^{-4}$ and $(1 - \kappa)^4 = 7.7 \cdot 10^{-4}$. *Non-circular check of Theorem 3.11(a):* the referee computed exact non-anchor probabilities by inclusion–exclusion over the support, deciding anchor status by feature search rather than by the events of Theorem 3.8, for induction, Separation and the template of Proposition 3.10(b); the bound held at every $N \in \{2, 4, 8, 12, 16, 24\}$ (for induction at $N = 24$: exact 0.00100, bound 0.00535).^{referee t7} The computations of Proposition 3.12(iii) were checked on every non-valid pair of five laws, and exact non-anchor probabilities for two small laws were cross-checked against an independent feature-template test with no disagreement.^{single e10}

B.6 Escalations

Lemma B.1 (Monotonicity). *[proved]^{single Lemma F.1} Let $D \subseteq D'$ be nonempty. (i) $C(D') \subseteq C(D)$ and $A(D') \subseteq A(D)$; a slot of D that is still in $A(D')$ is a slot of D' ; and $Y_\sigma(D) \subseteq Y_\sigma(D')$ whenever σ is a slot of both. (ii) If $C(D') = C(D)$ and $Y(D') = Y(D)$, then $E(D') \subseteq E(D)$, with the same maps $\bar{u}$. (iii) If $q \notin \text{Acc}(D)$, then $\Pi(D \cup \{q\}) \neq \Pi(D)$.*

Proof. (i) is immediate from the definitions: a slot of D' has its parent in $C(D') \subseteq C(D)$, and scopes are unions over the data. (ii) A feature of D' restricts to D , and its $\bar{u}$ is forced on $Y_\sigma(D) = Y_\sigma(D')$ (Lemma 3.2(d)). (iii) If Π is unchanged, then q has every Sym feature (C unchanged), every Scope feature (Y unchanged) and every Eq feature (E unchanged, with $\bar{u}$ forced by D). So $q \in \text{Feat}(D) = \text{Acc}(D)$ by Theorem 3.5. $\square$

Proof of Theorem 3.13. Let $D_j = \{d, q_2, \dots, q_j\}$. By Lemma B.1, each step $D_{j-1} \rightarrow D_j$ changes Π , and then it does one of the following: (α) shrink C ; (β) keep C and enlarge some scope (scopes only grow, and slots are unchanged when C is); (γ) keep C and every scope, and remove at least one pair from E (by (ii), E can only shrink then). (α) happens at most $|C(\{d\})| = n$ times. A position σ is a slot during one interval of steps: it becomes a slot when it leaves C , and it leaves A for ever when its parent leaves C . During that interval Y_σ only grows, inside $\{0, \dots, b(\sigma) - 1\}$; so (β) happens at most $\sum_\sigma b(\sigma)$ times. A pair (σ, r) can lie in E only while σ is a slot and $r \in A$, which is a single interval. Once removed, it never returns: if it were in E at times $t_1 < t_3$ but not at $t_2 \in (t_1, t_3)$, then σ is a slot and $r \in A$ at t_2 , and the feature at t_3 restricts to the data at t_2 (Lemma 3.2(d)), a contradiction. So (γ) happens at most once per ordered pair of incomparable positions of d . The second inequality uses $b(p) \leq b_{\max}$ and at most $n(n-1)$ ordered pairs. From no data the first query is escalated, and $b(p) \leq n-1$, so $\text{Esc}(\text{DT}^\circ; N) \leq 1 + N + N(N-1) + N(N-1) \leq 2N^2 + 1$.

The linear lower bound. With $k = N - 3$, the chain $S^k 0 = 0$, $S^k a = 0$, $S^{k-1} 0 = 0$, $\dots$, $0 = 0$, $0 = S0$, $\neg(0 = 0)$ has $N + 1$ members of size $\leq N$, and the successive acceptance sets are $\{S^k 0 = 0\}$, $\{S^k t = 0\}$, $\{S^{k-1} t = 0\}$, $\dots$, $\{t = 0\}$, $\{t = t'\}$ (t, t' closed): each query lies outside the previous set. All are instances of the universal template. Checked for $N = 5, 8, 12, 16$.^{single e8} $\square$

Proof of Proposition 3.14. (i) d_1 and d_2 differ exactly at the m left-hand sides (roots 0 and a). So $C(\{d_1, d_2\})$ is everything else, and the left-hand sides are slots with empty scope. Each $q_{j,k}$ agrees with d_1 off the slots, so C does not change along the chain. At slot j , $q_{j,k}$ has the bound variable with index k free (relative to the slot, which lies under the m binders and no others). Earlier data have at slot j only 0, a , or other bound variables $k' < k$. So $q_{j,k}$ violates $\text{Scope}(\text{slot } j)$ of its predecessors, and $q_{j,k} \notin \text{Feat} = \text{Acc}$. (Without the hard half of Theorem 3.5: T_0 of the predecessors is a covering DT° template that misses $q_{j,k}$.) The size is m binders, $m - 1$ conjunctions and $3m$ symbols in the atoms: $5m - 1$. The chain has $2 + m^2$ members. For $N \geq 4$, $m = \lfloor (N + 1)/5 \rfloor \geq 1$ gives size $5m - 1 \leq N$; the upper bound is Theorem 3.13.

(ii) The motive occurs four times in $\text{Ind}(\varphi)$: as $\varphi(0)$, $\varphi(x)$, $\varphi(Sx)$ and in the conclusion; since x does not occur, all four copies are φ . After $\text{Ind}(d_1)$ and $\text{Ind}(d_2)$ the left-hand sides in all four copies are slots, and their scopes contain no index bound by the motive's own m binders: x does not occur, and d_1, d_2 have no indices at the slots. $\text{Ind}(q_{j,k})$ puts the index k (bound by the motive's binders) at slot j of each copy, which no predecessor has. The size is $4(5m - 1) + 5 = 20m + 1$. For $m > 20c + 1$, $1 + m^2 > c(20m + 1)$. $\square$

Computed. Chains of length 6, 18, 38, 66 for $m = 2, 4, 6, 8$; for the universal template and $N \in \{4, 5, 9, 14, 19, 23, 24, 29, 34\}$ exactly $2 + \lfloor (N + 1)/5 \rfloor^2$ escalations with sizes $\leq N$; for the induction target, first data of size 41, 61, 81, 101 followed by 5, 10, 17, 26 escalations. Every step was certified twice, by the feature verifier and by an explicit covering T_0 that misses the query; the referee's independent code agrees.^{single e8, e12; referee t5, t5b}

Proof of Proposition 3.15. (i) For $D \ni d$ define $p \sim_D p'$ iff $p = p'$, or p, p' are positions of every datum and $d'|_p = d'|_{p'}$ for all $d' \in D$. This is an equivalence relation on the positions of d , and it refines as D grows. Consider a (γ) -step $D \rightarrow D' = D \cup \{q\}$; it changes neither C nor the scopes, hence neither the slots nor A . Suppose it removes (σ, r) with $Y_\sigma(D) = \emptyset$. Then $\text{Eq}(\sigma, r, \emptyset)$ was a D -feature, i.e. $d'|_r = d'|_\sigma$ for all $d' \in D$, so $\sigma \sim_D r$. It is not a D' -feature, although σ is still a slot, r still admissible and $\text{FV}(q|_\sigma) \subseteq Y_\sigma(D') = \emptyset$; so $q|_r \neq q|_\sigma$ and $\sigma \not\sim_{D'} r$. This is a strict refinement. An equivalence relation on n points refines strictly at most $n - 1$ times.

(ii) If d has no binder, every slot σ has $\text{b}(\sigma) = 0$, since its ancestors lie in C and carry the symbols of d . So all scopes are empty: there are no (β) -steps, and every (γ) -step is of the kind in (i). With at most n (α) -steps, $m - 1 \leq n + (n - 1)$. From no data add 1 for the first escalation; the lower bound $N + 1$ is the binder-free chain above. $\square$

Computed. 40 greedy adversarial chains from binder-free first data of 8–18 positions: no chain exceeded $2n - 1$, and every (γ) -step strictly refined $\sim$.^{single e13}

Lemma B.2 (Triangle constraints on equation kills). [*proved; computed*]^{single Lemma F.10} Let $D \subseteq D'$ with $C(D') = C(D)$ and $Y(D') = Y(D)$, and write $\bar{u}_{ab}$ for the forced map of a pair $(a, b) \in E$. (a) If $(k, i), (i, j) \in E(D')$, $k \perp j$, and k, j have the same sort, then $(k, j) \in E(D')$ with $\bar{u}_{kj} = \bar{u}_{ki}[\bar{u}_{ij}]$ on Y_k . (b) If $(k, i), (k, j), (i, j) \in E(D)$ and $(k, i), (k, j) \in E(D')$, then $(i, j) \in E(D')$. (c) If $(i, k), (k, j), (i, j) \in E(D)$ and $(i, k), (k, j) \in E(D')$, then $(i, j) \in E(D')$. (This lemma was suggested by the referee and added in the revision.) Hence a (γ) -step that removes (i, j) also

removes one of $(k, i), (k, j)$ for every common in-neighbour k , and one of $(i, k), (k, j)$ for every intermediate k .

Proof. First, $\bar{u}_{ki}[\bar{u}_{ij}]$ is defined on Y_k : for $y \in Y_k$, $\bar{u}_{ki}(y)$ is read off a datum d with $y \in \text{FV}(d|_k)$, so its free indices lie in $\text{FV}(d|_i) \subseteq Y_i$, the domain of $\bar{u}_{ij}$. (a) k is a slot and $j \in A$. For $d' \in D'$, $d'|_j = (d'|_i)[\bar{u}_{ij}] = ((d'|_k)[\bar{u}_{ki}])[\bar{u}_{ij}] = (d'|_k)[\bar{u}_{ki}[\bar{u}_{ij}]]$, and $\text{FV}(d'|_k) \subseteq Y_k$. (b) By (a) on D and forcing, $\bar{u}_{kj} = \bar{u}_{ki}[\bar{u}_{ij}]$ on Y_k . Let $q \in D' \setminus D$. The maps of (k, i) and (k, j) on D' are the forced maps on D (the domains are unchanged). So $q|_i = (q|_k)[\bar{u}_{ki}]$ and $q|_j = (q|_k)[\bar{u}_{kj}] = ((q|_k)[\bar{u}_{ki}])[\bar{u}_{ij}] = (q|_i)[\bar{u}_{ij}]$; moreover $\text{FV}(q|_i) \subseteq Y_i(D') = Y_i(D)$. So $\text{Eq}(i, j, \bar{u}_{ij})$ holds on D' . (c) Apply (a) to D' ; its map $\bar{u}_{ik}[\bar{u}_{kj}]$ equals $\bar{u}_{ij}$ on D by forcing. $\square$

Where the difficulty of Conjecture 3.16 sits. In a phase with C and the scopes fixed, “same value in every datum” is an equivalence relation on the nodes $A(D) \cup \{(\sigma, \bar{u}_{\sigma r})\}$ (the value of $(\sigma, \bar{u})$ in d being $(d|_\sigma)[\bar{u}]$); it refines with each datum, and (σ, r) survives iff $(\sigma, \bar{u}_{\sigma r}) \equiv r$. So a phase has fewer (γ) -steps than nodes: linearly many if every slot uses $O(1)$ distinct maps, but there can be order n^2 nodes, and up to $n + \sum b$ phases. *Computed:* 587 composition instances and 935 instances each of (b) and (c), 240 of them in steps that removed some pair, with no failure. single e13

B.7 Unions

Proof of Corollary 3.17. (i) If $w_0, w_1, \dots$ and $L_n = \text{inst}(T_n)$ witness infinite elasticity, then $w_n \notin \text{Acc}(\{w_0, \dots, w_{n-1}\})$ for $n \geq 1$, so the sequence is bounded by Theorem 3.13 with $d = w_0$. For thickness: $P := \lambda z.(0 = 0)$ puts $\forall x(0 = 0) \wedge (0 = 0)$ in $\text{inst}(\forall x P(x) \wedge P(t))$ for every closed t , and $P := \lambda z.(z = z)$ gives $\forall x(x = x) \wedge (t = t)$, which lies in the t -template only (matching forces $P = \lambda z.(z = z)$ from the pattern occurrence). (ii) is the cited preservation result together with “finite elasticity gives anchors” and the eventual exactness of the cautious verifier once an anchor is in the data (Claude (Anthropic), 2026).

(iii) With a k -ary relation symbol P , a unary function symbol g and a constant c , the chain of Claude (Anthropic) (2026) for the term signature transfers verbatim: query $P(g^{a_1}c, \dots, g^{a_k}c)$ for $a \in \{0, \dots, n\}^k$, $n + 1 = \lfloor (N - 1)/k \rfloor$, by non-increasing $\sum_j a_j$. Over $=, +, S, 0$, let $n + 1 := \lfloor (N - 1)/k \rfloor - 1 \geq 1$ (so $N \geq 2k + 1$), and query

$$q_a := (S^{a_1}0 + (S^{a_2}0 + \dots + S^{a_k}0)) = 0, \quad a \in \{0, \dots, n\}^k,$$

ordered by non-increasing $\sum_j a_j$. Each has size $k + 1 + \sum_j (a_j + 1) \leq k + 1 + k(n + 1) \leq N$, and all are instances of the target $(x_1 + \dots + x_k) = 0$ with 0-ary term metavariables x_j . Every earlier query a' has $a'_j \geq a_j + 1$ for some j ; otherwise $a' \leq a$ componentwise with $a' \neq a$, so $\sum a' < \sum a$, contradicting the order. Hence the earlier queries lie in $\bigcup_j \text{inst}(T_j)$ with $T_j := (x_1 + \dots + S^{a_j+1}(x_j) + \dots + x_k) = 0 \in \text{DT}^\circ$, and this union misses q_a . So $q_a \notin \text{Acc}_k(\text{earlier queries})$; the first query is escalated because $\text{Acc}_k(\emptyset) = \emptyset$. This gives $(n + 1)^k$ escalations. With a k -ary P the same proof works with overhead 1 instead of $k + 1$. $\square$

Computed. For $(k, n + 1) = (1, 6), (2, 3), (2, 4), (3, 2)$: 6, 9, 16, 8 escalations on sentences of size $\leq N = 8, 9, 11, 10$, each step certified by the explicit k -union and, for the first 12 steps, by the brute-force union verifier. single e16

Proof of Proposition 3.18. Let $\bigcup_{l \leq k} \text{inst}(T_l) \supseteq D$, $T_l \in \text{DT}^\circ$. Assign each $\theta \in \Theta_i$ to some l with $T_i^* \theta \in \text{inst}(T_l)$ (matchers are unique, so data and substitutions correspond). This splits Θ_i into

at most k parts, and into at least one nonempty part, since $\Theta_i \neq \emptyset$. Suppose no part's data were an anchor for $\text{inst}(T_i^*)$. By the “if” half of Theorem 3.8 (which needs no (Rich)), each nonempty part then fails (R*), (N) or (U), i.e. lies in a member of $\text{Fail}(T_i^*)$ (all roots at σ equal some h ; or z_m unused; or, with (R*) and (N) holding, a coincidence $\bar{u}$ at a non-valid pair). For a ground T_i^* a nonempty part is $\{T_i^*\} = \text{inst}(T_i^*)$, an anchor. So Θ_i would be covered by at most k members of $\text{Fail}(T_i^*)$, contrary to the hypothesis. Hence some part, contained in $\text{inst}(T_l)$, is an anchor, and $\text{inst}(T_l) \supseteq \text{inst}(T_i^*)$. This holds for every i . $\square$

Computed. With target $\text{inst}(T_{\text{Ind}}) \cup \{\text{Ext}\}$ and $k = 2$, three induction data and no Extensionality datum, the union verifier rejects Extensionality; with Extensionality among the data it accepts it; fresh induction instances are accepted in both cases. This is the failure of the first version of the proposition, which lacked $\Theta_i \neq \emptyset$. single e15

Proof of Proposition 3.19. A k -union $\bigcup_{l \leq k} \text{inst}(T_l)$ contains D iff some partition of D into at most k nonempty blocks $B_1, \dots, B_j$ has each block covered by its own T_l ($l \leq j$), the remaining templates being unconstrained. Suppose that for some such partition $q \notin \text{Acc}(B_l)$ for every $l \leq j$. Choose T_l covering B_l with $q \notin \text{inst}(T_l)$, and for $l > j$ a ground template other than q (possible since $\text{Acc}(\emptyset) = \emptyset$). This union contains D and misses q , so $q \notin \text{Acc}_k(D)$. Conversely, if every partition has a block B with $q \in \text{Acc}(B)$, let $U = \bigcup_l \text{inst}(T_l) \supseteq D$; assigning each datum to a template that covers it gives a partition, some block B of it has $q \in \text{Acc}(B) \subseteq \text{inst}(T_l) \subseteq U$. Each test $q \in \text{Acc}(B)$ is polynomial by Corollary 3.6. $\square$

B.8 Complexity of the version space

Proof of Proposition 3.20. Each datum has $4 + (4n - 1) + 1 = 4n + 4$ positions. $C(D_n) = \forall x(\Box = \Box) \wedge C_n$; the two slots have column $(x, 0)$ and scope $\{x\}$. The D_n -features are the Sym and Scope features, the two slot-to-slot equations (identity map), and, for each of the $2n$ zeros of C_n and each slot, “content = slot content with $x := 0$ ”. There are no formula slots, so every metavariable is term-valued with pattern occurrences at the slots, and by Theorem 3.3(b) a saturated template has at each zero either the rigid 0 or a derived occurrence. Every covering template is $\succeq$ one of the templates T_A (A a set of zeros): $f(x)$ at both slots (merging two metavariables at the slots is a specialization that still covers) and $f(0)$ exactly at the zeros in A . For $A \neq B$, the instance $f := \lambda z.Sz$ of T_A has $S0$ exactly at the positions of A , so T_A and T_B have incomparable instance sets. Hence every T_A is minimal ($T \preceq T_A$ covering implies $T \succeq T_B \preceq T_A$ for some B , forcing $B = A$ and $T \equiv T_A$), and $\text{Min}(D_n) = \{T_A\}$, with 4^n elements. By Theorem 3.5, $\text{Acc}(D_n) = \{\forall x(t = t) \wedge C_n : \text{FV}(t) \subseteq \{x\}, t[0/x] = 0\} = D_n$, since a term with root S or $+$ keeps its root under $[0/x]$, so $t \in \{x, 0\}$. $\square$

Computed. $|\text{Min}(D_n)| = 4, 16, 64$ for $n = 1, 2, 3$ from the saturated templates, and 4, 16 for $n = 1, 2$ by brute force (also by the referee). single e2; referee t2

Proof of Proposition 3.21. $(\Leftarrow) T_{\text{all}} \in \text{DT}^\circ$, and it covers D with $\theta_d(M_K) := (d|_s)[\bar{z}/Y_s]$, since $d|_x = (d|_s)[\bar{u}_{s,x}]$. Let $q = T_{\text{all}}\theta$. It has every Sym feature, and every Scope feature: $\text{FV}(q|_s) \subseteq Y_s$, and $\text{FV}(q|_x) \subseteq \text{FV}(\bar{u}_{s,x}) \subseteq Y_x$, because $\bar{u}_{s,x}$ is read off data at x . Let $\text{Eq}(\sigma, r, \bar{u})$ be a D -feature. By (i), r is a slot, and σ, r lie in one component K , with source s (put $\bar{u}_{s,s} := \text{id}$). Then $q|_r = \theta(M_K)[\bar{u}_{s,r}]$ and $(q|_\sigma)[\bar{u}] = \theta(M_K)[\bar{u}_{s,\sigma}[\bar{u}]]$. On D both $\bar{u}_{s,r}$ and $\bar{u}_{s,\sigma}[\bar{u}]$ solve the equation of the pair (s, r) , so they coincide on Y_s by forcing, and $q|_r = (q|_\sigma)[\bar{u}]$. So $\text{inst}(T_{\text{all}}) \subseteq \text{Feat}(D) = \text{Acc}(D) \subseteq \text{inst}(T_{\text{all}})$. *Syntactic leastness:* let U cover D and take a saturated U' with $U \succeq U'$

(Theorem 3.3(a)). An occurrence of U' in $C(D)$ would be a derived occurrence with an Eq-feature landing in $C(D)$, which (i) excludes; so all occurrences of U' are at slots, every slot is one, and the skeleton of U' is $C(D)$. Map each metavariable L of U' , with pattern slot σ_L and pattern arguments $\bar{y}$, to $\lambda\bar{z}.(T_{\text{all}}|_{\sigma_L})[\bar{z}/\bar{y}]$. Each other occurrence $L(\bar{t})$ at r gives the D -feature $(\sigma_L, r, \bar{y} \mapsto \bar{t})$, so r lies in the component of σ_L , and $T_{\text{all}}|_r = (T_{\text{all}}|_{\sigma_L})[\bar{y} \mapsto \bar{t}]$ by forcing. So this substitution maps U' to T_{all} , and $U \succeq U' \succeq T_{\text{all}}$.

($\Rightarrow$) Let T_ℓ be least. $T_\ell \preceq T_0(D)$ makes every position of $C(D)$ rigid in T_ℓ . Suppose an Eq-feature $\varphi = (\sigma, r, \bar{u})$ had $r \in C(D)$. Since T_ℓ covers D , $\text{skel}(T_\ell) \subseteq C(D)$, so the slot σ (whose parent is rigid) carries an occurrence $N(\bar{t})$ in T_ℓ . From $T_\ell = T_\varphi\tau$ we get $T_\ell|_\sigma = \tau(M_\sigma)[Y_\sigma]$, so $\tau(M_\sigma) = \lambda\bar{z}.N(\bar{t}')$, and then $T_\ell|_r = \tau(M_\sigma)[\bar{u}] = N(\bar{t}'[\bar{u}])$ is an occurrence at a rigid position, a contradiction. This proves (i). So the occurrences of T_ℓ are exactly the slots. The same argument shows that a feature $\sigma \rightarrow r$ makes σ and r occurrences of the same metavariable. For a metavariable N with pattern occurrence at s , the pair (s, x) is a feature for every other occurrence x of N , because T_ℓ covers D . So the components are the occurrence sets of the metavariables, each with the source s : this is (ii). $\square$

Proof of Corollary 3.22. By Theorem 3.8 (here (Rich) is used), (R*), (N) and (U) hold. So $C(D) = \text{skel}(T^*)$, $\text{slots}(D) = \text{Occ}(T^*)$, $Y_\sigma(D) = Y_\sigma^*$, and every Eq-feature comes from a valid pair: r is an occurrence of the same metavariable. Thus no feature lands in $C(D)$, which is (i). The slot graph has edges only between occurrences of one metavariable, and a pattern occurrence s of M has $s \rightarrow x$ for every occurrence x of M (the pair is valid and T^* covers D), which is (ii). T_{all} puts $M_K(Y_s)$ at s and $M_K(\bar{u}_{s,x})$ at x , where by forcing and (N) $\bar{u}_{s,x}$ is the map $\bar{y} \mapsto \bar{t}_x$; so T_{all} is T^* up to renaming and permutation of argument places. The construction of Proposition 3.21 is polynomial by Corollary 3.6. $\square$

B.9 Computations

All scripts are in `research/tracks/single/` and are run there with `python3 <script> [args]`; outputs are the `.out` files of the same name. The brute-force enumerator `dtenum.py` does not use the theory: it enumerates prefixes of the common prefix, fillers from a term pool and all sharings, and keeps the determinate covering templates. Agreement with it is therefore an independent check within its bounds. The referee's code in `referee_checks/` is a second, fully independent implementation.

Scripts and what they check: `e1_matching.py` (matching: unique matcher and recovery 1500/1500, generality by freezing 400/400, linear time up to $|s| = 3.3 \cdot 10^6$); `e2_examples.py` (Theorem 3.3, Example 3.4, Proposition 3.20, Proposition 3.10, Corollary 3.17(i)); `e3_finitary.py` 120 7 and 120 8 low (Theorems 3.3 and 3.5, Proposition 3.21 against brute force: 81/81 and 22/22 cases, Acc = Feat on 5804 and 1817 queries); `e4_anchor_random.py` 70 S 2, S = 11, 12, 13, and 60 14 2 (Theorem 3.8: prediction = feature truth 248/248; brute force decided 149 and agreed on 144, the other 5 being artifacts of its bounds); `e5_specializations.py` (Corollary 3.9: first-order 571/571; induction 495, Separation 495, Replacement 498, $\in$ -induction 491); `e6_escalation.py` 60 5 and `e6b_escalation_structured.py` 1 (Theorem 3.13, Conjecture 3.16); `e7_rates.py` (circular, see above) and `e10_rates_revision.py` (Theorem 3.11, Proposition 3.12, ground templates); `e8_counterexamples.py` (Proposition 3.10; the chains of Section 3.6); `e9_union.py` and `e15_ground_union.py` (Propositions 3.18 and 3.19); `e12_c11_refuted.py` (Proposition 3.14); `e13_escalation_structure.py` 3 40 (Proposition 3.15, Lemma B.2); `e16_union_lower_bound.py` (Corollary 3.17(iii)). The referee's scripts `t1`–`t10` check everything above except the items added in the revision, with full

agreement (Feat = brute-force intersection on 125,109 queries; anchor prediction 540/540).
Scripts `e11` and `e14` check the complexity results of track single, Theorem H, which are presented in Section 6.

C Proofs for Section 4 (universal axioms)

This appendix gives the proofs for Section 4, in its notation. The default is the λ convention: a 0-ary metavariable takes a term or formula without bound variables (parameters allowed). We write bound variables as names; where indices matter (syntax (dB)) we say so. A *value* is what a substitution assigns to a metavariable. The proofs were transcribed from the research records and checked again; where the presentation differs from the record we say so.

C.1 Anchors

The proof of Proposition 4.4(a) rests on one identity, which generalizes the two-point identity of the untagged analysis ^{untagged Lemma 5.5, step 1} from one metavariable to several.

Lemma C.1 (Identity lemma). *[proved]^{untagged Lemma 5.5; cases A2'} Let Z be a set of 0-ary metavariables (of term or formula sort), and let G, K be terms or formulas, possibly with binders, bound variables and parameters, in which members of Z occur only as leaves. Let $\theta_1, \dots, \theta_N$ assign to each $z \in Z$ a value of its sort without bound variables, such that (R) for each $z \in Z$ the roots of $\theta_1(z), \dots, \theta_N(z)$ are not all equal, and (D) for distinct $z, z' \in Z$ of the same sort some j has $\theta_j(z) \neq \theta_j(z')$. If $G\theta_j = K\theta_j$ for all j , then $G = K$.*

Proof. Induction on $|G| + |K|$. Substituting values for metavariable leaves changes nothing else; values contain no bound variables, so no shifting or capture occurs under binders. *Case $G = z \in Z$.* If $K = z$, we are done. If $K = z' \in Z$ with $z' \neq z$, then $\theta_j(z) = \theta_j(z')$ for all j , which is impossible for different sorts and contradicts (D) otherwise. Otherwise the root of K is rigid: a function or relation symbol, a connective, a quantifier, a constant, a parameter or a bound variable. Then every $K\theta_j$ has that root, so all $\theta_j(z) = G\theta_j$ have the same root, contradicting (R). *Case $K \in Z$.* Symmetric. *Otherwise* both roots are rigid. Comparing $G\theta_1 = K\theta_1$ shows that they are the same symbol with the same binding structure, and the children of G and K satisfy the hypothesis. By induction the children are equal, so $G = K$. $\square$

Proof of Proposition 4.4. (a) Let $\sigma \in \text{FO}$ and $D = \{\sigma\theta_1, \dots, \sigma\theta_N\}$ with values without bound variables satisfying (R) and (D); let $T \in \text{DT}^\circ$ cover D . Write $D_j := \sigma\theta_j$, and for a position c of σ let $\kappa(c) := \sigma|_c$.

Step 1 (skeleton). All data carry σ 's symbol at each rigid position of σ . At a metavariable position of σ , carrying z , the roots of the $\theta_j(z)$ are not all equal by (R). So the common prefix $C(D)$ of the data (Section 3) is the set of rigid positions of σ , and its slots are the metavariable positions of σ . By the rigid-prefix lemma (Section 3), the rigid skeleton of T lies in $C(D)$ with the same symbols, and every occurrence position of T is in $C(D)$ or is a slot. In particular every occurrence position c of T is a position of σ , and $\kappa(c)$ is defined.

Step 2 (bodies). Let M be a metavariable of T of arity n , with a pattern occurrence at c_1 , say $M(u_1, \dots, u_n)$ with pairwise distinct bound variables u_m in scope at c_1 . By the matching theorem (Section 2), its value in D_j is $\lambda\bar{h}.(D_j|_{c_1})[h_m/u_m]$, and $D_j|_{c_1}$ has no free bound variable outside $\{u_1, \dots, u_n\}$. Since $D_j|_{c_1} = \kappa(c_1)\theta_j$ and the values contain no bound variables, the free bound variables of $\kappa(c_1)$ are those of $D_j|_{c_1}$, hence among the u_m . Let $B_M := \kappa(c_1)[h_m/u_m]$. It

is a template body: it has no free bound variable, and its metavariables are the members of Z in $\kappa(c_1)$, all at leaves. Substituting θ_j commutes with abstracting the u_m , so $\lambda\bar{h}.B_M\theta_j$ is M 's value in D_j .

Step 3 (other occurrences). Let $M(\bar{u}^s)$ be any occurrence of M , at c_s . By preservation (Section 2), $D_j|_{c_s} = (B_M\theta_j)[\bar{u}^s] = (B_M[\bar{u}^s])\theta_j$. The second equality holds because plugging the metavariable-free arguments $\bar{u}^s$ into holes and substituting values for metavariable leaves act on different leaves, and values without bound variables need no shifting under the binders of B_M . Also $D_j|_{c_s} = \kappa(c_s)\theta_j$. So $G := B_M[\bar{u}^s]$ and $K := \kappa(c_s)$ contain the members of Z only as leaves and agree under every θ_j . By Lemma C.1, $B_M[\bar{u}^s] = \kappa(c_s)$.

Step 4. Let ρ be the template substitution $M \mapsto \lambda\bar{h}.B_M$ for all metavariables M of T . On the rigid skeleton of T , $T\rho$ and σ have the same symbols (Step 1). At every occurrence position c_s , $(T\rho)|_{c_s} = B_M[\bar{u}^s] = \kappa(c_s) = \sigma|_{c_s}$ (Step 3). Every position of T outside argument lists is in the skeleton or is an occurrence, so $T\rho = \sigma$ and $T \succeq \sigma$.

For the instance schema take $\sigma = \sigma_\varphi$ and $D \subseteq \text{inst}_o(\varphi)$: the values are terms over $L \cup \text{Par}$, without bound variables. Every covering $T \in \text{DT}^\circ$ satisfies $T \succeq \sigma_\varphi$, hence $\text{inst}(T) \supseteq \text{inst}(\sigma_\varphi)$, so D is an anchor for $\text{inst}(\sigma_\varphi)$. Since σ_φ covers D and lies below every covering template, $\text{Min}(D) = \{\sigma_\varphi\}$ up to $\equiv$.

(b) If (R) or (D) fails, $\text{lgg}(D) \not\equiv \sigma_\varphi$ by Plotkin's recovery lemma. If $\text{lgg}(D)$ contained $\text{inst}_c(\varphi)$, Proposition 4.2 would give $\text{lgg}(D) \succeq \sigma_\varphi$, and $\sigma_\varphi \succeq \text{lgg}(D)$ holds because σ_φ covers D ; so $\text{lgg}(D)$ misses a closed instance. It is a legitimate first-order pattern under the λ convention: its metavariables sit at disagreement positions of D , which lie at or below metavariable positions of σ_φ , where every datum has a closed term; so its values on D are closed terms, and it covers D . It lies in $\text{FO} \subseteq \text{DT}^\circ$.

(c) In $T = (f(S0) + f(0) = f(S0))$ no bound variable is in scope, so f has no pattern occurrence and $T \in \text{SO}^\circ \setminus \text{DT}^\circ$. $f := \lambda h.0$ gives $0 + 0 = 0$, and $f := \lambda h.h$ gives $S0 + 0 = S0$. If $f := \lambda h.\beta$ gave $SS0 + 0 = SS0$, then $\beta[0] = 0$ and $\beta[S0] = SS0$. A body whose root is S , $+$, $\cdot$ or a parameter keeps that root after plugging, so $\beta[0] = 0$ forces $\beta \in \{0, h\}$, and then $\beta[S0] \in \{0, S0\}$.

The examples with binders mentioned after the proposition are checked in the same way cases q1_dt; referee r5. (i) $\varphi = \forall y (x + y = y + x)$, data at 0 and $S0$, $T_1 = \forall y (G(0, S0) + G(y, y) = y + F(y))$: take $G := \lambda h_1 h_2. h_1$, $F := \lambda h.0$, resp. $G := \lambda h_1 h_2. h_2$, $F := \lambda h. S0$. The occurrence $F(y)$ is a pattern occurrence, but G has none. For the instance at $SS0$, $G(y, y) = y$ forces G 's body to be a hole, so $G(0, S0) \in \{0, S0\} \neq SS0$. (ii) $\varphi = (x + y = y + x)$, data $(0, S0)$ and $(S0, 0)$, $T_2 = (G(0, S0) = G(S0, 0))$: take $G := \lambda h_1 h_2. (h_1 + h_2)$, resp. $\lambda h_1 h_2. (h_2 + h_1)$. For the instance $0 + SS0 = SS0 + 0$ the body would be $\beta_1 + \beta_2$ with $\beta_1[0, S0] = 0$, forcing $\beta_1 \in \{0, h_1\}$, and $\beta_1[S0, 0] = SS0$, which is impossible. $\square$

Remarks. (1) With one metavariable, Lemma C.1 is the two-point identity, and Proposition 4.4(a) is the two-point lemma of the untagged analysis, including a 0-ary formula metavariable (Section 6). (2) The record proves (a) for closed data by a direct case analysis of the positions of $\kappa(c_1)$ and $\kappa(c_s)$ cases Prop A2'; the referee checked that analysis line by line, including hole positions that receive closed-term arguments at non-pattern occurrences. The proof above reorganizes it around Lemma C.1, which the untagged record proves for one metavariable, and allows parameters in the values; it is also the "if" half of the first-order corollary of the anchor theorem of Section 3. We found no error in either record.

C.2 Rates

Proof of Theorem 4.5. The data are i.i.d. $\sim \Lambda$. The failure event is the union of the events $\neg(R_i)$ and $\neg(D_{ii'})$.

(a) $\neg(R_1)$ is the disjoint union over f of A_f : “all N roots are f ”, with $\Pr[A_f] = p_f^N$.

(b) Let $A = \bigcup_f A_f$ (all roots of t_1 equal), $B = \bigcup_g B_g$ (all roots of t_2 equal, to g), and E : “ $t_1 = t_2$ in every datum”. The A_f are pairwise disjoint, and so are the B_g . $\Pr[A] = \sum_f a_f^N$, $\Pr[B] = \sum_g b_g^N$, $\Pr[E] = c^N$, $\Pr[A \cap B] = \sum_{f,g} d_{fg}^N$. On E the roots of t_1 and t_2 coincide, so $\Pr[A \cap E] = \sum_f e_f^N = \Pr[B \cap E]$, and $A_f \cap B_g \cap E = \emptyset$ for $f \neq g$ while $A_f \cap B_f \cap E = A_f \cap E$; so $\Pr[A \cap B \cap E] = \sum_f e_f^N$. Inclusion–exclusion gives

$$\Pr[A \cup B \cup E] = \sum_f a_f^N + \sum_g b_g^N + c^N - \sum_{f,g} d_{fg}^N - 2 \sum_f e_f^N + \sum_f e_f^N,$$

which is the formula. For independent components with term law μ and root law p : $a_f = b_f = p_f$, $d_{fg} = p_f p_g$ (so $\sum_{f,g} d_{fg}^N = S^2$), $c = \sum_t \mu(t)^2$ and $e_f = c_f$.

(c) Union bound: $P_N \leq \sum_i \Pr[\neg(R_i)] + \sum_{i < i'} \Pr[\neg(D_{ii'})]$ with $\Pr[\neg(R_i)] = \sum_f p_{i,f}^N$ and $\Pr[\neg(D_{ii'})] = c_{ii'}^N = (1 - r_{ii'})^N$. Let G be a best split for i . The events “all roots of t_i are f ”, for $f \in G$, are disjoint subsets of “all roots of t_i lie in G ”, so $\sum_{f \in G} p_{i,f}^N \leq \Lambda(\text{root}(t_i) \in G)^N$; likewise for the complement. Both $\Lambda(\text{root}(t_i) \in G)$ and $\Lambda(\text{root}(t_i) \notin G)$ are at most $1 - \rho_i$, so $\sum_f p_{i,f}^N \leq 2(1 - \rho_i)^N \leq 2(1 - \rho)^N$, and $c_{ii'}^N \leq (1 - \rho)^N$. Summing gives $(2k + \binom{k}{2})(1 - \rho)^N$, and $1 - \rho \leq e^{-\rho}$. $\square$

C.3 Parameters, guards and capture

Proof of Proposition 4.6. (a) $\varphi(\bar{p})$ substitutes distinct parameters for the free occurrences of the distinct variables $\bar{x}$; as φ has no parameters, the universal closure of $\varphi(\bar{p})$ is $\forall \bar{x} \varphi$ up to the names of bound variables. A member $\varphi(\bar{t})$ of $\text{inst}_o(\varphi)$ follows from $\forall \bar{x} \varphi$ by $\forall$ -elimination (the terms contain no bound variable, so the substitution is free), and its closure follows by generalization over its parameters. Conversely $\forall \bar{x} \varphi$ follows from $\varphi(\bar{p})$ by generalization over $\bar{p}$.

(b) Let $\mathcal{H}$ be $H_1(\text{FO})$ or $H_1(\text{DT}^\circ)$ and $\tau \in \mathcal{H}$ with $\text{inst}(\tau) \supseteq \text{inst}_c(\varphi)$. With $\theta(z_i) := t_i$ and $\theta'(z_i) := t'_i$ from (Rich_c) , the closed instances $\sigma_\varphi \theta, \sigma_\varphi \theta'$ satisfy (R) and (D), so $\tau \succeq \sigma_\varphi$ by Theorem 4.3 (for FO) or Proposition 4.4(a) (for DT°). As $\sigma_\varphi \in \text{FO} \subseteq \text{DT}^\circ$ contains $\text{inst}_c(\varphi)$, $\text{cl}_{\mathcal{H}}(\text{inst}_c(\varphi)) = \text{inst}(\sigma_\varphi)$, which is $\text{inst}_o(\varphi)$ under the λ convention with parameters. The inclusion $\text{inst}_c(\varphi) \subseteq \text{inst}_o(\varphi)$ is strict: every z_i occurs in σ_φ , so the text of $\varphi(\bar{p})$ determines the value p_i at z_i , which is not closed. Once the data contain an anchor, the cautious verifier accepts the closure (Section 2), in particular $\varphi(\bar{p})$. In (N) and (dB) the same argument gives $\text{cl}_{H_1(\text{FO})}(\text{inst}_c(\varphi)) = \text{inst}(\sigma_\varphi)$, since Theorem 4.3 holds in every syntax, and $\text{inst}(\sigma_\varphi) \supseteq \text{Cap}(\varphi)$ by Proposition 4.7(a).

(c) An instance of σ_φ whose values are all closed is a closed instance; closed values contain no variable, so there is no capture. Hence $\text{inst}(\sigma_\varphi, \{\text{closed}(z_i)\}_i) = \text{inst}_c(\varphi)$ in every syntax. By the most-specific-guard rule the learner’s hypothesis on D is $\text{lgg}(D)$ with every guard of the family that holds on all data. On closed data the values of every metavariable of $\text{lgg}(D)$ are subterms of closed terms, so closed is kept for all of them. Then $\varphi(\bar{p})$ is not accepted: at a metavariable position q of σ_φ the hypothesis either has a metavariable at or above q , whose value in $\varphi(\bar{p})$ contains a parameter and violates closed, or is rigid at q with the root shared by all data’s closed terms, which differs from a parameter. A capture instance is excluded the same way, because its value at some such position contains a bound variable. A datum with a parameter at z_i violates closed for the metavariable of that column and deletes it. $\square$

Proof of Proposition 4.7. (a) Let $\sigma_\varphi\theta$ be an instance (a sentence, or a formula whose free names are parameters). Suppose no value is captured. In (N) the substitution is then free for every x_i , and $\sigma_\varphi\theta = \varphi(\bar{\theta})$ is a genuine instance: closed if the values are variable-free, and otherwise at terms whose variable names are bound at no occurrence. Such names are free in the result, so they are parameters (impossible if data must be sentences). In (dB), an index $\#m$ in a value at an occurrence o refers to an enclosing binder if m is less than the binder depth of o (then the value is captured), and is loose otherwise (then the result is not well formed). So a non-captured value has no index, and $\sigma_\varphi\theta$ is a closed or parameter instance. Otherwise $\sigma_\varphi\theta \in \text{Cap}(\varphi)$. The reverse inclusions hold by definition. Disjointness: every z_i occurs in σ_φ , so the text $\sigma_\varphi\theta$ determines each $\theta(z_i)$ (the subtree at an occurrence of z_i), and whether it is captured is a property of that text. The values of a genuine open instance contain no bound variable, so they are not captured.

(b) (N), data are sentences: if a name y is bound at every free occurrence of x_i , then $\theta(z_i) := y$ (other values closed) gives a sentence in $\text{Cap}(\varphi)$. Conversely, if a sentence $\sigma_\varphi\theta \in \text{Cap}(\varphi)$ has the captured name y in $\theta(z_i)$, then every copy of $\theta(z_i)$, one at each free occurrence of x_i , contains y , and since the result has no free names, y is bound at each of them. (N), free names read as parameters: $\theta(z_i) := y$ with y bound above some occurrence is an admissible datum (copies of y elsewhere are read as a parameter); conversely capture needs a binder above an occurrence. (dB): if every free occurrence of x_i has binder depth ≥ 1 , then $\theta(z_i) := \#0$ is well formed at all of them and captured; conversely a captured value contains some $\#m$, and well-formedness at each occurrence o requires m to be less than the depth of o , so every depth is ≥ 1 .

(c) $\exists y \neg(y = y)$ is false in every structure, while $\exists y \neg(y = t)$ holds in every structure with at least two elements, whatever t denotes. $\forall x \exists y (y = Sx)$ is logically valid (take $y := Sx$), while $\exists y (y = Sy)$ is false in $\mathbb{N}$, where S has no fixed point.

(d) By Proposition 4.6(b), $\text{cl}_{H_1(\text{FO})}(\text{inst}_c(\varphi)) = \text{inst}(\sigma_\varphi) \supseteq \text{Cap}(\varphi)$; the anchor argument of Theorem 4.3 uses only closed instances, so it holds in (N) and (dB). Once the data contain an anchor the cautious verifier accepts the closure (Section 2). With φ as in (c), the accepted capture instance $\exists y \neg(y = y)$ is false in every structure.

(e) Closed values satisfy both guards, so closed data keep both, and $\text{inst}(\sigma_\varphi, \{\text{closed}, \text{nocap}\}) = \text{inst}_c(\varphi)$ because closedness implies nocap. A genuine parameter instance satisfies nocap and violates closed, so only $\text{closed}(z_i)$ is deleted; $\text{inst}(\sigma_\varphi, \{\text{nocap}(z_i)\}_i)$ is the set of instances with no captured value, which by (a) is $\text{inst}_c(\varphi) \cup \text{inst}_o(\varphi) = \text{inst}_o(\varphi)$. Genuine data never violate nocap, so it is never deleted, and capture instances, which violate it, are never accepted. $\square$

C.4 The ω -gap: details

Example 4.9(a). Closed terms over $\{0, 1, +, -, \cdot\}$ denote integers, by induction on terms. For an integer n , $n \cdot n \neq 2$, so $\mathbb{R} \models \neg(t \cdot t = 1 + 1)$ for every closed t ; $\sqrt{2}$ refutes $\forall x \neg(x \cdot x = 1 + 1)$. In a real closed field every positive element has a square root, and $1 + 1 > 0$, so $\text{RCF} \vdash \exists x (x \cdot x = 1 + 1)$. The rationals form an ordered field without $\sqrt{2}$, so $\text{OF} \not\models \exists x (x \cdot x = 1 + 1)$. Here $M_0 \cong \mathbb{Z}$, which is not elementary in $\mathbb{R}$, as Proposition 4.8 requires.

Example 4.9(b). $\emptyset$ is hereditarily finite, and HF is closed under $\{a, b\}$ and $a \cup b$, so every closed term denotes an HF set. Conversely, by $\in$ -induction every HF set is denoted: $\{a\} = p(a, a)$ and $\{a_1, \dots, a_n\} = u(p(a_1, a_1), \{a_2, \dots, a_n\})$. A set x with $\text{Ind}(x)$ contains $\emptyset$, $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}$, $\dots$, which are pairwise distinct, so it is infinite; hence $V \models \neg \text{Ind}(t)$ for every closed t . The axiom of Infinity provides an inductive set, so $\forall x \neg \text{Ind}(x)$ is false and refuted by ZF. The record remarks that each closed instance is probably provable in very weak set theories; only its truth is used here.

Example 4.9(c). Assuming PA consistent, $\text{PA} \not\models \text{Con}(\text{PA})$, so $\text{PA} + \neg \text{Con}(\text{PA})$ has a model M .

A closed term t denotes a standard number n , and $\neg\text{Prf}(t, \ulcorner \perp \urcorner)$ is a true Δ_0 sentence because n codes no proof of $\perp$. True Δ_0 sentences are provable in $\mathbf{Q}$ (see e.g. Hájek and Pudlák, 1993), so they hold in M , while $M \models \exists x \text{Prf}(x, \ulcorner \perp \urcorner)$. So M_0 , the standard part, is not elementary in M .

Example 4.10: the model of $\mathbf{Q}$ cases Ex. A8; q1_qmodel; referee r1 (4). The domain is $\mathbb{N} \cup \{a, b\}$ with $a, b \notin \mathbb{N}$; n, m range over $\mathbb{N}$. The operations are standard on $\mathbb{N}$, and otherwise:

- successor: $Sa = a, Sb = b$;
- addition: $n + a = n + b = b$; $a + n = a, b + n = b$; $a + a = a + b = a$; $b + a = b + b = b$;
- multiplication: $x \cdot 0 = 0$ for all x ; $a \cdot n = b \cdot n = b$ for $n \geq 1$; $n \cdot a = n \cdot b = a$ for $n \geq 1$; $0 \cdot a = 0 \cdot b = 0$; $a \cdot a = a \cdot b = b \cdot a = b \cdot b = a$.

($\mathbf{Q}$'s axioms do not mention $<$, which may be interpreted arbitrarily.) We check Q1–Q7.

- Q1, Q2, Q3: S is injective, 0 is not a successor, and every $x \neq 0$ is a successor ($n + 1 = Sn$, $a = Sa, b = Sb$).
- Q4, $x + 0 = x$: standard on $\mathbb{N}$; $a + 0 = a$ and $b + 0 = b$.
- Q5, $x + Sy = S(x + y)$. For $y = m$: if $x \in \{a, b\}$ then $x + m = x$ and $Sx = x$; otherwise standard. For $y \in \{a, b\}$: $Sy = y$, and every $x + y$ lies in $\{a, b\}$, which S fixes.
- Q6, $x \cdot 0 = 0$: by definition.
- Q7, $x \cdot Sy = x \cdot y + x$. For $y = m$ and $x = a$: $a \cdot (m + 1) = b$, and $a \cdot m + a$ is $0 + a = b$ if $m = 0$ and $b + a = b$ if $m \geq 1$. For $x = b$: $b \cdot (m + 1) = b$, and $b \cdot m + b$ is $0 + b = b$ or $b + b = b$. For $y \in \{a, b\}$: $Sy = y$, so we need $z + x = z$ for $z = x \cdot y$. If $x = 0$, $z = 0$ and $0 + 0 = 0$. If $x = n \geq 1$, $z = a$ and $a + n = a$. If $x = a$, $z = a$ and $a + a = a$. If $x = b$, $z = a$ and $a + b = a$.

In this model $0 + a = b \neq a$, and every closed instance $0 + t = t$ holds, since closed terms denote standard numbers. So $\mathbf{Q}$ together with all closed instances of $0 + x = x$ does not prove $\forall x (0 + x = x)$. The script `q1_qmodel.py` checks all seven axioms on all tuples from $\{0, \dots, 40\} \cup \{a, b\}$, and the referee's independent code on $\{0, \dots, 60\} \cup \{a, b\}$.

C.5 Consequences for a learner

Proof of Proposition 4.12. (1) Anchors: Theorem 4.3 and Proposition 4.4; rates: Theorem 4.5. Without a closedness guard and with parameters admissible, the closure of $\text{inst}_c(\varphi)$ contains $\varphi(\bar{p})$ (Proposition 4.6(b)); the guard makes $\text{inst}_c(\varphi)$ realizable and is learned from closed data (Proposition 4.6(c)). In (N) and (dB), capture instances are accepted without nocap whenever $\text{Cap}(\varphi) \neq \emptyset$ (Proposition 4.7(b), (d)), and the guard removes them (Proposition 4.7(e)). Under the λ convention $\text{Cap}(\varphi)$ does not arise. (2) The statement “for every φ whose closed instances all hold in M , $M \models \forall \bar{x} \varphi$ ” is condition (i) of Proposition 4.8, hence equivalent to $M_0 \preceq M$; Example 4.9 gives the negative cases. (3) Example 4.10. (4) If the target is $\text{inst}_o(\varphi)$, it equals $\text{inst}(\sigma_\varphi)$ under the λ convention and is realizable, with anchors as in Proposition 4.4(a); in (N) and (dB) it is realizable as σ_φ with the guards $\text{nocap}(z_i)$ (Proposition 4.7(e)). It contains $\varphi(\bar{p})$, from which $\forall \bar{x} \varphi$ follows by generalization (Proposition 4.11). $\square$

Proof of Proposition 4.13. (a) Every closed instance is true in M , so an evaluation of one returns “true” and refutes nothing. (b) A coherence refutation is a derivation of $\perp$ from $\Gamma \cup \{\forall \bar{x} \varphi\}$, which by completeness exists iff this set is inconsistent. RCF proves $\exists x (x \cdot x = 1 + 1)$; $\mathbb{Q}$ is a model of $\text{OF} + \forall x \neg(x \cdot x = 1 + 1)$; ZF proves $\exists x \text{Ind}(x)$ (Appendix C.4). (c) If M is term-generated,

$M_0 = M$, so by Proposition 4.8 a false $\forall \bar{x}\varphi$ has a false closed instance. Enumerating closed tuples and deciding each instance finds one at a finite stage. (d) $\neg\exists y \neg(y = y)$ is logically valid. In the model of Example 4.10, $Sa = a$, so $\mathbf{Q} \not\models \forall y \neg(y = Sy)$. $\mathbf{PA}$ proves it by induction on y : $0 \neq S0$ by Q1, and $y \neq Sy$ implies $Sy \neq SSy$ by Q2. A capture instance is neither a closed nor an open instance (Proposition 4.7(a)), so no evaluation of closed instances concerns it. $\square$

C.6 The untagged results

We use from Sections 2 and 6: the oracle is one-sided (Ref_d contains no true sentence); a set X of data is *d-coherent* iff some $T \in \text{Min}(X)$ has no instance in Ref_d ; and every covering DT° template lies above a minimal one, the finitariness theorem of Section 3.

Proof of Proposition 4.14. (a) Two data with different heads satisfy (R) ((D) is vacuous for one variable), so by Proposition 4.4(a) every covering DT° template is $\succeq \sigma_\varphi$ and $\text{Min}(D) = \{\sigma_\varphi\}$. Coherence of D is therefore the statement that σ_φ is not *d-refuted*. (b) φ has no parameters, so any parameter a is fresh for it, and $\varphi(a) \in \text{inst}_o(\varphi) = \text{inst}(\sigma_\varphi)$. Its closure is $\forall x\varphi$ (Proposition 4.11). So if all instances are true, $\forall x\varphi$ is true; conversely, if $\forall x\varphi$ is true, every instance is true by $\forall$ -elimination. (c) If $\forall x\varphi$ is true, then $\text{inst}(\sigma_\varphi)$ contains no false sentence, so by one-sidedness no instance of σ_φ is ever refuted. For $X \subseteq D$, σ_φ covers X and lies above some $T_0 \in \text{Min}(X)$ with $\text{inst}(T_0) \subseteq \text{inst}(\sigma_\varphi)$, so T_0 is not refuted and X is coherent. Hence any two clusters can be merged, and agglomeration stops only with the single cluster D , whose verifier accepts $\text{inst}(\sigma_\varphi) \ni \varphi(a)$ by (a). Target-blindness: the practice whose targets are the n ground sentences and the practice whose target is σ_φ can produce the same data, and the oracle, a fixed function of the query, answers the same; so no learner can be exact for the second and target-sound for the first. (d) $\text{Min}(D) = \{\sigma_\varphi\}$ has a *d-refuted* instance, so D is not *d-coherent* and is never formed. A subset whose terms all have the same head has other minimal templates, e.g. $\varphi(Sz)$ when all heads are S , which need not be refuted. (e) If every false instance of σ_φ lies in Ref_∞ , then σ_φ is never refuted iff it has no false instance iff $\forall x\varphi$ is true, by (b). For quantifier-free φ over L_A , Lemma C.2 applied to $T = \sigma_\varphi$ shows that $\mathbf{R}\text{-}\Delta_0$ has this property. For quantified sentences it does not: $\mathbf{R}\text{-}\Delta_0$ verifies a universal only by an equational derivation from true closed equations. The record's example is the false sentence $(\neg S0 = 0 \wedge \forall x (\neg x = 0 \rightarrow \neg Sx = 0)) \rightarrow \forall x \neg x = 0$, a wrong-base variant of induction; refuting it requires verifying the universal in the antecedent, which needs the axiom $\forall x \neg(Sx = 0)$ untagged referee U7; u8. At a finite budget d , a false universal whose least counterexample exceeds d is not *d-refuted*; such merges belong to the residue of Section 6. The facts about $\mathbb{R}$ are those of Appendix C.4. $\square$

Lemma C.2 (Quantifier-free arithmetic templates). *[proved]^{untagged Prop 5.13 (U20)} Let $T \in \text{DT}^\circ$ over L_A have no binder in its rigid part, so all its metavariables are 0-ary; formula metavariables may still take quantified closed formulas. If T has a false instance, it has a false quantifier-free instance, which $\mathbf{R}\text{-}\Delta_0$ refutes at some finite depth. So for $\mathbf{R}\text{-}\Delta_0$ at unbounded depth, every never-refuted template of this kind is truth-sound.*

Proof. Let $s = T\theta$ be false. Its universal closure is false in $\mathbb{N}$, so some assignment $\bar{n}$ of numerals to the parameters of s makes $s[\bar{n}]$ false. Let θ' agree with θ on term metavariables, and for a formula metavariable F let $\theta'(F) := (0 = 0)$ if $\theta(F)[\bar{n}]$ is true in $\mathbb{N}$ and $(0 = S0)$ otherwise. Then $s' := T\theta'$ is quantifier-free, since T 's rigid part, the term values and the new formula values are. Moreover $s'[\bar{n}]$ has the truth value of $s[\bar{n}]$: the rigid part has no binders, so it is a Boolean combination of equations between terms built from rigid symbols and unchanged term values, and of occurrences of formula metavariables, whose replacements have the same truth values at $\bar{n}$. So s' is false. At

depth at least $\max(\bar{n}, |s'|)$, $R\text{-}\Delta_0$ instantiates the parameters of s' (which are among those of s) by $\bar{n}$ and evaluates the resulting closed quantifier-free sentence, so $s' \in \text{Ref}_\infty$. $\square$

So in arithmetic, with $R\text{-}\Delta_0$ at unbounded depth, an unsound template that is never refuted must have a quantifier in its rigid part. This narrows, but does not settle, the open question of a natural permanent unsound residue (Section 6).

C.7 Computations

The computations cited in Section 4 are the following scripts of the track “cases” (directory `research/tracks/cases`): `q1_lgg.py` for Proposition 4.2, Theorem 4.3 (141 736 data sets), Proposition 4.6(c), Theorem 4.5 and Proposition 4.4(c), about 85 s; `q1_dt.py` with the bounded search `dt_search.py` for Proposition 4.4, 7 s; `q1_capture.py` for Proposition 4.7; `q1_qmodel.py` for Example 4.10; and the referee’s independent `r1_q1.py` and `r5_q1_dt.py` in its subdirectory `referee_code`. Experiment E1 is `experiments/e1_universal.py`, run from `code` (48 s, output `code/results/e1_universal.md`). The untagged computation is `u6_forall_nontrans.py` in `research/tracks/untagged/code`. Each script is run as `python3 script` from its directory and writes an output file named after it; the numbers quoted are those recorded in the final research notes.

D Proofs for Section 5

This appendix gives the proofs for Section 5, in the order of that section, and lists the computational evidence behind it (Appendix D.6). Proofs are adapted from the research records named in the source tags and were re-checked in transcription.

D.1 Preliminaries

For a set D of sentences, $C(D)$ is the set of positions at which all data have the same symbol, with all ancestors also in $C(D)$; a *slot* of D is a position outside $C(D)$ whose parent is in $C(D)$. For a template T , $\text{skel}(T)$ is the set of positions outside metavariable occurrences and their arguments.

Lemma D.1 (Rigid prefix). *[proved]^{prior induction C2(a); single Lemma P} Let $T \in \text{SO}^\circ$ and $D \subseteq \text{inst}(T)$. Every datum has, at every position of $\text{skel}(T)$, the symbol that T has there. Hence $\text{skel}(T) \subseteq C(D)$, and every occurrence of a metavariable of T lies in $C(D)$ or is a slot of D .*

Proof. In SO° all occurrences are rigid and have metavariable-free arguments, so an instance replaces each occurrence by a plugged body and leaves $\text{skel}(T)$ unchanged. The parent of an occurrence lies in $\text{skel}(T)$. $\square$

Over L_∞ with parameters, a metavariable-free argument term is a variable or a parameter, i.e. a leaf. Hence plugging arguments into the holes of a body β does not change its tree shape, and every non-hole position of β keeps its symbol in $\beta[\bar{u}]$. We read bound variables in locally nameless style, so that “ v is bound above c ” and “ v is bound inside the subtree at c ” are well defined. $\top := \forall w(w = w)$ and $\perp := \neg \top$ are closed.

D.2 Encodings: proofs for Section 5.2

Proof of Theorem 5.3. (a) Every occurrence in Table 4 applies P to distinct variables bound above it.

(b) “If”. Let $N := N_1 = \dots = N_m$ and $\sigma := F[Z, \dots, Z]$ with the guards $G = \{\text{fresh}(v, Z) : v \text{ frame-bound, } v \notin N\}$. Let $\sigma\theta$ satisfy G , with $\psi := \theta(Z)$. The frame-bound names free in ψ lie in N ; its other free names are parameters. Let β be ψ with its free occurrences of the names in N replaced by the holes, in argument order. At every occurrence the arguments are the same names, each bound there by the binder that the template intends, so $\beta[\bar{a}_k] = \psi$ for all k and $\sigma\theta = T^*[P := \lambda\bar{z}.\beta]$. Conversely $T^*[P := \lambda\bar{z}.\beta] = F[\psi, \dots, \psi]$ with $\psi = \beta[\bar{a}_1]$, whose free frame-bound names lie in N , so it satisfies G .

“Only if”. Suppose $N_k \neq N_l$ and $\text{inst}(\sigma, G) = \text{inst}(T^*)$ for a first-order schema σ , all of whose metavariables occur, and freshness guards G . Let s_1, s_2 be the instances with bodies $\varphi_1 := \bigwedge_{i \leq n} (z_i = z_i)$ and $\varphi_2 := \neg\varphi_1$, and $L := \text{lbg}(s_1, s_2)$. Frame positions are common. At each P -position the roots differ, so L has a metavariable there. The columns at occurrences k and l differ, because φ_1 mentions every argument and the tuples differ at some place. So L has distinct metavariables $Z_k \neq Z_l$ there. Since $s_1, s_2 \in \text{inst}(\sigma)$, $\sigma \succeq L$, say $L = \sigma\rho$. Let s_3 be L with $Z_k := \top$ and every other metavariable $:= \perp$; then $s_3 = \sigma(\rho\theta_3)$. For each metavariable Y of σ , the free names of $\rho\theta_3(Y)$ are those of the rigid part of $\rho(Y)$, which are also free in $\rho\theta_1(Y)$, the value of Y in s_1 (first-order matching is unique). Freshness guards are preserved when free names are removed, and s_1 satisfies G ; so $s_3 \in \text{inst}(\sigma, G)$. But $s_3 \notin \text{inst}(T^*)$. If $\beta[\bar{a}_k]$ is the closed $\top$, then β has no hole, since a hole would leave the variable $a_{k,i}$ free in the slot; so $\beta = \top$, and occurrence l would also read $\top$, not $\perp$.

(c) The same proof with indices in place of names. Loose-index guards are also preserved when loose indices are removed. At every occurrence the loose indices of the value must be smaller than its binder depth, so well-formedness implies $\text{loose}(Z) \subseteq \{0, \dots, d-1\}$ and nothing more. Since $I_1 = I_k$ for all k , $I_1 \subseteq \{0, \dots, d-1\}$; so the guard $\text{loose}(Z) \subseteq I_1$ is implied iff $I_1 = \{0, \dots, d-1\}$. $\square$

For Table 5: in de Bruijn notation Sep reads $P(\#0, \#2)$ at depth 3, so the guard “ $\#1$ (that is, y) not loose” is needed. Sep_J reads $P(\#0)$ at depth 3. All three occurrences of EInd read $P(\#0)$, at depths 2, 1, 1. Coll and Repl_{∃!} read $P(\#1, \#0, \#2)$ in the antecedent and $P(\#1, \#0, \#3)$ in the consequent. Repl_{sp} reads $P(\#1, \#0, \#2)$, $P(\#2, \#0, \#3)$, $P(\#1, \#0, \#3)$. Repl_J reads $P(\#2, \#1)$, $P(\#2, \#0)$, $P(\#0, \#1)$. With $\varphi(x, y)$ only, Coll reads $P(\#1, \#0)$ twice at depths 3 and 4. There $\#2$ is A in the antecedent and Y in the consequent, so the guard $\text{loose}(Z) \subseteq \{\#0, \#1\}$ is needed: without it, $Z := (\#0 = \#2)$ reads $y = A$ in the antecedent and $y = Y$ in the consequent, and the instance forces $Y \in Y$.

Proof of Proposition 5.4. ^{cases Prop B1 α} The hypothesis class is first-order schemas over the named signature, with formula metavariables and *name metavariables* (at binder-name positions and variable leaves), and guards $\text{fresh}(v, Z)$ and $\text{distinct}(v, w)$. “If”: let $b_1, \dots, b_K$ be the frame’s binders. Let σ be the frame with the name of b_k (at the binder and at every leaf it binds) replaced by a name metavariable v_k , and every occurrence of P replaced by one formula metavariable Z . Its guards are $\text{distinct}(v_k, v_l)$ for $k \neq l$, and $\text{fresh}(v_k, Z)$ for every b_k that is not an argument binder of P . An instance gives the binders distinct names and Z a formula ψ whose free names are argument names or names the frame does not use. Let β be ψ with the argument names replaced by holes. Every occurrence applies P to the same binders, hence to the same names, so every slot reads $\beta[\text{names}] = \psi$, and the instance is an α -variant of $T^*[P := \lambda\bar{z}.\beta]$. The converse is

read off in the same way. “Only if”: if occurrences k, l differ at argument place i (binders $b \neq b'$), fix one naming with all frame binders distinct and argue as in Theorem 5.3. The slot texts at k and l differ in s_1 (one mentions the name of b , the other that of b'), and distinct-guards only involve name positions, whose values in s_3 are those in s_1 . The ZF classification: Sep and Sep_J have one occurrence. EInd applies P to the binders $\forall y, \forall x$ (first) and $\forall x$ (second). In Coll and Repl_{∃!} the antecedent’s x, y and the consequent’s x, y have different binders. Repl_{sp} and Repl_J differ even in names. $\square$

Proof of Proposition 5.5. Frame positions are common columns. By (R) every P -position is a disagreement column, and the lgg merges equal columns. So positions k, l share a metavariable iff $\varphi_j[\bar{a}_k] = \varphi_j[\bar{a}_l]$ for all j . At a free occurrence of z_i the two texts carry $a_{k,i}$ and $a_{l,i}$, so the texts are equal iff the tuples agree at every place used by φ_j . For the last claim, let U be the set of places used by some body. All positions share one metavariable Z , so for $i \in U$ all occurrences carry the same name (index) a_i . The value of Z in datum j is $\psi_j = \varphi_j[\bar{a}_1]$, whose free frame variables are exactly the a_i with z_i free in φ_j . So the most specific guards say that the value’s free frame variables lie in $\{a_i : i \in U\}$. For an instance $F[\psi, \dots, \psi]$ respecting them, let β be ψ with each free a_i ($i \in U$) replaced by z_i . Then $\beta[\bar{a}_k] = \psi$ for every k , so the instance is $T^*[P := \lambda \bar{z}. \beta]$. $\square$

D.3 First-order failures: proofs for Section 5.3

The false instances of Proposition 5.6, complete list cases §2.4. Each is an instance of the computed lgg of the two named data, respects its most specific guards, and is not an instance of the schema.

- (1) EInd, named; data EInd($x = x$), EInd($\neg x \in a$). The columns are ($y = y, \neg y \in a$) at occurrence 1 and ($x = x, \neg x \in a$) at occurrences 2 and 3, so the lgg is $\forall x(\forall y(y \in x \rightarrow Z_1) \rightarrow Z_2) \rightarrow \forall x Z_2$. Take F_1 : $Z_1 := \neg y = y$, $Z_2 := \forall w \neg w \in x$. Its antecedent is $\forall x(\text{empty}(x) \rightarrow \text{empty}(x))$, which is valid, so F_1 is equivalent to the Π_1 sentence $\forall x \forall w \neg w \in x$. The Δ_0 matrix of that sentence fails at the hereditarily finite sets $x = \{\emptyset\}$, $w = \emptyset$; by Δ_0 -absoluteness F_1 is false. ZF refutes it from “some set has an element”.
- (2) Repl_J, named; data Repl_J($x \in y$), Repl_J($\neg x = a$). The lgg is $\forall x \forall y \forall u (Z_1 \wedge Z_2 \rightarrow y = u) \rightarrow \forall X \exists Y \forall y (y \in Y \leftrightarrow \exists x (x \in X \wedge Z_1))$. Take $Z_1 := y = y$, $Z_2 := \neg u = u$. The antecedent is valid, and at $X = \{\emptyset\}$ the consequent gives a set Y of all sets. Then $Y \in Y$, which contradicts Foundation; without Foundation, Separation yields Russell’s set.
- (3) Repl_J, de Bruijn: three blocks. $Z_1 := \perp$, $Z_2 := \perp$, $Z_3 := (y = y)$ give the same false consequent.
- (4) Coll, de Bruijn; data Coll($y = x$), Coll($\neg x = A$). The two occurrences decouple. $Z_1 := y = y$, $Z_2 := \perp$ give $\forall A(\forall x \in A \exists y y = y \rightarrow \exists Y \forall x \in A \exists y (y \in Y \wedge \perp))$, which is equivalent to $\forall A \neg \exists x x \in A$ and is refuted by the HF witness $A = \{\emptyset\}$.
- (5) Repl_{∃!}, de Bruijn; the same data. $Z_1 := y = x$ (and $\exists! y (y = x)$ is valid), $Z_2 := \perp$: equivalent to $\forall A \neg \exists x x \in A$.
- (6) Repl_{sp}, de Bruijn; the same data, three blocks. $Z_1 := y = x$, $Z_2 := \perp$, $Z_3 := \perp$. The antecedent $\forall x \in A \exists y (y = x \wedge \forall u (\perp \rightarrow u = y))$ is valid, and the instance is equivalent to $\forall A \neg \exists x x \in A$.

- (7) Guard violations. Sep without $\text{fresh}(y, Z)$: Russell’s instance, in both encodings (Proposition 5.9). Coll, $\text{Repl}_{\exists!}$, Repl_{sp} , named, without $\text{fresh}(Y, Z)$: $Z := (y = Y)$ gives

$$\forall A (\forall x \in A \exists y (y = Y) \rightarrow \exists Y \forall x \in A \exists y (y \in Y \wedge y = Y)).$$

Its antecedent holds, because Y is a parameter there, and its consequent forces $Y \in Y$ for $A \neq \emptyset$, contradicting Foundation. (For Repl_{sp} add $\neg u = u$ at the uniqueness occurrence.) Coll with $\varphi(x, y)$ only, de Bruijn, without its guard: $Z := (\#0 = \#2)$, as in Appendix D.2.

- (8) Named, α -varied. For Coll, $\text{Repl}_{\exists!}$ the blocks $(y = x)$, $\perp$, and for Repl_{sp} the blocks $(y = x)$, $\perp$, $\perp$, each give an instance equivalent to $\forall A \neg \exists x x \in A$.

Lemma D.2 (Deep leaf). *[proved]^{cases Lemma C1} Let s be a sentence of $L_{\in}$, ℓ a variable leaf of s , and τ a first-order schema with $s \in \text{inst}(\tau)$ and $|\tau| < \text{depth}(\ell)$. Then τ has a formula metavariable Z at a proper prefix q of ℓ with $\text{depth}(q) < |\tau|$. If $s|_q$ occurs in s only at q , then Z occurs in τ only at q , and $s[q \leftarrow v] \in \text{inst}(\tau)$ for every closed formula v . This remains true with freshness guards (named), loose-index guards (de Bruijn), and distinct-guards on name metavariables (α -varied named).*

Proof. Follow the path to ℓ in τ . Since τ has fewer than $\text{depth}(\ell)$ nodes, the path leaves τ at a leaf of τ at depth $< |\tau|$. That leaf is not a constant, because s continues below it, so it is a metavariable Z at a proper prefix q of ℓ . In $L_{\in}$ every proper prefix of a variable leaf is a formula position (binder-name children are not on the path), so Z is a formula metavariable. If Z also occurred at q' , then $s|_{q'} = \theta(Z) = s|_q$, so $q' = q$. Changing $\theta(Z)$ to v changes s only at q . A closed v has no free names and no loose indices, and the other metavariables keep their values, so all guards still hold. $\square$

Proof of Theorem 5.7. Let $\{\tau_1, \dots, \tau_r\}$ cover the schema. Take n with $2n > \max_i |\tau_i|$, a genuine instance s_n and a designated leaf ℓ below; some τ_i covers s_n . By Lemma D.2, once we check that every subterm along the path to ℓ occurs only once in s_n , τ_i contains $s_n[q \leftarrow v]$ for some proper prefix q of ℓ and both $v \in \{\top, \perp\}$. It therefore suffices that, for every proper prefix q of ℓ , one of $\top, \perp$ makes $s_n[q \leftarrow v]$ false. “Make the slot false” means: if q lies inside the slot below e negations, choose v with $\neg^e v \equiv \perp$; the path inside the slot passes only through $\neg, \forall$ and $\wedge$, and $\forall w v \equiv v$.

(i) EInd, named. $\varphi_n(x) := \neg^{2n} \forall w \neg (w \in x)$ (“ x is empty”); $s_n = \text{EInd}(\varphi_n)$; ℓ is the y of occurrence 1. For q the root, take $v := \perp$. For q the antecedent or its $\forall x$ -body, take $v := \top$: then s_n is equivalent to $\forall x \varphi_n(x)$, “every set is empty”, which is false. For q the formula $\forall y (y \in x \rightarrow \varphi_n(y))$ or its body, take $v := \perp$: the antecedent becomes $\forall x (\perp \rightarrow \varphi_n(x))$, true, and the conclusion is false. For q inside $\varphi_n(y)$, make the slot false: then $\forall y (y \in x \rightarrow \perp)$ says that x is empty, the antecedent reads $\forall x (\text{empty}(x) \rightarrow \text{empty}(x))$, which is true, and the conclusion is false. Every path subterm inside the slot contains y , which occurs only in occurrence 1 and in the frame atom $y \in x$, so all path subterms are unique.

(ii) Repl_J , both encodings. $\varphi_n(x, y) := \neg^{2n} (x \in y)$; ℓ is the u of occurrence 2, $\varphi_n(x, u)$. The consequent $C := \forall X \exists Y \forall y (y \in Y \leftrightarrow \exists x \in X x \in y)$ is false: at $X = \{\emptyset\}$, Y would contain every y with $\emptyset \in y$, and $y_0 := \{\emptyset, Y\}$ gives $Y \in y_0 \in Y$, contradicting Foundation. (Without Foundation, $R := \{y \in Y : y \notin y\}$ and $y_1 := R \cup \{\emptyset\}$ give $y_1 \in y_1 \leftrightarrow y_1 \notin y_1$.) For q the root take $\perp$. For q the antecedent, its $\forall x, \forall y, \forall u$ levels or its implication, take $\top$: the antecedent is true and C is false. For q the conjunction, take $\perp$: $(\perp \rightarrow y = u)$ is true. Inside $\varphi_n(x, u)$, make the slot false. Uniqueness: the path subterms contain u (named), and in de Bruijn the designated occurrence’s arguments $(\#2, \#0)$ differ from $(\#2, \#1)$ and $(\#0, \#1)$.

(iii) Coll, $\text{Repl}_{\exists!}$, Repl_{sp} , de Bruijn. $\varphi_n(x, y, A) := \neg^{2n}(y = x \wedge x \in A)$; ℓ is the A in the consequent's occurrence. The antecedent is true for every A , since $y := x$ is the unique witness. Every proper prefix q is the root ($v := \perp$), the $\forall A$ -body ($v := \perp$, giving $\forall A \perp$), or a subformula of the consequent containing ℓ . Above the slot, $v := \perp$ turns the consequent into $\exists Y \perp$, $\exists Y \forall x \perp$, or $\exists Y \forall x (x \in A \rightarrow \perp)$ (that is, “ A is empty”); inside the slot (including the conjunction and the atom $x \in A$), make the slot false. In each case the consequent is false for $A \neq \emptyset$, so $\forall A(\text{true} \rightarrow \text{false for nonempty } A)$ is false. Uniqueness: in de Bruijn the slot's atom $x \in A$ reads $\#1 \in \#3$, while the frame atoms read $\#0 \in \#2$ (consequent) and $\#1 \in \#2$ (antecedent). The argument tuple $(\#1, \#0, \#3)$ (resp. that of Repl_{sp} 's third occurrence) differs from the others.

(iv) Coll, $\text{Repl}_{\exists!}$, Repl_{sp} , named under α -variation. The same φ_n , with s_n an α -variant of the instance in which all frame binders have distinct names. The designated leaf is the y of the atom $y = x$ in the consequent's occurrence. (The A -leaf of (iii) is not usable: in the named encoding the slot atom $x \in A$ also occurs as a frame atom.) The case analysis is that of (iii); the only new prefix is the atom $y = x$ itself, where $v := \perp$ makes the slot false. Every path subterm inside the slot contains the consequent's own names for x and y , which occur nowhere outside the consequent, and inside the consequent the frame atoms are $x \in A$ and $y \in Y$, never $y = x$. $\square$

Proof of Proposition 5.8. (a) Let $L_S\theta$ satisfy the guards. The antecedent implies $\forall x \in A \exists y Z_1$ (drop the second conjunct). Z_1 's value may mention x, y, A and parameters, but not Y (guard). The name u is not bound at either occurrence of Z_1 , so a free u would be a parameter at both. Collection, with these parameters, gives $\exists Y \forall x \in A \exists y \in Y Z_1$, and ZF proves Collection. Strictness: $Z_1 := (y = x)$, $Z_2 := \perp$ respects the guards. It is not an instance of Repl_{sp} : a body β with $\beta[x, y, A] = (y = x)$ is $z_2 = z_1$, and then $\beta[x, u, A] = (u = x) \neq \perp$. (b) “If” is the proof of (a), which uses only Collection. “Only if”: let $\text{Coll}(\varphi)$ be a Collection instance, with Y not free in φ . Renaming parameters gives an equivalent universal closure, so we may assume $u \notin \text{FV}(\varphi)$. Then $Z_1 := \varphi$, $Z_2 := \perp$ satisfy the guards, and the instance $\forall A(\forall x(x \in A \rightarrow \exists y(\varphi \wedge \forall u(\perp \rightarrow u = y))) \rightarrow \exists Y \forall x(x \in A \rightarrow \exists y(y \in Y \wedge \varphi)))$ is logically equivalent to $\text{Coll}(\varphi)$, because $\forall u(\perp \rightarrow u = y)$ is valid. $\square$

D.4 Anchors for pattern targets: proofs for Section 5.4

Proof of Theorem 5.10. “If”, for $H = \text{SO}^\circ$ (hence for all smaller H). Let $T \in \text{SO}^\circ$ cover D , with $D_j = T\eta_j$.

Step 1 (skeleton). The root of $\varphi_j[\bar{a}_k]$ is $\text{root}(\varphi_j)$. By (R) every slot root of T^* is a disagreement position, while frame positions are common. By Lemma D.1, $\text{skel}(T)$ is a prefix of the frame, and T 's occurrences lie at frame positions or at slot roots. For such a position c put $\kappa(c) := T^*|_c$, so that $D_j|_c = \kappa(c)[P := \lambda \bar{z}. \varphi_j]$.

Step 2 (one body per metavariable). Let M have occurrences $M(\bar{u}^1)$ at $c_1, \dots, M(\bar{u}^r)$ at c_r . The arguments are bound variables in scope or parameters, possibly repeated. Let $\beta_j := \eta_j(M)$, so that $\beta_j[\bar{u}^s] = D_j|_{c_s}$.

(2a) *Shape.* Let p be a slot root of $\kappa(c_1)$. Then $D_j|_{c_1}$ has the formula symbol $\text{root}(\varphi_j)$ at p , so p is a non-hole position of β_j , and $D_j|_{c_s}$ also has $\text{root}(\varphi_j)$ at p . If $\kappa(c_s)$ had a frame symbol at p , all φ_j would share it. If p lay strictly inside a slot of $\kappa(c_s)$ with root p' , the non-leaf frame symbol of $\kappa(c_1)$ at p' would be a symbol of β_j , hence of $D_j|_{c_s}$ at p' , which is $\text{root}(\varphi_j)$, for all j . Both contradict (R). By symmetry, $\kappa(c_1)$ and $\kappa(c_s)$ have slots at the same positions and the same symbols at non-leaf frame positions. Frame leaves may differ.

(2b) *Frame leaves.* Let q be a frame leaf of $\kappa(c_1)$ carrying the variable v . If v is bound inside $\kappa(c_1)$, then $\beta_1|_q$ is that variable (holes become $\bar{u}^1$, which are bound above c_1 or are parameters),

so $\kappa(c_s)|_q$ is the variable bound by the corresponding binder of $\kappa(c_s)$; keep v . If v is bound above c_1 , then $\beta_1|_q$ is a hole h_m with $u_m^1 = v$ (β_1 has no free bound variable, and a parameter is not v); record m . By (2a), q is a frame leaf of $\kappa(c_s)$ as well, and $\kappa(c_s)|_q = D_1|_{c_s|_q} = u_m^s$.

(2c) *Slot arguments*. Let $\kappa(c_1)$ have the slot $P(\bar{a})$ at p and $\kappa(c_s)$ the slot $P(\bar{b})$ at p . For an argument place i , use (N_i) to pick j_i and a free occurrence q' of z_i in φ_{j_i} . At $p \cdot q'$, $D_{j_i}|_{c_1}$ carries a_i . If a_i is bound inside $\kappa(c_1)$, then $\beta_{j_i}|_{p \cdot q'}$ is that variable, so b_i is the variable bound by the corresponding binder of $\kappa(c_s)$; keep a_i . Otherwise $\beta_{j_i}|_{p \cdot q'} = h_m$ with $u_m^1 = a_i$; record m . Since $D_{j_i}|_{c_s|_p} = \varphi_{j_i}[\bar{b}]$ carries b_i at q' , $b_i = u_m^s$.

Let B_M be $\kappa(c_1)$ with every recorded leaf or argument replaced by its recorded hole. Every variable bound above c_1 that occurs in $\kappa(c_1)$ does so at a frame leaf or as a slot argument, so B_M is closed except for holes and the metavariable P . $B_M[\bar{u}^1] = \kappa(c_1)$, and $B_M[\bar{u}^s] = \kappa(c_s)$ for all s by (2a)–(2c). In particular every recorded u_m^s is a bound variable, even when $\bar{u}^s$ contains parameters or repetitions; any valid choice of β_1 and β_{j_i} works. (A term metavariable at a frame leaf is the case where $\kappa(c)$ is a variable.)

Step 3. Let $\sigma(M) := \lambda \bar{h}. B_M$ for every metavariable M of T . Each occurrence $M(\bar{u}^s)$ becomes $B_M[\bar{u}^s] = \kappa(c_s)$, and T 's rigid part is F 's, so $T\sigma = T^*$. Hence $T \succeq T^*$ and $\text{inst}(T) \supseteq \text{inst}(T^*)$.

“Only if” (*witnesses in PAT*). If all roots are f , replace each $P(\bar{a}_k)$ by $\Phi_f(\bar{a}_k)$, where $\Phi_\in(\bar{a}) = g_1(\bar{a}) \in g_2(\bar{a})$ and $\Phi_=(\bar{a}) = g_1(\bar{a}) = g_2(\bar{a})$ (term metavariables, whose values are projections or parameters), $\Phi_-(\bar{a}) = \neg Q(\bar{a})$, $\Phi_\circ(\bar{a}) = Q_1(\bar{a}) \circ Q_2(\bar{a})$ for binary $\circ$, and $\Phi_Q(\bar{a}) = Qw.R(\bar{a}, w)$ for $Q \in \{\forall, \exists, \exists!\}$. All occurrences are pattern occurrences. The template covers D , since atoms of $L_\in$ have variables or parameters as arguments, and it misses $T^*[P := \lambda \bar{z}. \psi]$ for any ψ with another root. If (N_i) fails, replace each $P(\bar{a}_k)$ by $P'(\bar{a}_k$ without its i -th entry). This covers D and misses $T^*[P := \lambda \bar{z}. (z_i = z_i)]$. $\square$

Lemma D.3 (Facing). *[proved]^{cases Thm E} In the setting of Theorem 5.11, let $T \in \text{SO}^\circ$ cover D , let M be a metavariable of T with occurrences $M(\bar{u}^s)$ at c_s and bodies β_j , and assume (R_P) for all P . (1) $\kappa(c_1)$ and $\kappa(c_s)$ have slots at the same positions and the same symbols at non-leaf frame positions. (2) If $\kappa(c_1)$ has a P -slot $P(\bar{a})$ at p and $\kappa(c_s)$ a Q -slot $Q(\bar{b})$ at p , then $\text{root}(\varphi_{P,j}) = \text{root}(\varphi_{Q,j})$ for all j . If moreover $\bar{u}^1$ is a pattern tuple and all $(N_{P,i})$ hold, there is a map $\pi : [n_P] \rightarrow [n_Q] \cup \text{Par}$, independent of j , with $\varphi_{Q,j} = \varphi_{P,j} \circ \pi$ for all j . If also $\bar{u}^s$ is a pattern tuple and all $(N_{Q,i})$ hold, π is a bijection onto $[n_Q]$.*

Proof. (1) is (2a) above, using (R) for the metavariable at each slot; the root claim of (2) is its first sentence. Let $\gamma_j := \beta_j|_p$, so that $\varphi_{P,j}[\bar{a}]$ and $\varphi_{Q,j}[\bar{b}]$ are γ_j with its holes replaced by $\bar{u}^1$, resp. $\bar{u}^s$. Define $\pi(i)$ for $i \leq n_P$: by $(N_{P,i})$ pick j and a free occurrence q' of z_i in $\varphi_{P,j}$; the leaf of $\varphi_{P,j}[\bar{a}]$ at q' is a_i . (α) If a_i is bound above c_1 , then $\gamma_j|_{q'} = h_m$ with $u_m^1 = a_i$, and m is unique because $\bar{u}^1$ is a pattern tuple. The leaf of $\varphi_{Q,j}[\bar{b}]$ at q' is u_m^s , which is free relative to p ; so it is some $b_{i'}$ or a parameter, and we put $\pi(i) := i'$, resp. that parameter. (β) If a_i is bound by a binder at relative position r between c_1 and p , then $\gamma_j|_{q'}$ is the variable bound at r , and the leaf of $\varphi_{Q,j}[\bar{b}]$ at q' is the variable bound at r in $\kappa(c_s)$, which lies above p ; so it is some $b_{i'}$, and $\pi(i) := i'$. In both cases $\pi(i)$ depends only on a_i (through m or r), not on j or q' . Now compare the two plugged texts leaf by leaf through γ_j . Constants, parameters and variables bound inside γ_j agree. A hole h_m gives u_m^1 , a free leaf of the slot and hence some a_i , and $u_m^s = b_{\pi(i)}$ (or the parameter $\pi(i)$). A variable bound between c_1 and p gives a_i and $b_{\pi(i)}$. Hence $\varphi_{Q,j} = \varphi_{P,j} \circ \pi$. If $\bar{u}^s$ is also a pattern tuple, every u_m^s is a bound variable, so π maps into $[n_Q]$. It is injective: distinct a_i give distinct m or distinct r , hence distinct variables, and the two cases give variables bound above, resp. below, c_s . Every $b_{i'}$ free in some $\varphi_{Q,j}[\bar{b}]$ arises in this way, so under $(N_{Q,i})$ π is onto. $\square$

Proof of Theorem 5.11. “If” parts. Let T in the class cover D . Step 1 of Theorem 5.10 holds verbatim, using (R_P) for every P . For each metavariable M of T choose c_1 to be a pattern occurrence in PAT and DT° , and any occurrence in SO° . Then every slot of $\kappa(c_1)$ faces, in each $\kappa(c_s)$, a slot of the *same* metavariable. In SO° this follows from (D^{root}) and the root claim of Lemma D.3; in DT° from (D^{DT}) and the map; in PAT (where $\bar{u}^s$ is a pattern tuple too) from (D^{PAT}) and the bijection. Steps 2b, 2c and 3 of Theorem 5.10 then apply slot by slot, with $(N_{P,i})$ for the metavariable of each slot, and give $T \succeq T^*$.

“Only if”. If (D_{PQ}^{PAT}) fails via a bijection π , let T' be T^* with every $Q(\bar{b})$ replaced by $P(b_{\pi(1)}, \dots, b_{\pi(n_P)})$, a pattern occurrence. Then $T'[P := \lambda \bar{z}. \varphi_{P,j}] = D_j$, because $\varphi_{P,j} \circ \pi$ plugged with $\bar{b}$ is $\varphi_{Q,j}[\bar{b}]$. T' misses every instance of T^* in which the P - and Q -bodies have different roots. So D is not an anchor in PAT or in any larger class. If $(D_{P \rightarrow Q}^{DT})$ fails via π , the same construction with arguments $b_{\pi(i)}$ or the parameter $\pi(i)$ gives $T' \in DT^\circ$ (P keeps its pattern occurrences, and the new occurrences have metavariable-free arguments), which covers D and misses the same instances. In (c), an SO° anchor is a DT° anchor, hence satisfies (D^{DT}) by (b). Non-reversibility is the example T_A of Section 5.4. There (D^{PAT}) and (D^{DT}) hold: the only map fitting the first datum is the swap, which fails on the second. The template $\forall x \forall y (G(x, y, x) \rightarrow G(x, x, y))$ covers the data with $G := \lambda h_1 h_2 h_3. h_3 \in h_2$, resp. $\lambda h_1 h_2 h_3. \neg h_1 = h_2$. It misses $\forall x \forall y (x \in y \rightarrow \neg x = x)$, because a body with $\beta[x, y, x] = (x \in y)$ is $h_1 \in h_2$ or $h_3 \in h_2$, so that $\beta[x, x, y] \in \{x \in x, y \in x\}$. Without (R_P) or $(N_{P,i})$, the witnesses of Theorem 5.10 applied to P work in every class. $\square$

Proof of Theorem 5.13. By Theorem 5.10, “not an anchor” is $\bigcup_f A_f \cup \bigcup_i B_i$, where A_f says that all roots are f and B_i that z_i is never free. In inclusion–exclusion over these events, every intersection containing two distinct A_f is empty. The intersection of A_F (or of no A , $F = *$) with B_i , $i \in I$, is the event that all N samples lie in $C_{F,I}$, of probability $\mu(C_{F,I})^N$, and its sign is $(-1)^{|F \neq *| + |I| + 1}$. The upper bound is the union bound. $\square$

D.5 PA induction: proofs for Section 5.5

Write $\text{Ind}(\varphi) = \text{imp}(U, V)$ with $U = \text{and}(a, s)$, $s = \forall x W$, $W = \text{imp}(b, c)$ and $V = \forall x d$. The *motive slots* a, b, c, d hold $\varphi[0/x]$, φ , $\varphi[Sx/x]$, φ , and the *five-node frame* consists of the positions of imp , and , s , W , V . Two facts are used throughout prior induction C3, Lemma A: (i) substituting terms for x changes neither the root of a formula nor its *logical size* ℓ (the number of atoms, connectives and quantifiers); (ii) if x is free in φ and $\varphi[u/x] = \varphi[v/x]$, then $u = v$. Also $St \neq x$, $St \neq 0$ and $St \neq t$ for every term t . With $L = \ell(\varphi) \geq 1$, the slots have logical sizes a, b, c, d : L ; W : $2L + 1$; s : $2L + 2$; V : $L + 1$; U : $3L + 3$; and the whole sentence $4L + 5$. These six values are pairwise distinct.

Proof of Proposition 5.14. prior induction B1, B3 (a) For $\varphi_1 = (x + 0 = x)$ and $\varphi_2 = (0 + x = x)$ the a -column is $(0 + 0 = 0, 0 + 0 = 0)$ and stays. The b -column $(x + 0 = x, 0 + x = x)$ gives $z_{(x,0)} + z_{(0,x)} = x$. The c -column $(Sx + 0 = Sx, 0 + Sx = Sx)$ gives $z_{(Sx,0)} + z_{(0,Sx)} = Sx$. The d -column equals the b -column. The four index tuples are distinct. The displayed instance $(z_1 = z_2 = z_3 := 0, z_4 := S0)$ is false: $0 + 0 = x$ forces $x = 0$, and then $0 + S0 = 1 = Sx$, so the premise holds, while $\forall x(0 + 0 = x)$ fails at $x = 1$. A single instance I is covered by the ground schema I . For $\text{Ind}(x = x)$, $\text{Ind}(Sx = Sx)$ the lgg is $(t = t) \wedge \forall x(u = u \rightarrow Su = Su) \rightarrow \forall x(u = u)$, and every sentence instance has the valid conclusion $\forall x(u = u)$. (b) By fact (i) the a, b, c columns all have mixed roots, so each becomes a metavariable indexed by its column. If some φ_j has x free, then $\varphi_j[0/x]$, φ_j , $\varphi_j[Sx/x]$ are pairwise distinct, so the three columns and their metavariables are distinct; the d -column equals the b -column. The displayed instance has a true premise and

the false conclusion $\forall x(0 = S0)$. The lgg differs from L_∞ only if all roots are equal (probability $\sum_f p_f^N \leq p_{\max}^{N-1}$) or no motive has x free (probability $(1 - q)^N$). $\square$

Proof of Theorem 5.15. prior induction B2 Let $\varphi_n := (g_n(x) = x)$ and $n < m$, $d := m - n$, so that $g_m(t) = g_n(g_d(t))$ as terms. In the b -slot both left sides begin with n copies of $0 + \cdot$, so the lgg descends n times and meets the column $(x, g_d(x))$. Its roots are a variable and $+$, giving $b := z_{(x, g_d(x))}$, so the slot is $g_n(b) = x$. Likewise the a -slot meets $(0, g_d(0))$ and gives $g_n(a) = 0$, and the c -slot meets $(Sx, g_d(Sx))$ and gives $g_n(c) = Sx$. The d -slot repeats b . The tuples indexing a, b, c have first entries $0, x, Sx$, so the metavariables are distinct, and the lgg is L_n . (For any subset of F' with least index n the same descent gives L_n .) For $M \geq n$, $I_M^* \in \text{inst}(L_n)$: put $a, b := g_{M-n}(0)$ and $c := g_{M-n}(S0)$, and use $g_n(g_{M-n}(t)) = g_M(t)$. I_M^* is false: $g_M(0)$ denotes 0 and $g_M(S0)$ denotes 1; the base holds, $g_M(0) = x$ forces $x = 0$ and then $g_M(S0) = Sx$, while $\forall x(g_M(0) = x)$ fails at 1. It is refuted in $\mathbf{Q}$ from the closed facts $g_M(0) = 0$ and $g_M(S0) = S(g_M(0))$, which $\mathbf{Q}$ proves by Σ_1 -completeness, and $\neg(g_M(0) = S0)$. Assume $g_M(0) = x$; by congruence and transitivity $g_M(S0) = Sx$. Discharging and generalizing gives the step clause, modus ponens gives $\forall x(g_M(0) = x)$, and instantiating at $S0$ contradicts $\neg(g_M(0) = S0)$. Now a schema covering $\text{Ind}(\varphi_n)$ and $\text{Ind}(\varphi_m)$ is $\succeq L_n$ and contains I_n^* . A union of k sound schemas covering $k+1$ members of F' would put two of them into one schema. If a finite union $\bigcup \text{inst}(\tau_i)$ contained F' , some τ_i would cover two members of F' and hence contain the false, $\mathbf{Q}$ -refutable sentence I_n^* ; so no such union lies between F' and $\text{Th}(\mathbb{N})$, or between F' and a consistent extension of $\mathbf{Q}$. The motives of F' are atomic, so the same holds for open induction. $\square$

Guards prior induction B2 remark. Data from F' instantiate b and c by terms containing x , and a by closed terms, so the strongest occurrence and freshness guards learnable from F' are $x \in \text{FV}(b)$, $x \in \text{FV}(c)$ and “ a closed”. The padded instance $a := 0$, $b := 0 \cdot x$, $c := S0 + 0 \cdot x$ satisfies them and has the truth values of I_n^* .

Proof of Theorem 5.16. prior induction referee R1 In a sentence instance, $z_1 := t$ is closed (it occurs outside the binder) and $z_2 := u$ has $\text{FV}(u) \subseteq \{x\}$. Let k be the value of t , and let $\hat{u}(g)$ be u with x replaced by a new constant g and each maximal closed subterm replaced by its numeral.

Case T: $\hat{u}(x)$ is literally $\underline{k} + x$. In every L_A -algebra M containing $\mathbb{N}$ as a subalgebra, $t + a = u(a)$ for every a , so the conclusion and hence the instance hold.

Case W: otherwise. Let $s := \underline{k} + g$ and $r := \hat{u}(g)$, so $s \neq r$. Consider ground terms over the numerals and g , with the rules $R_{\mathbb{N}}$ that evaluate $+$ and $\cdot$ on numerals, and one more rule ρ : $\rho := r \rightarrow s$ if s is a proper subterm of r , and $\rho := s \rightarrow r$ otherwise. *Termination*: if r is a numeral, use the lexicographic measure (number of g 's, number of closed non-numeral positions). If $\rho = s \rightarrow r$ with $|r| \geq |s|$, use (closed non-numeral positions, occurrences of s); no ρ -step creates a new occurrence of s , since an occurrence straddling the rewritten position would force $r = g$ or r closed. Otherwise use (closed non-numeral positions, size). In the first case a ρ -step removes a g and an $R_{\mathbb{N}}$ -step lowers the second component. In the other two cases an $R_{\mathbb{N}}$ -step lowers the first component, and a ρ -step keeps it (both sides contain g and have no closed non-numeral subterm, and the context above still contains g) and lowers the second. *Confluence*: there are no critical pairs. The left sides of $R_{\mathbb{N}}$ are closed, with only numerals as proper subterms; the left side of ρ contains g , is $R_{\mathbb{N}}$ -normal, and is ground. So $R_{\mathbb{N}} \cup \{\rho\}$ is convergent (critical pair lemma and Newman's lemma). Let $M_{t,u}$ be the ground terms modulo convertibility. Distinct numerals are distinct normal forms, so $\mathbb{N}$ is a subalgebra of $M_{t,u}$, and $M_{t,u} \models s = r$, i.e. $t + g = u(g)$. The step clause fails at g . If $\rho = s \rightarrow r$, the term $\underline{k} + Sg$ is normal, with root $+$, while the normal form

of $Su(g)$ is Sr or a numeral. If $\rho = r \rightarrow s$, the normal form of $Su(g)$ is Ss or s , and $\underline{k} + Sg$ is normal and differs from both. So the instance holds in $M_{t,u}$, its antecedent being false.

Amalgamation. Take one $M_{t,u}$ for each Case-W instance, rename them apart over $\mathbb{N}$, and let M be the free completion of their union: an operation whose arguments all lie in one M_i is computed there (this is well defined on $\mathbb{N}$, a subalgebra of each), and any other application is a new formal element. Each M_i is a subalgebra of M . Closed terms evaluate in $\mathbb{N}$, so M satisfies every true closed quantifier-free sentence. Each Case-W instance keeps its quantifier-free witness g , and each Case-T instance holds in every algebra containing $\mathbb{N}$. A refutation by sound rules would make some false closed quantifier-free sentence true in M , but such sentences have the same truth value in M as in $\mathbb{N}$. For Δ_0 premises with primitive $<$, interpret $a < b$ as “ $a, b \in \mathbb{N}$ and $a < b$ ”; bounded quantifiers with standard bounds then range over standard elements, so Δ_0 truth is preserved. $\square$

Proof of Theorem 5.18. prior induction C3 We write the proof for parameter-free motives, as in the prior record. (With parameters, the statement for DT° is the one-formula-metavariable case of the general anchor theorem of Section 3, single Cor D.3; we also checked that the steps below go through unchanged.) “*If*”. Fix $\varphi^\circ \in D$ with x free ($\mathbb{N}$) and let $T \in DT^\circ$ cover D . *Step 1.* By (R) and fact (i) the four motive slots have different roots in different data, so by Lemma D.1 $\text{skel}(T)$ is a prefix of the five-node frame. The occurrences of T sit at some of the positions of $\text{Ind}, U, V, s, a, W, b, c, d$, at most one per position. *Step 2.* Every metavariable has a pattern occurrence at one of these formula positions, so all metavariables are formula-valued, and all arguments are metavariable-free terms. Only x is in scope at W, b, c, d , and nothing at the others. So every metavariable is 0-ary, or unary with a pattern occurrence $M(x)$ at W, b, c or d . *Step 3.* Let $\kappa(\cdot)$ denote slot contents in $\text{Ind}(\varphi^\circ)$. A 0-ary metavariable takes a closed value, so it cannot occur at W, b, c, d , whose contents have x free. It cannot occur at two of Ind, U, V, s, a , whose contents have pairwise different logical sizes; so it occurs exactly once. A unary M with a pattern occurrence at π has the value $\lambda x. \kappa(\pi)$, and each occurrence $M(t)$ at σ needs $\kappa(\pi)[t/x] = \kappa(\sigma)$; by fact (i), σ lies in π ’s logical-size class, so $\pi = \sigma = W$ or $\pi, \sigma \in \{a, b, c, d\}$. If M has a pattern occurrence at b or d , take π there, so $\kappa(\pi) = \varphi^\circ$; fact (ii) forces $t = x$ at b, d , $t = Sx$ at c , and $t = 0$ at a . Otherwise M ’s pattern occurrences are at W (and then M occurs only there) or at c ; in the latter case $\kappa(c)[t/x] = \varphi^\circ[St/x]$, and occurrences at b, d or a would need $St = x$ or $St = 0$, which is impossible. So M occurs only at c , as $M(x)$. *Step 4.* Map a 0-ary metavariable at slot ρ to $T_{\text{Ind}}|_\rho$; M with a pattern occurrence at b or d to $\lambda y. P(y)$; M with pattern occurrences only at c to $\lambda y. P(Sy)$; and M_W to $\lambda y. (P(y) \rightarrow P(Sy))$. By Step 3 every occurrence of T becomes T_{Ind} ’s subformula at its position, so $T\sigma = T_{\text{Ind}}$ and $T \succeq T_{\text{Ind}}$. “*Only if*”. If all motives are closed, $Z_1 \rightarrow \forall x Z_2$ (size 4) covers D and misses $\text{Ind}(x = x)$, whose conclusion is not $\forall x Z_2$ with Z_2 closed. If all roots are f , use $(\Phi_f \wedge Z_1) \rightarrow Z_2$, where the filler of the a -slot is $\Phi_f = z_1 R z_2$ for an atomic root $R \in \{=, <\}$ (z_i 0-ary term metavariables), $\neg Z$, $Z' \circ Z''$, or $\text{Qy } K(y)$ with $K(y)$ a pattern occurrence. Each has size ≤ 7 , covers every $\text{Ind}(\varphi)$ with root f (because $\varphi[0/x]$ has root f), and misses $\text{Ind}(\neg x = 0)$ (if f is atomic) or $\text{Ind}(x = x)$. These witnesses lie in DT_s° for every $s \geq 7$. $\square$

Proof of Proposition 5.19. prior induction C4, C6.2; referee C (a) Every frame generalization is $\succeq T_{\text{Ind}}$ by the substitution of Step 4 above, so it contains $\text{inst}(T_{\text{Ind}})$. Conversely, $\text{inst}(T_{\text{Ind}})$ contains anchors, and Steps 1–3 show that every template of DT° covering an anchor is a frame generalization. The count: the prefixes of the five-node frame are $\emptyset$, $\{\text{imp}\}$, $\{\text{imp}, U\}$, $\{\text{imp}, V\}$, $\{\text{imp}, U, V\}$, $\{\text{imp}, U, s\}$, $\{\text{imp}, U, s, V\}$, $\{\text{imp}, U, s, W\}$ and the full frame. The motive slots

present are none, none, a , d , $\{a, d\}$, a , $\{a, d\}$, $\{a, b, c\}$ and $\{a, b, c, d\}$. The admissible partitions number 1, 1, 1, 1, 2, 1, 2, 4 and 13 (for the full frame: 4 with b, d together, 9 with b, d apart), 26 in all. For an anchor, the covering templates are these 26, and T_{Ind} (full frame, one class) is below all of them. (b) Sizes count symbols, an occurrence $M(\bar{t})$ counting $1 + \sum |t_i|$. “If”: $T_8 = (P(0) \wedge Z) \rightarrow \forall x P(x)$ (size 8), $T_{11} = (Z \wedge \forall x (P(x) \rightarrow P(Sx))) \rightarrow Z'$ (size 11) and $T_{12} = T_{\text{pat}}$ (size 12) contain $\text{inst}(T_{\text{Ind}})$. A sentence in $\text{inst}(T_{12})$ has the form $\alpha \wedge \forall x (\beta \rightarrow \gamma) \rightarrow \forall x \beta$ with β a motive; T_{11} adds $\gamma = \beta[Sx/x]$ and T_8 adds $\alpha = \beta[0/x]$. So the intersection is $\text{inst}(T_{\text{Ind}})$. “Only if” ($s \leq 11$): sharing a metavariable between d and b needs W and V rigid (five rigid nodes) and fillers of sizes $\geq 1, 2, \geq 2, 2$, so size ≥ 12 ; sharing between d and c needs ≥ 13 . So templates of size ≤ 11 impose at most $a = d[0/x]$, $a = b[0/x]$, $c = b[Sx/x]$ and a frame prefix. The sentence $s^* = (0 = 0 \wedge \forall x (x = x \rightarrow Sx = Sx)) \rightarrow \forall x (x = 0)$ satisfies all three equations, but $b \neq d$; so it is no induction instance, and it is false. Combine with the closure criterion of Section 2. $\square$

Proof of Proposition 5.17. prior induction C9 $\text{PAT} \subseteq \text{DT}^\circ$, and D is an anchor in DT° (Theorem 5.18). So the covering pattern templates are the frame generalizations of Proposition 5.19(a) without non-pattern occurrences: the a -slot gets a 0-ary metavariable or is cut, and the c -slot gets its own $M_c(x)$ or is cut. The most specific of them is the full frame with b, d sharing $P(x)$, which is T_{pat} ; all others are above it. The displayed instance ($Z := 0 = 0$, $P, Q := \lambda x. x = 0$) has a true premise and a false conclusion, and its c -slot $x = 0$ is not $P(Sx)$. $\square$

Proof of Proposition 5.20. prior induction C7 “Only if”. If (R) fails, use the root-specialized templates above. If (B) fails, let $\Theta = \{\theta_1, \dots, \theta_k\}$ be the parent terms of the free occurrences of x in the motives of D ; each is $\neq x$ with $\text{FV} \subseteq \{x\}$. $T_\Theta := Z \rightarrow \forall x P(\theta_1, \dots, \theta_k)$ covers D : in a motive, replace each maximal occurrence of some θ_i in which x is free by the hole z_i (closed motives are their own bodies). It misses $\text{Ind}(x = x)$, because in its instances every free x of the conclusion’s body lies inside some plugged $\theta_i \neq x$. “If”. Let $\varphi^\circ \in D$ have an unshielded free occurrence of x at position q , and let $T \in \text{SO}^\circ$ cover D . By (R) and Lemma D.1 the skeleton is a frame prefix, and all metavariables are formula-valued. For each M fix a body β_M° solving all its occurrences in $\text{Ind}(\varphi^\circ)$. Plugging terms preserves logical size, so the occurrences of M lie in one class: $\{a, b, c, d\}$, $\{W\}$, or a single closed depth-0 slot. *Key fact.* If M occurs at b or d with arguments $\bar{t}$, then β_M° has a hole z_m exactly at q , with $t_m = x$. The x at q must come from a plugged argument at some position $p \leq q$; if $p < q$, the subterm at p would be a proper term $\neq x$ with $\text{FV} \subseteq \{x\}$ containing the occurrence, contradicting unshieldedness. The same holds at W , at position $1 \cdot q$. At c , position q holds Sx , so $\beta_M^\circ|_q$ is z_m with $t_m = Sx$ or $S(z_m)$ with $t_m = x$; the hole cannot sit higher, because q ’s parent is a predicate (a formula position) or a function symbol whose subterm contains a variable bound inside φ° , which no argument contains. At a , position q holds 0: either a hole with argument 0 or the constant 0 of the body. *Substitution.* (i) M occurring at b or d , with hole z_m at q : $M \mapsto \lambda \bar{z}. P(z_m)$. Every occurrence $M(\bar{u})$ at a motive slot has $u_m = \kappa(\cdot)|_q \in \{0, x, Sx\}$, which is T_{Ind} ’s argument there. (ii) M at c but not at b, d : with z_m at q , $M \mapsto \lambda \bar{z}. P(z_m)$ (arguments 0 at a , Sx at c); with $S(z_m)$ at q , $M \mapsto \lambda \bar{z}. P(Sz_m)$, and then M does not occur at a , since $S(u_m) = 0$ is impossible. (iii) M only at W : $M \mapsto \lambda \bar{z}. (P(z_m) \rightarrow P(Sz_m))$. (iv) M only at a : $M \mapsto \lambda \bar{z}. P(0)$. (v) M only at a depth-0 slot ρ : $M \mapsto \lambda \bar{z}. T_{\text{Ind}}|_\rho$. Then $T\sigma = T_{\text{Ind}}$. $\square$

Proof of Proposition 5.21. prior induction C5, P1, P2, P6 The metavariables P, α, β receive $\varphi, \varphi[0/x]$ and $\varphi[Sx/x]$. By fact (i) the three (R) events of Plotkin’s lemma coincide with (R). The three (D) events coincide with (N), because $\varphi \neq \varphi[0/x]$ iff $\varphi \neq \varphi[Sx/x]$ iff $\varphi[0/x] \neq \varphi[Sx/x]$ iff x is free in φ : at a free occurrence the three formulas carry $x, 0$ and Sx . The rate is inclusion–exclusion over “all roots are f ” (disjoint) and “all motives closed”. If the induction variable is a metavariable X ,

its column consists of the induction variables, which adds the event that they are not all equal. Genuine data satisfy the guard “all Sub premises true”, so by the most specific guard rule the verifier accepts $\text{inst}(\text{lbg}D)$ intersected with the Sub-true steps. A Sub-true instance of σ_{Sub} has $\alpha = P[0/x]$ and $\beta = P[Sx/x]$ with P a formula, i.e. it is a genuine induction. $\square$

Proof of Proposition 5.22. prior induction B6 (a) In first-order logic with equality, $\forall x(x = t \rightarrow \varphi) \leftrightarrow \varphi[t/x]$ when $x \notin \text{FV}(t)$. Apply this with $t = 0, y, Sy$; since $\text{FV}(\varphi) \subseteq \{x\}$, y is not free in φ , and the step clause becomes $\forall y(\varphi[y/x] \rightarrow \varphi[Sy/x])$, an α -variant of $\forall x(\varphi \rightarrow \varphi[Sx/x])$. A sentence instance $\text{IndEq}(\psi)$ forces $\text{FV}(\psi) \subseteq \{x\}$, since the first conjunct would leave any other free variable free. Under universal closure, $P := (x = 0 \vee \neg x = y)$ gives a false instance: at $y = 1$ the premises hold (the step’s hypothesis holds only at $y = 0$, and its conclusion always holds), while $\forall x(x = 0 \vee \neg x = 1)$ fails at $x = 1$. (b) The four occurrences of P carry the same ψ_j in each datum and all other positions agree, so the lgg is IndEq applied to the anti-unifier of the ψ_j -column, computed once (equal columns get equal results). It is $\text{IndEq}(P)$ iff the ψ_j do not all have the same root; (D) is vacuous with one metavariable. (c) By (a) the instances give every parameter-free induction instance. For $\varphi(x, \bar{p})$ apply parameter-free induction to $\psi(x) := \forall \bar{p}[\varphi(0, \bar{p}) \wedge \forall z(\varphi(z, \bar{p}) \rightarrow \varphi(Sz, \bar{p})) \rightarrow \varphi(x, \bar{p})]$. Here $\psi(0)$ is trivial, $\psi(x) \rightarrow \psi(Sx)$ uses the step clause, and $\forall x \psi(x)$ gives induction for φ at every $\bar{p}$. Conversely, PA proves every instance. The trick raises quantifier complexity, which is why parameter-free IS_n^- is weaker than IS_n (Kaye et al., 1988). $\square$

Recorded motives in Lean and Metamath prior induction C12, P8, P9; referees A, C. In Lean 4 (version 4.32.0, checked locally by the prior track and its referee), **Nat.rec** takes the motive $\{\text{motive} : \text{Nat} \rightarrow \text{Sort } u\}$ as an *implicit* argument. The **induction** tactic elaborates to **Nat.recAux** (**fun** $n \Rightarrow \dots$), so the kernel term records the motive as a λ . For eliminators marked **elab_as_elim** the elaborator infers the motive by abstracting the discriminants, using **kabstract**. For closed $\forall$ -form steps the recorded motive is the unique matcher of T_{Ind} , so recording it changes neither anchors nor rates. In CIC proof terms, though, it is recoverable only up to conversion. Writing the motive as a λ -redex $\text{app}(\lambda x.P, t)$, with trusted β , gives a first-order pattern in one metavariable whose lgg is exact iff the motives’ roots are not all equal. But a *named* outer binder $\forall n$ captures a parameter n in P : $P := (x = 0 \vee \neg x = n)$ yields, after β , a sentence with a true antecedent and the false consequent $\forall n(n = 0 \vee \neg n = n)$. So the binder must be nameless or guarded by $n \notin \text{FV}(P)$.

Metamath’s **set.mm** states induction (**finds**, **nnind**) with implicit-substitution hypotheses and distinct-variable conditions (Megill and Wheeler, 2019). Transcribed to PA, the premises are H1 $x = 0 \rightarrow (\varphi \leftrightarrow \psi)$, H2 $x = y \rightarrow (\varphi \leftrightarrow \chi)$, H3 $x = Sy \rightarrow (\varphi \leftrightarrow \theta)$, H4 ψ and H5 $\chi \rightarrow \theta$, and the conclusion is φ . This is a first-order pattern in $\varphi, \psi, \chi, \theta$. A set of guards from $\{x \notin \text{FV}(\psi), x \notin \text{FV}(\chi), x \notin \text{FV}(\theta), y \notin \text{FV}(\varphi), x \neq y\}$ makes every instance a derived rule of PA iff it contains $y \notin \text{FV}(\varphi)$ and one of $x \notin \text{FV}(\chi), x \notin \text{FV}(\theta)$ [proved]. *Sufficiency*: fix a valuation v and let $X = \{a : \varphi(v[x:=a])\}$. H1 and H4 give $0 \in X$. For $a \in X$ put $w = v[x:=a, y:=a]$ and $w' = v[x:=a+1, y:=a]$. Then $\varphi(w)$ holds since $y \notin \text{FV}(\varphi)$, and H2 gives $\chi(w)$. Then H5 gives $\theta(w')$, either at w' (if $x \notin \text{FV}(\chi)$) or at w (if $x \notin \text{FV}(\theta)$). H3 at w' gives $\varphi(w')$, so $a+1 \in X$, and induction gives $X = \mathbb{N}$. *Necessity*: $\varphi = \neg x = SSy$, $\psi = \neg 0 = SSy$, $\chi = \neg y = SSy$, $\theta = \neg Sy = SSy$ violates only $y \notin \text{FV}(\varphi)$, has PA-provable premises, and concludes the false $\neg x = SSy$. The instance $\varphi = (x = 0)$, $\psi = (0 = 0)$, $\chi = \theta = (x = 0)$ violates only the two x -guards. So **set.mm**’s conditions, read as “not free”, are sound but not minimal. Learned from canonical uses, the most specific guards include the spurious $y \notin \text{FV}(\psi)$, which only a

non-canonical use removes.

D.6 Computational evidence

Everything above is proved. The computations corroborate it and found no counterexample. Scripts and outputs are in `research/tracks/cases/` and `research/prior/induction/`. Each record also has an independent referee implementation that imports none of the author’s code. Bounded searches are evidence, not proof.

- *Encodings and first-order failures* cases z1, z4, z5, z8; referee r2. The criterion of Theorem 5.3 was computed from the frames. The witness s_3 was checked for all 7 non-first-order cases. The block structure of Proposition 5.5 matched on all pool pairs with (R) and on 2417 random pairs. Every exhibited false sentence is an instance of the computed lgg and not of the schema. For $n \leq 10$ the path subterms in Theorem 5.7(i)–(iii) are unique, and 2136 random small generalizations τ of s_n contain both $s_n[q \leftarrow \top]$ and $s_n[q \leftarrow \perp]$. The pure-logic steps of the case analysis hold on all $\in$ -structures of size ≤ 3 . The guard criterion of Proposition 5.9(b) matched on all pool pairs and on 1537 random pairs.
- *Anchors, pattern lgg and rates* cases z2, z3, z6, z7, z9; referee r3, r4. For each of the seven schemas, all 120 pairs of a 16-body pool were checked with a reduced DT° enumeration, which is complete for anchor-hood: $\text{anchor} \iff (\text{R}) \wedge (\text{N})$ on 120/120. Full DT° and SO° enumerations on the smaller frames, and bounded $\text{DT}^\circ/\text{SO}^\circ$ searches on all frames (7×20 pairs), agree, with identical SO° and DT° verdicts. The referee’s SO° search, with repeated and parameter arguments, found 0 disagreements in 460 runs. An n -ary pattern anti-unifier with a permutation merge is exact iff $(\text{R}) \wedge (\text{N})$ on all pairs and 400 random triples per schema; the referee’s implementation agrees on 3500 data sets, and 10 500 random instances of its lggs are genuine. We did not run the published implementation of Baumgartner et al. (2017). For several metavariables, the pattern-lgg criterion held on 3×1500 random data sets and the DT° criterion on 2×150 pairs. Every SO° pair with (D^{root}) was an anchor, and none violating (D^{DT}) was. The few pairs with (D^{DT}) but not (D^{root}) were SO° anchors under the bounded search, which suggests the conjecture of Section 5.4. The exact rates agree with Monte Carlo.
- *Induction* prior induction b1–b4, c0–c5; referees B, C. An exactly false lgg instance was found for 2357 of the 2556 pairs of quantifier-free motives of size ≤ 5 ; the smallest unsound pair has motive sizes $3 + 3$. $\text{lgg} = L_n$ for all $0 \leq n < m \leq 10$. For K_0 , a congruence-closure test on 2224 instances finds the step clause refutable over $\mathbb{N}$ exactly in the 12 Case-T instances. For all 351 pairs of a 27-motive pool, all SO° templates of size ≤ 14 (arity ≤ 2 , arguments of size ≤ 3 ; 4.6 million covering templates) were classified: DT° anchor $\iff (\text{R}) \wedge (\text{N})$, and SO° anchor $\iff (\text{R}) \wedge (\text{B})$, on 351/351. The referee found no disagreement on 176 further data sets. With bounded arity, though, some non-(B) sets are SO° anchors, since Proposition 5.20 needs unbounded arity. The 26 generalizations and the threshold 12 were reproduced. The Sub-lgg and DT° verdicts coincide on all pairs and 2925 triples. IndEq and Ind agree on 30 000 random pairs of a motive and a finite structure.
- *Experiment E2* experiments E2: see Section 5.5 and Section 7.

E Proofs and details for Section 6

Notation is that of Section 6.1. “Template” means a template in DT° ; “covering X ” means having every element of X as an instance. The finitariness facts of Proposition 6.2 are used throughout.

E.1 The setting

Proof of Lemma 6.3. (a) $R^* = \bigcup_i \text{inst}(\sigma_i) \in H_k$ because $k \geq k'$; it contains D and avoids N . Larger D or N shrink the version space, so they enlarge its intersection. (b) Each displayed union is in H_k (at most k blocks), contains D and avoids N , so it lies in $\text{VS}_k(D, N)$; hence $\text{Acc}_k(D, N)$ is contained in the displayed intersection. Conversely let $h = \bigcup_l \text{inst}(\tau_l) \in \text{VS}_k(D, N)$. Assign each datum to the first slot covering it; this partitions D into at most k nonempty blocks $B \subseteq \text{inst}(\tau_l)$. By Proposition 6.2(b), $\tau_l \succeq T_B$ for some $T_B \in \text{Min}(B)$, and $\text{inst}(T_B) \subseteq \text{inst}(\tau_l) \subseteq h$ avoids N , so $T_B \in \text{Min}_N(B)$ and $h \supseteq \bigcup_B \text{inst}(T_B)$. So every member of the version space contains a displayed union, and the displayed intersection is contained in $\text{Acc}_k(D, N)$. Decidability: there are finitely many partitions, each $\text{Min}(B)$ is finite and computable, and avoidance of a finite N and membership of a query are decided by matching (Section 3). $\square$

With $N = \emptyset$ and $\text{Acc}(B) := \bigcap \text{Cov}(B) = \bigcap_{T \in \text{Min}(B)} \text{inst}(T)$ (and $\text{Acc}(\emptyset) := \emptyset$), part (b) says, since the choices T_B are independent across blocks:

$$q \in \text{Acc}_k(D) \iff \forall \pi \exists B \in \pi : q \in \text{Acc}(B), \quad (1)$$

where π ranges over the partitions of D into at most k nonempty blocks.

Lemma E.1 (monotonicity of coherence). *[proved]^{untagged} Lemma 5.1* X is d -coherent iff some $T \in \text{Cov}(X)$ is not d -refuted. d -coherence is downward closed ($\emptyset \neq Y \subseteq X$ and X d -coherent imply Y d -coherent) and antitone in d .

Proof. If $T \in \text{Cov}(X)$ is not d -refuted, then $T \succeq T_0 \in \text{Min}(X)$, and $\text{inst}(T_0) \subseteq \text{inst}(T)$ is not d -refuted either; the converse holds as $\text{Min}(X) \subseteq \text{Cov}(X)$. A covering template of X covers Y . And $\text{Ref}_d \subseteq \text{Ref}_{d+1}$. $\square$

E.2 Positive data: bounds, spare slots and head classes

Proof of Theorem 6.5. (a) $h \subseteq R^*$ and $N \cap R^* = \emptyset$ give $h \cap N = \emptyset$, and $D \subseteq h \in H_k$. (b) Let $h = \bigcup_l \text{inst}(\tau_l) \in \text{VS}_k(D, N)$ and $L_j := \{l : \tau_l \text{ covers a datum of } D_j\}$. Each L_j is nonempty because D_j is. By cross-separation a slot covering data of two targets would meet N , so the L_j are pairwise disjoint and $|L_i| \leq k - (k' - 1) = m$. Every datum of D_i is covered by a slot in L_i , and these slots avoid N , so by hypothesis $\bigcup_{l \in L_i} \text{inst}(\tau_l) \supseteq \text{inst}(\sigma_i)$; hence $h \supseteq \text{inst}(\sigma_i)$. For the second claim, $h := \bigcup \text{inst}(F) \cup \bigcup_{l \neq i} \text{inst}(\sigma_l)$ uses at most $m + k' - 1 = k$ slots, contains D , avoids N (the σ_l lie in R^*) and misses q . If the instance sets are pairwise disjoint and the first condition fails, some admissible family misses some $q \in \text{inst}(\sigma_i)$, which lies in no other instance set, so the second claim applies: the condition is necessary and sufficient. $\square$

Proof of Theorem 6.6. (a) Let $h \in \text{VS}_k(D, N)$ with slots τ_l . If some slot contains $\text{inst}(\sigma_i)$ we are done. Otherwise, by (Abs), no slot covers both a datum of D_i and a datum of $\bar{D}_i$. The slots covering $\bar{D}_i$ -data form an N -avoiding cover of $\bar{D}_i$ by templates not containing $\text{inst}(\sigma_i)$, so there are at least c_i of them, and they are disjoint from the slots covering D_i . So at most $k - c_i$ slots cover D_i , and by hypothesis their union contains $\text{inst}(\sigma_i)$. (If $k \leq c_i$, this shows that some slot

of every h contains $\text{inst}(\sigma_i)$.) (b) $\bigcup \text{inst}(F \cup G)$ has at most k slots, contains D , avoids N and misses q . $\square$

Proof of Corollary 6.7. (Abs). At the top level of a sentence no bound variable is in scope, so a metavariable whose pattern occurrence is not under a binder is 0-ary. A Q-axiom with root $\neg$ or $=$ and an induction instance (root $\rightarrow$) have no common root, so their minimal covering template is the bare F_0 . Q2 and Q7 have root $\rightarrow$; their antecedents ($=$, resp. $\neg$) differ in head from the induction antecedent ($\wedge$), and their consequents ($=$, resp. $\exists$) from the induction consequent ($\forall$); the two slots are closed and not equal in either datum, so the minimal covering template is $F_0 \rightarrow F_1$. This holds for every motive (a remark of the referee). Every covering template lies above one of these (Proposition 6.2), and both contain $\text{inst}(T_{\text{Ind}})$. $c = 4$ with $N = \emptyset$. Any template covering two of Q1, Q2, Q7, or one of them and one of Q3–Q6, lies above F_0 or $F_0 \rightarrow F_1$ (compare the roots and, for Q2–Q7, the heads of antecedents and consequents), hence contains $\text{inst}(T_{\text{Ind}})$. So Q1, Q2 and Q7 need a slot each and Q3–Q6 at least one; $z_0 = z_1$ covers Q3–Q6 and contains no induction instance. $c = 7$ with $N = \{0 = S0, (p + 0) = 0, (p \cdot 0) = S0\}$. Q3 and Q4 have the lgg $p + z_0 = z_1$ (common prefix $=, +, p$; closed slots), which has the instance $(p + 0) = 0 \in N$; Q5 and Q6 have $p \cdot z_0 = z_1 \ni (p \cdot 0) = S0$; the other pairs among Q3–Q6 have $z_0 = z_1 \ni 0 = S0$. All covering templates of these pairs lie above the lggs, so Q3–Q6 need a slot each. (F_0 is now refuted, $F_0 \rightarrow F_1$ is not, so (Abs) still holds.) Then induction gets $k - 7 = 1$ slot, and by Theorem 6.6 it is learned exactly iff its data contain an anchor, which for T_{Ind} is (R)+(N). *The four-slot criterion.* Let q be an induction instance whose motive has x free and has a main connective that occurs in no datum. By the anchor theorem for T_{Ind} , a block of induction data is a non-anchor iff its motives all have the same root f or are all vacuous; then $T_{\text{Ind}}[P := \lambda x.f(\dots)]$, resp. $T_{\text{Ind}}[P := \lambda x.A]$, covers the block and misses q . If $\chi \leq 4$, these templates for an optimal partition, together with the cover $\{Q1, Q2, Q7, z_0 = z_1\}$ of the Q-data, form a hypothesis with 8 slots missing q (Theorem 6.6(b)). If $\chi > 4$, every family of at most four templates covering the induction data has, by partition into first covering members, an anchor block, whose template contains $\text{inst}(T_{\text{Ind}})$ (Theorem 6.6(a)). $\square$

Corollary E.2 (thresholds in terms of failure families). *[proved]^{untagged} Cor 3.3* Call T a proper specialization of σ if $\text{inst}(T) \subsetneq \text{inst}(\sigma)$. A failure family $\text{Fail}(\sigma)$ is a set of proper specializations such that every finite non-anchor $X \subseteq \text{inst}(\sigma)$ lies in $\text{inst}(F)$ for some $F \in \text{Fail}(\sigma)$. (Examples: for first-order patterns the head specializations $\sigma[x \mapsto f(\bar{z})]$ and the identifications $\sigma[y \mapsto x]$, by Plotkin's (R)+(D); for T_{Ind} the eight root specializations $T_{\text{Ind}}[P := \lambda x.f(\bar{P}(x))]$, $f \in \{=, \neg, \wedge, \vee, \rightarrow, \leftrightarrow, \forall, \exists\}$, and the vacuous one $T_{\text{Ind}}[P := \lambda x.A]$.) Let $m = k - k' + 1$. (a) If $D_i \subseteq \bigcup_{j \leq m} \text{inst}(F_j)$ with $F_j \in \text{Fail}(\sigma_i)$, and some $q \in \text{inst}(\sigma_i) \setminus \bigcup_j \text{inst}(F_j)$ lies in no other $\text{inst}(\sigma_l)$, then $q \notin \text{Acc}_k(D, N)$ for every sound N . (b) Suppose N cross-separates D , N target-separates D_i (it meets every template that covers a datum of D_i and whose instance set neither contains nor is contained in $\text{inst}(\sigma_i)$), and the instance sets are pairwise disjoint. Then $\text{inst}(\sigma_i) \subseteq \text{Acc}_k(D, N)$ iff D_i is not covered by m proper specializations of σ_i whose union misses an instance of σ_i . If moreover no m members of $\text{Fail}(\sigma_i)$ cover $\text{inst}(\sigma_i)$, this holds iff D_i is not covered by m members of $\text{Fail}(\sigma_i)$.

Proof. (a) Apply Theorem 6.5(a) to $h = \bigcup_j \text{inst}(F_j) \cup \bigcup_{l \neq i} \text{inst}(\sigma_l)$, which has k slots and satisfies $D \subseteq h \subseteq R^*$. (b) If exactness for i fails, then by Theorem 6.5(b) (which uses cross-separation) some N -avoiding family of at most m templates covers D_i and misses some $q \in \text{inst}(\sigma_i)$. No member contains $\text{inst}(\sigma_i)$, so by target-separation each member covering a D_i -datum is a proper specialization. Each block $X_j = D_i \cap \text{inst}(T_j)$ is covered by a template not containing $\text{inst}(\sigma_i)$,

so it is not an anchor and lies in some $\text{inst}(F_j)$, $F_j \in \text{Fail}(\sigma_i)$. Conversely, a cover by proper specializations that misses q gives $q \notin \text{Acc}_k(D, N)$ by Theorem 6.5(b) and disjointness, and a cover by m failure templates misses some q under the extra hypothesis, so (a) applies. $\square$

The first version of the research record added that Plotkin’s failure family is m -noncovering for every m , “because parameters are infinitely many names”. The referee showed this false once data are read under universal closure (as here and in the code): renamed copies are the same datum, and the head classes are finitely many (Proposition E.3). The claim was withdrawn. In the same way, on positive data ($N = \emptyset$) target i is learned exactly if every partition of D_i into at most k blocks (at most $k - c_i$ under (Abs)) has an anchor block, and not if D_i is covered by $k - k' + 1$ proper specializations missing an instance of σ_i that lies in no other target (untagged Prop 6.1). For PA the gap between $k - c$ and $k - k' + 1$ is the whole story: $c = 1$ for Sub-encoded steps, $c = 4$ for raw sentences, and $c = 7 = k' - 1$ with three negatives. The classical sufficient condition “not covered by k failure sets” is vacuous for T_{Ind} at $k \geq 8$, because the eight root specializations cover every induction instance.

Head classes. In the rest of this subsection sentences are identified up to bijective renaming of parameters, as the closure reading demands and as the implementation does (parameters are renamed canonically by first occurrence).

Proposition E.3 (finitely many head classes). *[proved; computed]^{untagged Prop 3.4} Let $\sigma = \varphi(z)$ have a single metavariable, a 0-ary term metavariable z occurring in φ ; let F be the function symbols (constants included) and $c_1, \dots, c_p$ the parameters of φ . Every closed term t has one head class $\text{head}(t) \in F \cup \{c_1, \dots, c_p\} \cup \{\nu\}$, where ν means “a parameter other than the c_j ”. Put $\sigma_f := \varphi(f(u_1, \dots, u_r))$ (fresh 0-ary u ’s), $\sigma_{c_j} := \varphi(c_j)$, $\sigma_\nu := \varphi(b)$ for a parameter b not in φ . Then (a) $\text{inst}(\sigma) = \bigcup_g \text{inst}(\sigma_g)$ over the $h = |F| + p + 1$ classes, and (b) no σ_g is redundant.*

Proof. (a) For an instance $\varphi(t)$: if $t = f(\bar{t})$ it is $\sigma_f[\bar{u} := \bar{t}]$; if $t = c_j$ it is σ_{c_j} ; if t is a parameter $b' \notin \{c_j\}$, the renaming swapping b and b' maps $\varphi(b)$ to $\varphi(b')$, since φ contains neither, so the two are the same datum. (b) Suppose a renaming π maps $\varphi(t)$ to $\varphi(t')$. At the rigid occurrences of c_j in φ it fixes c_j , so it maps other parameters to other parameters; at the position of z it maps t to t' . So $\text{head}(t) = \text{head}(t')$, and σ_g covers only instances whose z -value has class g . $\square$

Computed (untagged u12(a)): the 5 head-class templates of $z+0 = z$ and the 6 of $c_1+z = z+c_1$ (one parameter) cover 2000/2000 random canonical instances each, and each template is needed.

Proposition E.4 (threshold for one-variable instance schemas). *[proved; computed]^{untagged Prop 3.5} Assume (H1) $\sigma_i = \varphi(z)$ with φ quantifier-free and parameter-free and z a 0-ary term metavariable occurring in it; (H2) every datum of D_i has a z -value with at most one parameter, and T is the set of these values with the parameter renamed a ; (H3) N cross-separates D , the instance sets are pairwise disjoint, every D_j is nonempty, and $m = k - k' + 1$. Let $\text{cl}(T)$ be the set of head classes occurring in T and $h = |F| + 1$. (a) If $m < |\text{cl}(T)|$, then $\text{inst}(\sigma_i) \subseteq \text{Acc}_k(D, N)$. (b) If some class is missing from T , then $\text{inst}(\sigma_i) \subseteq \text{Acc}_k(D, N)$ iff $m < |\text{cl}(T)|$. (c) If all h classes occur and $m = h$, then $\text{inst}(\sigma_i) \subseteq \text{Acc}_k(D, N)$ iff for each $f \in F$ of arity $r \geq 1$ the block T_f of terms with head f has $\text{lgg} T_f = f(x_1, \dots, x_r)$ with distinct variables, i.e. Plotkin’s (R)+(D) holds for T_f (each argument column has two different heads, the parameter counting as one head, and no two columns are equal).*

Proof of Proposition E.4. Step 0 (least covering template of a block). Let $\emptyset \neq B \subseteq D_i$ have z -values $\{t_b\}$. Every $T \in \text{Cov}(B)$ satisfies $\text{inst}(T) \supseteq \text{inst}(\varphi(\text{lgg}\{t_b\}))$, the lgg (Plotkin’s) taken with

a as a constant. Indeed: the data are quantifier-free, so T has no binder in its rigid part, no bound variable is in scope, and every metavariable is 0-ary: T is a first-order pattern. It covers each datum through a representative $\varphi(t_b[n_b/a])$. By Plotkin's theorem, $T \succeq \text{lgg}\{\varphi(t_b[n_b/a])\} = \varphi(G)$ with $G := \text{lgg}\{t_b[n_b/a]\}$ (lgg commutes with the parameter-free context φ : rigid positions carry constant tuples, and the z -positions carry the same tuple, hence the same variable). If all n_b equal some n , $G = \text{lgg}\{t_b\}[n/a]$, the same up to renaming. Otherwise G has no rigid parameter (a rigid n at position q would force $t_b|_q = a$ and $n_b = n$ for all b), and $G \succeq \text{lgg}\{t_b\}$: the rigid symbols of G are function symbols at positions where all t_b agree, and two variable positions of G with the same tuple $(t_b[n_b/a]|_q)_b$ also have the same tuple $(t_b|_q)_b$ (undo the renamings), so mapping each variable of G to the corresponding subterm of $\text{lgg}\{t_b\}$ is well defined and turns G into $\text{lgg}\{t_b\}$. So $\text{inst}(T) \supseteq \text{inst}(\varphi(G)) \supseteq \text{inst}(\varphi(\text{lgg}\{t_b\}))$. The template $\varphi(\text{lgg}\{t_b\})$ covers B , lies below σ_i , and so avoids every sound N .

Step 1 (reduction). By Theorem 6.5(b), exactness for i holds iff every N -avoiding family of at most m templates covering D_i has union $\supseteq \text{inst}(\sigma_i)$. Partitioning D_i by first covering member and using Step 0, this holds iff for every partition π of D_i (equivalently of T) into at most m blocks, $\bigcup_{B \in \pi} \text{inst}(\varphi(\text{lgg}T_B)) = \text{inst}(\sigma_i)$. Since z occurs in the parameter-free φ , $\varphi(t)$ determines t up to renaming, so this says that every closed term is, up to renaming, an instance of some $\text{lgg}T_B$: $\bigcup_{B \in \pi} \text{inst}(\text{lgg}T_B) = G_{\text{cl}}$, the closed terms modulo renaming.

Step 2 (mixed blocks). A block with two head classes has a variable as lgg , which covers G_{cl} . So failing partitions are head-pure, and a head-pure partition has at least $|\text{cl}(T)|$ blocks. This proves (a).

Step 3 (a missing class). If a class g is missing from T , the head partition ($|\text{cl}(T)|$ blocks) has lggs that are constants, the parameter a , or terms with heads in $\text{cl}(T)$, so no term of class g is covered and it fails. With (a), this proves (b).

Step 4 ($m = h$, all classes present). The only head-pure partition with at most h blocks is the head partition, and it covers G_{cl} iff each block covers its class. A constant class (a 0-ary f , or ν) has one term, which its block covers. For f of arity $r \geq 1$, $\text{inst}(\text{lgg}T_f) \supseteq f(G_{\text{cl}}^r)$ iff $\text{lgg}T_f = f(x_1, \dots, x_r)$ with distinct variables: a non-variable argument misses terms with another head there, and a shared variable misses terms with different subterms in the two arguments. Plotkin's construction gives a variable in argument j iff column j has two different heads, and the same variable in two arguments iff the columns are equal. $\square$

Proposition E.5 (closed form for unary signatures). *[proved; computed]^{untagged Prop 3.6} Assume (H1)–(H3) with function symbols a finite set C of constants and one unary S (so (H2) is automatic and $h = |C| + 2$). Put $T_0 := T$, $T_{j+1} := \{u : Su \in T_j\}$, and let r be the least j with $|\text{cl}(T_j)| < h$. Then $\text{inst}(\sigma_i) \subseteq \text{Acc}_k(D, N)$ iff $m < \kappa(T) := (h - 1)r + |\text{cl}(T_r)|$.*

Proof of Proposition E.5. Let $\kappa(T)$ be the least number of blocks of a partition of T whose lggs fail to cover G_{cl} ; by Steps 1–3, exactness holds iff $m < \kappa(T)$, and failing partitions are head-pure. If $r = 0$ (a class is missing), $\kappa(T) = |\text{cl}(T_0)|$ by Step 3. If all h classes are present, each of the $h - 1$ constant classes (C and ν) is a single term and forms a block of a head-pure partition, and the other blocks partition the S -class. Since lgg of an S -block is S applied to the lgg of the arguments and $\text{inst}(Ss) = S(\text{inst}s)$, the S -blocks cover $S(G_{\text{cl}})$ iff the corresponding blocks of T_1 cover G_{cl} . So $\kappa(T) = (h - 1) + \kappa(T_1)$, and induction on r gives $\kappa(T) = (h - 1)r + |\text{cl}(T_r)|$. $\square$

Computed (untagged u12): for $z + 0 = z$ and 60 random data sets, exactness at $m = h = 5$, decided by brute force over all partitions with coverage of all 608 canonical instances of depth ≤ 2 (decisive, since every failure in Step 4 is witnessed at depth ≤ 2), agreed with (c) in 60/60

cases (18 exact), and with $m = |\text{cl}(T)| - 1$ the target was exact in 60/60; the closed form agreed with brute force on all 254 term sets of size ≤ 7 over $\{S^j b : j \leq 3, b \in \{0, \nu\}\}$ and all 2509 sets of size ≤ 6 over $\{S^j b : j \leq 3, b \in \{0, c, \nu\}\}$.

E.3 Self-generated negatives

Corollary E.6 (self-generated negatives). *[proved]^{untagged} Cor 2.2* Assume (Dec) and fix an effective enumeration of $\mathcal{S}$. (a) If every $T \in \text{Min}(\{a, b\})$ with $a \in A_i, b \in A_j, i < j$, is d -refuted, then one refuted instance n_T per such T gives a set $N_\times$ satisfying (X) of Theorem 6.9, with $|N_\times| \leq \mu \sum_{i < j} |A_i| |A_j|$, where $\mu = \max |\text{Min}(\{a, b\})|$ over cross anchor pairs. (b) The set $N(D, d)$ of first d -refuted instances of all d -refuted $T \in \text{Min}(\{a, b\})$, $a, b \in D$, is computable from D without labels, is contained in Ref_d , and contains a choice of $N_\times$. Hence $\text{Acc}_{k'}(D, N(D, d))$ is sound at all times, decidable, and equal to R^* as soon as D contains anchors A_i satisfying the hypothesis of (a).

Proof. (a) If T' covers $a \in A_i$ and $b \in A_j$, then $T' \in \text{Cov}(\{a, b\})$, so $T' \succeq T$ for some $T \in \text{Min}(\{a, b\})$ and $n_T \in \text{inst}(T) \subseteq \text{inst}(T')$. (b) $\text{Min}(\{a, b\})$ is computable and “ d -refuted” decidable; the first refuted instance is found by enumeration. Within-target pairs may contribute further refuted sentences, which are false and harmless. Apply Lemma 6.3 and Theorem 6.9. $\square$

Remark E.7 (combining with the k -union learner). *[proved]^{untagged} Rem 5.3* Let $k \geq k'$ be known and accept q iff $q \in \text{Acc}_d \cap \text{Acc}_k(D, N(D, d))$. This is target-sound at all times, by Lemma 6.3, with or without separation. Under $\text{RS}_d(D)$ with $k = k'$, once every D_i contains an anchor, it equals R^* : DTRC is exact by Theorem 6.11(iii), and $\text{RS}_d(D)$ makes every minimal template of every cross pair d -refuted, so $N(D, d)$ satisfies (X) and Theorem 6.9 applies. (The research record says “exact whenever DTRC is exact and $k = k'$ ”; DTRC can be exact without anchors, because refuted templates of whole clusters drop out, while $N(D, d)$ contains only refutations of pair templates, so the anchor hypothesis is needed for the argument given.)

For PA (Table 12) the four sentences $0 = S0, 0 = 0 \rightarrow 0 = S0, p + 0 = 0, p \cdot 0 = S0$ refute every cross template; since the Q/induction templates F_0 and $F_0 \rightarrow F_1$ are absorbing, three suffice (Corollary 6.7).

E.4 dtrc: proofs

Proof of Theorem 6.11. (i) By Proposition 6.2(c) some $T_0 \in \text{Min}(X)$ has $\text{inst}(T_0) \subseteq \text{inst}(\sigma_i) \subseteq R^* \subseteq \text{Tr}$, so $\text{inst}(T_0) \cap \text{Ref}_d = \emptyset$ by (OS). (ii) Call a cluster pure if it lies inside one D_i ; initially all are. If $C \subseteq D_i$ and $C' \subseteq D_j$ are pure with $i \neq j$, every $T \in \text{Cov}(C \cup C')$ covers a datum of D_i and one of D_j , so it is d -refuted by $\text{RS}_d(D)$: the union is not coherent and is never formed. So all clusters stay pure. The process stops after finitely many merges. At termination, two final clusters inside one D_i would have a coherent union by (i), contradicting termination. So each nonempty D_i is exactly one final cluster, whatever the policy. (iii) For the final cluster D_i , the T_0 of (i) lies in $\text{Min}_d^+(D_i)$, so $\text{Acc}_d(D_i) \subseteq \text{inst}(T_0) \subseteq \text{inst}(\sigma_i)$ and $\text{Min}_d^+(D_i) \neq \emptyset$. If $A_i \subseteq D_i$ is an anchor, every $T \in \text{Min}(D_i)$ covers A_i , so $\text{inst}(T) \supseteq \text{inst}(\sigma_i)$ and $\text{Acc}_d(D_i) \supseteq \text{inst}(\sigma_i)$. (iv) Under global separation, (ii) and (iii) hold for every $D \subseteq R^*$. So DTRC fails to be exact only if some D_i lacks an anchor, which is the failure event of the tagged learner given the labels. Averaging the tagged bound over the binomial number of target- i data gives the rate. (v) A pair answer depends only on the two clusters. There are $n(n-1)/2$ initial pairs; there are at most $n-1$ merges, and after each only pairs with the new cluster (at most n) need retesting. For the online policy under

$\text{RS}_d(D)$, by induction on the data there is at most one cluster per target at all times: a datum of target i joins the first coherent cluster, cross unions are incoherent, and the cluster of target i is coherent with it by (i). So each datum is tested against at most k' clusters, and the final pass tests the $k'(k' - 1)/2$ pairs of pure clusters. $\square$

Proof of Theorem 6.12. (a) A pure final cluster $C \subseteq D_i$ contributes $\text{Acc}_d(C) \subseteq \text{inst}(\sigma_i)$, as in the proof of Theorem 6.11(iii). A mixed final cluster C is d -coherent or an incoherent singleton contributing $\emptyset$ (Proposition 6.18), and it contributes $\text{Acc}_d(C) \subseteq \text{inst}(T)$ for each $T \in \text{Min}_d^+(C)$, which is part of $\text{Res}_d(D)$ (take $X = C$). (b) $\text{Acc}_d(C) \subseteq \text{inst}(T) \subseteq \text{Tr}$. (c) By definition; $\text{Res}_d(D)$ decreases in d because coherence and non-refutation are antitone in d (Lemma E.1). (d) In the example of Section 6.5 all data are quantifier-free, so every minimal template is a Plotkin lgg (Proposition E.9(b)). $\{g, 0 + 0 = 0\}$ has the lgg $0 + z = z$ (the columns $(S0, 0)$ at the second argument and at the right-hand side are equal), which is true and never refuted; $\{g, 0 + 0 = 0, S0 + 0 = S0\}$ has $z_0 + z_1 = z_2$ and $\{S0 + 0 = S0, g\}$ has $z_0 + z_1 = S0$, both refuted by $0 + 0 = S0$. So in the order $g, 0 + 0 = 0, S0 + 0 = S0$ the final clusters are $\{g, 0 + 0 = 0\}$ and $\{S0 + 0 = S0\}$; $p + 0 = p$ is in neither acceptance set, while $0 + p = p$ is accepted. Computed in untagged u9 (Example B), with the referee's variant with overlapping instance sets ($\sigma_2 = 0 + z = z$) behaving the same way in 2 of 3 orders. $\square$

Proof of Lemma 6.14. Two-point identity. Let $G(z), K(z)$ be expressions in which z occurs only as a leaf, with $G(t_j) = K(t_j)$ for $j = 1, 2$. Then $G = K$. Induction on G and K : if $G = z$ and $K \neq z$, then K has a rigid head h (a symbol or an index) that survives substitution, and $K(t_j) = t_j$ forces $h = \text{head}(t_1)$ and $h = \text{head}(t_2)$, a contradiction; symmetrically for $K = z$. If neither is z , their heads agree (compare at t_1), and we recurse into the arguments; binders are harmless, because z is not a bound variable and the t_j are closed. *Skeleton.* The two data agree everywhere except at and below the z -positions of φ , where their heads differ. By the rigid-prefix lemma of Section 3 the rigid skeleton of a covering template T lies in the common prefix of the data, so it avoids the z -positions and everything below them; every occurrence of T is at a position of φ . *Bodies.* For each metavariable M of T choose a pattern occurrence $M(\bar{y})$ at position p . For any closed t put $\theta_t(M) := \lambda \bar{y}.(\varphi(t)|_p)$. This is a legal body: t is closed, so the bound variables free in $\varphi(t)|_p$ are those free in $\varphi(t_1)|_p$, which are among $\bar{y}$ because T covers $\varphi(t_1)$. *Every occurrence fits.* For an occurrence $M(\bar{u})$ at position p' put $G(z) := (\lambda \bar{y}.\varphi(z)|_p)(\bar{u})$ (plugging) and $K(z) := \varphi(z)|_{p'}$. G contains z only as a leaf, since $\bar{u}$ is metavariable-free. T covers both data with the unique matchers $\theta_{t_1}, \theta_{t_2}$, and substituting a closed t_j for z commutes with abstraction and plugging, so $G(t_j) = K(t_j)$; hence $G = K$ and $G(t) = K(t)$. With the skeleton, $T\theta_t = \varphi(t)$, so $\text{inst}(T) \supseteq \text{inst}(\varphi(z))$. $\square$

Proof of Proposition 6.15. (a) $\varphi(z)$ covers D , and by Lemma 6.14 every covering template contains $\text{inst}(\varphi(z))$; so some covering template is unrefuted iff $\varphi(z)$ is (Lemma E.1). (b) In closure-normal form a parameter is read universally; it must be fresh, so that $\varphi(a)$ is not an instance at a special parameter. (c) If $\forall x \varphi$ is true, every instance is true, so every subset of D is coherent (Theorem 6.11(i) with the single target $\varphi(z)$), no two clusters can remain at termination. In the single final cluster every unrefuted minimal template contains $\text{inst}(\varphi(z))$ (Lemma 6.14), and one of them is contained in it (Proposition 6.2(c)), so $\text{Acc}_d = \text{inst}(\varphi(z))$. The oracle is a fixed function of the query, so the practices “ n ground instances” and “ $\varphi(z)$ ” produce the same run. (d) By (a). (e) The first claim is the definition of completeness for false instances plus Proposition 6.16. $\text{R-}\Delta_0$ verifies universal statements only by equality reasoning from true closed equations, so a false sentence whose refutation needs a true universal premise that this reasoning cannot verify is

never refuted; the noisy-PA computation (Example E.8) contains such a sentence, a wrong-base induction variant whose refutation needs the universal premise $\forall x(\neg x = 0 \rightarrow \neg Sx = 0)$, an instance of Q1. At a finite budget, a merge through a false universal whose least counterexample lies beyond the budget is part of Res_d (Theorem 6.12). In $\mathbb{R}$ with $0, 1, +, \cdot$ every closed term denotes a natural number (induction on terms) and no natural number squares to 2, so every closed instance of $\neg(z \cdot z = 1 + 1)$ is true, while $\neg(p \cdot p = 1 + 1)$, read universally, is false; a decision procedure for real closed fields would refute it. $\mathbf{Q} \vdash 0 + \underline{n} = \underline{n}$ for each n but $\mathbf{Q} \not\vdash \forall x(0 + x = x)$ is shown in Section 4. $\square$

Proof of Theorem 6.17. Construction. By the MRDP theorem (Davis et al., 1961; Matiyasevich, 1993) there are polynomials $P(e, \bar{x})$, $Q(e, \bar{x})$ with natural coefficients, $\bar{x} = x_1 \dots x_k$, such that $e \in K \iff \exists \bar{x} \in \mathbb{N}^k P(e, \bar{x}) = Q(e, \bar{x})$, K the halting set. Let P_e, Q_e be the terms with the numeral $\underline{e}$ substituted, and put

$$H_e(z) := \exists x_1 \dots \exists x_k (z = SS(x_1 + \dots + x_k) \wedge P_e(\bar{x}) = Q_e(\bar{x})),$$

with a 0-ary term metavariable z under k binders (z is closed, so no capture). Let $T_e := \neg H_e(z)$, $A_e := \neg H_e(0)$, $B_e := \neg H_e(S0)$, $q_e := \neg H_e(SS0)$. Practice $\mathcal{P}_e$ has the single target T_e ; practice $\mathcal{P}'_e$ has the ground targets A_e, B_e ; both have the data law uniform on $\{A_e, B_e\}$. *Facts.* (1) A_e, B_e are true, since $0 \neq SSu$ and $S0 \neq SSu$; so $\mathcal{P}'_e$ is valid. (2) $\mathcal{P}_e$ is valid iff $e \notin K$: its instances, including the parameter instance read universally, are all true iff no solution exists. (3) $\{A_e, B_e\}$ is an anchor for T_e (Lemma 6.14: heads 0 and S). (4) If $e \in K$ with solution $\bar{x}^*$, let $\bar{c}$ be the numeral of $2 + \sum x_i^*$. The instance $\neg H_e(\bar{c})$ is false, and $\text{R-}\Delta_0$ refutes it at depth $d_e := \max(|\neg H_e(\bar{c})|, \max_i x_i^*)$: refuting $\neg\psi$ means verifying $\psi = \exists \bar{x}(\dots)$, done by the witnesses $\bar{x}^*$ and closed evaluation of two atoms; no universal has to be verified. By (3) every template covering A_e and B_e contains $\neg H_e(\bar{c})$, so $\mathcal{P}'_e$ is separated at depth d_e and $\text{Res}_\infty(\mathcal{P}'_e) = \emptyset$. (5) If $e \notin K$, T_e is truth-sound and never refuted, and $\text{Res}_\infty(\mathcal{P}'_e) \supseteq \text{inst}(T_e) \ni q_e$; $\mathcal{P}'_e$ is separated at no depth, so (b) asks nothing of it. (6) If $e \notin K$, $\mathcal{P}_e$ has a single target, is separated vacuously, and its data contain the anchor $\{A_e, B_e\}$ with positive probability. *Reduction.* Let $S := \{e : \exists t \text{ Pr}[q_e \in \text{Acc}_{t'} \text{ for some } t' \leq t] > \frac{1}{2}\}$. If $e \notin K$, (b) for $\mathcal{P}_e$ gives eventual acceptance of $q_e \in R_{\mathcal{P}_e}^*$ with probability $\geq 1 - \delta > \frac{1}{2}$; by continuity along the increasing events “accepted by time t ”, some single t has probability $> \frac{1}{2}$, so $e \in S$. If $e \in K$, (a) for $\mathcal{P}'_e$ (with $R^* = \{A_e, B_e\}$, $\text{Res}_\infty = \emptyset$, $q_e \notin R^*$) gives $\text{Pr}[q_e \text{ ever accepted}] \leq \delta < \frac{1}{2}$, so $e \notin S$. Hence $S = \omega \setminus K$. The learner, the data law and the oracle are computable uniformly in e (Ref_d consists of the sentences of size $\leq d$ refuted with numeral bounds d , decidable uniformly in d), and $\mathcal{P}_e, \mathcal{P}'_e$ have the same data law and oracle; so the probability that q_e is accepted by time t is computable from (e, t) (for randomized learners, approximable from below), and S is Σ_1 . But $\omega \setminus K$ is not Σ_1 . $\square$

The first version of this construction used Kleene’s T predicate, whose bounded universal quantifiers $\text{R-}\Delta_0$ cannot verify; the referee found this gap, and the MRDP form above repairs it. Alternatively, keep the Kleene- T form and take as oracle the full evaluation of closed Δ_0 sentences with bounded quantifiers as primitives, which is also sound and decidable per depth.

Proof of Proposition 6.18. Non-transitivity and policy dependence: the example after the proposition (all data quantifier-free, so the minimal templates are Plotkin lggs). DTRC merges only coherent unions, so its clusters stay coherent, apart from a singleton $\{m\}$ with $m \in \text{Ref}_d$, which no merge involves since every covering template of a set containing m has m as an instance. Under $\text{RS}_d(D)$ the subsets of the D_i are coherent (Theorem 6.11(i)) and every other set is not,

so the pairwise-coherence graph is a disjoint union of cliques D_i , and all three procedures return $\{D_i\}$. $\square$

Proof of Proposition 6.19. (a) $m \in \text{inst}(T)$ for every $T \in \text{Cov}(X)$ with $m \in X$. (b) A template covering clean $a \in D_i$ and $b \in D_j$ covers $\{a, b\}$ and is d -refuted. (c) By definition; Example E.8 gives an instance. (d) Example E.8. (e) First, the definition of $\text{Acc}_d^e(C)$ agrees with the finite form

$$\text{Acc}_d^e(C) = \bigcap_{E' \subseteq C, |E'| \leq e} \bigcap_{T \in \text{Min}_d^+(C \setminus E')} \text{inst}(T) :$$

every unrefuted T with $|C \setminus \text{inst}(T)| \leq e$ covers $C \setminus E'$ for $E' := C \setminus \text{inst}(T)$ and lies above some $T_0 \in \text{Min}(C \setminus E')$, which is unrefuted; conversely each such T_0 is an unrefuted template missing at most e data ($C \setminus E' \neq \emptyset$ since $|C| > e$). Soundness: take E' to be the mistakes; $C \setminus E' \subseteq D_i$, so Proposition 6.2(c) gives an unrefuted $T_0 \in \text{Min}(C \setminus E')$ with $\text{inst}(T_0) \subseteq \text{inst}(\sigma_i)$. Exactness: for each E' , $C \setminus E'$ contains an anchor, so every member of $\text{Min}(C \setminus E')$ contains $\text{inst}(\sigma_i)$. *Robust DTRC.* Refutable and noise-separated mistakes never join a cluster with clean data, so by (b) and Theorem 6.11(ii) the clean data form the clusters D_i , and clusters of mistakes have multiplicity $< s$ and are not asserted. Each asserted D_i has no mistakes, so (e) applies with zero mistakes. If noise-separation is dropped, each asserted cluster still has clean data of one target (by (b)) and, by assumption, at most e mistakes, so (e) gives soundness and the stated exactness per cluster. $\square$

Example E.8 (noisy PA). [computed; reproduced by the referee]^{untagged u8} To 60 clean PA data add three refutable mistakes (e.g. $p + 0 = Sp$), two unrefutable false ones (J_0^* , and the wrong-base variant $(\neg S0 = 0 \wedge \forall x(\neg x = 0 \rightarrow \neg Sx = 0)) \rightarrow \forall x \neg x = 0$, whose refutation needs Q1), and two true non-targets ($0 + p = p$, $\neg Sp = p$). The refutable mistakes stay incoherent singletons; the other mistakes stay singletons of multiplicity 1, since their merges with the diverse induction cluster are refuted, so with $s = 2$ none is asserted. *Absorption:* J_0^* together with the non-diverse data $\text{Ind}(S^n 0 + x = S^n x)$, $n < 3$, is coherent, with the single unrefuted minimal template $(f(0) + 0 = f(0) \wedge \forall x(f(0) + x = f(x) \rightarrow f(0) + Sx = Sf(x))) \rightarrow \forall x(f(0) + x = f(x))$, which is K_0 -shaped and not below T_{Ind} (unique among 999 covering templates of size ≤ 20 , by the referee's enumeration). The plain verifier accepts the false J_0^* and $J_1^* = (S0 + 0 = S0 \wedge \forall x(S0 + x = S0 \rightarrow S0 + Sx = SS0)) \rightarrow \forall x(S0 + x = S0)$; the trimmed verifier with $e = 1$ rejects both and accepts the genuine $\text{Ind}(S^3 0 + x = S^3 x)$. *Fragmentation:* in the order $\text{Ind}(0 + x = x)$, J_0^* , $\text{Ind}(S0 + x = Sx)$, then three diverse inductions, DTRC returns the unsound cluster $\{\text{Ind}(0 + x = x), J_0^*, \text{Ind}(S0 + x = Sx)\}$ and a cluster of the diverse inductions; with the diverse data first it puts all five inductions together and leaves J_0^* alone. Merging most specific or largest clusters first is a heuristic remedy, not a theorem.

E.5 Complexity of untagged verification and of the refutation test

Proposition E.9 (cost of a test). [proved; computed]^{untagged Prop 5.10} (a) Given $\text{Min}(X)$, a d -coherence test needs at most $|\text{Min}(X)|$ refutation checks. (b) If X is quantifier-free, every covering template is a first-order pattern and $\text{Min}(X) = \{\text{lgg}(X)\}$ (Plotkin's lgg with 0-ary term and formula variables), computable in polynomial time. (c) If $X \subseteq \text{inst}(\sigma)$ contains an anchor of σ , then, under (Rich), $\text{Min}(X) = \{\sigma\}$, computable from X in polynomial time. (d) For one negative n : some $T \in \text{Min}(X)$ avoids n iff $n \notin \text{Acc}(X)$, decidable in polynomial time. (e) In every coherence test of the PA runs (63 + 102 + 98 tests) and ZF runs (164 + 131 + 147 tests) of Section 6.11, $|\text{Min}(X)| = 1$; the worst case met was in low-diversity induction pairs ($|\text{Min}|$ up to 27).

Proof. (a) By definition. (b) The data have no binders, so neither has the rigid part of a covering template; no bound variable is in scope, so pattern occurrences, and hence all metavariables, are 0-ary. Plotkin's lgg is least among first-order patterns (Plotkin, 1970; Reynolds, 1970). (c) X is itself an anchor (its covering templates cover the anchor), so the anchor-determines-template result of Section 3 (single Cor G.4) makes σ the least covering template, computed by its polynomial lgg construction. (d) If $n \notin \text{Acc}(X) = \bigcap \text{Cov}(X)$, some covering T misses n , and so does a minimal T_0 below it; the converse is trivial. $\text{Acc}(X)$ is decided in polynomial time by the feature verifier (single Thm C). (e) Computed (untagged u3, u5, u0). $\square$

Features (recalled from Section 3). For a nonempty finite set B of sentences: the *common prefix* $C(B)$ contains the root iff all $d \in B$ have the same root symbol (an index counts as a symbol), and $p \cdot i$ iff $p \in C(B)$ and all $d|_{p \cdot i}$ have the same root symbol; the *slots* are the positions outside $C(B)$ whose parent is in $C(B)$ (or the root, if it is not in $C(B)$); $A(B) := C(B) \cup \text{slots}(B)$; $Y_\sigma(B) := \bigcup_{d \in B} \text{FV}(d|_\sigma)$, the indices free at σ . For positions $\sigma \perp r$ of the same sort and a sentence d , the *matcher* e_d , if it exists, is the unique partial map on $\text{FV}(d|_\sigma)$ with $d|_r = (d|_\sigma)[e_d]$, where $[\bar{u}]$ replaces each free index y by u_y , shifted under the binders inside (*forcing*). The *features* of a sentence q are: $\text{Sym}(p)$, $p \in C(B)$ ($q|_p$ has the common root symbol); $\text{Scope}(\sigma)$, σ a slot ($\text{FV}(q|_\sigma) \subseteq Y_\sigma(B)$); and $\text{Eq}(\sigma, r, \bar{u})$, $\sigma \perp r$, same sort, $\bar{u}$ a map from $Y_\sigma(B)$ to terms in scope at r ($\text{FV}(q|_\sigma) \subseteq Y_\sigma(B)$ and $q|_r = (q|_\sigma)[\bar{u}]$). $\text{Feat}(B)$ is the set of sentences having every feature that all $d \in B$ have. The feature theorem of Section 3 (single Thm C) states $\text{Acc}(B) = \text{Feat}(B)$, decidable in polynomial time. With (1), “ $q \in \text{Acc}_k(D)$ ” is in coNP for every k : a partition into at most k blocks B with $q \notin \text{Acc}(B)$ is a certificate of non-membership.

Lemma E.10 (failure types). [*proved; computed*]^{single Lemma H.1} Fix q . Say that d agrees with q above p if every strict ancestor of p is a position of d with the same root symbol as in q (then p is a position of d). For nonempty finite B , $q \notin \text{Acc}(B)$ iff B fits one of the following types, indexed by positions of q :

- $\text{Sym}(p, h)$, $h \neq \text{root}(q|_p)$: every $d \in B$ agrees with q above p and has root h at p ;
- $\text{Scope}(\sigma, y)$, $y \in \text{FV}(q|_\sigma)$: every $d \in B$ agrees with q above σ , and $y \notin \text{FV}(d|_\sigma)$;
- $\text{Eq}(\sigma, r)$, $\sigma \perp r$ of the same sort: every $d \in B$ agrees with q above σ and above r and has a matcher e_d ; the e_d are pairwise compatible; and either e_q does not exist or it is incompatible with some e_d , $d \in B$.

There are at most $|q|(|\Sigma_D| + 2b_{\max}(q)) + |q|^2$ types (Σ_D the non-index symbols of the data, $b_{\max}(q)$ the binder depth of q). For each type, “ B fits” is a conjunction of: $B \subseteq$ a fixed allowed set; B a clique of a fixed compatibility graph (Eq-types); B meets a fixed witness set (only for Eq-types where e_q exists).

Proof. ($\Rightarrow$) q lacks some B -feature; take the first applicable case. If q lacks $\text{Sym}(p)$ for some $p \in C(B)$, let p' be the topmost position on the path to p where q 's root differs from B 's common root h' ; its strict ancestors agree with q , so p' is a position of q , and B fits $\text{Sym}(p', h')$. If q has every Sym feature but lacks $\text{Scope}(\sigma)$, the strict ancestors of σ lie in $C(B)$ with q 's symbols; any $y \in \text{FV}(q|_\sigma) \setminus Y_\sigma(B)$ gives the type $\text{Scope}(\sigma, y)$. If q has every Sym and Scope feature but lacks $\text{Eq}(\sigma, r, \bar{u})$, each e_d exists and, by forcing, is the restriction of $\bar{u}$ to $\text{FV}(d|_\sigma)$, so the e_d are pairwise compatible; also $\text{FV}(q|_\sigma) \subseteq Y_\sigma(B) = \text{dom } \bar{u}$ and $q|_r \neq (q|_\sigma)[\bar{u}]$. So either e_q does not exist, or $e_q(y) \neq \bar{u}(y) = e_d(y)$ for some y and some d with $y \in \text{FV}(d|_\sigma)$.

($\Leftarrow$) By induction on the size of $q|_\sigma$ (resp. $q|_p$). The named positions lie in $A(B)$, since their strict ancestors lie in $C(B)$ with q 's symbols. $\text{Sym}(p, h)$: $p \in C(B)$, and q lacks $\text{Sym}(p)$.

Scope(σ, y): if σ is a slot, q lacks Scope(σ). If $\sigma \in C(B)$ with common root g and q 's root differs, Sym(σ) fails. Otherwise g is not an index ($g = y$ would put y into every $FV(d|_\sigma)$; another index would give $FV(q|_\sigma) = \{g\} \not\subseteq y$), so y occurs, shifted if g binds, in some child $q|_{\sigma \cdot i}$, and B fits Scope($\sigma \cdot i, y'$); apply induction. Eq(σ, r): if σ is a slot, $\bar{u} := \bigcup_d e_d$ is well defined on $Y_\sigma(B)$ by compatibility, Eq($\sigma, r, \bar{u}$) is a B -feature, and q has it only if e_q exists and $e_q \subseteq \bar{u}$, which the type excludes. If $\sigma \in C(B)$ with common root g and q 's root differs, Sym(σ) fails. If g is an index (free at σ), then $d|_r = e_d(g)$ for each d , and compatibility makes all $d|_r$ equal to one term u , so r and the positions of u below it lie in $C(B)$; e_q exists with $e_q(g) = q|_r$, and the type forces $q|_r \neq u$, so a Sym feature fails inside u . If g is a non-index symbol, every $d|_r = (d|_\sigma)[e_d]$ has root g , so $r \in C(B)$, and if $q|_r$ has another root, Sym(r) fails. Otherwise pass to the children: the restricted matchers $e_d^{(i)}$ exist and are pairwise compatible (a local index bound by g is mapped to itself). The failure of e_q shows at some child i and some index z free in $q|_{\sigma \cdot i}$: q 's child matcher does not exist, maps z to a wrong value, or two children of q disagree on z . If some $d \in B$ has z free in $d|_{\sigma \cdot i}$, compatibility fixes the correct value and B fits Eq($\sigma \cdot i, r \cdot i$) (in the disagreement case, at the child with the wrong value); otherwise B fits Scope($\sigma \cdot i, z$). Apply induction.

The count: Sym-types are indexed by a position of q and a symbol of the data or one of fewer than $b(p)$ indices, Scope-types by a position and an index, Eq-types by a pair of positions. The conditions are read off the definitions. $\square$

Computed (single e11): on 600 random cases (mixtures of 1–3 random DT^o templates with low-diversity bodies, 3–6 data), “ $q \notin \text{Acc}(B)$ ” agreed with “ B fits a type” on all 118,818 pairs (B, q) , with 15,220 Eq-types having non-trivial compatibility graphs.

Proof of Theorem 6.20(a). $k = 1$ is the feature theorem. For $k = 2$: by (1), $q \notin \text{Acc}_2(D)$ iff $q \notin \text{Acc}(D)$ or there are types τ_1, τ_2 and a partition $D = B_1 \sqcup B_2$ with B_l fitting τ_l . Fix (τ_1, τ_2) and, for Eq-types that need one, witnesses $w_l \in B_l$ (polynomially many tuples). With variables x_d (“ $d \in B_1$ ”), “ B_l fits τ_l ” is a conjunction of clauses with at most two literals: $\neg x_d$ for d not allowed by τ_1 ; x_d for d not allowed by τ_2 ; $\neg x_d \vee \neg x_{d'}$ for d, d' incompatible for τ_1 ; $x_d \vee x_{d'}$ for d, d' incompatible for τ_2 ; and $x_{w_1}, \neg x_{w_2}$. 2-SAT is solvable in linear time (Aspvall et al., 1979). The algorithm first checks $q \in \text{Acc}(D)$; after that, a solution with an empty block would make D fit a type, i.e. $q \notin \text{Acc}(D)$, so every solution is a genuine 2-partition. $\square$

Proof of Theorem 6.20(b). Membership in coNP was noted above. Hardness: reduction from graph k -colourability, NP-complete for each fixed $k \geq 3$ (Karp, 1972; Garey and Johnson, 1979). Let $G = (V, E)$ with $V = \{1, \dots, n\}$, $E = \{e_1, \dots, e_m\}$, each edge oriented (tail, head). Under m binders $\forall y_1 \dots \forall y_m$ put $\text{tup}_u := t_{u,1} + (t_{u,2} + (\dots + (t_{u,m} + 0)))$ with $t_{u,i} = y_i$ if $u \in e_i$ and 0 otherwise; let c_u map y_i to 0 if u is the tail of e_i and to $S0$ if it is its head; and put

$$d_u := \forall \bar{y} (S^u(\text{tup}_u) = 0 \wedge S^u(\text{tup}_u[c_u]) = 0), \quad q := \forall \bar{y} (S^n(0) = 0 \wedge S^{n+1}(0) = 0),$$

of total size $O(n(n + m))$. Let σ_w, r_w be the positions at S -depth w in the left-hand sides of the two conjuncts. *Claim:* for nonempty $B \subseteq D$, $q \notin \text{Acc}(B)$ iff the vertices of B form an independent set. Then by (1), $q \notin \text{Acc}_k(D)$ iff V splits into at most k independent sets, iff G is k -colourable. *Independent* $\Rightarrow q \notin \text{Acc}(B)$. If $|B| = 1$, $\text{Acc}(\{d_u\}) = \{d_u\} \not\subseteq q$. If $|B| \geq 2$, let w be the least vertex of B ; all $d_u \in B$ have S above depth w on both sides, and at depth w the datum d_w has root $+$ and the others S , so σ_w, r_w are slots. For $u \in B$, $d_u|_{r_w} = (d_u|_{\sigma_w})[c_u]$. As no y_i is free in two members (independence), the restrictions of the c_u are compatible, and their union $\bar{u}$ makes Eq($\sigma_w, r_w, \bar{u}$) a B -feature. But $q|_{\sigma_w} = S^{n-w}(0)$ and $q|_{r_w} = S^{n+1-w}(0)$ are closed and different. An edge $\{u, v\} \subseteq B \Rightarrow q \in \text{Acc}(B)$. Acc is monotone in the data, so it suffices

that $q \in \text{Feat}(\{d_u, d_v\})$. With $w = \min(u, v)$, $C(\{d_u, d_v\}) = \forall^m(S^w(\square) = 0 \wedge S^w(\square) = 0)$ with slots σ_w, r_w ; q has every Sym feature ($n > w$) and every Scope feature (its contents there are closed). Eq-features with source σ_w : for (σ_w, r_w) , the edge's index y_e is free at σ_w in both data and $c_u(y_e) \neq c_v(y_e)$, so by forcing there is no common $\bar{u}$; for the other admissible term positions $p \perp \sigma_w$ (the right-hand 0's and the S -positions of the second conjunct above r_w), the datum with vertex w has root $+$ at σ_w , and after any substitution, but 0 or S at p . Eq-features with source r_w have empty scope (the c_u are closed), so they are equalities; the datum with vertex w has root $+$ at r_w , the candidates carry 0 or S , except σ_w , which contains a free index while r_w is closed. So the only features are Sym and Scope features, which q has. $\square$

Proof of Theorem 6.20(c). (i) Reduction from Set Cover (Karp, 1972): universe $\{1, \dots, n\}$, sets $F_1, \dots, F_t$, bound k . Let $d_u := \bigwedge_{j \leq t} (c_{uj} = 0)$ with $c_{uj} = 0$ if $u \in F_j$ and $S0$ otherwise, and $q := \bigwedge_{j \leq t} (S0 = 0)$. If some u lies in no F_j , then $d_u = q \in D$, there is no cover, and $q \in \text{Acc}_k(D)$, consistently. Otherwise read off the types of q (Lemma E.10); let ℓ_j be the left-hand side of conjunct j , where data have 0 or S and q has S . The type $\text{Sym}(\ell_j, 0)$ has allowed set exactly $\{d_u : u \in F_j\}$. There are no other Sym-types (below an S at ℓ_j data and q both have 0) and no Scope-types (no indices). All matchers are empty maps, so an Eq-type needs $q|_\sigma \neq q|_r$; the pairs with $q|_\sigma \neq q|_r$ that data can satisfy are $(\ell_j, \text{a 0-position})$, its converse, and $(\ell_j \cdot 1, \ell_{j'})$ with $u \notin F_j$ (and its converse); each forces $d|_{\ell_{j'}} = 0$ for some j' , so its allowed set lies inside some $\{d_u : u \in F_{j'}\}$. Formula-sort pairs are void. So a block fails iff it lies inside some $\{d_u : u \in F_j\}$, and $q \notin \text{Acc}_k(D)$ iff k of the sets cover the universe. All data are quantifier-free, so all covering templates are first-order patterns. (ii) For binder-free D every matcher is the empty map, so Eq-types have no incompatible pairs and need no witness unless e_q exists, in which case it is compatible with every e_d and the type is void. So every type has the form “ $B \subseteq$ allowed set”, there are polynomially many, and $q \notin \text{Acc}_k(D)$ iff D is covered by at most k allowed sets, which for fixed k is decided by trying all k -tuples. $\square$

Proof of Theorem 6.20(d). In NP. Some $T \in \text{Min}(X)$ avoids R iff some covering template does (it lies above a minimal one), iff some saturated covering template does (minimal templates are saturated, Section 3). A saturated template is determined by an occurrence set inside $A(X)$ and a pattern slot per metavariable, a certificate of polynomial size, checked by matching. NP-hard (from CNF-SAT; Garey and Johnson, 1979). Take variables $x_1, \dots, x_n$, distinct parameters $a_1, \dots, a_n$, and $X = \{d_1, d_2\}$ with $d_1 := \forall x \bigwedge_j (x = a_j)$, $d_2 := \forall x \bigwedge_j (a_j = x)$. The common prefix is everything except the left-hand sides σ_j (column (x, a_j) , scope $\{x\}$); the right-hand sides z_j carry the rigid a_j . Besides Sym and Scope features, the X -features are exactly $\text{Eq}(\sigma_j, z_j, x \mapsto a_j)$ ($\sigma_i \rightarrow \sigma_j$ fails on d_2 as $a_i \neq a_j$; $\sigma_i \rightarrow z_j$ would need $a_i = a_j$; there are no formula slots). So the saturated covering templates are $T_A := \forall x \bigwedge_j (f_j(x) = \zeta_j)$, $A \subseteq [n]$, with $\zeta_j = f_j(a_j)$ for $j \in A$ and $\zeta_j = a_j$ otherwise; they are pairwise incomparable ($f_j := \lambda z. Sz$ separates them), so $\text{Min}(X) = \{T_A\}$. For a clause C (w.l.o.g. not containing both x_j and $\neg x_j$) put $n_C := \forall x \bigwedge_j \gamma_j$ with $\gamma_j = (x = a_j)$ if x_j does not occur in C , $\gamma_j = (Sx = a_j)$ if $x_j \in C$, and $\gamma_j = (Sx = Sa_j)$ if $\neg x_j \in C$. By matching, where $\gamma_j = (x = a_j)$ the matcher takes $f_j := \lambda z. z$ and both choices of ζ_j fit; where the left side is Sx it takes $f_j := \lambda z. Sz$, and then $\zeta_j = a_j$ fits only $Sx = a_j$ while $\zeta_j = Sa_j$ fits only $Sx = Sa_j$. So $n_C \in \text{inst}(T_A)$ iff the assignment “ x_j true iff $j \in A$ ” falsifies C , and some $T \in \text{Min}(X)$ avoids $R := \{n_C : C \in \varphi\}$ iff φ is satisfiable. The size is $O(n|\varphi|)$. *Polynomial cases.* Whether X has a least covering template is decidable in polynomial time, and then it is computed (single Prop G.3), and $\text{Min}(X)$ is that template; under (Rich) an anchor gives one (single Cor G.4). If $r \in R \cap \text{Acc}(X)$, every covering template has r as an instance. $\square$

In this reduction d_1 is false, so the hardness does not transfer to DTRC’s use (true data, false negatives in a fixed world); that case is open. Computed (single e11, e14): the 2-SAT verifier equals the brute-force partition verifier on all 5,918 queries; for 12 graphs (including K_3 , C_5 , K_4 , the Petersen graph) $q_G \in \text{Acc}_k(D_G)$ iff $\chi(G) > k$ for $k = 1, \dots, 4$; on 25 random set systems $q \in \text{Acc}_k(D)$ iff no k sets cover; for $n \leq 6$, $|\text{Min}(X)| = 2^n$ in the SAT reduction, and the test equals satisfiability on 360/360 random CNFs. These results were added to the single track’s record after its referee round and were not refereed independently.

E.6 MDL

For a finite set H of templates and data D , each datum d is assigned a covering template $T_d \in H$ (the cheapest if several), with unique matcher θ_d , and $L(H, D) = \sum_{T \in H} L_{\text{tmpl}}(T) + \sum_{d \in D} [L_{\text{idx}}(T_d) + L_{\text{body}}(\theta_d \mid T_d)]$. Without the body term this is the parent report’s observation that MDL without likelihood picks the bare metavariable (Claude (Anthropic), 2026). The split hypothesis is $H_F = \{Q1, \dots, Q7\} \cup \{T_f : f \in F\}$ with F the main connectives of the induction data and $T_f := T_{\text{Ind}}[P := \lambda x.f(P_1(x), \dots)]$ (for “=”: $\lambda x.t_L(x) = t_R(x)$; for $\forall, \exists$: $\lambda x.\forall y P_1(x, y)$). For a datum $\text{Ind}(\varphi)$ with root f the matcher of T_f has bodies of total size $|\varphi| - 1$; under T_{Ind} the body is $\lambda x.\varphi$.

Proposition E.11 (a naive code over-splits). *[proved]^{untagged Prop 6.2} Let $L_{\text{tmpl}}(T) = \lambda|T|$ and $L_{\text{body}}(\theta) = \lambda \sum_M |\theta(M)|$ with $\lambda = \log_2 A$, A the number of symbols; let L_{idx} be the two-part maximum-likelihood code of the label sequence (empirical entropy plus $\frac{|H|-1}{2} \log_2 n + O(1)$, e.g. the KT code (Krichevsky and Trofimov, 1981)); let n_{Ind} be the number of induction data and $\hat{H}$ the empirical entropy of their main connectives. Then*

$$L(H_F, D) - L(H_{\text{true}}, D) = \lambda \left(\sum_{f \in F} |T_f| - |T_{\text{Ind}}| \right) + \frac{|F| - 1}{2} \log_2 n - n_{\text{Ind}}(\lambda - \hat{H}) + O(1).$$

Since $\hat{H} \leq \log_2 |F| < \lambda$, the difference tends to $-\infty$ linearly in n_{Ind} . With a uniform index code ($\log_2 |H|$ bits per datum) the same holds once the induction frequency exceeds $\log_2((7 + |F|)/8)/\lambda$. Applied recursively, naive MDL keeps splitting each class by its next symbol while the class has enough data.

Proof. Template part: as displayed. Body part: each induction datum costs λ fewer bits under its T_f , for every root, $\forall$ and $\exists$ included. Index part: the split labels refine the true labels by the root of the induction data, so their empirical entropy is $n\hat{H}(\text{true labels}) + n_{\text{Ind}}\hat{H}$ (chain rule), and the parametric term grows by $\frac{|F|-1}{2} \log_2 n$. Ground axioms cost the same under both hypotheses. For the uniform code, the index cost grows by $n \log_2((7 + |F|)/8)$ and the body cost falls by $n_{\text{Ind}}\lambda$. $\square$

Proposition E.12 (well-specified codes gain at most $O(\log n)$). *[proved, using standard coding facts cited from memory]^{untagged Prop 6.3} Let the data be i.i.d. from a law P_0 (template index and instance law), and let the code for H_{true} be a Bayesian mixture over an r -parameter family containing P_0 with regret at most $\frac{r}{2} \log_2 n + C$ (Rissanen, 1986; Clarke and Barron, 1990). Then for every fixed split hypothesis and $K > 0$, $P[L(H_F, D) \leq L(H_{\text{true}}, D) - \frac{r}{2} \log_2 n - K - C'] \leq 2^{-K}$, for a constant C' .*

Proof. The data part of $L(H_F, \cdot)$ is a prefix code for the data sequence given the template list, i.e. a sub-probability Q ; by the no-hypercompression inequality (see Grünwald, 2007), $P[-\log_2 Q(D) \leq -\log_2 P_0(D) - K] \leq 2^{-K}$. The data part of $L(H_{\text{true}}, \cdot)$ is at most $-\log_2 P_0(D) + \frac{r}{2} \log_2 n + C$, and the template parts are constants; take C' to absorb C and the template parts. $\square$

The referee checked both proofs; the regret and no-hypercompression facts are standard but were cited from memory in the research record and not re-checked there.

Table 11: $L(H_{\text{root}}) - L(H_{\text{true}})$ in bits on PA data (Q and induction, induction frequency 0.5); negative means that MDL splits induction by main connective. NAIVE: Proposition E.11 with $A = 23$; PC: sequential KT per template with context (parent symbol, child index); DPC: KT with context (depth below the motive root, parent, child index). G1: a natural usage law (random motives with quantifiers; scope-dependent terms); G2: a law for which DPC is well specified (symbols depend only on depth, parent and index). Computed (untagged u7), reproduced exactly by the referee.

usage law	n	NAIVE	PC	DPC
G1 natural	1000	−223	+2325	+2701
	16000	−16466	−1199	−3036
	64000	−68243	−19783	−34412
G2 well specified for DPC	1000	−915	+1141	+1428
	16000	−21467	+1053	+2822
	64000	−87156	−1522	+3564

E.7 Natural examples: details

Proposition E.13 (a passing run is DTRC with a fixed oracle). *[proved]^{untagged Prop 5.12} The implementation keeps a global set N of refuted sentences; a template is reported refuted iff it covers some $n \in N$ or its own budgeted search finds a refuted instance, which is then added to N . Every coherence decision is logged. A run on data D (fixed order and policy) ending with N_{final} passes the audit if every “coherent” decision had a minimal template covering no element of N_{final} , and every asserted template avoids N_{final} . If the sentence refuters are sound, then $\text{Ref}_d := N_{\text{final}}$ satisfies (OS) and (Dec), and the run’s final clusters and asserted templates coincide with those of DTRC_d with this fixed oracle on the same data, order and policy.*

Proof. (OS) holds because N_{final} consists of sentences refuted by sound procedures; (Dec) holds by matching. The merge policy is a deterministic function of the sequence of coherence answers, so it suffices that each answer agrees with the fixed oracle: “incoherent” answers agree because $N \subseteq N_{\text{final}}$ at every time (each refuted minimal template covers its refuting sentence), and “coherent” answers agree by the audit. The asserted templates are the members of $\text{Min}(C)$ not reported refuted: every excluded member covers an element of N_{final} , and every included one avoids N_{final} by the audit. $\square$

Refuter soundness is supported by code reading and by the referee’s fuzzing (0 of 3000 true arithmetic and 1206 true set-theoretic sentences refuted). Every DTRC run of the record passed the audit on the first pass (24 runs checked), and the re-audit after the separation check found no inconsistent decision. Caveats: completeness of the computed minimal templates was checked against brute-force enumeration (by the author on induction pairs; by the referee’s independent enumerator on all PA and ZF cross pairs, several within-schema sets, the K_0 pair and the J_0^* clusters), always with no covering template outside the computed ones; internal caps never changed a result; templates are matched literally, not modulo renaming of their parameters, which can only under-accept.

Table 12: PA cross merges: the unique minimal covering template of each cross pair and an R- Δ_0 refutation (computed, untagged u1; the referee’s independent enumeration of all covering templates of size ≤ 14 agrees). Q_i^s is the instance schema of Q_i (closed terms for the parameters), an alternative presentation of Q .

cross pair	minimal template	refuting instance
different roots (Q1–Q3 etc.); Q_i –Ind, root $\neq \rightarrow$	F_0	$0 = S0$
Q2–Q7, Q2–Ind, Q7–Ind	$F_0 \rightarrow F_1$	$0 = 0 \rightarrow 0 = S0$
Q3–Q5, Q3–Q6, Q4–Q5, Q4–Q6	$z_0 = z_1$	$0 = S0$
Q3–Q4	$p + z_0 = z_1$	$p + 0 = 0$ at $p := 1$
Q5–Q6	$p \cdot z_0 = z_1$	$p \cdot 0 = S0$ at $p := 0$
Q_i^s – Q_j^s (21 pairs, 4 samples each)	various	all refuted

In the instance-schema presentation, one $Q2^s$ sample lacked an S -free instance and gave the sound specialization $SSz_0 = Sz_1 \rightarrow Sz_0 = z_1$. The PA near-miss set had 104 sentences (50 step-by-two and 50 wrong-base induction variants, 4 mutated axioms); 13 variants are genuine inductions (vacuous motives) and were accepted, the other 91 rejected. Held-out data come from other seeds; the referee found no leakage.

Example E.14 (ZF cross merges). [computed; independently confirmed by the referee]^{untagged u4, u13} The ZF targets in closure-normal form: Ext $\forall x(x \in p \leftrightarrow x \in p') \rightarrow p = p'$; Pair $\exists c(p \in c \wedge p' \in c)$; Union $\exists b \forall c(c \in b \leftrightarrow \exists d(d \in p \wedge c \in d))$; Power $\exists b \forall c(c \in b \leftrightarrow \forall d(d \in c \rightarrow d \in p))$; Inf (empty set and successor written out with $\in$); Found $\exists y(y \in p) \rightarrow \exists y(y \in p \wedge \forall z(z \in y \rightarrow \neg z \in p))$; Separation $\exists b \forall c(c \in b \leftrightarrow c \in p \wedge P(c))$; Replacement $\forall x(x \in p \rightarrow \exists y(P(x, y) \wedge \forall z(P(x, z) \rightarrow z = y))) \rightarrow \exists b \forall y(y \in b \leftrightarrow \exists x(x \in p \wedge P(x, y)))$; $\in$ -induction $\forall x(\forall y(y \in x \rightarrow P(y)) \rightarrow P(x)) \rightarrow \forall x P(x)$. All 36 cross pairs (one instance per schema) have a unique minimal covering template: F_0 or $F_0 \rightarrow F_1$ in 22 pairs, refuted by an HF counterexample ($\neg p = p$, resp. $p = p \rightarrow \neg p = p$, at $p := \emptyset$); in 14 pairs a template keeping quantifier structure, refuted by logic: $\exists x F_0(x)$ (Pair–Union, Pair–Power, Pair–Sep, Union–Inf, Power–Inf, Inf–Sep; instance $\exists x \neg p = p$), $\exists x(F_0(x) \wedge F_1(x))$ (Pair–Inf), $(\forall x F_0(x)) \rightarrow F_1$ (Ext–Rep, Ext–EInd), $F_0 \rightarrow \exists x F_1(x)$ (Found–Rep), $(\forall x(F_0(x) \rightarrow F_1(x))) \rightarrow F_2$ (Rep–EInd), and naive comprehension $\exists x \forall y(y \in x \leftrightarrow F_0(y))$ (Union–Power, Union–Sep, Power–Sep), refuted through Russell’s instance (Herbrand: $y := x$). The referee’s independent enumerator listed all 598 covering templates of size ≤ 14 of 63 ZF cross pairs (two samples per schema): none outside the computed minimal ones, all refuted with the global-negative-set oracle (with the first implementation, 120 of them had been reported unrefuted, all above refuted naive comprehension). Within each schema, three random instances have the schema as unique minimal template (confirmed by the referee for Separation, $\in$ -induction and Replacement).

Example E.15 (DTRC on ZF and ZFC). [computed]^{untagged u5, u10} ZF (three seeds of 45 data): besides the numbers in Section 6.11, 60/60 held-out instances of each schema were accepted, separation on the data was verified on all 467, 371 and 348 cross pairs, and $|\text{Min}| = 1$ in all 442 tests. ZFC: in closure-normal form, with p the family,

$$\begin{aligned} \text{AC}(p) = & [\forall y(y \in p \rightarrow \exists z z \in y) \wedge \forall y \forall w((y \in p \wedge w \in p \wedge \neg y = w) \rightarrow \neg \exists z(z \in y \wedge z \in w))] \\ & \rightarrow \exists c \forall y(y \in p \rightarrow \exists z(z \in y \wedge z \in c \wedge \forall u((u \in y \wedge u \in c) \rightarrow u = z))). \end{aligned}$$

AC is true in V and not refuted. Its 12 cross pairs (two samples per schema) each have one minimal template: $F_0 \rightarrow F_1$ (with Ext and EInd) and F_0 (with Pair, Union, Power, Inf, Sep),

refuted by an HF counterexample; $F_0 \rightarrow \exists x F_1(x)$ (with Found) and $F_0 \rightarrow \exists x \forall y F_1(y, x)$ (with Rep), refuted by logic. On ZFC data (three seeds of 50): 10 pure clusters, schema clusters exact, AC accepted, 180/180 held-out schema instances, separation verified on 614, 294 and 428 cross pairs. *Weak Union and Power* (part B): with logic and HF only, the two weak forms merge into the universal-set cluster and $\exists x \forall y (y = y \rightarrow y \in x)$ is accepted (8 clusters); with the closure of Russell’s Separation instance designated (bootstrapped coherence), 9 pure clusters and the universal set rejected. Over the UnionW–PowerW pair, the referee’s enumerator found 81 of 91 covering templates unrefuted, all above the universal-set schema, as Proposition 6.21 predicts.

K_0 . $K_0 = (z_1 + 0 = z_1) \wedge \forall x (z_1 + x = z_2 \rightarrow z_1 + Sx = Sz_2) \rightarrow \forall x (z_1 + x = z_2)$ is the first-order lgg (named encoding) of $\text{Ind}(0 + x = x)$ and $\text{Ind}(S0 + x = Sx)$. The prior work’s referee proved that the true closed quantifier-free sentences of $\mathbb{N}$ together with all sentence instances of K_0 are consistent (by building, for each instance, a model of the quantifier-free diagram in which the step clause fails, and amalgamating); so no sound first-order refutation from quantifier-free truths exists, and in particular $R\text{-}\Delta_0$ and logic refute none of its instances, among them the false J_0^* . In DT° the pair has two minimal covering templates (computed; the referee’s enumerator found no other among 6258 covering templates of size ≤ 24): $T_{\text{Ind}}[P := \lambda x. (f(0) + x = f(x))]$ and the K_0 -shaped template of Example E.8. The intersection verifier excludes J_0^* (it is not an instance of the first). The K_0 -shaped template is refuted by coherence with designated Q (a derivation found in the prior work: the descent through Q3 is unblocked once Q3 is designated).

Computations. The scripts are in `research/tracks/untagged/code` (Python 3.11, no external packages; `shrun_all.sh`, about 15 minutes), with outputs `<script>.out`: u0 (minimal templates vs brute force), u1 (Table 12), u2 ($c = 4/7/7$), u3 (DTRC on PA), u4, u5, u5b (ZF cross merges, DTRC on ZF, weak forms), u6 ($\forall x \varphi$ from numerals; non-transitivity), u7 (Table 11), u8 (Example E.8), u9 (clean fragmentation), u10 (ZFC), u12 (head classes), u13, u14 (monotone oracle, audit). Theorem H’s checks are `research/tracks/single/e11_union_complexity.py` and `e14_dtrc_step2.py`; the referee’s independent code is in `research/tracks/untagged/referee_code`.

Credits. Gold’s theorem (Gold, 1967); lgg and witness events (Plotkin, 1970; Reynolds, 1970); finite elasticity of bounded unions (Wright, 1989; Motoki et al., 1991); minimal multiple generalizations for unions of pattern languages (Arimura et al., 1994) (overlap with Theorem 6.20 not checked); MRDP (Davis et al., 1961; Matiyasevich, 1993); Δ_0 absoluteness (Kunen, 1980; Jech, 2003); MDL and coding (Rissanen, 1978, 1986; Clarke and Barron, 1990; Krichevsky and Trofimov, 1981; Grünwald, 2007); the size principle (Tenenbaum and Griffiths, 2001); conceptual clustering (Michalski and Stepp, 1983), in whose spirit the clustering is. Covering problems for first-order patterns, relevant to the open spare-slot thresholds, are decidable by results on complement problems and ground reducibility (Lassez and Marriott, 1987; Comon and Jacquemard, 2003) (cited from memory in the research record, not re-checked there). As far as the record knows, the refutation test, the separation theorem, the depth impossibility, the head-class thresholds and the implementation audit are new.

F Implementation and experiments: proofs and details

This appendix proves the statements of Section 7: the normal form for minimal covering templates (Appendix F.1), the soundness of the oracles (Appendix F.2), the guarantees for DTRC and the

separation of the mixtures (Appendix F.3), and the facts about hard pairs and false merges (Appendix F.4). It ends with further numbers (Appendix F.5) and the commands that reproduce them (Appendix F.6). We use the de Bruijn conventions and the substitution facts of Appendix A: plugging is compositional, injective in every hole that occurs in the body, and preserves the root symbol of a body whose root is not a hole; formula bodies never have a hole as root.

F.1 The normal form

Proof of Proposition 7.2. Parameters are compared by name throughout; “covers” means literal covering.

(a) Let N be a normal configuration. An own occurrence $M_\sigma(\bar{y}_\sigma)$ is a pattern occurrence: its arguments are distinct variables bound above σ . Put $\theta_d(M_\sigma) := \lambda\bar{z}.\beta_d(\sigma)$. This is a body, since $\bar{y}_\sigma$ lists every index free in $d|_\sigma$, and abstraction inverts plugging, so the occurrence produces $d|_\sigma$. A derived occurrence $M_{c(\pi)}(\bar{t})$ at π produces $\beta_d(c(\pi))[\bar{t}] = d|_\pi$ by the definition of $\rightarrow$; its arguments are subterms of the data with no metavariables, and their free indices are free in $d|_\pi$ (every hole of $\beta(c(\pi))$ occurs in some $\beta_d(c(\pi))$), so they refer to binders above π . Positions not at or below A carry labels common to all data. Hence $N\theta_d = d$ for every d . Every metavariable has its pattern occurrence at its own slot, so $N \in \text{DT}^\circ$; term metavariables are own only at term slots of $\mathcal{T}(D)$, whose columns are closed, so they are 0-ary and $N \in \text{DT}_F^\circ$.

(b) Let $T \in \text{DT}_F^\circ$ and $d = T\theta_d$ for all $d \in D$. Each of the following steps replaces T by $T\sigma$ for a template substitution σ (so the new template is $\preceq T$), keeps it in DT_F° , and keeps it covering D .

Step 0 (where occurrences sit). By the preservation lemma of Section 2, every position of the skeleton of T carries the same label in T and in every datum. So the skeleton lies in the common part of $C(D)$, and every occurrence position, whose parent is a skeleton position or which is the root, is a common position or a slot of $C(D)$. No occurrence lies strictly below an F-slot ν . Otherwise ν is a skeleton position. Some slot σ below ν has an open column; it is not a skeleton position, because the data disagree there, so some occurrence sits at a position π with $\nu < \pi \leq \sigma$. Then π lies inside the atom at ν , so the occurrence is a 0-ary term metavariable, and $d|_\pi = \theta_d(f)$ is closed for every d . But for some d the subterm $d|_\sigma$ of $d|_\pi$ contains an index bound above σ , hence above ν (atoms contain no binders), hence above π ; so $d|_\pi$ is not closed. Consequently every occurrence of T sits at a node of $\mathcal{T}(D)$; and every slot of $\mathcal{T}(D)$, F-slots included, lies at or below an occurrence (an F-slot in the skeleton would have an occurrence strictly below it).

Step 1 (push down). Let M have a pattern occurrence $M(\bar{z})$ at a node p of $\mathcal{T}(D)$ that is not a slot. Then all data carry the same label h at p , and $\theta_d(M) = \lambda\bar{z}.\beta_d$ with β_d the abstraction of $d|_p$ over $\bar{z}$. Specialize $M := \lambda\bar{z}.B$:

- h a connective or quantifier: $B = h(N_1(\bar{z}_1), \dots)$ with fresh formula metavariables; $\bar{z}_i = \bar{z}$, or under a quantifier the newly bound variable followed by $\bar{z}$;
- h a relation symbol, p not an F-slot: every slot below p has a closed column; B is the common part of the β_d with a fresh 0-ary term metavariable at each such slot;
- h a function symbol (M is then a 0-ary term metavariable): $B = h(f_1, \dots, f_m)$ with fresh 0-ary f_i ;
- h a leaf: B is h if h is 0 or a parameter (the same name in all data), and the hole z_k if h is a bound variable, which must then be among the arguments $\bar{z}$ of M ; M disappears.

The new metavariables take the bodies read off the β_d (children of β_d , or the closed columns at the slots), so the result covers D ; at M ’s other occurrences $M(\bar{t})$ the plugged body is $B[\bar{t}]$,

which agrees with $\beta_d[\bar{t}]$ on the common part, and on the closed parts the values coincide. The new metavariables have pattern occurrences at p 's children or at the slots below p , and new term metavariables are 0-ary. Each step adds the node p of $\mathcal{T}(D)$ to the skeleton, and the skeleton stays inside $\mathcal{T}(D)$ by Step 0, so the process terminates. Afterwards every pattern occurrence sits at a slot of $\mathcal{T}(D)$.

Step 2 (own slots and arguments). For each metavariable M choose a pattern occurrence $M(\bar{y})$, at a slot σ . Every index free in the column at σ is among $\bar{y}$, since the matcher reads $\theta_d(M)$ off this occurrence and bodies have no free index. If some y_m is free in no $d|_\sigma$, the hole z_m occurs in no $\theta_d(M)$; specialize $M := \lambda \bar{z}. M'(\bar{z} \setminus z_m)$. Repeating, the arguments are exactly the indices of $\bar{y}_\sigma$; reordering them gives an equivalent template. So M is M_σ up to renaming, with $\theta_d(M_\sigma) = \beta_d(\sigma)$. Distinct metavariables get distinct own slots, since a position carries one occurrence. Other occurrences of M , pattern or not, are treated as derived.

Step 3 (forced arguments). Let $M_\sigma(\bar{u})$ be an occurrence at π . Then $\beta_d(\sigma)[\bar{u}] = d|_\pi$ for all d . Every hole of M_σ occurs in some $\beta_d(\sigma)$ (Step 2), and plugging is injective in a hole that occurs, so $\bar{u}$ is determined: $\sigma \rightarrow \pi$ and $\bar{u} = \bar{t}(\sigma, \pi)$. This also shows the uniqueness of $\bar{t}(\sigma, \pi)$ claimed in Definition 7.1. Now T is a configuration: S' is the set of own slots, $A = \text{Occ}(T)$ is an antichain of nodes of $\mathcal{T}(D)$ that covers every slot (Step 0), and each derived occurrence names its source. Only the independence of S' may fail.

Step 4 (independence). Let $\sigma \in S'$ be derivable from $\rho \in S' \setminus \{\sigma\}$, with $\bar{t} = \bar{t}(\rho, \sigma)$. The free indices of $\bar{t}$ are free in some $d|_\sigma$ (every hole of $\beta(\rho)$ occurs in some $\beta_d(\rho)$), so they are among $\bar{y}_\sigma$, and $\lambda \bar{z}. M_\rho(\bar{t}')$, with $\bar{t}'$ the abstraction of $\bar{t}$ over $\bar{y}_\sigma$, is a template body. Specialize M_σ to it. An occurrence $M_\sigma(\bar{u})$ at π becomes $M_\rho(\bar{t}'[\bar{u}])$, and $\beta_d(\rho)[\bar{t}'[\bar{u}]] = (\beta_d(\rho)[\bar{t}'])[\bar{u}] = \beta_d(\sigma)[\bar{u}] = d|_\pi$; so by Step 3 it is the derived occurrence $M_\rho(\bar{t}(\rho, \pi))$. M_ρ keeps its own occurrence, and $|S'|$ drops by one. Iterating gives a normal configuration $N \preceq T$.

(c) There are finitely many own sets, antichains and choices, and $\bar{t}(\sigma, \pi)$ is unique, so there are finitely many normal configurations; let $\text{Min}_F^{\text{lit}}(D)$ be the minimal ones, one per $\equiv$ -class. A covering T is $\succeq N$ for a normal N by (b), and N lies above a minimal normal configuration. If a covering T' satisfied $M \succ T'$ for some $M \in \text{Min}_F^{\text{lit}}(D)$, then by (b) $T' \succeq N_1$ for a normal N_1 , and $M \succ N_1$ (if $N_1 \succeq M$ then $T' \succeq M$), contradicting the minimality of M among normal configurations. The last clause is (b) applied to T^* . $\square$

For parameter-free T^* literal covering and covering up to renaming coincide, so $D \subseteq \text{inst}(T^*)$ gives a member of $\text{Min}_F^{\text{lit}}(D)$ below T^* experiments Cor E1-pf. For templates with rigid parameters this fails (Appendix A), and $\text{Min}_F(D)$ is computed over alignments: the alignment proposition of Appendix A, whose finiteness hypothesis (F) for DT_F° is Proposition 7.2(c), gives (a)–(c) for $\text{Min}_F(D)$.

Computed checks experiments E0, E0b; referee §3. E0 compares the code with the prior referee's bounded enumerator of covering templates (de Bruijn levels, imported read-only) on 48 arithmetic data sets: 40 pairs of induction instances and 8 others. It enumerates 3703 covering DT_F° templates of size ≤ 14 and 2515 DT° templates of size ≤ 13 . Every enumerated template lies above a computed minimum, every computed minimum within the size bounds is enumerated, and none lies strictly below one. The coverage is narrow: no $<$, $\in$, $\leftrightarrow$ or parameters, and 41 of the 48 sets have $|\text{Min}| = 1$. E0b draws a random generating template T^* with the referee's generator and 2–3 of its instances, 200 data sets per language and with or without a rigid parameter in T^* , and checks: (a) the computed minima cover D , lie in DT_F° and are pairwise incomparable; (b) some minimum is $\preceq T^*$; (c) along random specialization walks from T^* (≤ 8 steps), every covering template reached lies above a minimum and none strictly below one.

language	rigid param. in T^*	Min	data sets	minima	(a) fail	(b) fail	walk templates	(c) fail
PA	no	literal	200	200	0	0	380	0
PA	no	aligned	200	200	0	0	380	0
PA	yes	literal	200	200	0	17	317	6
PA	yes	aligned	200	203	0	0	320	0
ZF	no	literal	200	200	0	0	408	0
ZF	no	aligned	200	206	0	0	408	0
ZF	yes	literal	200	204	0	36	381	27
ZF	yes	aligned	200	220	0	0	419	0

The literal failures are exactly those predicted for rigid parameters. The aligned set also differs from the literal one on 38/200 (PA) and 72/200 (ZF) parameter-free data sets: the data’s bodies carry parameters, and alignment finds more specific templates with rigid parameters.

F.2 Soundness of the oracles

Proof of Proposition 7.3. Both evaluators work on a formula together with an environment. The universal closure of a datum is evaluated by treating its parameters as universally quantified. We prove, by induction on the formula: a verdict True (False) is correct for every admissible interpretation of the environment.

PA. The environment maps variables to numerals or to *symbols*, which stand for arbitrary natural numbers; “admissible” means any values of the symbols. Atoms: at numerals the evaluation is exact. With symbols, terms built from $0, S, +, \cdot$ denote polynomials with nonnegative integer coefficients; $s = t$ holds for all values iff the polynomials p_s, p_t are equal, since polynomials that agree on $\mathbb{N}^k$ are equal; if $p_s - p_t$ has all coefficients ≥ 0 and a positive constant term then $s > t$ for all values; the other classifications are symmetric, and everything else is unknown. Connectives: Kleene’s strong tables are monotone in the information order, so known values stay correct. Supervaluation: for fixed values of the symbols, the actual truth values of the unknown leaves form one of the enumerated assignments, and the known leaves have their computed values by induction; so a verdict common to all assignments is correct. Quantifiers: bounded quantifiers with numeral bounds are enumerated exactly. For an unbounded $\forall x$, a verdict for the body with x a fresh symbol holds for all values of x ; a numeral counterexample is a counterexample; and in the case split every number is one of $0, \dots, K-1$ or $x' + K$ with $x' \in \mathbb{N}$, so if all cases are True then $\forall x$ is true and if one case is False there is a counterexample. One-point forms $(\forall x(x = t \rightarrow \varphi))$ with x not in t are equivalent to $\varphi[t/x]$. $\exists$ is dual. Stripping a vacuous quantifier is valid in a nonempty domain. A local budget overrun returns unknown, which is always sound.

ZF. The environment maps variables to HF sets or to generics. A *realization* ν maps each generic G to a set $\nu(G) \notin E_G$, where E_G is the transitive closure of the concrete values in scope when G was introduced, with $\nu(G) \neq \nu(H)$ for every generic H in scope at that time. Two generics in one environment were introduced in nested scopes, so the later one was introduced with the earlier in scope. Atoms: between HF sets, membership and equality are exact (HF sets are standard; equality by Extensionality). $G \in c$ is False if $c \in E_G$ or every element of c lies in E_G : in both cases $c \subseteq E_G$ (E_G is transitive) and $\nu(G) \notin E_G$. $G = c$ is False if $c \in E_G$. $G \in G$ is False by Foundation. $G = H$ is False for distinct generics, by the freshness condition. Every other mixed atom is unknown. Connectives and supervaluation: as for PA, for each fixed realization. Unbounded $\forall x$: the cases are $x \in E$ (the transitive closure of the concrete values in scope), x equal to an in-scope generic, and x a fresh generic H with $E_H = E$. Fix a realization ν and a set a . Either $a \in E$, or $a = \nu(G)$ for an in-scope G , or ν extended by $H \mapsto a$ is a realization. So if

every case is True under every realization, the body holds for every a . If some case is False under every realization, there is a counterexample; for the fresh case this needs some set outside E and different from the finitely many values of the in-scope generics, and such a set exists because no set contains all sets. Extra HF search values are concrete counterexamples or witnesses. $\exists$ is dual. Memoization is keyed on the formula and the environment’s values, and generic identifiers are unique, so a verdict is reused only for the same environment.

Constant propagation (both). The rewrite rules ($A \wedge \top \equiv A$, $A \rightarrow \perp \equiv \neg A$, $\neg\neg A \equiv A$, $Qx \top \equiv \top$, ...) are logical equivalences in every structure with a nonempty domain; $t = t$ is logically true; closed arithmetic atoms are evaluated exactly; $t < t$ is false in $\mathbb{N}$; $x \in x$ is false in V by Foundation. So the simplified formula has the same truth value as the original in $\mathbb{N}$ (resp. V) under every environment.

The ZF argument uses Foundation, Extensionality on HF sets, the existence of HF sets, and that no set contains all sets; not Infinity. $\square$

The proof states the freshness condition on generics relative to the generics *in scope* when a generic is introduced; the record says “created before”. Only generics of one environment are ever compared, so the two readings agree.

Independent checks experiments §3.2; recheck R3, R4, R11. The referee’s evaluators were copied unchanged and re-run against the revised oracles. The independent PA evaluator is three-valued and decides only through exact bounded quantifiers, counterexamples and witnesses. The independent ZF evaluator evaluates exactly in finite structures $V_4 \cup \{\text{extra objects}\}$ in which the HF sets are standard, the extras are never elements of HF sets, membership into and among the extras is arbitrary but irreflexive, and equality is identity. These structures satisfy exactly what the proof uses, so every True or False verdict must hold in them. R3 compared the verdicts logged during DTRC runs on the mixtures with mistakes and on the near-miss sets: 2197 PA verdicts and 413 ZF verdicts of quantifier depth ≤ 4 plus 179 of depth 5–6, with 0 contradictions. R4 compared 3000 random PA and 600 random ZF sentences, with 0 contradictions. R11 is a mutation test of the ZF check: dropping the in-scope-generic case is detected 39/600 times and dropping the fresh-generic case 292/600 times; mutations of the generic-versus-HF atom rules are not detected, because those rules rarely decide a verdict, and the referee verified them by hand.

F.3 Proofs for dtcr and the separation of the mixtures

Proof of Theorem 7.5. No datum is discarded: genuine data are true and the oracle is sound.

Invariant: every cluster is contained in some D_i . It holds for singletons. When clusters $A \subseteq D_i$ and $B \subseteq D_j$ are tested, the test fails if $i \neq j$, by (S), and succeeds if $i = j$, by Proposition 7.4(b) applied to $A \cup B \subseteq \text{inst}(T_i)$. So merges preserve the invariant. A pair is marked failed only if it is cross-label, and every later pair of clusters containing its two parts is cross-label as well; so the inheritance of failures never blocks a within-label merge.

The final clusters are the nonempty D_i . Each merge reduces the number of clusters, so the agglomeration terminates. Every pair of current clusters stays in the implementation’s queue (with minimum score 0) until it is tested or inherits a failure. At the end, therefore, every pair of final clusters has failed, directly or by inheritance, and is cross-label. With the invariant, the final clusters are exactly the nonempty D_i .

Soundness. By Proposition 7.4(b) and its proof there is $T_0 \in \text{Min}_F(D_i)$ with $T_0 \preceq T_i$, and no sound refuter refutes it. So T_0 is among the intersected templates, and $\text{Acc}(D_i) \subseteq \text{inst}(T_0) \subseteq \text{inst}(T_i)$.

Exactness. If D_i contains an anchor of T_i , every member of $\text{Min}_F(D_i)$ covers the anchor, so its instance set contains $\text{inst}(T_i)$. The set of unrefuted members is nonempty (it contains T_0), so $\text{Acc}(D_i) \supseteq \text{inst}(T_i)$.

(H-ref). If verdicts do not depend on the call history, the final verification of D_i sees the same verdicts as the tagged refutation-filtered verifier on D_i , so the acceptance sets coincide. $\square$

Proof of Proposition 7.7. Call a cluster *pure* if it is contained in some $\text{inst}(T_l)$. All clusters are pure at all times: a test whose union lies in no $\text{inst}(T_l)$ fails by (S^+) , and a successful merge or addition therefore has a pure union. Inheritance of failures is correct: if a test on $A \cup X$ failed, then $A \cup X$ lies in no $\text{inst}(T_l)$ (otherwise it passes, by Proposition 7.4(b)), and neither does any superset.

l is injective. If two final clusters $C \neq C'$ of the agglomeration had $l(C) = l(C') = l$, then $C \cup C' \subseteq \text{inst}(T_l)$ would pass the merge test, and the pair is never marked failed (by the inheritance remark); so the phase would not have ended.

The sharing pass. Let C' be the growing cluster, initially the phase-one cluster C with $l(C) = l$. For $d \in D \cap \text{inst}(T_l)$, $C' \cup \{d\} \subseteq \text{inst}(T_l)$ passes by Proposition 7.4(b). For $d \notin \text{inst}(T_l)$, $C' \cup \{d\}$ lies in no $\text{inst}(T_j)$: not in $\text{inst}(T_l)$ because of d , and not in $\text{inst}(T_j)$ for $j \neq l$ because $C \subseteq C'$ is contained in exactly one target's instance set, by (P); so the test fails by (S^+) . By induction C' stays inside $\text{inst}(T_l)$, and it ends as $C \cup (D \cap \text{inst}(T_l)) = D \cap \text{inst}(T_l)$, since genuine data are never discarded.

Acceptance. $\text{Acc}(C^+) \subseteq \text{inst}(T_l)$ by Proposition 7.6; with an anchor in C^+ , equality holds as in the proof of Theorem 7.5; and under (H-ref) the tagged learner with membership labels sees the same data C^+ and the same verdicts. $\square$

The record's statement of Proposition 7.7 adds that DTRC+share is exact whenever the membership-labelled tagged learner is. Without (H-ref) this follows only when that learner's exactness comes from an anchor (its exactness can also come from refutations that DTRC's refuter, with another call history, need not find), so we state anchor-exactness and the (H-ref) equality separately.

Lemma F.1 (Root symbols under derivation). *[proved]^{experiments Lemma E8.1} In $\mathcal{T}(D)$, if $\sigma \rightarrow \pi$ then for every $d \in D$: $d|_\pi = d|_\sigma$ if σ is a term slot, and $d|_\pi$ has the root symbol of $d|_\sigma$ if σ is a formula slot. Consequently a clash slot (a slot at which two data a, b have different root symbols $r_a \neq r_b$) derives only slots at which a and b have the root symbols r_a and r_b ; it derives no interior node.*

Proof. Term slots of $\mathcal{T}(D)$ have closed columns, so $\beta_d(\sigma) = d|_\sigma$ has no hole, and $\beta_d(\sigma)[\bar{t}] = d|_\sigma$. A formula body's root is not a hole, and plugging preserves the root of such a body. Interior nodes of $\mathcal{T}(D)$ carry a label common to all data. $\square$

Proof of Proposition 7.8. Write $d = T^*[z := t_d]$. Since T^* has no parameter and some datum has none, there is a single alignment and $\text{Min}_F(D) = \text{Min}_F^{\text{lit}}(D)$ up to renaming. All positions outside the z -positions are common.

$(\Leftarrow)$ If the roots of the t_d are not all equal, every z -position is a slot, with the same closed column $(t_d)_d$, and there is no other slot. An own set contains at most one of these slots, since each derives the others. By Lemma F.1 a derived occurrence at π needs $d|_\pi = t_d$ for all d , impossible at an interior node, where all data share a root. So the only normal configuration has one own z -slot and derived occurrences at the others: it is T^* .

($\Rightarrow$) If all t_d have root h , the z -positions are interior nodes with label h . Own slots lie strictly below z -positions and are term slots with closed columns; by Lemma F.1 such a slot derives only nodes π with $d|_\pi = d|_\sigma$, which excludes the z -positions and the nodes above them (a term never equals a proper subterm). So every normal configuration, hence every member of $\text{Min}_F(D)$, carries h rigidly at the z -positions, and rigid symbols survive instantiation. An instance of T^* whose substituted term has another root (0 if $h \neq 0$, $S0$ if $h = 0$) is therefore an instance of no member, and $\text{Acc}(D) \neq \text{inst}(T^*)$.

For i.i.d. terms the probability that all N roots coincide is $\sum_h p_h^N$. In E1 every parameter instance canonicalizes to the same datum, so among $N \geq 2$ distinct data some datum is parameter-free. $\square$

The mixture targets. In the implementation the ZF-mix targets are $\text{Ext} = \forall x \forall y (\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y)$, $\text{Pair} = \forall x \forall y \exists z \forall u (u \in z \leftrightarrow u = x \vee u = y)$, $\text{Union} = \forall x \exists y \forall z (z \in y \leftrightarrow \exists u (u \in x \wedge z \in u))$, $\text{Pow} = \forall x \exists y \forall z (z \in y \leftrightarrow \forall u (u \in z \rightarrow u \in x))$, $\text{Inf} = \exists x (\exists y (y \in x \wedge \forall z \neg z \in y) \wedge \forall y (y \in x \rightarrow \exists z (z \in x \wedge \forall u (u \in z \leftrightarrow u \in y \vee u = y))))$, $\text{Found} = \forall x (\exists y (y \in x \rightarrow \exists y (y \in x \wedge \forall z (z \in y \rightarrow \neg z \in x)))$, $T_{\text{Sep}} = \forall x \exists y \forall z (z \in y \leftrightarrow z \in x \wedge P(z, x))$, $\text{Repl}_{\text{sp}} = \forall x (\forall y (y \in x \rightarrow \exists z (P(y, z, x) \wedge \forall u (P(y, u, x) \rightarrow u = z))) \rightarrow \exists y \forall z (z \in y \leftrightarrow \exists u (u \in x \wedge P(u, z, x))))$, and T_{EInd} . The PA-mix targets are Q1–Q7 as in Section 2, T_{Ind} , and the three instance schemas.

Proof of Proposition 7.9. Cases A–D. Inspection of the targets shows, for every listed pair, a position p above all metavariable positions of both targets at which the two targets differ, while all positions not below p carry the same rigid symbols in both (in case D there are two such positions). In case A, p is the root: Q1–Q7 have root $\forall$, T_{Ind} root $\rightarrow$, the instance schemas root $=$; Inf has root $\exists$, T_{EInd} root $\rightarrow$, and the other ZF targets root $\forall$. In case B, p is the node right below the common prefix $\forall x$ or $\forall x \forall y$. There the roots differ, except for Q4–Q6 and Q5–Q7, where both data carry an equation whose term disagreement contains a bound variable, so p is an F-slot. In case C, p is the right side of $\leftrightarrow$ in the comprehension frame $\forall x \exists y \forall z (z \in y \leftrightarrow \cdot)$, where the targets read $\exists u(\dots)$, $\forall u(\dots)$, and $z \in x \wedge \varphi$. In case D the two positions are the antecedent under $\forall x$ ($\exists$ against $\forall$) and the body under the consequent's $\exists y$ ($\wedge$ against $\vee$).

Hence for instances a, b , $\mathcal{T}(\{a, b\})$ is the shared prefix with one slot (two in D), and its interior nodes carry the shared labels. Each slot is a clash slot, or in case B possibly an F-slot, whose root pair is $(=, =)$ while the interior nodes are quantifiers. By Lemma F.1 no slot derives an interior node, and in case D neither slot derives the other, since their root pairs $(\exists, \forall)$ and $(\wedge, \vee)$ differ. So the only normal configuration puts an own metavariable at each slot, applied to the variables free in its column. In C these are z and x (each of the three right sides mentions both, and a Separation body cannot mention y); in D, x for the antecedent and y, x for the consequent body. Parameters occur only inside the slots' columns, so every alignment gives the same template, and $\text{Min}_F(\{a, b\})$ is this single template. The instances are false: $\perp$; $\forall \bar{x} \perp$ (the domain is nonempty); Russell's $\forall x \exists y \forall z (z \in y \leftrightarrow \neg z \in z)$, which fails at $z := y$; and $\forall x (\top \rightarrow \exists y \perp)$.

Case E. All slots have closed term columns, so by Lemma F.1 $\sigma \rightarrow \pi$ requires identical columns. (i) $U_{\text{add}0} - U_{\text{mul}0}$: $a = (t + 0 = t)$, $b = (s \cdot 0 = 0)$. The left sides clash ($+$ against $\cdot$); the right sides t and 0 form a slot unless $t = 0$. No two of these columns are identical, and no column can be derived at a common node, so $\text{Min}_F = \{z_0 = z_1\}$ or $\{z_0 = 0\}$, with the false instance $1 = 0$. (ii) $U_{\text{mul}0} - U_{\text{add}}$: $a = (t \cdot 0 = 0)$, $b = (0 + u = u)$; symmetric to (i). (iii) $U_{\text{add}0} - U_{\text{add}}$: $a = (t + 0 = t)$ with $t \neq 0$ (otherwise $a \in \text{inst}(U_{\text{add}})$) and $b = (0 + u = u)$ with $u \neq 0$. The left sides share $+$, with slots σ_1 (column $(t, 0)$) and σ_2 (column $(0, u)$). A node π with $a|_\pi = t$ is σ_1 or the root of the right side, where b has $u \neq 0$; a node with $b|_\pi = u$ is σ_2 or the root of

the right side, where a has $t \neq 0$; slots inside the right side have columns of proper subterms. So σ_1, σ_2 derive no other node and are derived from none, and every normal configuration has the form $z_0 + z_1 = R$ with z_0, z_1 occurring only there and R built from the common prefix of t and u . Setting every metavariable of R to 0 gives a term r ; with $z_0 := S(r)$ and $z_1 := 0$ the instance $S(r) + 0 = r$ is false in $\mathbb{N}$ for every value of its parameters. $\square$

F.4 Hard pairs and false merges

Proof of Proposition 7.11. (a) Union_W and Power_W share the prefix $\forall a \exists b \forall x (\cdot \rightarrow x \in b)$ and clash at the antecedent ($\exists$ against $\forall$), whose column mentions x and a but not b . By Lemma F.1 this slot derives no other node, so the universal-set schema U is the unique normal configuration and the unique minimal covering template. Its instance with $P := \top$ simplifies to $\exists b \forall x (x \in b)$ (the quantifier $\forall a$ is vacuous). With no concrete value in scope, $E = \emptyset$ and there is no generic, so the only case for b is a fresh generic G ; then $\forall x (x \in G)$ is False in its case $x := G$, because $G \in G$ is False. So the oracle returns False, soundly by Proposition 7.3: in V , a set b with $\forall x (x \in b)$ would satisfy $b \in b$. The impossibility for oracles sound in $\text{HF} \cup \{u\}$ is proved in Section 6.

(b) Let $e = \text{EInd}(\varphi)$ and $e' = \text{EInd2}(\psi)$, where the roots of φ and ψ differ, φ does not begin with $\forall$, and φ or ψ uses its variable. The common prefix is $(\forall x (\forall y (y \in x \rightarrow \Box_1) \rightarrow \Box_2)) \rightarrow \forall x \Box_3$, with columns $(\varphi(y), \forall z (z \in y \rightarrow \psi(z)))$ at $\Box_1$ and $(\varphi(x), \psi(x))$ at $\Box_2$ and $\Box_3$. All three are clash slots. The columns at $\Box_2$ and $\Box_3$ are identical (each sits under one binder), so each derives the other. The root pairs of $\Box_1$ and $\Box_2$ are (root $\varphi, \forall$) and (root φ , root ψ). If they differ, Lemma F.1 excludes derivations between $\Box_1$ and $\Box_2, \Box_3$. If they coincide (ψ begins with $\forall$), a size argument excludes them: terms of L_ϵ are variables or parameters, so plugging does not change the size of a body, while $\forall z (z \in y \rightarrow \psi(z))$ is larger than $\psi(x)$ and $\psi(t)$. No slot derives an interior node (Lemma F.1). So the only normal configuration, up to renaming, has own slots $\Box_1, \Box_2$ and $\Box_3$ derived from $\Box_2$; it is T_{EE} : P_0 is unary, since only y occurs in the column at $\Box_1$ and $z \in y$ mentions it, and P_1 is unary, since some motive uses its variable. Its instance with $P_0 := \perp$, $P_1(x) := \forall z \neg z \in x$ has the premise $\forall x (\forall y (y \in x \rightarrow \perp) \rightarrow \forall z \neg z \in x)$, i.e. $\forall x (x = \emptyset \rightarrow x = \emptyset)$, which is true, and the conclusion $\forall x \forall z \neg z \in x$, which fails at $x = \{\emptyset\}$. If neither motive uses its variable, P_1 is 0-ary and the template is valid: when P_1 is false its premise $\forall x \neg \forall y (y \in x \rightarrow P_0(y))$ fails at $x = \emptyset$. This case, not covered by the record's statement, is why (b) assumes that some motive uses its variable.

The remaining claims are computed (E9, E9b), and the false and valid instances quoted after Proposition 7.11 are checked by the same kind of reading: with $P_0 := \perp$ and $P_1(x) := \forall y \in x \forall z \in y \perp$ the premise of the surviving template is $\forall x (\alpha(x) \rightarrow \alpha(x))$, and $\forall x \alpha(x)$ fails at $\{\{\emptyset\}\}$; the valid merges are covered by Lemma 7.13(5), and Separation with swapped conjuncts is equivalent to Separation. $\square$

The false templates met in E4(b). Each was found refuted by the post-hoc refuter; the false instances were checked by hand experiments §3.6.

- PA, numerals, seed 0, budget 80: $(\forall x (P_0(x, 0) \rightarrow P_1(x, 0)) \wedge \forall x (\forall y (P_0(y, x) \rightarrow P_1(y, x)) \rightarrow \forall y (P_0(y, Sx) \rightarrow P_2(y, x)))) \rightarrow \forall x \forall y (P_0(y, x) \rightarrow P_1(y, x))$. With $P_0 := \top$, $P_1(y, x) := (x = 0)$, $P_2 := \top$ the premises are $\forall x (\top \rightarrow 0 = 0)$ and $\forall x (\cdots \rightarrow \forall y (\top \rightarrow \top))$, both true, and the conclusion $\forall x (x = 0)$ is false.
- PA, numerals, seed 1, budget 80: $(\forall x (P_0(x, 0) \rightarrow P_1(x, 1)) \wedge \forall x (\forall y (P_0(y, x) \rightarrow P_1(y, x)) \rightarrow \forall y (P_0(y, Sx) \rightarrow P_1(y, Sx)))) \rightarrow \forall x \forall y (P_0(y, x) \rightarrow P_1(y, x))$. With $P_0 := \top$, $P_1(y, x) :=$

$\neg x = 0$ the premises $\neg 1 = 0$ and $\forall x(\neg x = 0 \rightarrow \neg Sx = 0)$ are true, and the conclusion $\forall x \neg x = 0$ fails at 0.

- ZF, budget 80, seeds 0–2: five merges of a Separation-capture mistake with Separation instances, e.g. $\forall a \exists b \forall x (x \in b \leftrightarrow x \in a \wedge \forall u P(u, x, b, a))$. With $P := \neg x \in b$ the body becomes $x \in b \leftrightarrow x \in a \wedge \neg x \in b$, contradictory for $x \in a$, so the sentence fails at $a = \{\emptyset\}$. The post-hoc witnesses of the other four reduce in the same way to $x \in b \leftrightarrow x \in a \wedge \neg x \in b$, or to $x \in b \leftrightarrow x \in a \wedge \neg w_0 \in b$, which fails at $a = \{\emptyset\}$, $w_0 = \emptyset$ (if $\emptyset \in b$ the right side is false at $x = \emptyset$; if not, it is true).
- PA, both regimes, seed 0, budgets 80 and 400: T_F of Proposition 7.12.

F.5 Further numbers

E2, further exhibits. Accepted sentences certified false by the oracle ($N = 2$): Replacement, first-order de Bruijn, $\forall x(\forall y(y \in x \rightarrow \exists z(y \in x \wedge \forall u(\neg \forall v(v = v) \rightarrow u = z))) \rightarrow \exists y \forall z(z \in y \leftrightarrow \exists u(u \in x \wedge \forall v(v = v))))$; $\in$ -induction, named, $(\forall x(\forall y(y \in x \rightarrow y \in w_0) \rightarrow \neg x \in w_0)) \rightarrow \forall x \neg x \in w_0$, false with $w_0 = \{\{\emptyset\}\}$; induction, first-order de Bruijn, $(0 = w_0 \wedge \forall x(x = 0 \rightarrow 0 = x)) \rightarrow \forall x(x = 0)$; induction, genuine pattern lgg, $(0 = 0 \wedge \forall x(w_0 = Sx + (0 + x) \rightarrow 0 = 1 \cdot Sx)) \rightarrow \forall x(w_0 = Sx + (0 + x))$. Per-probe counts are in `code/results/e2_schemas.md`.

E4(b), outcome of the mixed clusters. Table 13 lists, per data regime and refuter budget, the seeds in which a false template was accepted and the residual templates, checked with a post-hoc refuter (budget 4000 for PA, 1500 for ZF) and by hand (Lemma 7.13, Appendix F.4).

data	budget	seeds with a false accepted template	false templates	valid residual templates
PA, mixed	80, 400	0	T_F	$(z < 0 \wedge \dots) \rightarrow \dots$, $(0 = Sz \wedge \dots) \rightarrow$ $\dots$
PA, mixed	2000	none	–	the same, and $(P(0) \wedge \forall x(P(x) \rightarrow P(Sx))) \rightarrow$ $\forall x P(Sx)$
PA, numerals	80	0, 1	T_F ; two induction-step templates	false-antecedent forms with $x < 0 + 0$, $x < 0 \cdot 0$, $0 = Sz$; the $\forall x P(Sx)$ form; $0 + z = z$
PA, numerals	400	0	T_F	the same
PA, numerals	2000	none	–	the same, and the form with $x < 0$
ZF	80	0, 1, 2	five Separation-capture templates	$\forall x(P(x) \rightarrow x =$ $x) \rightarrow \forall x(x = x)$
ZF	400	none	–	the same

Table 13: E4(b): mixed clusters with injected mistakes experiments E4; code/results/e4_stress.md.

The fates of the mistakes per run (discarded as refuted, singleton, or merged), the ARI on clean data, held-out acceptance and the oracle-call counts are in `code/results/e4_stress.md`. Refuted mistakes number 2–3 per run.

F.6 Reproduction

All commands run from `axiom-schemas/code/`; the experiment scripts are in `experiments/`. `sh run_all.sh` runs the tests and E0–E9 (593s wall time on 4 shared cores); E10 and E9b take a few seconds. Outputs other than timings reproduce exactly. The results of the first version are kept in `code/results_v1/`.

The scripts are `e0_crossval.py` (run as `14 40 F` and `13 40 full`), `e0b_property.py` 200, `e1_universal.py`, `e2_schemas.py`, `e3_mix.py`, `e4_stress.py`, `e5_curves.py`, `e6_kunion.py`, `e7_blowup.py`, `e9_separation.py`, `e9b_nosimplify.py` and `e10_pairtable.py` 100, each writing `results/eN_*.md`; `python3 -m pytest -q tests` runs the 43 unit tests, and `sh experiments/recheck/run_recheck.sh` re-runs the referee’s oracle checks (outputs in `experiments/recheck/*.out`). Seeds are listed in each results file.

G How the results were established